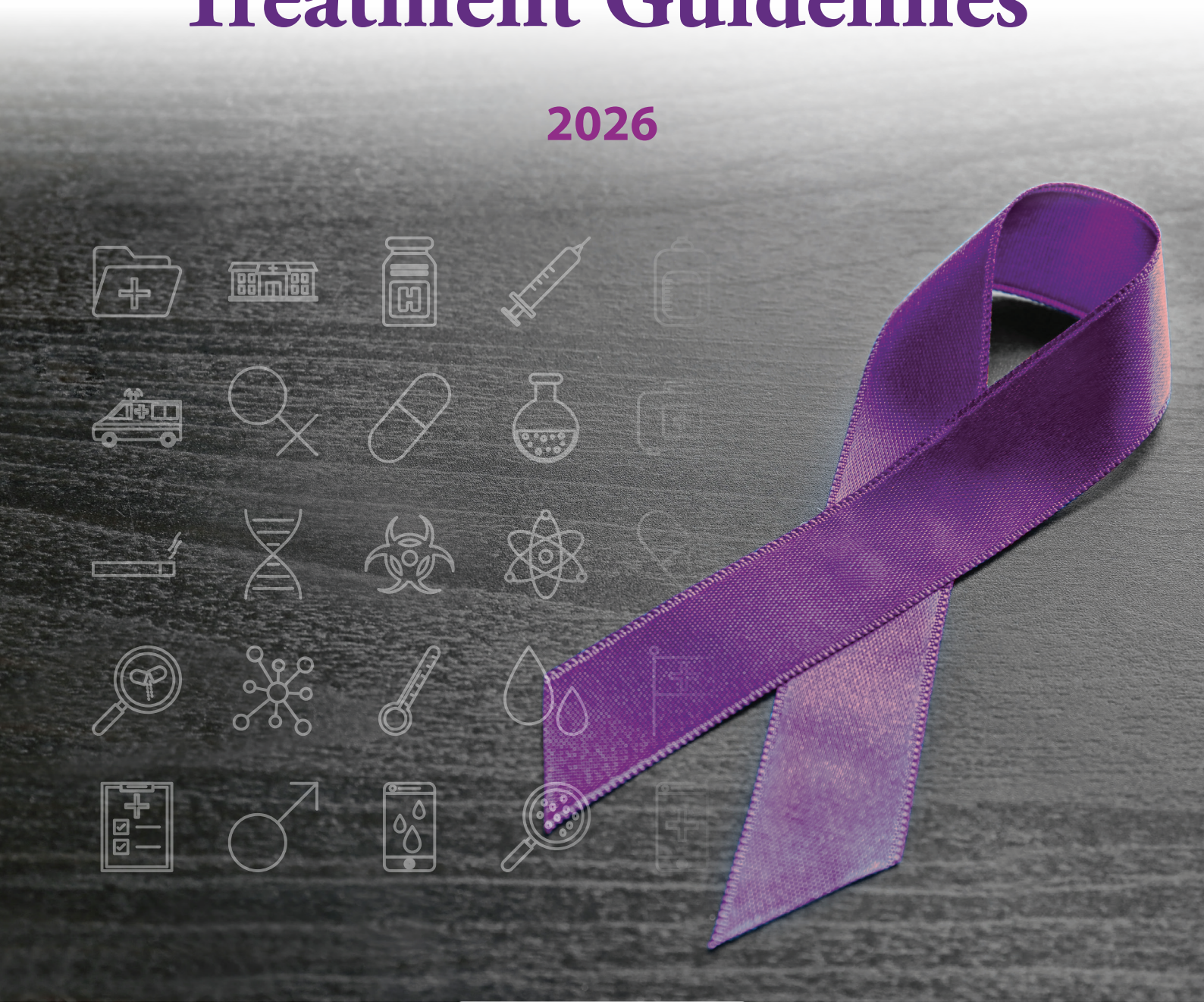
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National Cancer Treatment Guidelines

2026



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2026

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Kenya National Cancer Treatment Guidelines

Developed by the National Cancer Control Program, Division of Cancer & Non-Communicable Diseases, Ministry of Health

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Foreword



Dr. Patrick Amoth, CBS

Cancer remains one of the most formidable public health challenges facing Kenya today. As the third leading cause of death after infectious and cardiovascular diseases, cancer accounts for approximately 44,726 new cases and 29,317 deaths annually, according to GLOBOCAN 2022. These numbers represent not just statistics, but mothers, fathers, children, and community members whose lives are interrupted by this devastating disease.

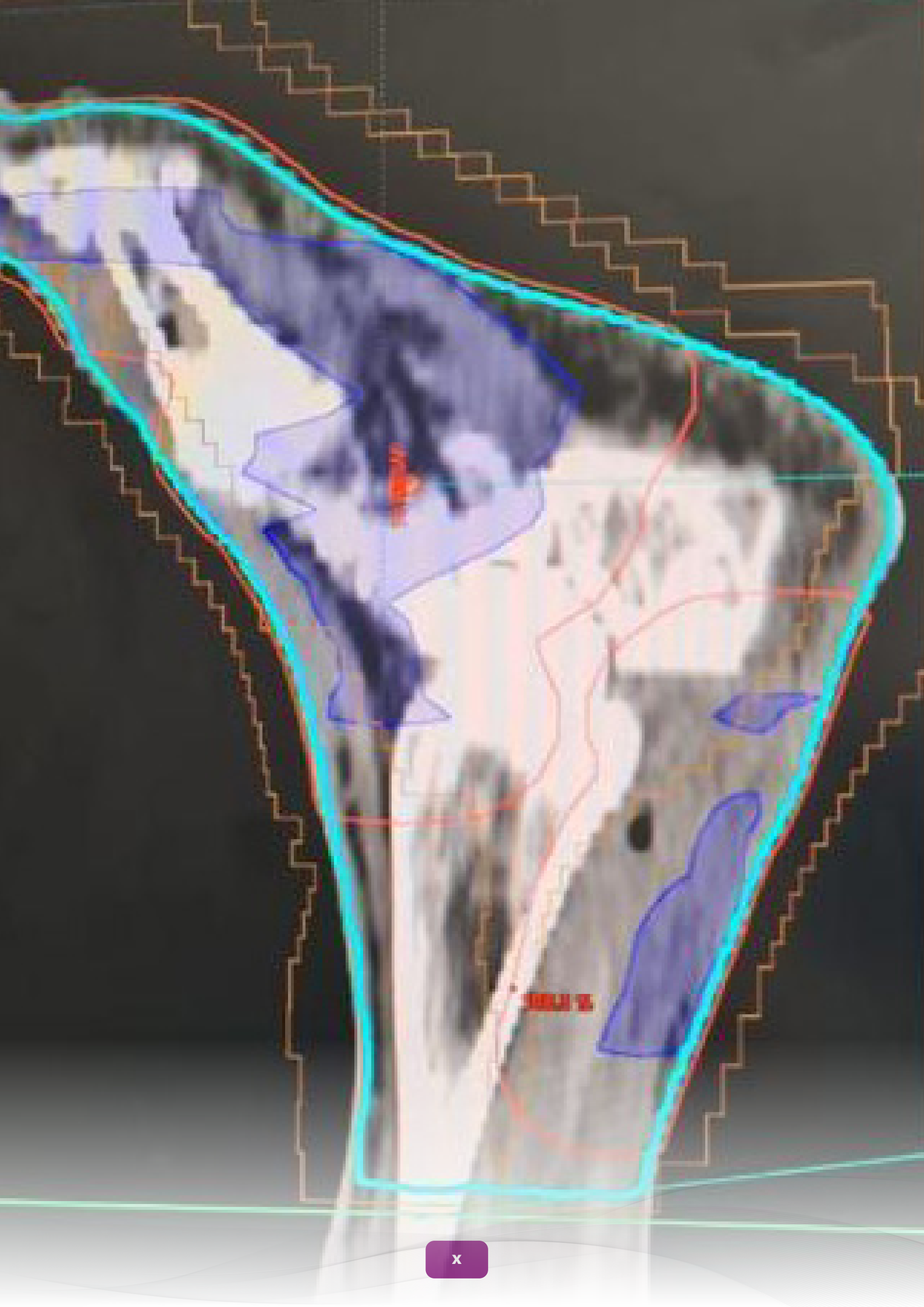
The Ministry of Health has consistently demonstrated its commitment to addressing the cancer burden through evidence based policy interventions. The first National Cancer Treatment Guidelines, published in 2019, provided a foundational framework that standardized cancer care across our health system. Since then, significant scientific advancements have emerged: new diagnostic tools, novel therapeutic agents, refined treatment protocols, and a deeper understanding of molecular oncology. The healthcare financing landscape has also evolved with the operationalization of the Social Health Authority (SHA), necessitating updated costing structures and treatment packages.

It is against this backdrop that the Ministry of Health presents the National Cancer Treatment Guidelines 2026. This revised edition represents a comprehensive update that incorporates the latest global best practices while remaining responsive to Kenya's unique epidemiological, resource, and health system contexts. Key enhancements include the integration of updated chemotherapy regimens, expanded paediatric cancer treatment protocols, refined guidance on oncologic emergencies, strengthened survivorship care frameworks, and aligned costing with current market realities under Universal Health Coverage (UHC).

As we implement these guidelines, I call upon all healthcare providers, county governments, professional associations, training institutions, and civil society organizations to champion their adoption. Regular updates, continuous professional development, and robust monitoring and evaluation mechanisms will ensure these guidelines remain living documents that evolve with emerging evidence.

Every Kenyan diagnosed with cancer deserves timely, affordable, and high quality care. These guidelines are our roadmap to that destination.

Dr. Patrick Amoth, CBS
Director General for Health
Ministry of Health, Kenya



Acknowledgement

The Ministry of Health expresses sincere gratitude to all who contributed to the revision of the National Cancer Treatment Guidelines. We thank the Cabinet Secretary, Principal Secretary Medical Services, Director General for Health, Director of Family Health, and Head of the Division of Cancer and Non Communicable Diseases for their strategic guidance.

We acknowledge financial and technical support from WHO, AstraZeneca, Roche, CHAI and the National Cancer Institute of Kenya (NCI-K). Special thanks to the National Cancer Control Program team led by Dr. Joan-Paula Bor. The coordinating role provided by the NCCP pillar 3 program officers; Dr. Joyfrida Chepchumba, Dr. Collins Masolo, and Mr. Isaac Kamau is greatly appreciated.



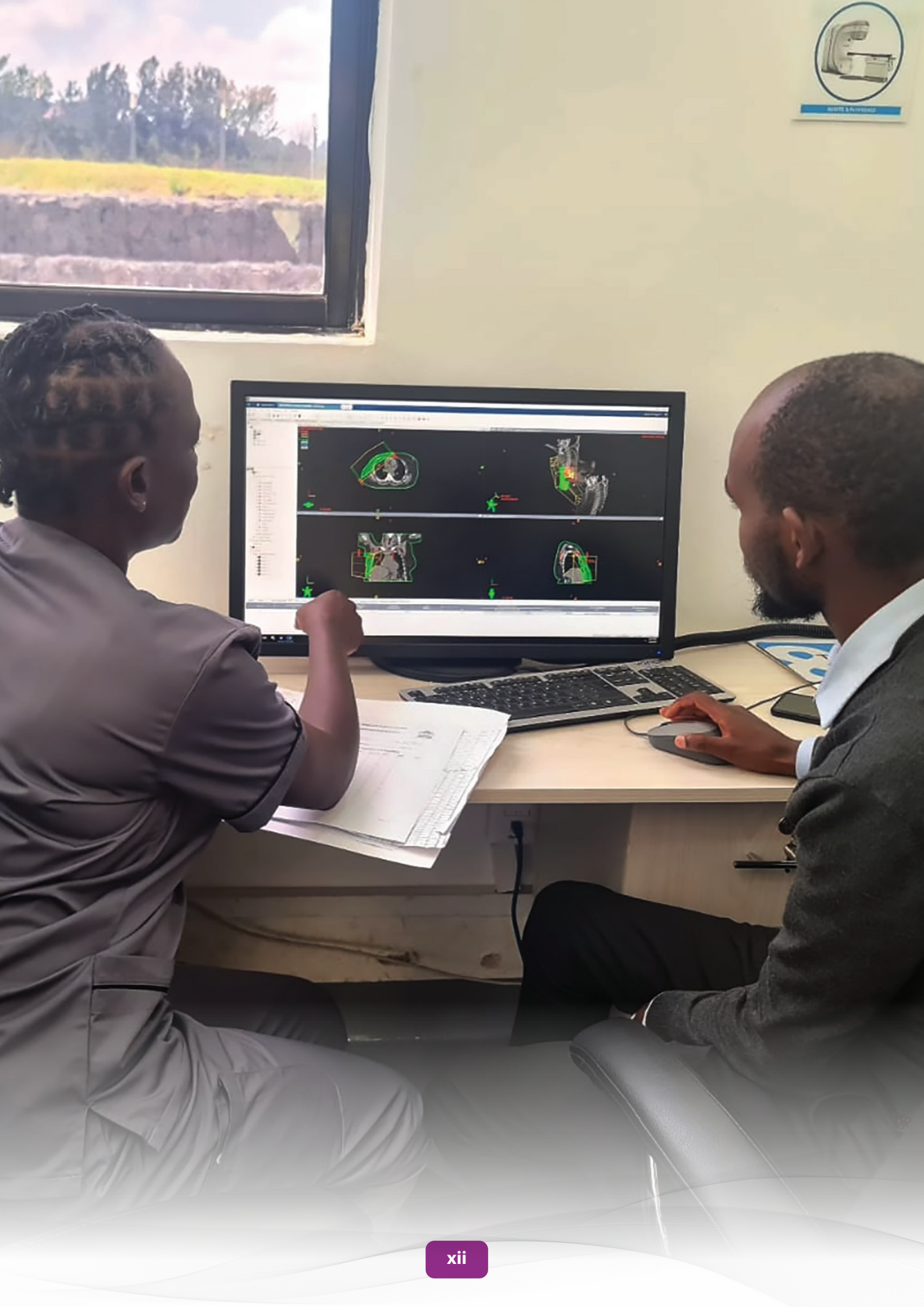
Dr. Issak Bashir, EBS

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In conclusion, the Ministry reiterates its appreciation to all individuals and organizations involved. Their collective efforts have produced a resource that will improve and standardize cancer treatment in Kenya.



Dr. Issak Bashir, EBS
Director – Directorate of Family Health
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Executive Summary

This document presents a comprehensive revision of the 2019 National Cancer Treatment Protocols, responding to scientific advances, novel therapies, SHA financing changes, and gaps in paediatric cancer care.



Dr. Gladwell Gathecha

Led by the National Cancer Control Program, the process included an independent evaluation of the 2019 Protocols, a comprehensive review by a multidisciplinary expert panel of oncologists, paediatric care teams, palliative care specialists, pharmacists, nurses, and MOH representatives, among other specialists through a series of stakeholder workshops and desk reviews, and a validation workshop.

Key updates include revised referral timelines:

- Immediate for emergencies
- Fourteen (14) days for cancer centre assessment
- Sixty (60) days for adults and thirty (30) days for children from first contact to diagnosis
- Thirty-one (31) days to treatment initiation

A strengthened multidisciplinary team framework is provided, alongside enhanced survivorship and end of life care.

Site specific protocols for breast, gastrointestinal, gynaecological, haematological, and childhood cancers now incorporate latest chemotherapy, immunotherapy, targeted agents, and ninth edition AJCC staging. A new chapter addresses oncologic emergencies including febrile neutropenia, spinal cord compression, tumour lysis syndrome, extravasation, and anaphylaxis. Palliative care sections include the updated WHO pain ladder, antiemetic protocols, and opioid SOPs. Mandatory safety standards for antineoplastic therapy administration are now included.

The revision has aligned the guidelines with current evidence for both adult and paediatric care. Expected outcomes include standardised cancer diagnosis and treatment across all facility levels, strengthened multidisciplinary care, improved equity and access, and integration of palliative care and survivorship.

Implementation requires commitment from healthcare providers, county and national governments, and partners. The Ministry will support infrastructure, training, monitoring, and regular updates.



Dr. Gladwell Gathecha
Ag. Head, Division of Cancer & Non-Communicable Diseases
Ministry of Health, Kenya



Chapter 1
Introduction

List of Abbreviations

Abbreviation	Definition
AYA	Adolescents and Young Adults
GLOBOCAN	Global Cancer Observatory
HPV	Human Papilloma Virus
IASP	International Association for the Study of Pain
IHC	Immunohistochemistry
LMIC	Lower Middle-Income Country
MDT	Multidisciplinary Team
NCD	Non-Communicable Diseases
NSAIDs	Non-Steroidal Anti-Inflammatory Drugs
PC	Palliative Care
WHO	World Health Organization

1.1 General Principles of Cancer Management

Cancer is the fourth leading non-communicable disease (NCDs) globally, after cardiovascular diseases, diabetes, and chronic respiratory diseases. Together, these NCDs account for over 60% of global mortality annually. Cancer alone is responsible for an estimated 10 million deaths each year, with approximately 75% of these occurring in low- and middle-income countries (LMICs). Without significant intervention, the global cancer burden is projected to rise by 60% over the next two decades, with LMICs expected to experience the greatest increase—up to 81% in new cases.

In Kenya, cancer is the third leading cause of death after infectious and cardiovascular diseases. According to GLOBOCAN 2022 estimates, Kenya reported an annual incidence of 44,726 cancer cases and 29,317 cancer-related deaths in a population of over 53 million. The five most common cancers in the country are breast, cervical, prostate, oesophageal, and colorectal cancers. Women are disproportionately affected, with a higher cancer incidence rate compared to men. The lifetime risk of developing cancer before the age of 75 is 18% for women and 14.3% for men. The leading childhood cancers in Kenya include leukemias, lymphomas, Wilms tumor, brain tumors and retinoblastomas. According to GLOBOCAN 2022, there were approximately 2,294 new cancer cases diagnosed among children and adolescents.

Cancer not only threatens individual health but also significantly impacts national development, contributing to reduced productivity, increased unemployment, and loss of human and capital resources. Addressing this growing burden requires a coordinated, patient-centered, and evidence-based approach across all levels of the health system.

Effective cancer care is complex and requires a specialized, multimodal approach involving a multidisciplinary team (MDT). This team includes but is not limited to oncologists, physicians, surgeons, radiologists, pathologists, oncology nurses, nutritionists/dietitians, dermatologists, plastic surgeons, dentists, and palliative care specialists. These professionals work collaboratively across the continuum of care—from prevention and early detection, through diagnosis, treatment, rehabilitation, survivorship, and end-of-life care. Clinical trials advance new treatments and provide alternative therapies for patients who would not otherwise have access to these drugs and procedures.

In line with the *Kenya National Cancer Control Strategy 2023–2027*, the review of the Kenya National Cancer Treatment Guidelines was done with contributions from leading experts and institutions providing cancer care in the country. These guidelines aim to standardize care, reduce disparities in service delivery, and improve clinical outcomes for patients nationwide. They will be reviewed regularly to incorporate updated evidence with guidance from the Ministry of Health, and recommendations from relevant national and international health organizations.

Cancer management must be person-centered with focus on their needs, preferences, beliefs and values of patients and their families. Listening to patient and caregiver feedback is essential for continuous improvement of services. Effective, compassionate, and clear communication between healthcare providers and patients is a cornerstone of quality cancer care and contributes significantly to patient satisfaction, adherence to treatment, and overall outcomes.

Prevention remains a key pillar of cancer control. The four shared modifiable risk factors for major NCDs—tobacco use, harmful alcohol consumption, unhealthy diet, and physical inactivity—are also strongly linked to cancer. National strategies must prioritize health promotion, risk reduction, vaccination (e.g., HPV and

Hepatitis B), and screening programs to reduce the incidence of preventable cancer. In general, screening may not be a reliable way to detect or prevent childhood cancer. The best way to improve treatment outcomes is through use of early warning signs and a high index of suspicion which leads to an early diagnosis and timely treatment initiation.

A core principle of the Kenya National Cancer Treatment Guidelines is equitable access to high-quality cancer services regardless of geographical location or socioeconomic status. Strengthening cancer care infrastructure, improving diagnostic capabilities, decentralizing treatment services, investing in healthcare worker training, healthcare safety, incorporation and use of e-health as an essential measure to achieving this goal; Refer to Cancer establishment guidelines

1.1.1 Early Diagnosis and Referral

This guideline recommends that cancer patients be referred appropriately in a timely manner to the appropriate level of care where the specific service required in the care pathway is available. The guideline also recommends that children be referred directly to a childhood cancer treatment center when cancer is suspected. Triage based on signs and symptoms is required before referral.

Designated clinicians should ensure that an appropriate diagnostic work-up is provided for patients suspected to have cancer. The definitive diagnosis of cancer is confirmed by histopathological examination of the biopsy specimen. Patients confirmed with cancer should be referred without delay to the appropriate multidisciplinary team (MDT) in the nearest facility providing cancer care. There should be pre-booking systems for appointments at both referring and receiving clinics, where each patient with a new cancer diagnosis should be reviewed by an oncologist. The referring clinician should also be informed of the diagnosis and decision made.

These guidelines recommend the following minimum timelines in order to minimize delays and improve outcomes:

1. Immediate referral will require admission acutely within a few hours of referral such as in oncological emergencies.
2. Urgent referral will require a 14-day standard from referral to assessment in a cancer center with rapid assessment by a designated clinician for children and adults.
3. A 60-day standard from first contact at a health facility to diagnosis for adults, and 30-day standard for children (<19years).
4. A 31-day standard from registration (at cancer center- post diagnosis) to start of therapy (including for recurrent disease) for children and adults.
5. All children diagnosed with cancer should be referred promptly to a designated centre of excellence for specialized cancer care within 14 days of diagnosis.

1.1.2 Inter-Professional and Patient Communication

Effective communication is essential throughout the patient's journey and must be timely, accurate, and concise. Clear and consistent information-sharing ensures continuity of care and supports informed decision-making by both patients and professionals.

Communication must be prioritized at the following stages of the patient journey: upon diagnosis, during multidisciplinary team (MDT) discussions, at assessment clinics, during routine clinic appointments and reviews, during treatment reviews, at decision points involving changes to the care plan and at initiation of end-of-life care planning.

At each key stage, the following information must be communicated and documented: manually and where available in digital medical records

- **Patient Information:** Details of what the patient has been told about their diagnosis, prognosis, and condition.
- **Educational Resources Provided:** Any written materials, digital resources, or support information shared with the patient and/or caretakers.
- **Planned Next Steps:** Dates for upcoming reviews, appointments, or treatment initiation.
- **Treatment Intent:** Clear explanation of whether the aim is curative, palliative, or supportive.
- **Medication Summary:** A current medication list, including any changes made and the rationale for those changes.
- **Points of Contact:** Contact details for the clinical team or service for further queries, clarification, or support.
- **Specialist Input:** Summary of specialist assessments and interventions undertaken.
- Treatment Plan Overview: A clear summary of the treatment plan, including creation and amendment dates.
- **Team Communication:** All relevant correspondence must be shared with appropriate team members responsible for any aspect of the patient's care.

1.1.3 Referral Pathway

Cancer patient referrals may originate from a variety of sources and flow both *upwards* (to higher levels of care for diagnosis and treatment) and *downwards* (to lower levels for follow-up, supportive, or palliative care). Effective communication, documentation, and coordination are essential at each step.

Referral Sources:

- **Accident & Emergency (A&E):** Patients presenting with acute symptoms or undiagnosed complications.
- **Internal Referrals:** From general medicine, surgery, or other hospital departments.
- **External Referrals:** From primary care centers, community health centers, or private clinics.

1.1.4 Patient Care Pathway

A coordinated, patient-centered cancer care pathway ensures timely, evidence-based interventions across the entire cancer journey—from initial assessment to survivorship or end-of-life care.

Overview of the Key Stages and their Core Components

a) Assessment and Diagnosis

- **Initial Consultation:** Patient attends a consultant-led clinic appointment.
- **Diagnostic Investigations:** Relevant tests are carried out to assess organ function (e.g., renal profile, full blood count), and imaging studies (e.g., CT, MRI, ultrasound). Histological diagnosis is confirmed through fine needle aspiration cytology (FNAC) and/or biopsy. Digital pathology and radiology workflows are recommended to improve efficiency of the diagnostic process
- **Case Discussion:** All findings should be reviewed collaboratively to ensure accurate diagnosis and individualized care.
- Diagnosis Disclosure: Diagnosis is communicated to the patient with empathy and clarity, including an opportunity to ask questions.

- Diagnosis entered in cancer registry

b) Multidisciplinary Team (MDT) Discussion

- **Case Presentation:** Referral and minimum required clinical information must be available for effective MDT discussion.
- **Treatment Recommendation:** The MDT outlines a personalized treatment strategy, which may include surgical and non-surgical options.
- **Special Population Consideration:** Cases involving pediatric or adolescent/young adult (AYA) patients must be referred to and discussed in the appropriate specialist MDT.
- **Communication:** Recommendations are shared with the patient and the referring clinician, ensuring alignment in care planning.

c) Treatment Planning

- **Patient and Family Engagement:** The proposed treatment plan is explained in depth to the patient and their family/next of kin, including expected outcomes, potential side effects, and planned reviews.
- **Emergency Contact Information:** Patients are provided with contact details for oncology emergencies (e.g., febrile neutropenia).
- **Holistic Needs Assessment:** A comprehensive review of physical, psychological, social, and spiritual needs is conducted.
- **Supportive Referrals:** Patients are referred to relevant oncology rehabilitation services, including physiotherapy, nutrition, psycho-oncology, social support, palliative care as appropriate.

d) Pre-Treatment Assessment

- **Clinical and Surgical Evaluation:** General and disease-specific surgical assessments are completed to ensure fitness for planned interventions.
- **Information Prescription:** Patients receive tailored information resources (written or digital) to support informed decision-making.

e) Treatment Phase

- **Modalities of Treatment:** Surgical intervention where indicated, radiotherapy tailored to disease type and stage, systemic therapy including chemotherapy, immunotherapy, targeted agents and best supportive care.
- **Rehabilitation and Recovery:** An aftercare plan is created in collaboration with the patient, addressing functional recovery and reintegration.
- **Treatment Summary:** A comprehensive End of Treatment Summary is completed upon discharge, outlining: treatments received, follow-up schedule, surveillance plans and ongoing care needs.
- **Palliative Care:** Early referral is made to palliative care centers and community support teams for enhancement of quality of life.
- **Communication with Referrer:** Updated clinical information is shared with the referring clinician to maintain continuity of care.

f) Follow-Up and Surveillance

- **Risk-Stratified Follow-Up:** A personalized follow-up plan is developed based on the patient's cancer type, treatment history, and recurrence risk.

- **Functional Monitoring:** Ongoing assessment of physical, cognitive, and psychosocial functioning.
- **Fertility and Pregnancy Planning:** Counseling and support are provided where fertility preservation or future pregnancy is relevant for children and adults.
- **Ongoing Rehabilitation:** Access to rehabilitation services continues as required.
- **MDT Re-evaluation Triggers:** Any new or concerning developments (e.g., recurrence, metastasis, complex symptoms) should prompt re-discussion at the MDT.

1.1.5 Multidisciplinary Team (MDT) Meetings

Multidisciplinary team (MDT) care is recognized globally as best practice in delivering high-quality cancer management. MDTs have been shown to improve patient outcomes by coordinating care among specialists. During MDT meetings, team members assess and document comorbidities, performance status, and cancer staging. They order necessary diagnostic tests, and collaboratively develop, review, and implement treatment plans. Patients should be informed that their case will be discussed at an MDT meeting and must be given the opportunity to consent.

Terms of Reference for MDTs

It is recommended that each MDT develops a clear and concise Terms of Reference (ToR) outlining how the team operates. At a minimum, the ToR should include the following:

a) A brief statement describing the MDT's objectives and role in patient care.
b) Team membership, roles and responsibilities of each member including identification of the chairperson and meeting coordinator.
c) Meeting frequency, venue, format, and typical duration. Regular meetings help reduce waiting times and improve resource use.
d) Record-keeping and data management.
e) Criteria for case referrals to the MDT and the process for discussion and documentation of cases.
f) Patient confidentiality and consent for the MDT.
g) Education and Training- MDTs may accommodate other health professionals for training purposes.
h) Use of Videoconferencing is encouraged to include regional or primary care providers in discussions.

Where appropriate, participation in clinical trials, including the process for identifying eligible patients, should also be addressed in the terms of reference.

Characteristics of Effective MDT Meetings

To ensure MDT meetings are effective, the following elements are essential:

a) **Expert-led Care:** Treatment decisions are made by specialists with expertise in the relevant cancer type.
b) **Patient-Centered Support:** Patients receive appropriate information and emotional support.
c) **Continuity of Care:** Clear communication ensures consistent care, regardless of provider or location.
d) **Collegial Team Dynamics:** Strong, respectful working relationships among team members.
e) **Educational Opportunities:** It supports ongoing learning.

f) **Efficient Use of Resources:** Tools like telehealth are used to optimize access and efficiency.
g) **Monitoring Changes in Care Plans:** Significant changes to treatment plans are tracked to allow for review and learning.
h) **Quality Improvement:** Routine auditing and monitoring of MDT processes support service improvement and sustainability.
I) **Accurate documentation of the MDT proceedings** (see appendix 8 for MDT reporting template)

1.2 Palliative Care, Supportive Care and End of Life Care

Palliative, supportive, and end-of-life care are essential components of comprehensive cancer care, extending beyond the treatment of disease to address the full spectrum of patient needs. These approaches aim to improve the quality of life for patients facing life-limiting illness and provide holistic support to their families and caregivers.

Palliative care is not limited to the final stages of illness—it is appropriate at any point in the patient journey and should be delivered alongside treatments such as chemotherapy or radiotherapy. It offers relief from pain and other distressing symptoms, integrates psychological and spiritual support, and helps individuals live as actively and meaningfully as possible, even in the face of serious illness.

Support systems are also extended to families, both during the patient's illness and into bereavement, recognizing the emotional and practical challenges they face. Delivered through a multidisciplinary team approach, palliative and supportive care services help manage complex symptoms, guide difficult decision-making, and ensure compassionate care planning at every stage.

By enhancing comfort, dignity, and overall well-being, these forms of care not only alleviate suffering but also positively influence the course of illness, making them integral to person-centered cancer care across all levels of the health system.

1.2.1 Palliative Care

World Health Organization defines palliative care (PC) as an approach that improves the quality of life of patients (adults and children) and their families facing the problems associated with life- threatening illnesses, through the prevention and relief of suffering by means of early identification and impeccable assessment and treatment of pain and other problems, physical, psychological and spiritual (WHO, 2002).

Lancet 2017 estimates that cancer is a major driver, accounting for 34% of the major need for palliative care. Palliative and supportive care must be multi-disciplinary. Integrating palliative care from the time of diagnosis improves patient and family experience, which ultimately leads to better patient reported outcomes (Lancet, 2018). The inclusion of PC as a fundamental health service and right is incorporated in

the Kenya Health Act (2017). However, according to the Kenya Palliative Care Policy 2021-2030, less than 10% of patients who require palliative care in Kenya do access it.

Components of Palliative Care

Palliative care aims to improve quality of life through symptom control, psychosocial support, spiritual care, and assistance with decision-making. The following components represent a comprehensive model of palliative care:

a) **Integrated Treatment Navigation:** Seamlessly incorporates palliative care into standard oncology care pathways. Ensures coordinated care across various specialties, improving treatment access, understanding, and adherence.

b) **Communication and Decision Support:** Facilitates clear, compassionate, and ongoing communication with patients, families, and caregivers. Provides support in discussing diagnosis, prognosis, treatment goals, and transitions in care.

c) **Pain and Symptom Management:** Effective control of pain and other physical symptoms such as nausea, fatigue, breathlessness, constipation, and insomnia. Incorporates the use of pharmacological and non-pharmacological interventions tailored to individual needs.

d) **Psychosocial Support:** Involves thorough psychosocial assessment to understand the patient's social support systems, coping mechanisms, fears, and emotional wellbeing. Offers access to psychological counseling, mental health services, and caregiver support.

e) **Spiritual Care:** Provides space for patients to explore existential or spiritual concerns. Involves chaplaincy services or culturally appropriate spiritual care as desired by the patient.

f) **Chronic Wound and Stoma Care:** Expert management of complex wounds and stomas to enhance comfort and dignity. Education for caregivers and patients on care techniques.

g) **Body Image and Sexuality Support:** Addresses changes in body image, self-esteem, and sexual health resulting from cancer or its treatments. Counseling and supportive therapies offered as part of holistic care.

h) **Anticipatory Symptom Management:** Proactive planning for likely disease-related complications to reduce distress and emergency hospital visits. Includes medication kits and clear action plans for managing symptoms at home.

i) **Ethical and Legal Support:** Guidance on ethical issues in care, including autonomy, consent, and equitable access. Promotes awareness of patients' rights to care as a component of health equity and human rights.

j) **Advance Care Planning:** Involves development of advance directives, allowing patients to express preferences for future care when they may no longer be able to decide. This may include writing a will, power of attorney – appointing a person to make decisions on the patient's behalf.

k) **Rehabilitation Services:** Aims to maintain or improve function, mobility, and independence through: Physical therapy, occupational therapy, speech therapy, nutrition, prosthetics and assistive devices enabling meaningful activity.

l) **Support Groups and Peer Support:** Provides emotional support and shared experiences through facilitated patient and caregiver groups. This enhances coping, reduces isolation, and builds community support networks.

m) **Continuity and Transition of Care:** Ensures smooth transition across care settings (hospital, hospice, home) hence reducing care fragmentation and improving both clinical and psychosocial outcomes for patients and families.

n) **Home-Based Palliative Care:** Delivers care in the comfort of the patient's home to enhance autonomy and dignity. Includes symptom management, nursing care, and psychosocial support coordinated with community services.

1.3 Pain Management

Pain is a complex and deeply personal experience that affects a significant number of patients requiring palliative care. According to the International Association for the Study of Pain (IASP, 2020), pain is defined as **"an unpleasant sensory and emotional experience associated with, or resembling that associated with, actual or potential tissue damage."** As famously stated by McCaffery (1968), **"Pain is whatever the experiencing person says it is, existing whenever and wherever the person says it does."** These definitions underscore the subjective and multidimensional nature of pain.

Importance of a Holistic Approach

Pain is one of the most common and distressing symptoms faced by individuals with life-limiting illnesses, particularly cancer. Up to 80% of cancer patients will experience moderate to severe pain during the course of their illness. Effective pain management must therefore adopt a holistic, person-centered approach—often referred to as the **"total pain"** model. This model addresses not only the physical dimension of pain but also its psychological, social, and spiritual components.

Guiding Principles for Management as Defined by the WHO Pain Relief Program include;

- Multidisciplinary involvement is essential for optimal pain control.
- The WHO Pain Ladder serves as a practical framework for the selection and escalation of analgesics, beginning with non-opioids and progressing to stronger opioids as needed.
- Regular reassessment and patient feedback are crucial to ensure ongoing effectiveness and to adjust treatment plans appropriately.



Figure 1.1: WHO Pain Relief Program

Pain Assessment: Key Elements from Patient History

A thorough pain assessment using an appropriate pain rating scale is the foundation of effective management. Important aspects to explore include:

- Impact on Function: Is the pain interfering with daily activities or mobility?
- Patient Perception: How does the patient describe and interpret their pain?
- Treatment Expectations: What outcomes does the patient hope to achieve through pain management?
- Emotional Response: Are there any fears or anxieties associated with the pain or its treatment?
- Past Experiences: What has been the patient's previous experience with pain, serious illness or pain medications?

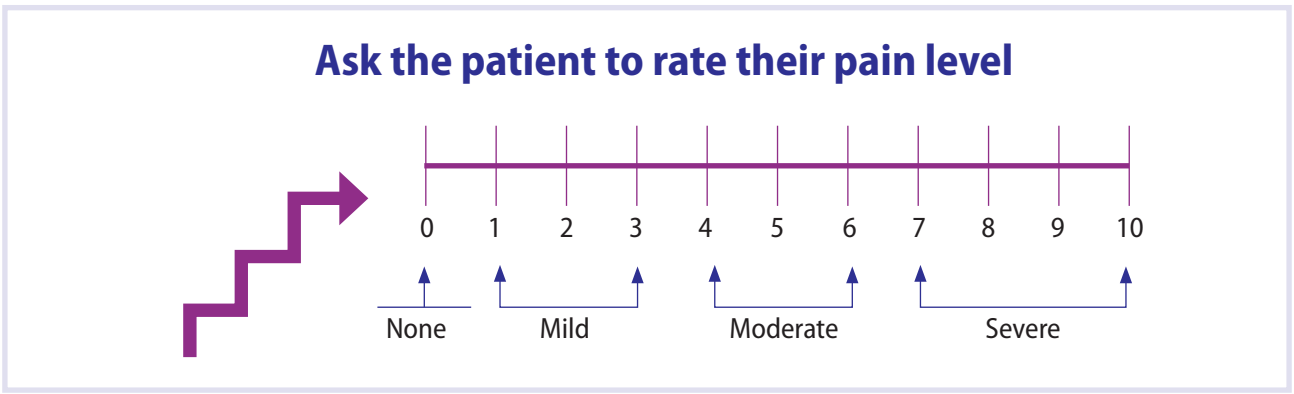
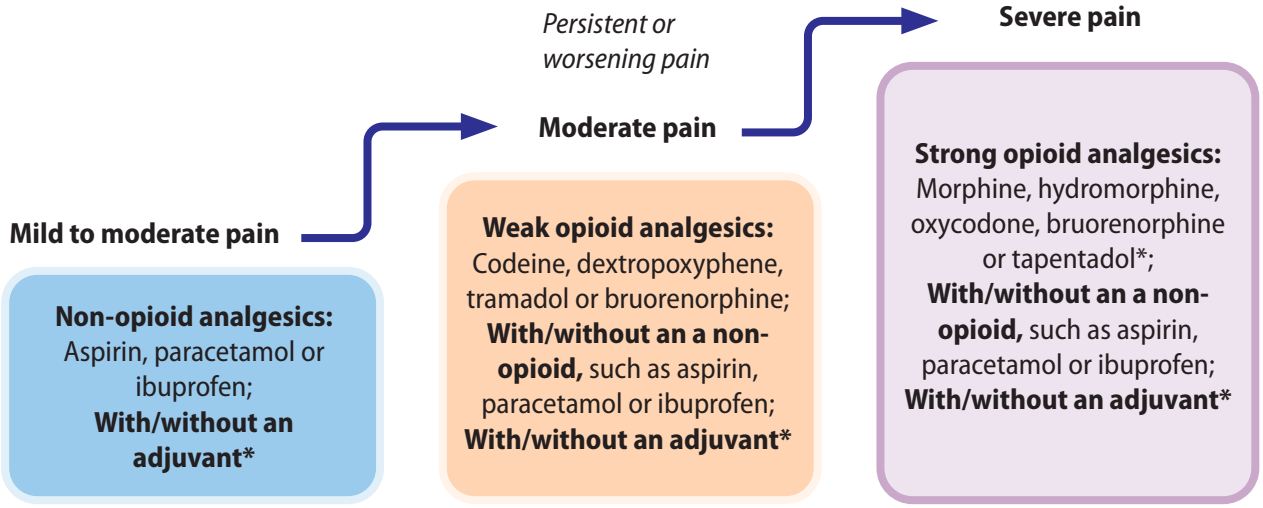


Figure 1.2: Numerical Pain scale

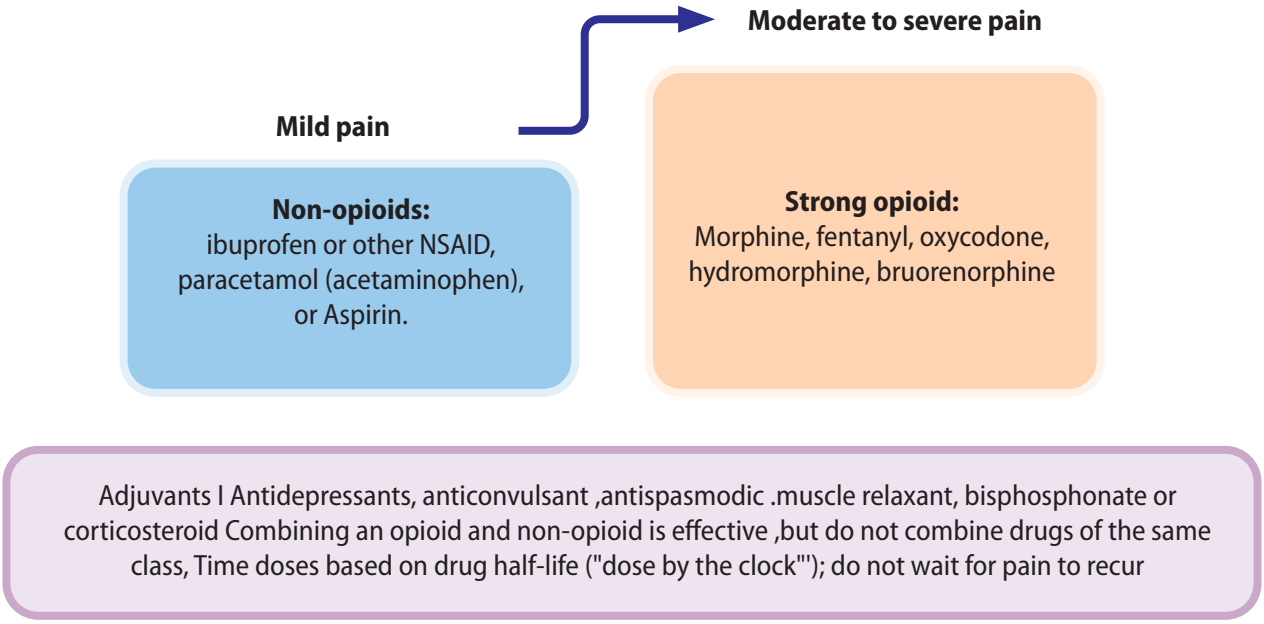
Treatment of Pain

- Non-pharmacological therapy – music therapy, aromatherapy, massage
- Pharmacological therapy – NSAIDs, opioids, adjuvants



[*Examples of adjuvants include corticosteroids, antidepressants, hypnotics and anticonvulsants/antiepileptic agents.]

Figure 1.3: Step WHO Adult Pain Management Ladder



NB: For children, moderate to severe pain management is through strong opioids

Figure 1.4: Step WHO Paediatric Pain Management ladder
NB: For children, weak opioids are not recommended, NB: Refer to opioid handling SOP appendix 9

1.3 Management of Other Symptoms

Symptom management in patients with cancer extends beyond pain control and must address a wide range of physical, emotional, and spiritual concerns. The presentation and severity of symptoms vary depending on multiple factors, and individualized care is essential for optimal outcomes.

Factors Influencing Symptom Presentation and Management

The nature and intensity of symptoms depend on:

- The specific organ(s) involved
- The stage of the disease (early vs. advanced)
- Local involvement – how the tumor or disease process affects adjacent structures or organs
- Presence of metastases – symptoms may arise in areas distant from the primary tumor
- Systemic manifestations – such as fatigue, cachexia, or paraneoplastic syndromes
- Psychosocial and spiritual dimensions – the patient's emotional wellbeing, social support systems, and spiritual needs must be considered alongside physical symptoms

Core Principles of Symptom Management

1. **Thorough Assessment is Essential Before Treatment:** Every symptom must be assessed and understood in context before initiating any intervention.
2. **Use of Standardized Assessment Tools:** Tools like the Edmonton Symptom Assessment Scale–Revised (ESAS-R) help quantify symptom severity and monitor changes over time.
3. **Regular and Ongoing Evaluation:** Set clear treatment goals tailored to the patient's needs and preferences, Monitor the effectiveness of each intervention, adjust treatment plans accordingly and Facilitate communication among care providers and with the patient and their family.

Symptom Management Process: A Structured Approach

For every symptom—whether physical (e.g., nausea, breathlessness, fatigue) or psychological (e.g., anxiety, depression)—follow this structured pathway:

- **Evaluation:** Accurately diagnose the underlying cause of the symptom.
- **Patient Explanation:** Clearly communicate findings and the rationale for proposed interventions to the patient and family.
- **Management:** Provide an individualized treatment plan, incorporating pharmacological and non-pharmacological strategies.
- **Monitoring:** Continually assess the response to treatment and make timely adjustments as needed.
- **Attention to Detail:** Consider all contributing factors and subtle changes in the patient's condition that may impact symptom control.

1.3.1 Supportive Care

Supportive care plays a vital role in enhancing the quality of life for individuals living with life-threatening illnesses such as cancer. Its primary aim is to prevent or address at the earliest opportunity the physical symptoms, treatment-related side effects, and the psychological, social, and spiritual challenges associated with the disease and its therapies.

Supportive care is integrated throughout the continuum of cancer care and is essential for maintaining treatment adherence, reducing complications, and preserving patient wellbeing.

Key Components of General Supportive Care include:

- **Vascular Access Management:** Care and maintenance of central venous catheters (e.g., chemo ports) and peripheral intravenous access to ensure safe and effective delivery of medications and fluids.
- **Hydration Therapy:** Prevention and treatment of dehydration to support organ function and overall health.
- **Blood and Blood Product Transfusion:** Administration of red blood cells, platelets, or plasma to manage anemia, thrombocytopenia, or bleeding complications.
- **Nutritional Support:** Dietary assessment and interventions, including enteral or parenteral nutrition, to counter malnutrition, maintain strength, and support healing.
- **Management of Acute Treatment-Related Side Effects:** Addressing common side effects from chemotherapy and radiotherapy, such as:
- Nausea and vomiting.
- Fever.
- Oral mucositis (painful inflammation and ulceration in the mouth).
- Constipation.
- Anorexia and cancer-related cachexia.
- Fatigue.
- Diarrhea.
- Dehydration.
- Premature menopause.
- Skin and hair changes (keratinization).
- Adverse drug reactions (SJS, TENS)

1.3.2 Nausea and Vomiting

Introduction

Nausea and vomiting are one of the most common and distressing side effects of cancer therapy and can have a significant impact on patients' quality of life during treatment.

Anti-cancer drugs vary in their ability to induce vomiting (emetogenicity potential) and this is important in understanding of the appropriate prophylactic regimen.

Risk factors

- Younger age
- Female sex
- Previous history of chemotherapy-induced nausea and vomiting
- Little or no previous alcohol use
- Prone to motion sickness
- History of pregnancy nausea/vomiting.
- Anxiety/high pre-treatment expectation of nausea

Types of Nausea and/or Vomiting

- **Chemotherapy Induced Nausea and Vomiting**

1. **Acute-Onset emesis**
   Occurs within the first 24 hours following administration of anti-cancer therapy. The severity most often reaches its peak approximately 5 to 6 hours after treatment initiation.
2. **Delayed-onset emesis**
   Occurs more than 24 hours after administration of anticancer therapy and may last up to 7 days. It is most frequently associated with highly emetogenic agents.
3. **Anticipatory emesis**
   Occurs before patients receive their next treatment with anticancer agents. It is primarily a conditioned response, typically occurring after a previous negative experience with anticancer agents.
4. **Breakthrough emesis**
   Occurs despite prophylactic antiemesis treatment and/or requires rescue with antiemetics.
5. **Refractory emesis**
   Development of nausea and/or vomiting that occurs during subsequent treatment cycles when antiemetic prophylaxis and/or rescue has not been effective in earlier cycles.

- **Radiation-Induced Nausea and/or Vomiting**

- Patients undergoing total body irradiation (TBI) have the highest risk of developing nausea and vomiting. Radiation directed to the upper abdomen carries a moderate emetogenic risk (30%–90%).
- The likelihood of nausea and vomiting also increases with higher daily fraction sizes, greater cumulative radiation doses, and larger irradiated volumes.

- **Exclude**

· Comprehensive clinical evaluation is very important to exclude alternative etiologies and guide optimal management. In patients with cancer, nausea and vomiting may result from causes other than chemotherapy-induced nausea and vomiting (CINV).
· The differential diagnosis includes partial or complete bowel obstruction, vestibular dysfunction, central nervous system involvement, metabolic abnormalities, gastroparesis, excessive secretions (in head and neck cancers), malignant ascites, cannabinoid hyperemesis syndrome, rapid opioid withdrawal, gastrointestinal dysmotility, concurrent radiotherapy, and pancreatitis, in addition to psychophysiological factors.

Classification of Emetogenicity Risk

The incidence of chemotherapy-induced emesis is determined mostly by the intrinsic emetogenic potential of the individual anticancer agents administered.

Hesketh and Grunberg classification divides chemotherapeutic agents into four levels based on risk of acute emesis: -

- High emetic risk – >90% of patients experience acute emesis
- Moderate emetic risk – >30% to 90% of patients experience acute emesis

- Low emetic risk – 10% to 30% of patients experience acute emesis
- Minimal emetic risk – <10% of patients experience acute emesis

Principles of Emesis Control

- Prevention of nausea and vomiting is the goal -patients need to be protected throughout the full period of risk.
- Choice of antiemetics used should be based on the emetic risk of therapy, prior experience with antiemetics and patient factors.
- For combination/multidrug protocols regimens, select therapy based on the drug with the highest emetogenic risk
- Prophylactic antiemetics should be administered 30-60 minutes prior to anti-cancer therapy.
- Consider the toxicity of the specific antiemetics
- Consider using proton pump inhibitors or H2 antagonists for patients with gastro-oesophageal reflux symptoms that may mimic nausea/vomiting.
- Consider potential drug interactions between anti-cancer/antiemetic therapies and various other drugs.
- Lifestyle measures may help alleviate nausea/vomiting

Prevention and Management of Acute and Delayed Emesis

Table 1.1: High Emetic risk parenteral anticancer drugs for Acute and Delayed Emesis

Acute	Delayed
DAY 1:	**DAY 2, 3 and 4**
<u>High Emetic Risk</u> **Option A** 5-HT3RA: Dolasetron, Granisetron, Ondansetron, Palonosetron * PLUS Dexamethasone 12 mg PO/IV once *PLUS NK 1 RA (choose one): Aprepitant PO/IV, Fosaprepitant IV *PLUS Olanzapine PO (clinician discretion) **Option B** • Netupitant 300mg/Palonosetron IV • Olanzapine 2.5-10 mg PO • Dexamethasone 12mg PO/IV	**Option A** -Olanzapine PO 5-10mg PO daily(bedtime) on days 2,3 and 4 - Aprepitant 80mg daily PO on days 2 and 3 (if Aprepitant is used on day 1) - ± Dexamethasone 8mg PO/IV daily on days 2,3 and 4 **Option B** Olanzapine PO .5mg-10mg daily on day2,3 and 4
<u>Moderate Emetic Risk (30 – 90% risk)</u> 5-HT3 RA (choose one) *PLUS Dexamethasone 8 mg PO/IV	-Dexamethasone 8 mg PO/IV daily on days 2 and 3 (Only required for moderate risk drugs with known delayed nauseas and vomiting. Maybe reduced or omitted)

Acute	Delayed
Low Emetogenic Risk (10-30 % risk) Single dose Dexamethasone 4 to 8 mg PO or IV OR 5HT3 RA^ (ondansetron 8 mg PO or IV or Granisetron 2 mg PO or 3 mg IV) OR Metoclopramide 10 mg PO or IV	Not routinely required
Minimal Emetogenic risk	No prophylaxis required.

*Avoid use of corticosteroids with cellular therapies
* Consider dexamethasone -sparing approach with Immune Checkpoint Inhibitors (ICI)

Emetic Risk in Oral Anticancer Drugs

- There is a lack of clinical trials evaluating the prevention and treatment of nausea and vomiting associated with oral anti-cancer drugs. The use of steroids is not recommended due to the continuous administration of many oral anti-cancer drugs.
- Oral anti-cancer drugs have been classified as moderate or high, and minimal or low emetic risk.
- When combined with intravenous drugs in a combination anti-cancer protocol, the antiemetic recommendations for the drug with the highest level of emetogenicity should be followed.
- If multiple oral anti-cancer drugs are combined, emetic risk may increase and require prophylaxis.

Table 1.2: Emetogenic Stratified Management Of Emesis

Risk Potential	Day of anticancer therapy
Moderate or high emetic risk	5HT3 RA (ondansetron 8 to 16 mg or Granisetron 2 mg) PO daily
Minimal or low emetic risk	Treat only on "as required" basis e.g. metoclopramide 10 mg PO three times daily

Breakthrough Emesis

Consider re-evaluation of the emetic risk, disease status, concurrent illnesses, and medications; and ascertain that the best regimen is being administered for the emetic risk.

The general principle of breakthrough treatment is to add one agent from a different drug class to the current regimen.

Table 1.3: Breakthrough emesis Options

Drug class	Drug	Recommended dose
Atypical antipsychotic	Olanzapine (preferred for high risk)	5mg PO daily (when not used during prophylaxis)
Benzodiazepines	Lorazepam	0.5 to 1mg po 6 hourly
Dopamine receptor antagonists (RA)	Metoclopramide Haloperidol	10mg PO/IV tds (max daily dose 30mg) 0.5 to 2 mg po/IV every 6 hours PRN
Phenothiazines	Prochlorperazine Promethazine	10 PO every 6hours 10 to 25 mg PO/IV every 4-6 hours

*Olanzapine maximum daily dose can be upto 10mg if it was initially given prophylactically.
* Caution when dopamine RA are given with olanzapine -sedation risk.

Anticipatory Emesis Prevention/Treatment

- Ensure optimal antiemetic therapy during every cycle of treatment.
- Non-pharmacological methods may be useful including behavioral and psychological techniques.
- Consider anxiolytic therapy: Lorazepam 0.5-1mg PO beginning on the night before treatment and then repeated the next day 1–2 hours before anticancer therapy begins or Olanzapine PO as a single dose.

Radiation-induced emesis prevention/treatment

Radiation therapy to the upper abdomen/localized or a total body irradiation, use the following agents; Start pre-treatment for each day of RT treatment (order does not imply preference):

- Granisetron PO
  OR
- Ondansetron PO
- ± Dexamethasone PO

References

NCCN Clinical Practice Guidelines in Oncology for Antiemesis V.2.2025. Accessed may 2025. www.nccn.org

1.4 Survivorship

An individual is considered a survivor from the time of cancer diagnosis, during and immediately after treatment, and through the balance of their life. It covers the physical, psychosocial and economic issues of cancer, beyond the diagnosis and treatment phases. It includes issues related to the ability to get health care and follow-up treatment, rehabilitation, surveillance for late effects of treatment, screening for recurrence & secondary cancers and quality of life. Family members, friends and caregivers are also considered part of the survivorship experience.

Care of the survivors should include:

1. Prevention of new and recurrent cancers and other late effects.
2. Surveillance for recurrence/spread of cancer and screening for subsequent primary cancers at least annually.
3. Assessment for late psychosocial and physical effects.
4. Interventions for consequences of cancer treatment such as pain and peripheral neuropathy management, lymphedema management, for those who received anthracyclines, assess for anthracycline-induced cardiotoxicity (evaluate for heart failure signs and symptoms, presence of risk factors), need for immunizations.
5. Possibility of fertility preservation should be explored before cancer treatment begins whenever possible in children, women and men of childbearing age.
6. Care coordination between specialists and primary caregivers with specific roles delineated to ensure the health needs of the cancer survivor are met.
7. Survivorship care planning: This plan will include a summary of treatment received, follow-up care, surveillance and screening recommended for the survivor, post-treatment needs of the survivor, healthy lifestyle behavior recommendations and information on treatment related side effects and anticipated health risks.

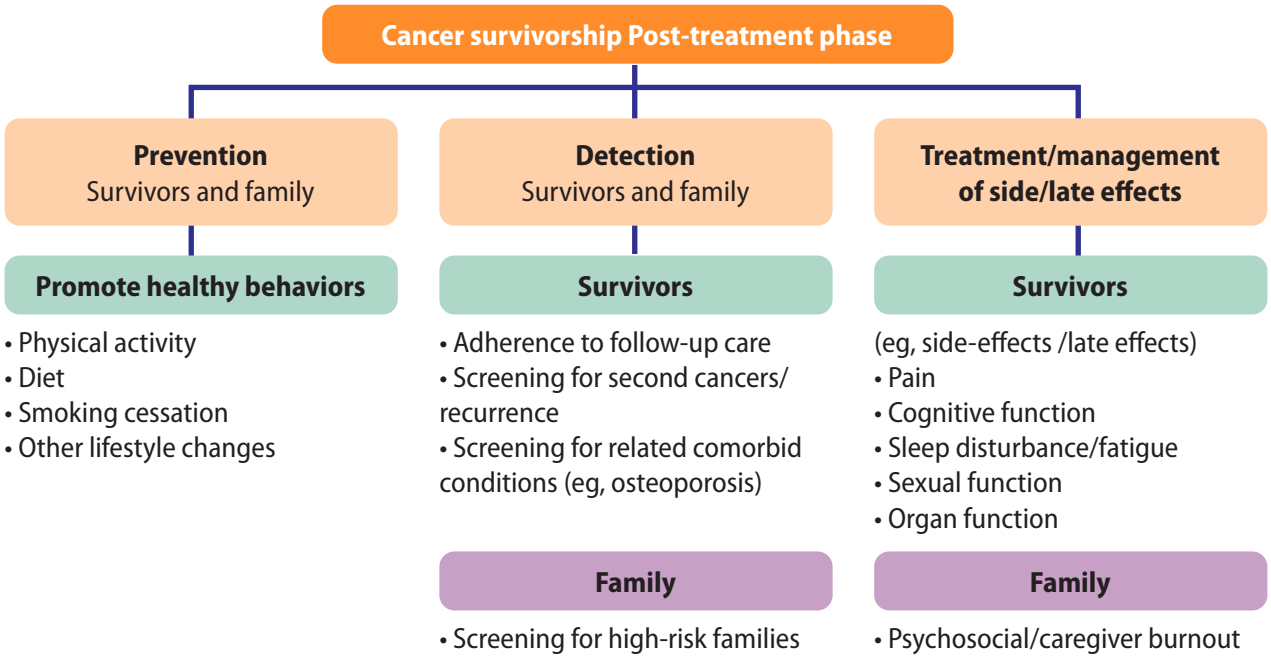


Figure 1.5: Aspects of cancer survivorship
(Grant, Marcia & Economou, Denice. (2008). The evolving paradigm of adult cancer survivor care. Oncology (Williston Park, N.Y.). 22.13-22, 27.)

1.4.1 Evaluating Success of Survivorship Care will include:

- Improved communication and care coordination: (perceived) patient-physician communication, physician-physician communication.
- Improved understanding of needed follow-up tests, their purpose and timing, and who will conduct them, potential late effects of illness and what symptoms might be important to report.
- Better adherence to recommended follow-up activities; fewer requests for unnecessary tests; reduced duplication of services.

- Improved ability to identify providers and resources to address persistent effects of cancer and its treatment.
- Improved overall survival with decreased cancer-related morbidity.
- Improved healthy lifestyle choices with better quality of life and function.
- Improved knowledge about and ultimately standardization of follow-up care behaviors.
- Improved ability to monitor survivors' health and implement changes in care in response to new information about treatment exposures and follow-up needs.
- Improved access to information necessary to guide follow-up care; less time spent searching for this.
- Enhanced quality of care delivery (such as compliance with evolving quality standards)

1.4.2 End of life Care

End of life care (EOL) is a key component of palliative care. It refers to the comprehensive support provided to individuals with advanced, progressive disease, when death is imminent, and life expectancy is limited to a short time such as advanced cancer. The focus is on comfort, dignity, and quality of life as the disease approaches its final stages. Care decisions during this time are often complex and require careful consideration of the patient's autonomy, ethical implications of continuing or discontinuing medical interventions, and the appropriate use of healthcare resources. In addition, optimal care is guided in part by autopsy-based cancer research. This is the best opportunity for sensitization concerning advanced consent for participation in postmortem studies.

1.4.3 Core Components of EOL include

- **Prognostication:** Open and compassionate communication with patients and their families about expected life expectancy, to enable informed planning and decision-making for the time remaining.
- **Pain and Symptom Control:** Effective management of physical discomfort and distressing symptoms to ensure patient comfort.
- **Spiritual Care:** Identification and support of the patient's spiritual or existential needs, respecting cultural and personal beliefs.
- **Anticipatory Symptom Management:** Proactive planning for symptoms that may arise in the final stages, with clear strategies and medications in place to address them promptly.
- **Ethical and Legal Considerations:** Support with legal documentation and ethical decision-making, including respect for the patient's wishes and rights regarding treatment and care.
- **Home-Based End-of-Life Care:** Providing care in the home setting, when possible, to allow patients to remain in a familiar and supportive environment.
- **Management of Palliative Care Emergencies:** Rapid intervention for urgent issues such as catastrophic bleeding, airway obstruction, or severe pain crises, always with a focus on comfort.
- **Goals-of-Care Discussions:** Clear, ongoing conversations with patients and families to align care plans with the patient's values and preferences.
- **Family Meetings and Conferences:** Structured meetings to facilitate communication, provide updates, and support shared decision-making.
- **Nutrition and Hydration Issues:** Thoughtful consideration of the need for and appropriateness of artificial nutrition and hydration in the final stages of life.
- **Surrogate Decision-Making:** When a patient loses decision-making capacity, a designated surrogate or legally authorized representative should be identified to act in their best interest.

- **Do Not Resuscitate (DNR) Orders:** Documentation of the patient's decision to forgo resuscitation efforts in the event of cardiopulmonary arrest.
- **Withholding or Withdrawing Treatment:** Ethical consideration and implementation of decisions to stop or refrain from initiating treatments that no longer provide benefit.
- **Futile Interventions:** Avoidance of treatments that do not contribute to the patient's comfort or meaningful extension of life.
- **Last Rites and Cultural Practices:** Facilitation of religious or cultural end-of-life rituals as desired by the patient and family.
- **Bereavement and Grief Support:** Psychological and emotional support for families during the patient's final days and after death, to assist with grieving and adjustment.
- In the event of death best standards in causes of death should be adhered.

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Chapter 2

Breast Cancer

List of Abbreviations

Abbreviation	Definition
AD	Axillary Dissection
AI	Aromatase Inhibitor
ALND	Axillary Lymph Node Dissection
BCS	Breast Conserving Surgery
BIRADS	Breast Imaging Reporting and Data System
BMI	Body Mass Index
BRCA	Breast Cancer Gene
CK 5/6	Cytokeratin Antibody 5/6
CT	Computerized Tomography
CXR	Chest X Ray
DCIS	Ductal Carcinoma In Situ
EBC	Early Breast Cancer
ECOG	Eastern Cooperative Oncology Group Performance Status
EGFR	Epithelial Growth Factor Receptor
ENE	Extra Nodal Extension
ER	Estrogen Receptor

Abbreviation	Definition
FNAC	Fine Needle Aspirate Cytology
HER2	Human Epidermal Growth Factor Receptor 2
HRT	Hormonal Replacement Therapy
LCIS	Lobular Carcinoma In Situ
LVI	Lympho-vascular Invasion
LHRH	Luteinizing Hormone Release Hormone
LVEF	Left Ventricular Ejection Fraction
MRI	Magnetic Resonance Imaging
MRMR	Modified Radical Mastectomy
OFS	Ovarian Function Suppression
PNI	Perineural Invasion
PR	Progesterone Receptor
RT	Radiotherapy
SLNB	Sentinel Lymph Node Biopsy
US	Ultrasound
WLE	Wide Local Excision

2.1 Introduction

Breast cancer is the leading cancer both in incidence and mortality among women globally. In Kenya, it is the most common cancer with an age standardized rate of 40.8 per 100,000, and the third leading cause of all cancer deaths annually. (GLOBOCAN, 2025). Breast cancer tends to occur at a relatively young age with a median age of 49 years in Kenya, majority (54%) under the age of 50, in comparison to United States countries 62years). About 90% of cases occur sporadically, while only 5-10% can be at- tributed to genetic predisposition. Invasive ductal carcinoma (IDC) is the most common histological type diagnosed, accounting for up to 75% of all breast cancers (Soares et al., 2020).

2.1.1 Clinical Evaluation

All women with breast lump should undergo a Triple assessment which includes:

- Clinical evaluation preferably by experienced clinician
- Diagnostic imaging-bilateral mammogram and or complimentary ultrasound as needed
- Core biopsy for Histopathology and Immunohistochemistry

2.1.2 Clinical History

Take a comprehensive history

- Lump duration, pain, nipple discharge
- Previous abnormal breast biopsy (history of atypical ductal hyperplasia or lobular carcinoma in situ).
- History of irradiation, especially chest wall irradiation.
- Lifestyle - obesity, physical inactivity, tobacco & alcohol use.
- Local: Symptoms such as breast lump, pain, ulceration, nipple changes/discharge, peau d'orange.

and inquire about the following risk factors:

- Family history of breast/ovarian cancer/previous positive BRCA gene test especially among first degree relatives.
- Reproductive history (estrogen exposure): null parity, early age at menarche (<11) and/or later age of menopause (>55), long menstrual cycles (>32 days), history of not breastfeeding, history of hormone replacement therapy (HRT), age at first live birth (<30), contraceptive history (type and duration of use).
- Systemic evaluation: Cough, difficulty in breathing, back pain,



Figure 2.1: Symptoms of breast cancer

Physical Examination

- Assess performance status
- Examine for breast mass, ulceration, skin dimpling (peau d'orange), nipple changes (inverted, redness, discharge and scars) and axillary & supraclavicular/infraclavicular and cervical lymphadenopathy. See Appendix 3 for breast examination.
- Describe the tumor size, location, asymmetry, chest wall fixation, skin & chest wall changes, and regional lymph node involvement.
- Assess the opposite breast.
- Conduct a complete systemic physical examination.

2.2 First-Line Diagnostic Workup – Triple Assessment

1. **Clinical Examination – as above**.
2. **Imaging (based on age & risk):**
   a. Age <30 years: Ultrasound breast.
   b. Age ≥30 years: Mammogram (bilateral, diagnostic views) ± targeted ultrasound.
   c. MRI breast if: occult lesion, multifocal/multicentric disease, dense breasts, or pre-neoadjuvant therapy response assessment.
3. **Pathology:**
   a) Image-guided core needle biopsy (preferred).
   b) Fine Needle Aspiration Cytology (FNAC) if core not feasible.
   c) Excision biopsy only if above inconclusive.

Diagnosis is confirmed only when all the above three assessments agree.

2.3 Histopathology & Receptor Testing

- Histological type and grade.
- Immunohistochemistry (IHC): ER, PR, HER2.
- Ki-67 for proliferation index.

2.3.1 Staging Workup (based on stage at diagnosis)

Early disease (Stage I–II, no symptoms):

- Chest X-ray.
- Ultrasound abdomen/liver.
- Locally advanced disease (Stage III) or high-risk biology:
- CT chest/abdomen ± pelvis OR PET-CT.
- Bone scan if bone pain, raised ALP, or suspicion.

Metastatic disease (Stage IV or symptomatic):

- CT chest/abdomen/pelvis or PET-CT.
- Bone scan.
- Brain MRI if neurological symptoms.

2.3.2 Baseline & Supportive Tests

- CBC, renal and liver function tests.
- Pregnancy test (if premenopausal and chemo planned).
- ECG/Echo if anthracyclines or trastuzumab planned.
- Viral serologies (HIV, Hepatitis B/C).
- Personalised testing: may include BRCA testing, Oncotype DX and MammaPrint

2.3.3 Documentation & MDT

- Summarize findings in a structured report.
- Discuss at Multidisciplinary Tumor Board (surgery, oncology, radiology, pathology, nursing).

2.3.4 Key Principle

Every suspected malignant breast lump should undergo Triple Assessment ➔ Histopathology confirmation ➔ Stage-appropriate systemic imaging ➔ MDT discussion before treatment.

Mammography and ultrasound findings,
category and recommendations are as tabulated below:

Table 2.1: Breast Imaging Reporting and Data System (BIRADS)

Assessment	Category	Recommendation
BIRADS 0	Need additional imaging evaluation	Add views or ultrasound
BIRADS 1	Negative	Annual mammography
BIRADS 2	Benign finding	Annual mammography
BIRADS 3	Probably benign finding - short interval followup suggested	Bilateral mammography 6 month followup is suggested
BIRADS 4	Suspicious abnormality	Biopsy should be considered
BIRADS 5	Highly suggestive of malignancy - appropriate action should be taken	Biopsy
BIRADS 6	Biopsy proven carcinoma	

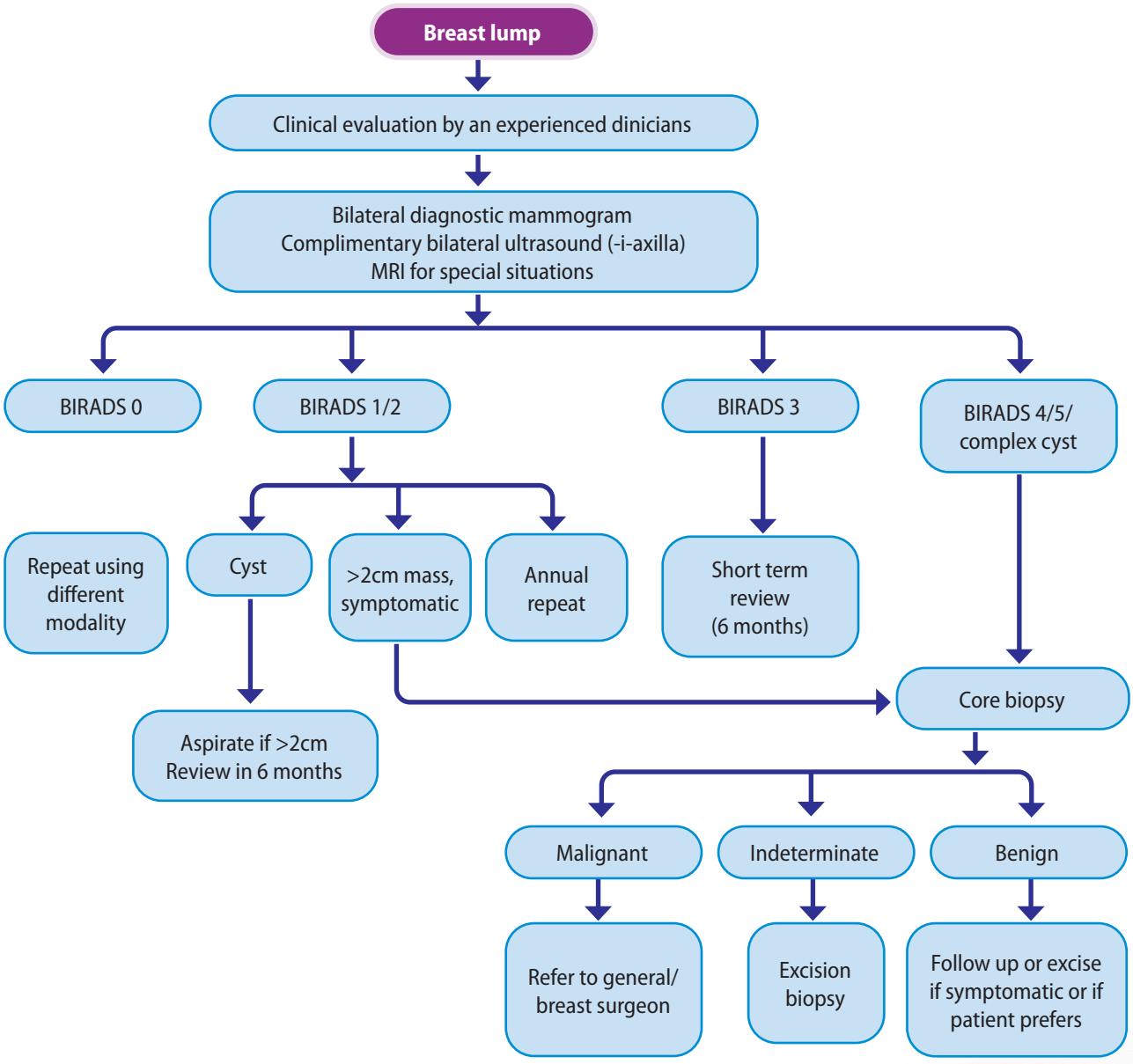


Figure 2.2: Diagnostic Algorithm: Workup of a Breast Lump

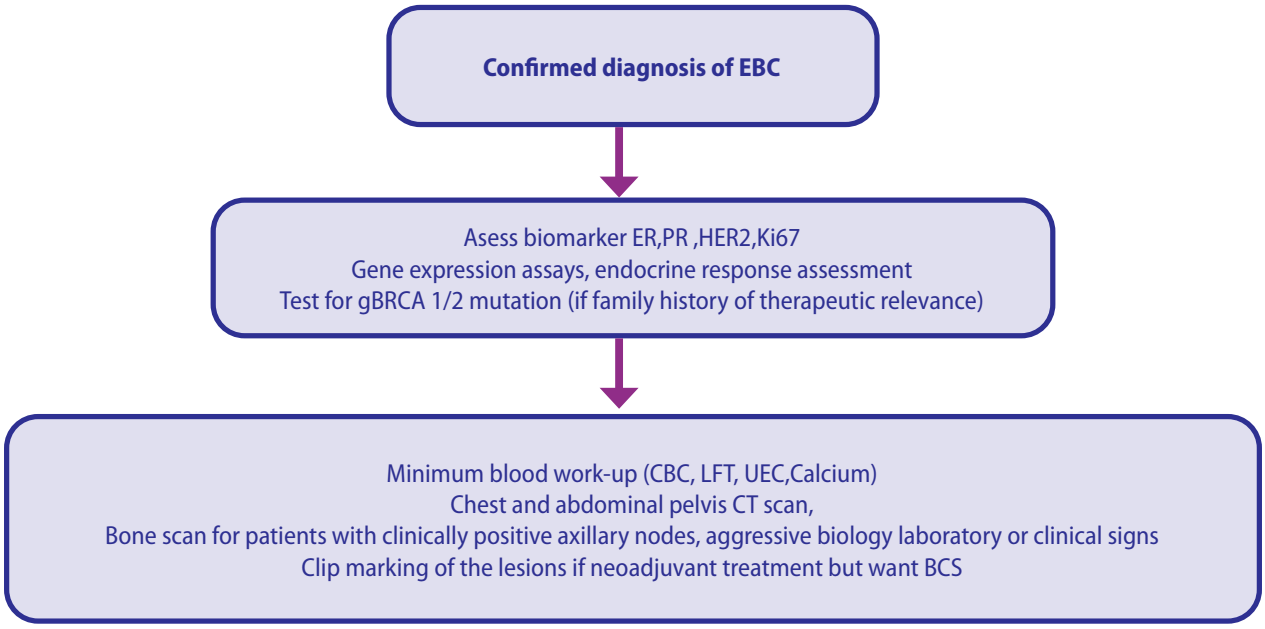


Figure 2.3: Pre-Treatment Assessment

All patients must be staged before treatment and staging is mainly by imaging and clinical examination.

- CT chest, abdomen and pelvis Or PET/CT may be done for specific cases
- MRI brain for patients with CNS symptoms

Other considerations

- Discuss fertility concerns with the patients
- Consider genetic counselling for patients at risk
- Pregnancy test for all patient of childbearing age
- Echocardiogram as indicated (doxorubicin /trastuzumab)

2.3.5 Staging

T-Staging – Clinical examination, imaging and surgical specimen

- The initial imaging should indicate the absence or presence of additional lesions in the ipsilateral breast, axilla and the contralateral breast, otherwise, review or repeat the imaging.

N-Staging - Clinical examination, imaging and surgical specimen

- Preoperative staging of N is clinical examination and CT scan results.

M- Staging - Imaging.

- Early breast cancer (T2 and below)- chest x-ray and abdominal ultrasound
- Locally advanced (T3 and 4) - CT Chest & Abdominopelvic, and bone scan (if necessary).
- Other tests as indicated by clinical evaluation, for instance CT head for CNS symptoms)

Table 2.2: AJCC staging of breast cancer

T-staging

T Status	Characteristics
Tx	Primary tumour cannot be assessed
T0	No evidence of primary tumour
Tis	Ductal carcinoma in situ
Tis (Paget)	Paget's disease not associated with parenchymal disease or ductal carcinoma in situ
T1	Tumour less than 20mm
T1mi	Tumour less than 1mm
T1a	Tumour more than 1mm but <5mm
T1b	Tumour >5 mm but <10mm
T1c	Tumour more than 10 but less than 20mm
T2	Tumour more than 20mmbut <50mm
T3	Tumour more than 50mm in the greatest dimension
T4	Tumour of any size with invasion of skin, chest wall or both
T4a	Invasion to chest wall (not pectoralis major)
T4b	Invasion to skin (ulceration, macroscopic satellite, peau de orange)
T4c	Invasion to both chest wall and skin
T4d	Inflammatory

Nodal status

Clinical		Pathology	
CNx	Regional nodes cannot be assessed (Previously removed)	pNx	Regional nodes cannot be assed for pathological studies
CN0	No evidence of nodal metastasis by imaging or clinical examination	pT0	No regional lymph node identified or TCs (isolated tumor cell clusters) only
		pNo(i+)	ITCs only
		pN0(mol+)	Detected by PCR, no ITC
CN1	Clinically mobile level I/Level II nodes	PN1	1-3 nodes
CN1mi	Micro metastasis, less than 2mm, larger than 0.2mm	pN1mi	Micro metastasis
cN2	Ipsilateral level I/II clinically fixed or matted	pN2	4-9 nodes
cN3	Ipsilateral infraclavicular nodes or supraclavicular nodes	pN3	More than 10 nodes axillary, infraclavicular node or supraclavicular node

M-status

M0	No clinical or radiological evidence of distant metastasis
M0(i+)	Above but with evidence of less than 0.2mm deposit or bone marrow, or circulating tumor cells without symptoms
cM1	Clinically or radiologically detected distant metastasis
pM1	Pathologically detected larger than 0.2mm

Table 2.3: Group Staging

T	N	M	Stage
Tis	N0	M0	0
T1	N0	M0	IA
T0	N1m1	M0	IB
T1	N1Mi	M0	IB
T0	N1	M0	IIA
T1	N1	M0	IIA
T2	N0	M0	IIA
T2	N1	M0	IIB
T3	N0	M0	IIB
T0	N2	M0	IIIA
T1	N2	M0	IIIA
T2	N2	M0	IIIA
T3	N1	M0	IIIA
T3	N2	M0	IIIA
T4	N0	M0	IIIB
T4	N1	M0	IIIB
T4	N2	M0	IIIB
Any T	N3	M0	IIIC
Any T	Any N	M1	IV

Table 2.4: Molecular Profiling

molecular intrinsic variant	ER	PR	HER2	Ki67	CK5/6	EGFR
Luminal A	Positive	Positive	Negative	<20 %	Negative	Negative
Luminal B	Positive	Positive	Negative	≥20%	Negative	Negative
Luminal B like	Positive	Positive	Positive	any Ki-67	Any	Any
HER2-enriched	Negative	Negative	Positive	Any Ki 67	Negative	Negative
Basal like	Negative	Negative	Negative	Any Ki67	Positive	Positive

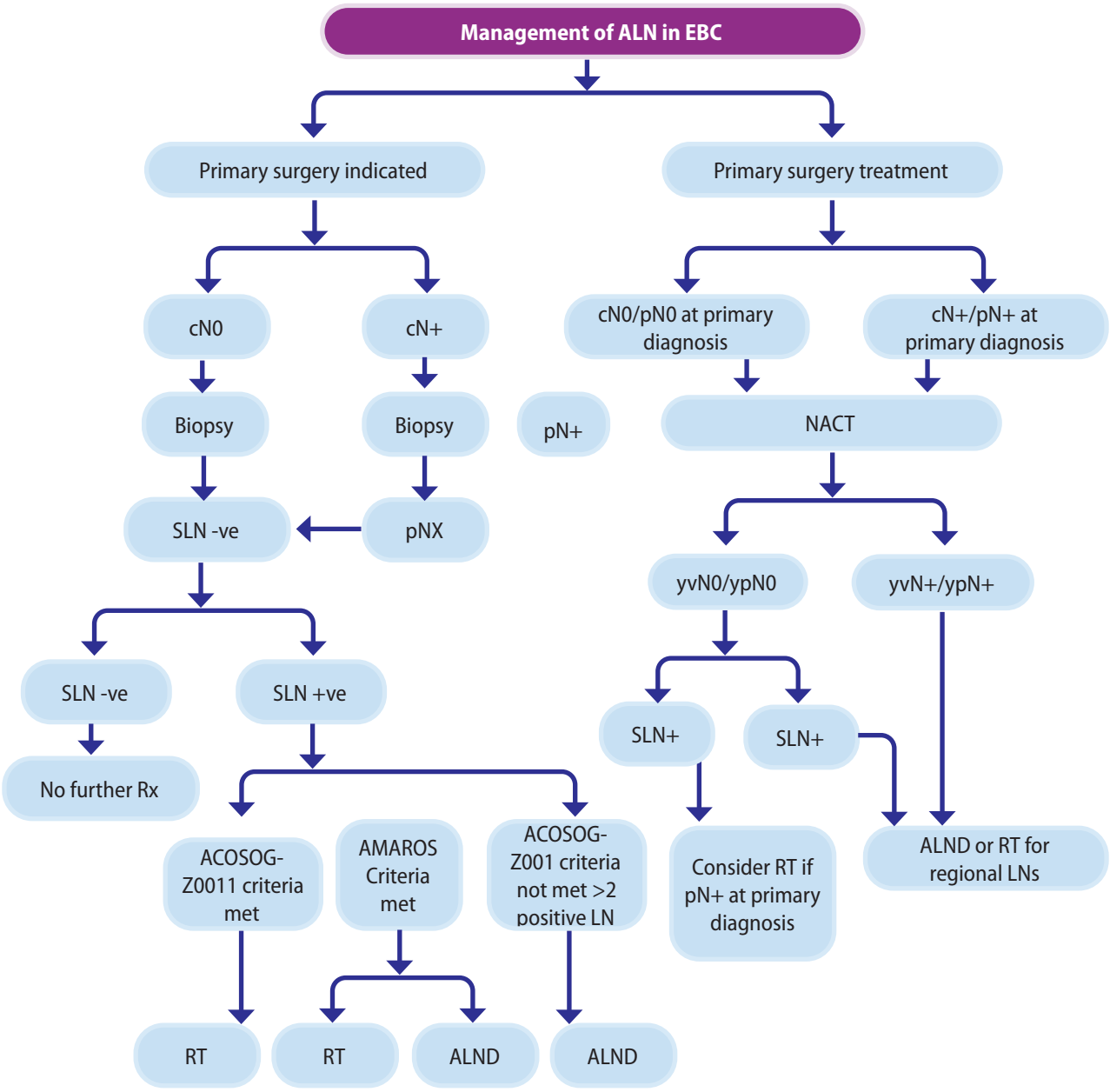


Figure 2.4: Algorithm for Axillary Staging (for early breast cancer)

2.3.6 Treatment

- All new breast cancer patients should be reviewed by a multi-disciplinary team (MDT) whose core members include designated breast surgeon(s), radiologist, pathologist, oncologist, Oncology nurse, nutritionists, Clinical Pharmacists, palliative care specialist, social worker and MDT coordinator
- Other disciplines, such as plastic and reconstructive surgeons, psychologists occupational/ physiotherapists may be included as and when need rises.
- Treatment should be guided by molecular subtypes and staging.
- ER low 1-9%, HER 2 negatives should be treated as TNBC but should receive adjuvant endocrine therapy.

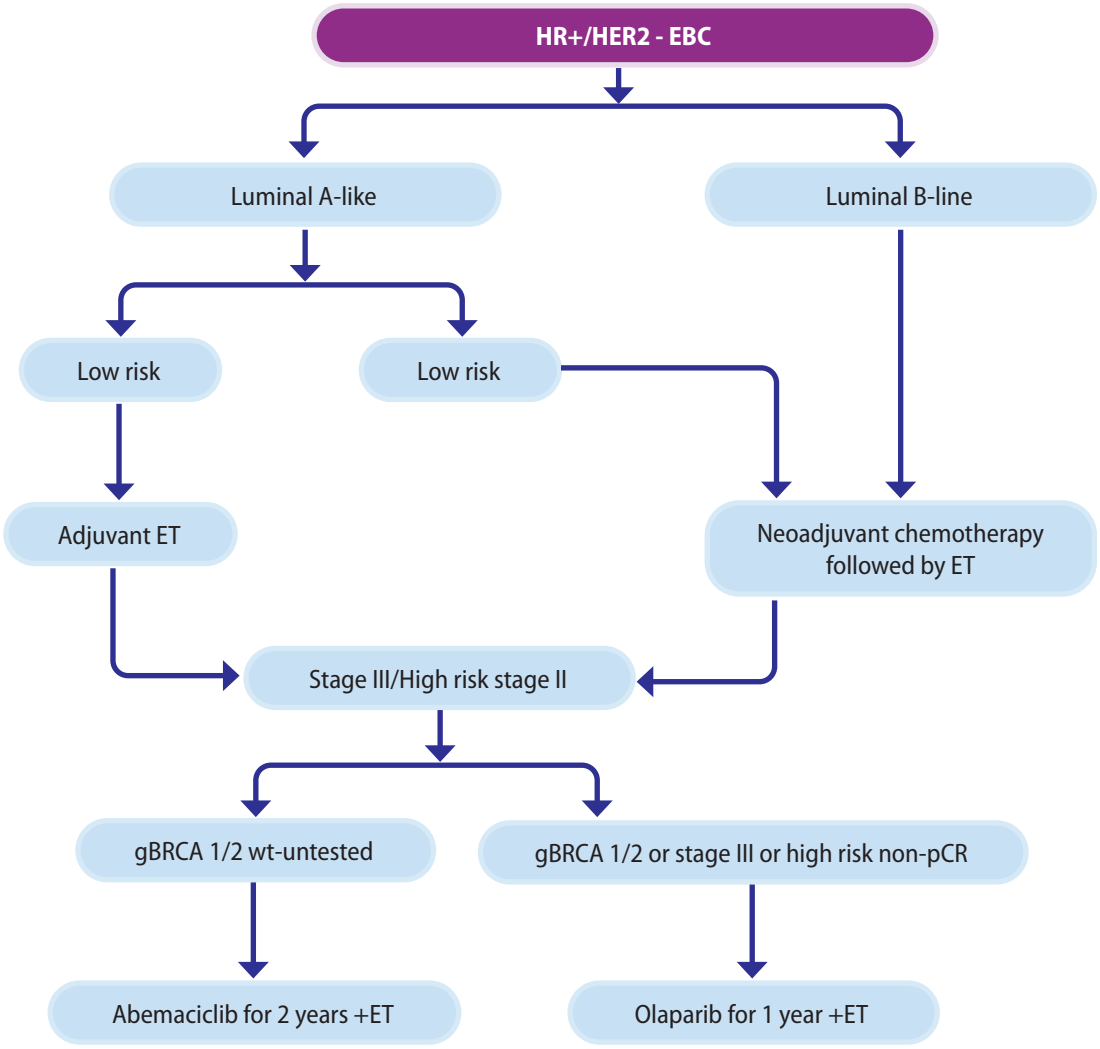


Figure 2.5: How To Treat The Different Subtypes

2.3.7 Adjuvant Radiotherapy

Indications for postoperative adjuvant radiotherapy to chest wall +/- nodal area include:

- All patients undergoing BCS
- Post mastectomy with the following:
- Tumor >5cm (T3)
- T4 tumors
- Chest wall involvement, skin involvement and Inflammatory breast cancer)
- Node positive >4 nodes
- 1-3 nodes with high-risk features (LVI, PNI, ENE)

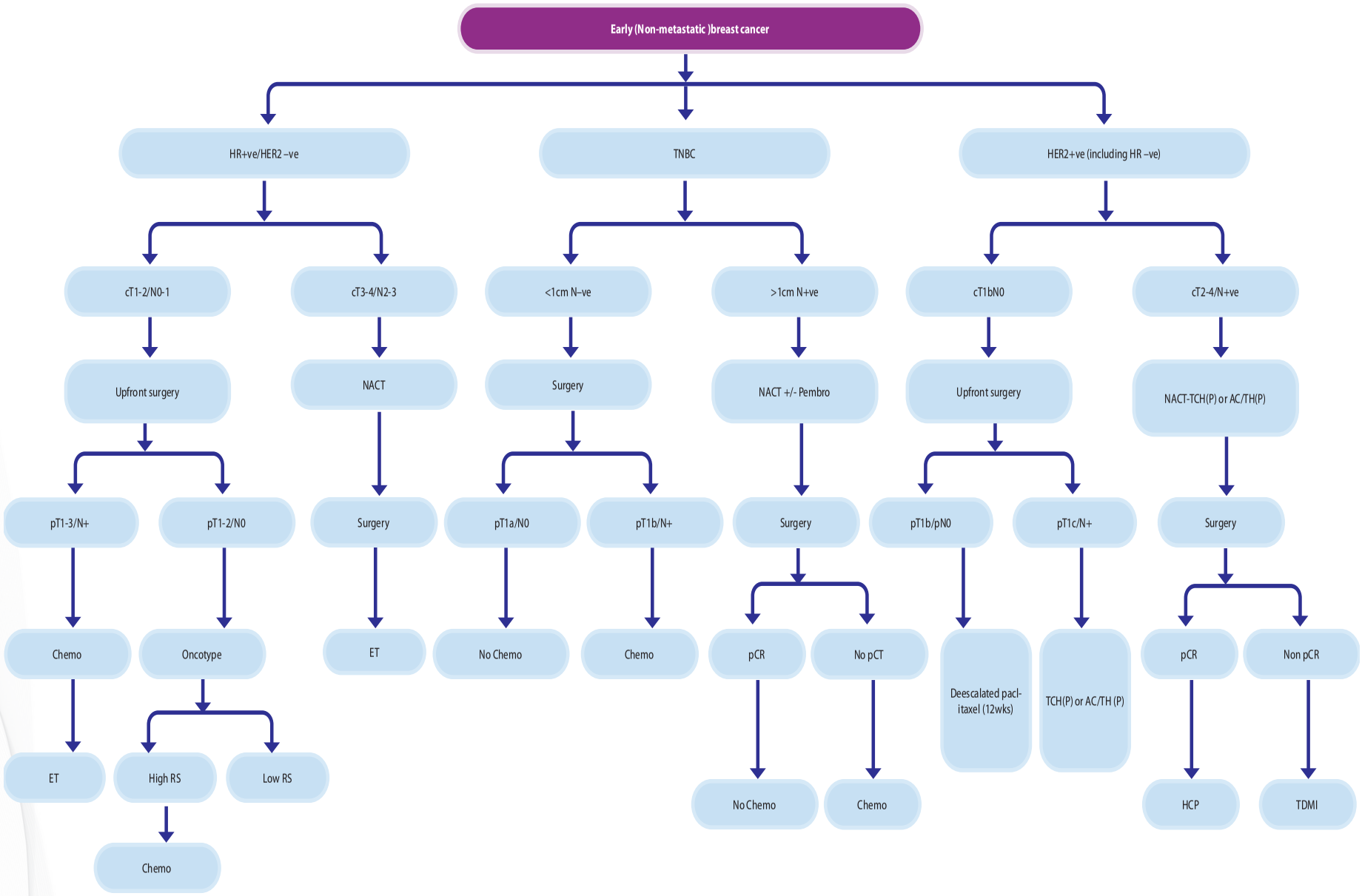
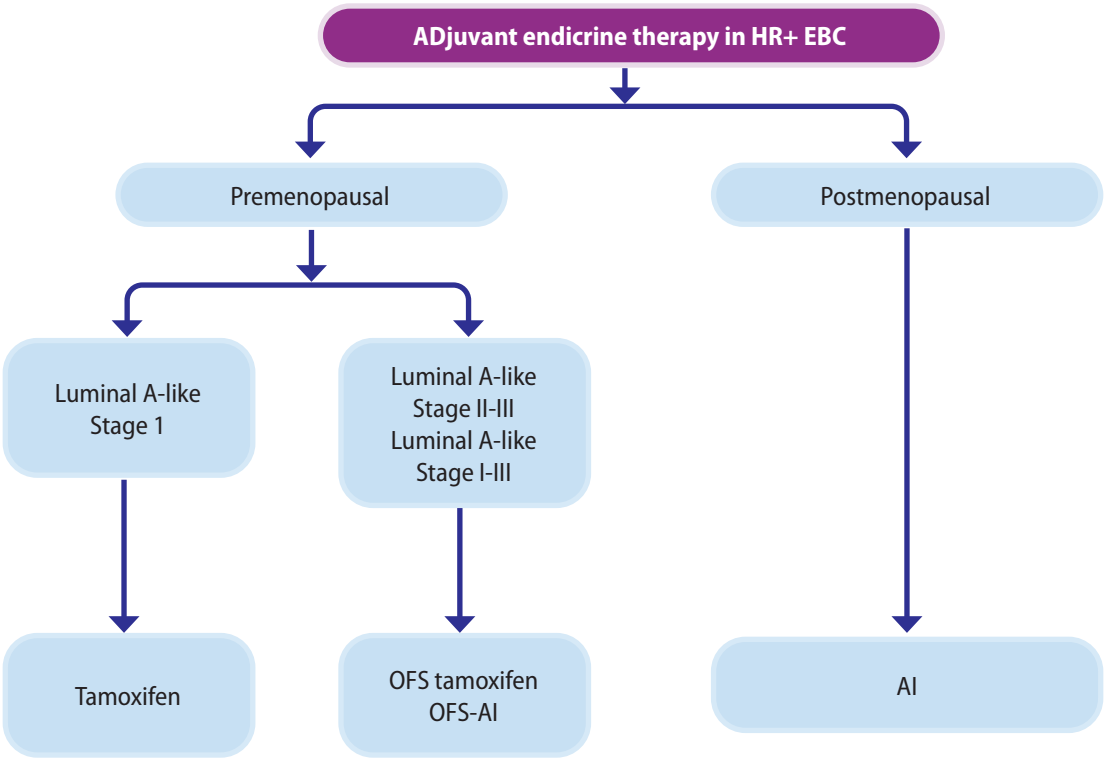


Figure 2.7: Adjuvant Endocrine Therapy



All patients with hormone receptor positive breast cancer should get endocrine therapy in the adjuvant setting usually after the completion of chemotherapy.

- Male patients should receive tamoxifen as their endocrine therapy
- Female patients who are premenopausal prior to chemotherapy should receive tamoxifen or aromatase inhibitors with ovarian function suppression.
- Ovarian function suppression (OFS) should be considered strongly in those patients with high-risk disease (tumour >2cm, node positive, grade 3 tumours and <35 years).
- MDT discussion is recommended when OFS is being considered.
- Patients who are post-menopausal should be offered aromatase inhibitors if there is no contraindication.
- Extended endocrine therapy (10 years) is indicated for patients with >4 nodes, large tumour, grade 3 disease
- Aromatase inhibitors increase the risk of osteoporosis; therefore, lifestyle advice, such as regular weight bearing exercise, smoking caseation, calcium supplements, should be discussed with the patients.
- Monitoring for bone loss should be done using bone density scan at baseline and every 1-2 years while on aromatase inhibitors.
- Patients who develop osteoporosis should be treated with bisphosphonates.
- Renal function and calcium test should be done prior to bisphosphonate therapy.

2.3.8 HER2 Directed Therapy

- All patients with HER2 positive disease should receive HER2 directed therapy .
- At the minimum, patients should receive trastuzumab and where possible dual antiHER2 therapy should be given as per the algorithm.
- Cardiac evaluation should be done at baseline and every 3 months while on HER2 directed therapy.
- Patients with Left ventricular ejection fraction (LVEF) less than 50% should not be given HER2 directed therapy ***The patient should then be referred to a cardiologist***.
- Any decline of LVEF of >10 points, treatment should be withheld and Echo repeated after 4 weeks and ***referred to a cardiologist***.

In both above scenarios, the patients should be referred to a Cardiologist.

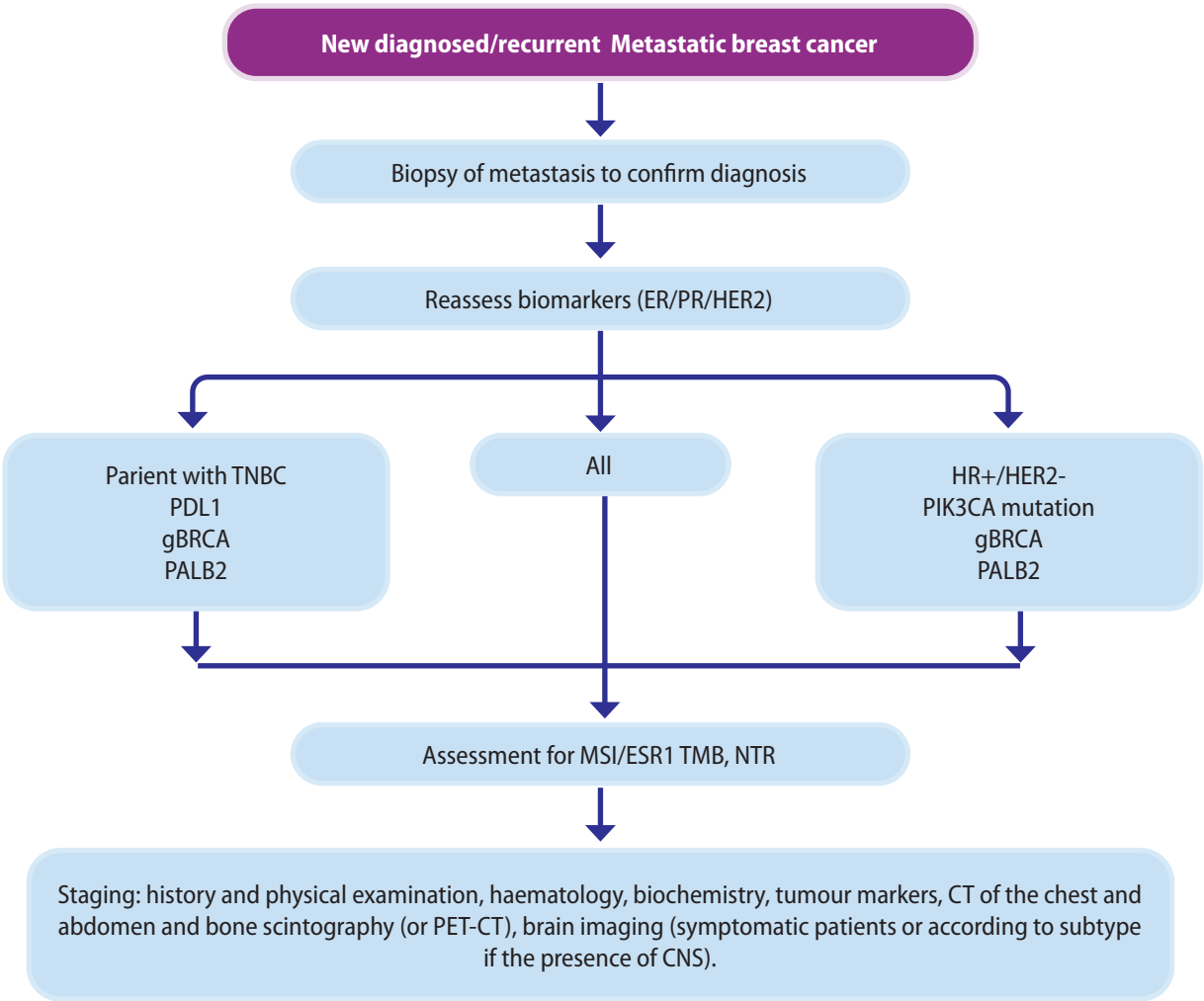


Figure 2.8: Diagnostic Approach To Metastatic Breast Cancer

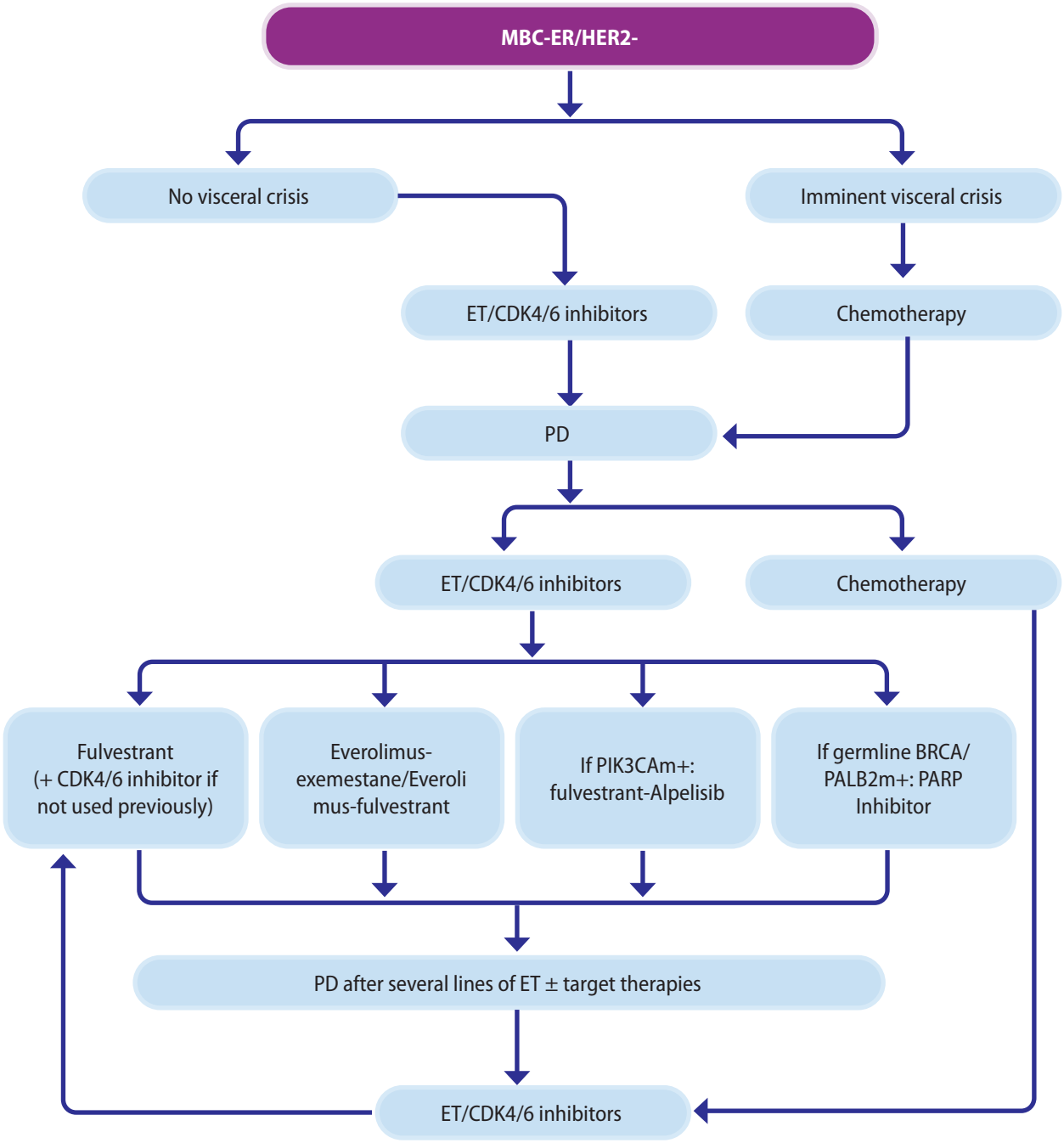


Figure 2.9: Treatment Approach for Metastatic HR+/HER2 negative

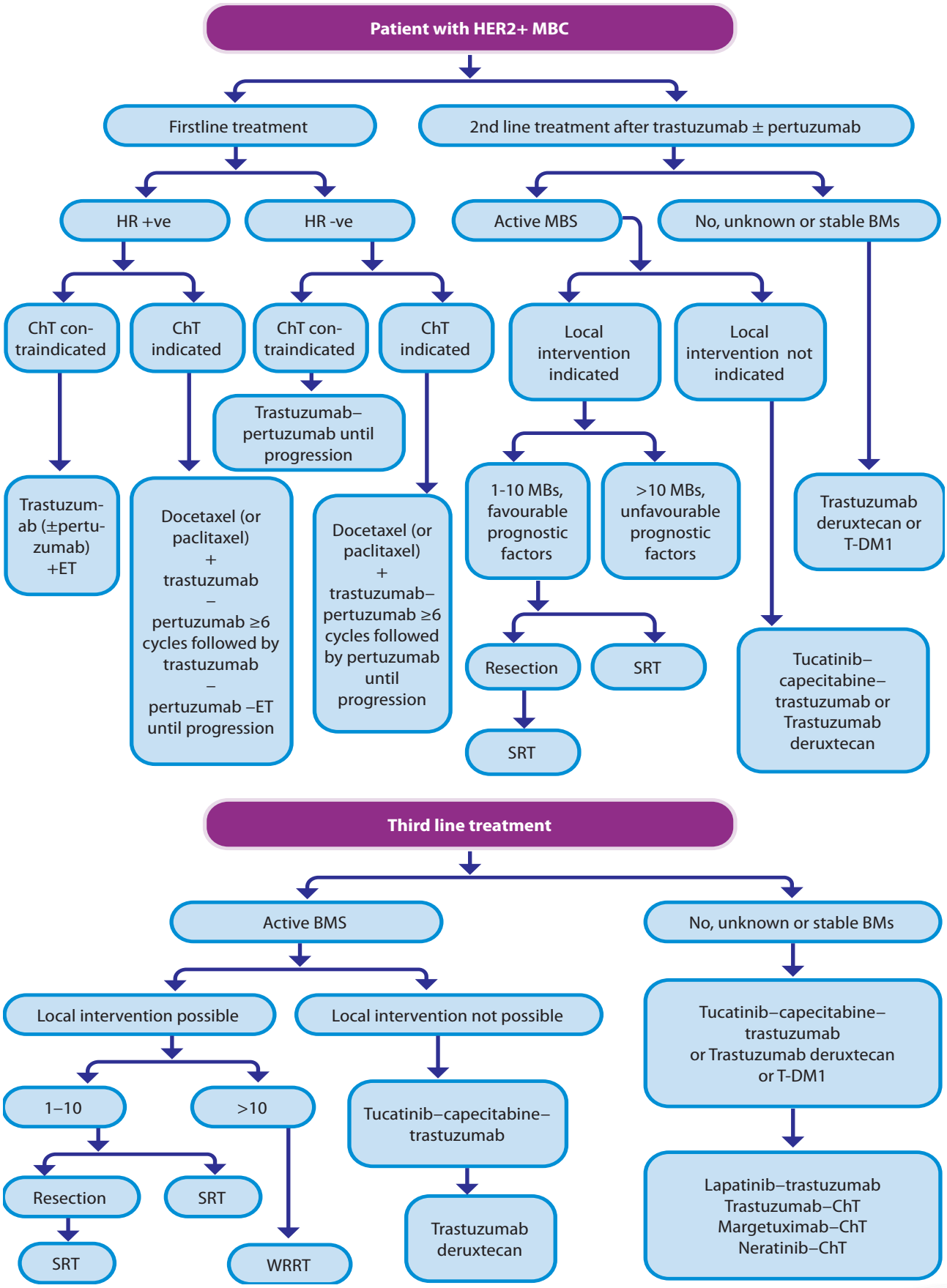


Figure 2.10: Treatment Approach for Metastatic HER2+ve

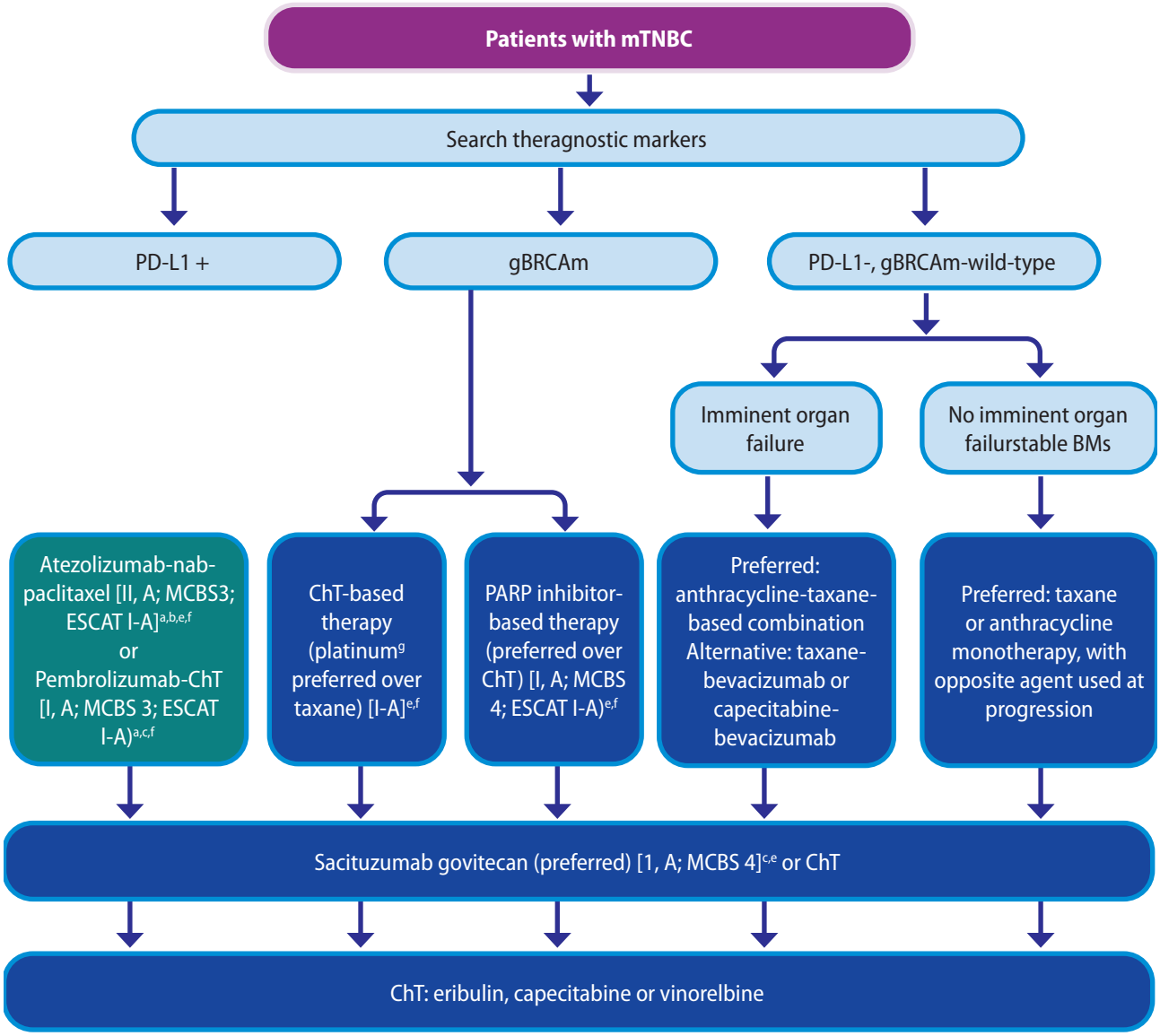


Figure 2.11: Treatment Approach for Metastatic TNBC
Adopted from ESMO clinical Practice guidelines

2.3.9 Chemotherapy

Adjuvant chemotherapy should be started as soon as clinically possible, usually within 6 weeks of completion of surgery. Choice of the regime should be left to the oncologist dependent on the patient characteristics, tumour characteristics, aim of treatment and system/social-eco- nomic characteristics, among others.

2.3.10 Chemotherapy Regimes

Table 2.5: Systemic Therapy For Locoregional Recurrent Unresectable Or Stage IV (M1) Disease

HR-Positive and HER2-Negative with Visceral Crisis or Endocrine Refractory		
Setting	Subtype/Biomarker	Regimen
First line	No germline **BRCA1/2** mutation and/or HER2 IHC 0+, 1+, or 2+/ISH negative	Systemic chemotherapy, or fam-trastuzumab deruxtecan
	Germline **BRCA1/2** mutation	PARPi (olaparib, talazoparib)
Second line	HER2 IHC 0+, 1+, or 2+/ISH negative	Fam-trastuzumab deruxtecan
	Not a candidate for fam-trastuzumab deruxtecan	Sacituzumab govitecan
		Systemic chemotherapy
		Targeted therapy
		For HER2 IHC 0, 1+, or 2+/ISH negative Datopotamab deruxtecan
Third line	Any	Systemic chemotherapy
	Biomarker positive (ie, MSI-H, NTRK, RET, TMB-H)	Targeted agents and emerging biomarker options

Table 2.6: Systemic Therapy For Locoregional Recurrent Unresectable Or Stage IV (M1) Disease

HR-Negative and HER2-Negative (Triple-Negative Breast Cancer; TNBC)		
Setting	Subtype/Biomarker	Regimen
First line	PD-L1 CPS ≥10 regardless of germline BRCA mutation status	Pembrolizumab + chemotherapy (albumin-bound paclitaxel, paclitaxel, or gemcitabine and carboplatin)
	PD-L1 CPS <10 and no germline BRCA1/2 mutation	Systemic chemotherapy
	PD-L1 CPS <10 and germline	● PARPi (olaparib, talazoparib)
	BRCA1/2 mutation	● Platinum (cisplatin or carboplatin)
Second line	Germline BRCA1/2 mutation	PARPi (olaparib, talazoparib)
	Any	Sacituzumab govitecan
		Systemic chemotherapy or targeted therapy
	No germline BRCA1/2 mutation and HER2 IHC 0+, 1+, or 2+/ISH negative	Fam-trastuzumab deruxtecan
Third line	Any	Systemic chemotherapy
	Biomarker positive (i.e., MSI-H, NTRK, RET, TMB-H)	Targeted agents and emerging biomarker options

Table 2.7: Systemic Therapy For Locoregional Recurrent Unresectable Or Stage IV (M1) Disease

HR-Positive or -Negative and HER2-Positive	
Setting	Regimen
First line	Pertuzumab + trastuzumab + docetaxel
	Pertuzumab + trastuzumab + paclitaxel
Second line	Fam-trastuzumab deruxtecan
Third line	Tucatinib + trastuzumab + capecitabine
	Ado-trastuzumab emtansine (T-DM1)
Fourth Line and Beyond (Optimal sequence is not known)	Trastuzumab + docetaxel or vinorelbine
	Trastuzumab + paclitaxel ± carboplatin
	Capecitabine + trastuzumab or lapatinib
	Trastuzumab + lapatinib (without cytotoxic therapy
	Trastuzumab + other chemotherapy agents
	Neratinib + capecitabine
	Margetuximab + chemotherapy (capecitabine, eribulin, gemcitabine, or vinorelbine)
	Abemaciclib in combination with fulvestrant and trastuzumab (for HR+ only)
	Targeted Therapy and emerging biomarker Options

Table 2.8: Recurrent Unresectable (Local Or Regional) Or Stage IV (M1) Disease

Systemic Chemotherapy for HR-Positive or -Negative and HER2-Negative		
Preferred Regimens	Other Recommended Regimens	Useful in Certain Circumstances
• Anthracyclines • Doxorubicin • Liposomal doxorubicin • Taxanes • Paclitaxel • Anti-metabolites Capecitabine Gemcitabine • Microtubule inhibitors • Vinorelbine • Eribulin	• Cyclophosphamide • Docetaxel • Albumin-bound paclitaxel • Epirubicin • Ixabepilone	• AC (doxorubicin/ cyclophosphamide) • EC (epirubicin/ cyclophosphamide) • CMF (cyclophosphamide/ methotrexate/fluorouracil) • Docetaxel/capecitabine • GT (gemcitabine/paclitaxel) • Gemcitabine/carboplatin • Carboplatin + paclitaxel or albumin-bound paclitaxel

Table 2.9: Targeted Therapies And Associated Biomarker Testing For Recurrent Unresectable (Local Or Regional) Or Stage IV (M1) Disease

Breast subtype	Biomarker	Detection	Agent
HR-positive, HER2-negative	PIK3CA activating mutation	NGS/PCR	Inavolisib + palbociclib + fulvestrant
			Alpelisib + fulvestrant
	PIK3CA or AKT1mactivating mutations or PTEN alterations		Capivasertib + fulvestrant
	ESR1 mutation		Elacestrant
Any	Germline BRCA1 or BRCA2 mutation	Germline sequence	Olaparib, Talazoparib
	Germline PALB2		Olaparib
	NTRK fusion	FISH, NGS, PCR	Larotrectinib, Entrectinib, Repotrectinib
	MSI-H/dMMR	IHC, NGS, PCR	Pembrolizumab, Dostarlimab
	TMB-H (≥10 mut/Mb)	NGS	Pembrolizumab
	RET-fusion		Selpercatinib

Table 2.10: Emerging Biomarkers And Novel Therapies For Patients With Stage IV (M1) Disease

Breast subtype	Emerging Biomarker	Detection	Agent
ER+/HER2- ER-/HER2-	HER2 activating Mutations	NGS/PCR	Neratinib ± fulvestrant
			Neratinib + trastuzumab/ fulvestrant
Any	Somatic BRCA1/2 Mutations	NGS	Olaparib
	FGFR1-3 fusion/ Mutation		Erdafitinib

2.3.11 Bone Resorptive Therapies

- Bisphosphonates (e.g. zolendronic acid) and denosumab in patients with skeletal metastases to reduce skeletal related events like compression or fragility fractures.
- Patients on bisphosphonates should be on calcium supplements with monthly monitoring of kidney function and calcium serum levels.
- Dosages should be adjusted according to the renal function.

2.3.12 Palliative Radiotherapy

Palliative Radiotherapy is given to patients with metastatic disease for symptomatic relief such as:

- **Bone metastasis:**
  - Radiotherapy along with bisphosphonates is the main stay of treatment for symptomatic bone metastasis, impending fracture or cord compression.
  - Established fractures in patients with more than 6 months of life expectancy should be fixed surgically and then given postoperative RT.

- **Brain metastasis:**
  - Most symptomatic patients would benefit from steroids and fractionated whole brain radiotherapy.
  - For single brain metastasis in patients with controlled primary and no other site of systemic disease, consider surgery or radiosurgery of the brain lesion followed by whole brain radiotherapy.

- **Spinal cord compression:**
  - Palliative radiotherapy with steroids is the mainstay of treatment.
  - Select cases with life expectancy of more than six months presenting with recent onset paraparesis or paraplegia and limited vertebral involvement should be considered for surgical decompression prior to palliative radiotherapy.

- Haemostatic Radiotherapy for bleeding wounds

Follow up and Surveillance

- Patients should be counselled on the diagnosis, treatment care plan and anticipated financial implications.
- The need for regular follow up for timely detection and salvage of recurrences, treatment of contralateral/metastatic disease and management of late sequelae of treatment including quality of life and survivorship plan should be emphasized.

History & Physical Exam

- Take a detailed history - metastatic symptoms, physical / psychosocial sequelae of treatment
- Perform systemic and local examination of the breast bed, axilla and contralateral breast
- Schedule visits every 4–6 months for the first 5 years after primary therapy
- Annually thereafter
- Patients with significant family history of cancer or known BRCA mutation should be kept on lifelong 6-monthly or annual follow up

Breast Imaging

- Annual mammography recommended
- First mammogram 6–12 months after completing breast-conserving therapy (BCT) with radiation
- No routine imaging of reconstructed breast
- MRI may be considered for high-risk patients (e.g., BRCA mutation, >20% lifetime risk)

Screening for Recurrence/Metastases

- Routine blood tests (e.g., alkaline phosphatase, LFTs) and tumour markers are not recommended
- No role for routine CT, MRI, PET, bone scans, or ultrasound in asymptomatic patients

Endocrine Therapy Monitoring

- Tamoxifen: yearly gynaecologic assessments; evaluate promptly for vaginal spotting (no routine biopsy/ultrasound)
- Aromatase inhibitors: monitor bone health with baseline and periodic bone mineral density testing; consider bisphosphonates/denosumab

Other Monitoring

- Cardiac monitoring for patients exposed to anthracyclines, HER2-targeted therapy, or left-sided chest radiation
- Lymphedema: patient education, monitoring, and referral as needed

Lifestyle & Support

- Encourage healthy lifestyle: regular activity, balanced diet, limited alcohol, smoking cessation, BMI 20–25.
- Screen for anxiety, depression, distress, and provide psychosocial support

Ensure coordination of care between oncology specialists and primary care, with survivorship plans in place

2.3.13 Supportive Care for Breast Cancer

1. **Lymphedema management**: often caused by the interruption of regional lymphatic drainage.

- Circumferential measurements of both extremities should be taken at the metacarpal-phalangeal joints, the wrists, 10 cm distal and 15 cm proximal to the lateral epicondyles at baseline and after treatment.
- A difference of 2 cm or greater at any point is clinically significant.
- Symptoms will include swelling on the same side of treatment, sensation of heaviness of the limb, fatigue, fullness/tightness of skin or pain.
- Risk factors for lymphedema include obesity (BMI> 30kg/m^2), localized infections and increased number of nodes removed.
- Lymphedema progresses through four stages.
  - Stage 0 indicates a normal extremity clinically, but with abnormal lymph transport (i.e., illustrated by lymphoscintigraphy).
  - Stage 1 is early edema, which improves with limb elevation.
  - Stage 2 represents pitting edema that does not resolve with elevation.
  - Stage 3 describes fibroadipose deposition and skin changes.

- Early detection is key because stages 0 and 1 are reversible.
- Options of care include external compression garments (such as lymphedema stockings), massage/ manual lymphatic drainage, elevation, exercise, psychosocial support and prompt treatment of infections.

2. **Breast reconstruction**:
   - Every patient undergoing mastectomy should be informed of the option of breast reconstruction for patients with early breast cancer or locally advanced breast cancer as appropriate.
   - For early breast cancer, there must be an option of breast conserving surgery.

3. **Consider Breast prostheses** for breast cancers survivors who have undergone mastectomy.
4. **Management of hormone-related symptoms:** menopause management in females by appropriate specialist.
5. **Symptom management** – pain, wound management, depression, anaemia, side effects of their medication, loss of appetite, vomiting, numbness.
6. **Psychological support** linkage with support groups.
7. **Sexuality and fertility support management** should be discussed and if needed patient sent to experts for management.
8. **Spiritual support**

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Chapter 3

Brain and Central Nervous System Tumors

List of abbreviations

Abbreviation	Definition
AIDS	Acquired Immune Deficiency Syndrome
CT	Computerized Tomography
CSI	Craniospinal Irradiation
GTR	Gross Tumor Resection
IGF-1	Insulin-like Growth Factor 1
MEN	Multiple Endocrine Neoplasia
MRI	Magnetic Resonance Imaging
MSR	Maximal Safe Resection
NF1/2	Neurofibromatosis Type 1 and 2
PNET	Primitive Neuroectodermal Tumor
RT	Radiotherapy
RPA	Recursive Partitioning Analysis
SRS	Stereotactic Radiosurgery
SPECT	Stereotactic Positron Emission Computed Tomography
PET	Positron Emission Tomography

3.1 Introduction

Brain tumors are heterogeneous and can either be primary or metastatic. The pathologic subtypes and presentation also vary depending on the age of the patient. Glioblastoma is the most common primary malignant brain tumor in adults. Brain metastases may arise in the setting of various other malignancies such as lung cancer, breast cancer, renal cancer or melanoma (etc.). Primary CNS tumors can arise in the brain or in the spine. Some primary tumors have potential to spread within the neural axis but they rarely spread to other areas of the body.

The annual global age-standardized incidence of primary malignant brain tumors is ~3.7/100,000 for males and 2.6/100,000 for females. Rates appear to be higher in more developed countries (males, 5.8 and females, 4.1/100,000) than in less developed countries (males 3.0 and females 2.1/100,000). In adults, two thirds of primary brain tumors arise from supratentorial region with gliomas, metastases, meningiomas, pituitary adenomas and acoustic neuro- mas accounting for 95% of all brain tumors. They account for approximately 2% of all cancer deaths.

3.1.1 Risk Factors

- Increasing age
- Male gender
- Familial syndromes such as NF1/2, von Hippel-Lindau disease, tuberous sclerosis, Li-Fraumeni, Turcot syndrome, nevoid basal cell carcinoma syndrome
- Environmental/occupational exposures - vinyl chloride
- Prior cranial irradiation
- Black ethnicity
- Immunosuppression due to AIDS is a well-recognized cause of cerebral lymphoma

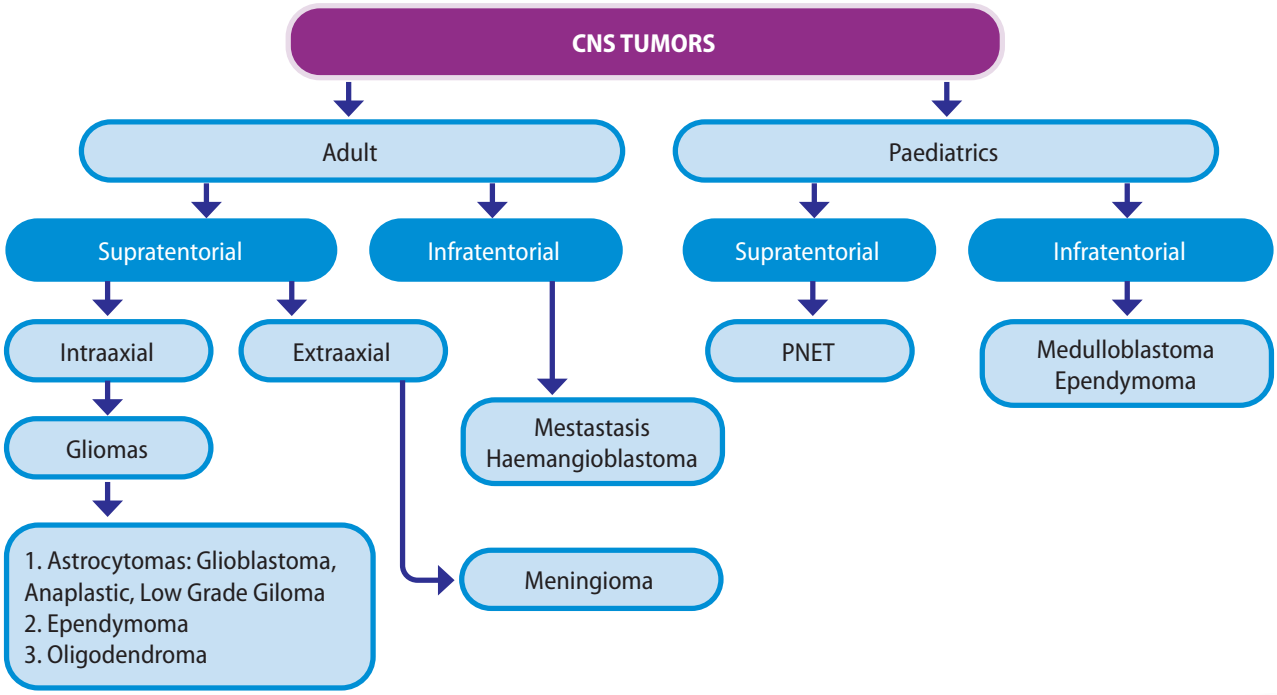


Figure 3.1: Common Types and Subtypes of CNS Tumors

3.1.2 Clinical Evaluation

The signs and symptoms of brain tumors are largely dependent on location and a high index of suspicion is invaluable for early diagnosis which ultimately impacts on patient outcomes. Clinical presentation varies depending on the location of tumor and the rate of growth.

Signs and Symptoms

Table 3.1: Signs and symptom of CNS tumors

Increased intracranial pressure	• Persistent headache • Vomiting • Visual blurring • Cushing's Triad (bradycardia, reduced respiration, systolic hypertension) in childhood tumors-hydrocephalus with papilledema, sunsetting eyes, increased head size, tense fontanelle
Brain Dysfunction	• Mental function deterioration • Memory disturbance • Personality changes • Seizures
Focal neurological deficits for supratentorial lesions	• Frontal lobe Personality changes, seizures, headache • Temporal lobe - Seizures, speech problems • Suprasellar - Endocrinopathies, visual acuity or field loss • Occipital- Cortical Visual Deficits • Thalamus - Motor sensory deficits • Focal neurological deficits for infratentorial lesions • Cerebellar Nystagmus, ataxia, hydrocephalus, resting tremor, incoordination • Brainstem - Cranial neuropathies, gait, swallowing incoordination, drooling, breathing dysfunction • Posterior fossa - nausea, vomiting, cerebellar signs, gait ataxia, incoordination

3.1.3 Diagnostic Evaluation

CNS tumors must be differentiated from other space occupying lesions that may have similar clinical presentation.

Diagnostic tools include:

1. Imaging tests: CT and MRI
   a. CT - used in initial evaluation of a symptomatic patient and can provide differential diagnosis and initiate immediate management
   b. MRI - superior soft tissue resolution
   c. Others - PET (and SPECT) - can help differentiate recurrence from radiation necrosis after treatment
2. Biopsy and histopathological evaluation

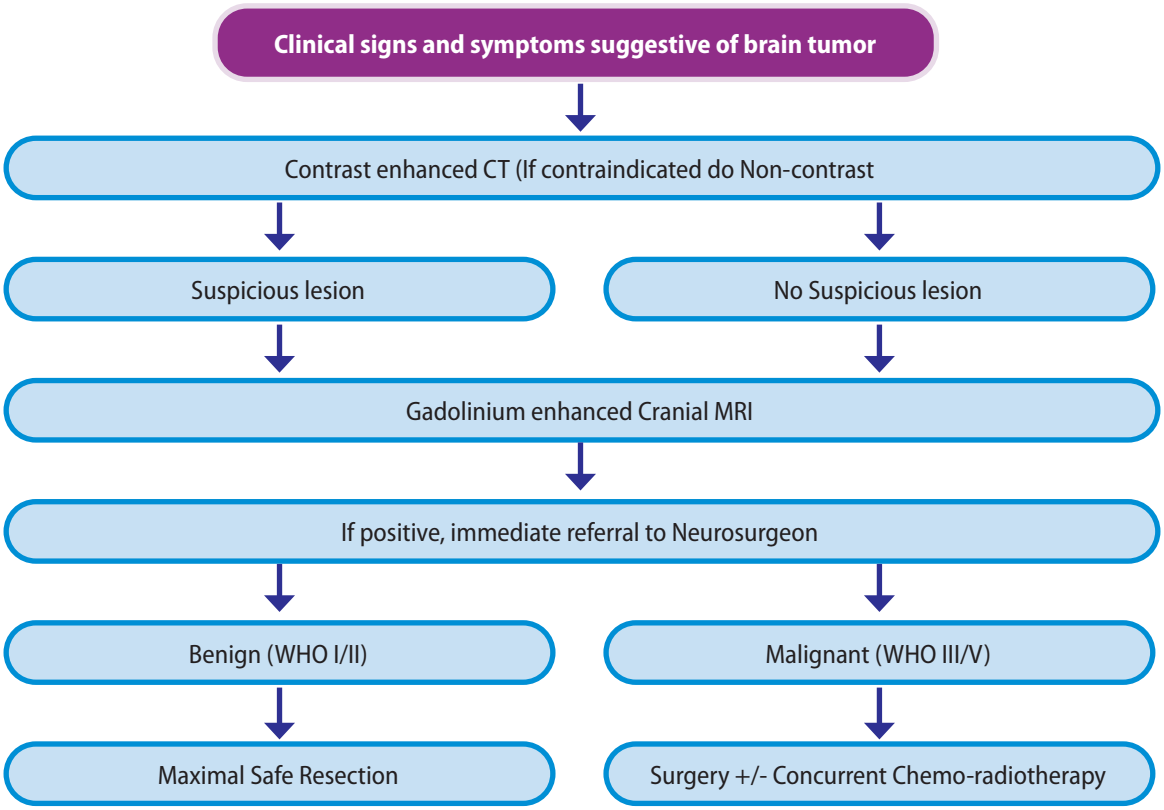


Figure 3.2: Diagnostic Imaging and Treatment Algorithm

Note: **MRI is the investigation of choice and is superior to CT scan due to better contrast resolution, further characterization and assessment of full extent of disease but if not accessible unenhanced computerized tomography (CT) scan is mandatory to evaluate**

3.1.4 Pathological Diagnosis

- Tumor tissue can be collected through biopsy or tumor resection.
- The 8th edition of the WHO 2021 Classification of Tumors of the CNS recommends that all suspicious brain lesions must be evaluated using relevant markers to include histopathologic and molecular information. Molecular information can be used to diagnose, in treatment planning and prognostication.
- Most CNS tumors are classified into 4 grades (i.e. WHO Grade I - IV) based on histopathologic presence or absence of atypia, endothelial proliferation, necrosis and mitotic index.

3.1.5 Management

CNS tumors should be managed in a multidisciplinary team setting.

The ability to perform a maximally safe resection is the most critical factor that determines the outcome in CNS tumors.

Decisions for surgery depend on factors such as: patient's age and performance status, location of the tumor, feasibility of decreasing mass effect with aggressive surgery, tumor resectability (number and location), recurrence vs primary disease.

Subsequent treatment is guided by the outcome of the surgery (subtotal vs gross total resection), the pathology, grade and molecular subtype of tumor, and patient factors such as age and performance status. The approach to treatment can be stratified as:

1. Initial (emergency) management of symptoms associated with the CNS tumor
2. Definitive management - which can consist of:

a. Surgery
b. Radiotherapy
c. Systemic therapy

Some tumors can spread within the CSF and that should be factored in the workup and treatment (pineoblastoma, medulloblastoma, PNET, CNS lymphoma, germ cell tumors, ATRT).

3.1.6 Indications for Urgent Referral of a Patient with a CNS Tumor

- Mass lesion on CT scan or suspected mass in a patient with headache, drowsiness, neurologic deficits or vomiting;
- Tumors associated with midline shift, hydrocephalus or severe deficits;
- Cauda equina syndrome;
- Gradual loss of movement/sensation in an arm or leg;
- Unsteadiness or imbalance, especially if associated with headache;
- A new epileptic seizure, intractable seizure;
- Loss of vision in one or both eyes (especially if more peripheral), double vision (especially if associated with headache);
- Hearing loss with or without dizziness;
- Speech difficulty of gradual onset;
- Nausea and vomiting, more severe in the morning.

The referring health care provider should inform the health care provider at the receiving facility about the referral and write a comprehensive medical report to accompany the patient including all scans done.

3.1.7 Immediate Management of Patients with CNS Tumors

- Raised intracranial pressure and cerebral edema - Dexamethasone 4-8mg TDS P.O/I. V with a proton pump inhibitor for gastric protection.
- Seizures - Levetiracetam
  NB: Prophylactic antiepileptic therapy is not recommended

3.1.8 Definitive Management of CNS Tumors

Treatment options include:

1. Surgery - gross total resection (maximal safe resection) is the goal

a. Post-op imaging with MRI - ideally within 48hrs after surgery

2. Radiotherapy - dose depends on tumor type

a. Adjuvant - for subtotal resection in lower grade tumors and for all high-grade tumors
b. Definitive - when surgery is not feasible
c. Can be via standard fractionated or hypofractionated EBRT or stereotactic RT for select cases

3. Systemic therapy
a. Chemotherapy - alkylating agents are the most effective for CNS tumors
b. Targeted therapy - e.g bevacizumab, BRAF/NTRK targeting
    i. Brain metastasis - other specific drugs depending on primary tumor biology

3.2 Low Grade Glioma (WHO Grade 1-2)

- Grade 1 gliomas are curable by surgery alone. This is through maximum safe resection; followed by post-op MRI within 48hrs (up to 72hrs).
- Molecular testing - IDH1/2, ATRX + TERT, 1p/19q codeletion, BRAF mutation are done on the specimen thus obtained.

3.2.1 Current Risk Stratification and Treatment Recommendation (NCCN):

Traditional poor prognostic factors in grade 2 gliomas include:

- Age >40yrs
- Tumor crossing the midline
- Tumor >6cm
- Neurologic deficits prior to surgery
- Astrocytoma histology (survival is better in oligodendroglioma)
- Subtotal resection

Treatment based on risk stratification

- Low risk - age <40yrs and gross total resection
- Surgery then close follow-up
- High risk - age >40yrs or subtotal resection
- Adjuvant radiotherapy (45 - 54Gy) then chemotherapy (PCV) or temozolomide

Adjuvant radiotherapy is indicated for patients with poor prognosis

3.2.2 Recurrent or Progressive Disease

- consider re-excision,
- RT if not previously given (consider biopsy prior),
- re-irradiation,
- systemic therapy or
- best supportive care.

Low Grade tumors may eventually evolve into higher grade tumors

3.3 High-Grade Glioma (WHO Grade 3 and 4)

3.3.1 Management of High-Grade Glioma (WHO Grade 3 and 4)

1. Neurosurgical review for maximum safe resection (extent of resection impacts outcomes). Unfortunately, nearly all high-grade gliomas recur.

a. Histopathological characterization and molecular testing - IDH1/2, MGMT, 1p19q codeletion

2. Adjuvant radiotherapy and chemotherapy is recommended after surgery for all patients with high grade glioma

- For **Grade 3 oligodendroglioma** and good performance status (KPS≥60) radiotherapy followed by PCV chemotherapy is an option (where feasible)
- For **WHO grade 3 or 4 IDH-mutant astrocytoma** and good performance status (KPS ≥60), radiotherapy with or without concurrent temozolomide then 12 cycles of adjuvant temozolomide are recommended:
- For **Glioblastoma** - treatment approach depends on patient's age, performance status (KPS) and MGMT status:
  - Age ≤70yrs and good KPS - standard brain radiotherapy with concurrent and adjuvant temozolomide (+/- alternating electric field therapy (AEF), where available)
  - Age >70yrs and good KPS with MGMT promoter methylation treatment approach can be:
    - Hypofractionated brain radiotherapy + concurrent and adjuvant temozolomide
    - Standard brain radiotherapy with concurrent and adjuvant temozolomide (+ AEF)
    - Age >70yrs and good KPS with MGMT-unmethylated (or indeterminate)
      - Hypofractionated RT (+ temozolomide) approach is preferred
      - Standard RT + temozolomide approach, based on patient preference after adequate information is provided

    - Patients with poor KPS - options include:
      - Hypofractionated brain RT (+/- temozolomide, based on the clinical scenario)
      - Temozolomide alone
      - Best supportive care

- Radiotherapy and chemotherapy dosage for high grade glioma
  - Standard brain RT - 60Gy in 30 daily fractions (or 59.4Gy @ 1.8Gy per fraction) over 6 weeks
  - Concurrent temozolomide 75mg/m2 during radiotherapy
  - Adjuvant temozolomide 150-200mg/m2 for 6 months (start 1 month after completion of radiotherapy)
  - Hypofractionated brain RT - 40Gy in 15 fractions, 34Gy in 10 fractions, or 25Gy in 5 fractions

Patient follow-up after treatment entails serial brain MRI scans starting 2-8 weeks after radiation then every 2-4 months for 3years then every 3-6 months

- **Pseudoprogression** - scans may appear worse during the first 3 or so months after RT (especially if tumor was MGMT methylated)
- May require close follow-up with repeat MRI, steroids, biopsy, MR spectroscopy/perfusion or brain PET-CT to rule out necrosis vs progression (vs pseudoprogression)
  - RT-induced necrosis is detected between 6-24 months after irradiation

- **Pseudoresponse** may be reported in some patients e.g while on bevacizumab due to reduced vascular permeability

Tumor recurrence - treatment approach depends on extent of disease and patient condition. Treatment efficacy after recurrence is minimal.

- Assess for resectability/operability
- Consider re-irradiation - SRS for low volume disease, IMRT (35Gy/10#)
- Systemic therapy - temozolomide (if long interval since prior treatment), carmustine/lomustine, bevacizumab, regorafenib, PCV
- Best supportive care
- Enrolment in clinical trial

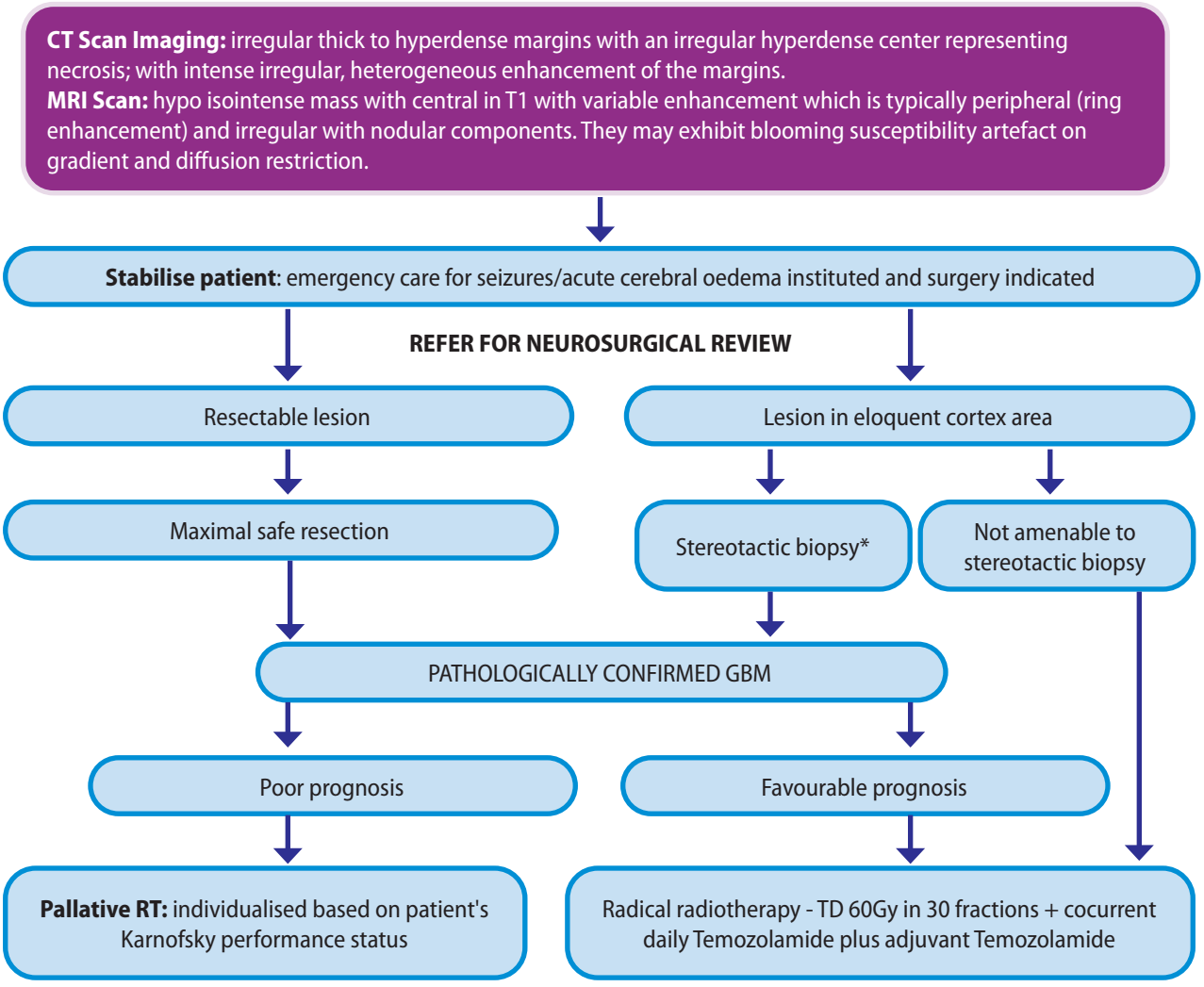


Figure 3.3: Algorithm for Management of Glioblastoma

3.4 Ependymomas - Intracranial and Spinal

- These constitute <2% of CNS tumors. The site of occurrence varies with age; in adults they occur most commonly in the spinal canal than intracranially.
- Imaging should include the entire neuraxis (brain and spine) to rule out drop metastases. CSF dissemination occurs in about 15% of intracranial ependymomas. Lumbar CSF cytology is done after spinal MRI (LP is contraindicated in patients with raised ICP/risk of herniation).
- Supratentorial ependymomas (most are high grade) have poorer prognosis than infratentorial ependymomas
- Molecular markers - ZFTA fusions/RELA, YAP1, MYCN-amplification, H3K27

3.4.1 Treatment Approach:

1. Surgery - gross total resection (maximal safe resection) improves survival
   a. Repeat MRI within 48hrs after surgery; if spinal MRI not done prior to surgery then it should be done 2-3 weeks after surgery
   b. Grade 1 subependymomas can be cured by surgery alone
   c. For most ependymomas local recurrence is the primary pattern of failure and CSF seeding is uncommon in the non-recurrent setting

2. Adjuvant therapy depends on: extent of resection, histology (and grade), staging results after craniospinal MRI and CSF cytology
   a. Radiation therapy - for intracranial ependymoma, myxopapillary ependymoma after subtotal resection or capsule violation
      i. No RT for totally resected grade 1 spinal ependymoma
      ii. Craniospinal RT - if CSF dissemination present
      iii. NB - prophylactic craniospinal irradiation (CSI) or whole brain RT does not improve survival or prevent spinal metastases.

3. Follow-up after treatment - depends on tumor grade, extent and location
   a. Localized disease - brain and spine MRI (+contrast) 2-3weeks post-operatively then 3-4monthly for one year then 6 monthly subsequently
   b. If recurrence - resection, RT, CSI, clinical trial, systemic therapy or best supportive care

3.5 Adult Medulloblastoma

- Medulloblastoma is rare in adults (is the most common brain tumor in children).
- WHO now defines medulloblastoma histopathologically and molecularly (WNT, SHH - TP53 wild type or TP53 mutant, non-WNT/non-SHH).
- Adult medulloblastoma is genomically different from pediatric medulloblastoma.
- Metastatic disease is less common in adults.
- MRI is the gold standard for assessment.
- Staging uses the modified Chang system

3.5.1 Treatment Approach:

1. Surgery - maximal safe tumor resection +/- placement of ventriculoperitoneal shunt for hydrocephalus
2. Adjuvant therapy - approach depends on the recurrence risk stratification

a. High risk - large cell medulloblastoma, disseminated disease, unresectable tumor, residual tumor >1.5cm^2 - Craniospinal RT and systemic therapy

b. Average risk - CSI +/- systemic therapy or reduced dose CSI + systemic therapy

3. Radiation therapy entails CSI and posterior fossa boost

a. 30-36Gy CSI ,then boost the primary site to 54-55.8Gy.

b. Lower dose entails CSI to 23.4Gy, then boost primary site to 54-55.8Gy.

4. Systemic therapy - optimal adjuvant chemotherapy for adult patients is unclear. Options include temozolomide, vismodegib or sonidegib for SHH-activated tumors (vincristine - severe toxicity in adults thus often avoided)High dose chemotherapy with autologous stem cell rescue if recurrence.

Follow-up - no optimal approach. Brain MRI every 3-6 months

3.5.2 Recurrent Disease/ Progression

- Surgery is indicated in selected patients to relieve symptoms, improve performance status and quality of life.
- Repeat radiotherapy may be considered, depending on size of lesion, previous dose and the interval since the last radiotherapy treatment.
- Chemotherapy may also be considered ; CNS active agents include Carmustine, Vincristine, Temozolomide, Irinotecan and Bevacizumab.
- Repeat scans should be done after 2-3 cycles to determine response.

3.5.3 Follow Up

A baseline cranial CT scan should be done at 4 months post-radiotherapy as a reference. Thereafter scans are usually done at 6 months and then annually, or if clinically indicated

3.6 Pituitary Gland Tumors

They include adenomas, carcinomas and other rarer types which constitute 10% of intracranial malignancies. They rarely metastasize but are frequently locally invasive. Morbidity is due to their mass effect as well as endocrine consequences. Etiology is unknown, but may rarely be associated with multiple endocrine neoplasia (MEN) type 1 syndrome.

3.6.1 Diagnosis

- Pituitary function: All patients should have baseline pituitary function measured which includes: Thyroid function tests, serum prolactin levels, cortisol levels, growth hormone levels and IGF-1 levels.
- Ophthalmology evaluation –Visual Acuity and Visual Field assessment should be done pre-operatively.
- Imaging as indicated above.

3.6.2 Management

- Surgery is the primary treatment for most pituitary tumors (except prolactinomas which may be managed medically).
- MRI is done 3 months after surgery to assess for residual disease.
- Radiotherapy is indicated for sub totally resected tumors, recurrent tumors, patients with persistently elevated circulating hormone levels, and medically inoperable patients. It is very effective for control of growth of pituitary tumors (>95%), but is less effective for decreasing circulating hormone levels of endocrinologically active tumors whose control may take years to achieve after radiotherapy. Consultation with the endocrinologist is therefore mandatory.

3.7 Meningiomas

They constitute 20% of all intracranial tumors with a male to female ratio of 1:2. Peak incidence occurs in the 7th decade. Possible etiologies include previous exposure to ionizing radiation, trauma, viral infections and exposure to sex hormones (approx. 75% are Progesterone Receptor positive). They are associated with neurofibromatosis - Type 2 and breast cancer.

3.7.1 Imaging

It is typically a well circumscribed, smooth contoured tumor and usually has homogeneously increased density on unenhanced CT and moderate to intense enhancement with contrast. 60% may have surrounding edema. Frequently (15-20%) associated with bony destruction/distortion or hyperostosis, but this usually does NOT represent bony invasion. It is often associated with linear meningeal thickening ("dural tail"), which frequently represents reactive change but may represent spread along the meningeal plane. Imaging may give an indication as to the grade of the meningioma with the lesions with predominant surrounding edema being higher grade.

3.7.2 Surgery

Both intracranial and spinal meningiomas are best managed with total excision if possibly achievable with acceptable morbidity. About a third are NOT fully resectable because of location, size and proximity to critical structures.

3.7.3 Radiotherapy

Post-operative radiotherapy significantly improves survival rates and is standard of care.

The dose of radiotherapy given is dependent on the grade of meningioma, extent of resection +/- recurrent vs primary disease and tumor location (proximity to dose limiting structures).
External beam radiotherapy is used for treatment with few indications for the use of stereotactic radiosurgery in small tumors located next to eloquent areas.

3.8 Brain Metastases

They are the commonest diagnosed brain tumors in adults. Tumors that commonly present with brain metastasis include, but are not limited to lung cancer, malignant melanoma, breast and prostate cancer.

The approach to management must factor in the type of primary tumor, prior treatments offered, extent of intracranial and intracranial disease, patient factors such as age and performance status and preferences, available treatment expertise/technology and collaboration between various multidisciplinary teams.

3.8.1 Definitive Management

1. Surgery can be offered to resect solitary brain metastasis or multiple brain metastases (where feasible) with a controlled primary tumor.
2. Radiotherapy: can be offered after surgery or as the primary treatment
   a. Stereotactic RT.
   b. EBRT - Whole brain radiotherapy can be offered but this results in significant neurocognitive decline that needs to be discussed with the patient before the treatment. Hippocampal sparing (with or without memantine) radiotherapy can be offered (where available and feasible) for select patients based on RTOG 0614/ NRGCC001 protocols

3. Systemic therapy - for definitive treatment of the primary tumor. Consider use of 'small molecules' such as tyrosine kinase inhibitors which may have better CNS penetration where applicable (guided by the primary tumor biology)

Table 3.2: Recursive Partitioning Analysis (RPA) Classification for Brain Metastases

RPA Class	Criteria	Median Survival (months)
Class I	● KPS ≥ 70 ● Age < 65 years ● Controlled primary tumor ● No extracranial metastases	~7.1
Class II	● KPS ≥ 70 ● Does not meet all Class I criteria (age ≥ 65, uncontrolled primary, or extracranial disease)	~4.2
Class III	● KPS < 70 (regardless of other factors)	~2.3

Adopted from the Radiation Therapy Oncology Group (RTOG) criteria (Gaspar et al., IJROBP, 1997)

3.8.2 Interpretation:

- **KPS** is the strongest prognostic factor.
- **Class I** patients are the best candidates for aggressive local therapy (e.g., surgery or SRS).
- **Class III** generally benefit from palliative or supportive care due to poor prognosis.

Table 3.3 Management Based On Recursive Partition Analysis Class Of Brain Metastasis

RPA Class	Surgery	SRS	Whole Brain Radiotherapy
I	Resection	Post resection/Unresectable	Discuss post resection
II	Consider	If amenable	Offer
III	Minimal role	No role	Offer

3.9 Primary Spinal Tumors

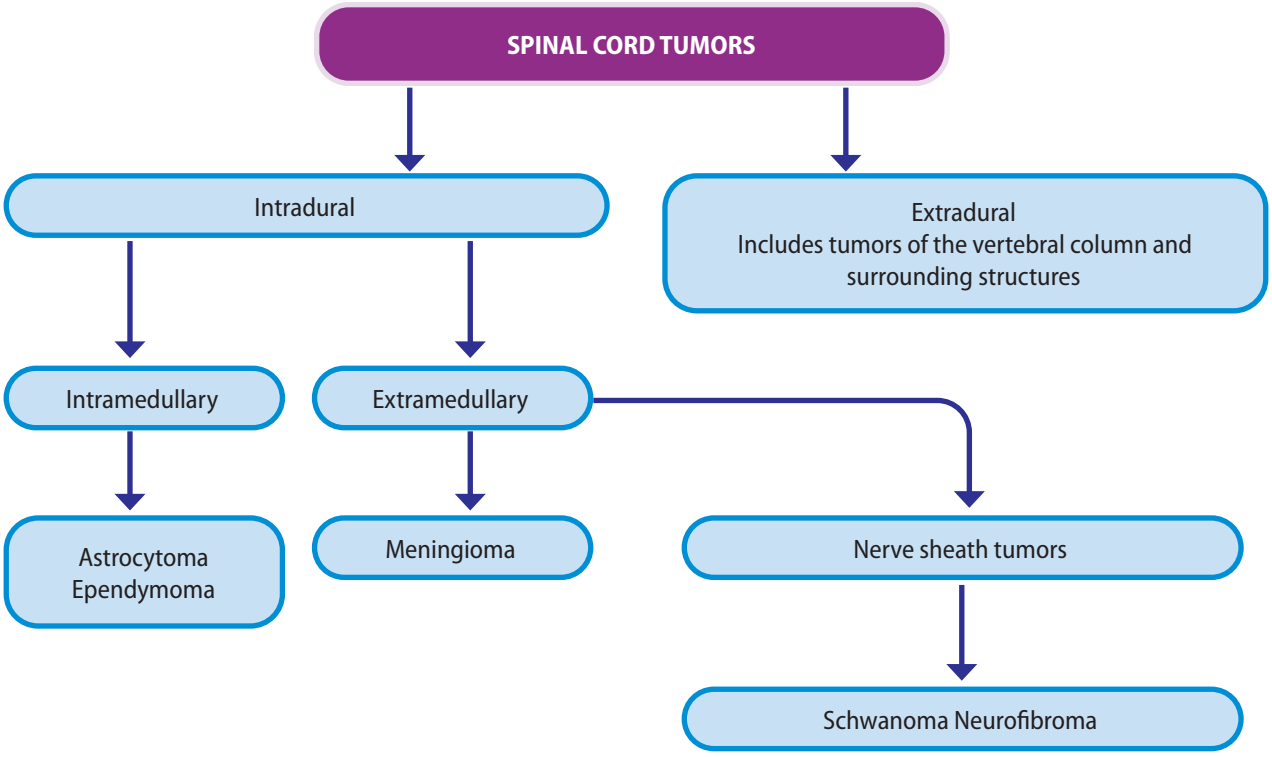


Figure 3.4: Flow diagram illustrating of diagnosis of spinal cord tumors

70% of primary spinal cord tumors are low grade and slow growing so patients **may suffer for months to years with symptoms of back pain** before a diagnosis is made. Patients with Neurofibromatosis type I, II and Von Hippel-Lindau syndrome are more susceptible to developing particular spinal tumors especially-astrocytomas, peripheral nerve sheath tumors, ependymomas and intramedullary haemangioblastomas.

3.9.1 Clinical Presentation

Local symptoms corresponding to level of lesion:

- Pain
- Segmental/ nerve root weakness
- Sensory deficit (dermatomal)
- Distal symptoms corresponding to long tract involvement:
- Paresis (diffuse)
- Sensory deficits
- Autonomic dysfunction

3.9.2 Management of Spinal Cord Tumors

1. Observation - for asymptomatic patients especially grade I lesions without immediate intervention.
2. Surgery – aim for maximal safe resection) if the patient is symptomatic and the lesion is amenable to surgery.
3. Radiotherapy – can be considered based on the tumor biology, grade and extent of resection and MDT recommendation
4. Systemic therapy – guided by tumor biology and MDT

3.10 Spinal Cord Metastasis

These account for 95% of extradural tumors and are mostly bone metastasis commonly from breast, prostate, lung and renal cancers. Patients can be asymptomatic or present with pain.

Three types of pain patterns have been described;

1. **Local pain** – this is due to tumor growth, presents as a deep ache that improves on steroid use
2. **Mechanical back pain**- varies with movement and spinal position is due to Spinal column instability
3. **Radicular pain** – sharp and stabbing pain. Due to nerve root compression. Spinal cord compression accounts for 5-10% of presentation in oncology patients and can be debilitating especially with prolonged symptoms.

3.10.1 Imaging

MRI of Entire spine is mandatory as multiple areas on the spine that may be involved.

3.10.2 Management of Spinal Cord Metastasis

Palliative therapy is mostly instituted and depends on performance status, duration of cord compression, site(s) of involvement, structural integrity of the spine and available resources.

Treatment options include:

a. **Surgery** –for patients with a life expectancy of >three (3) months with solitary Metastasis and for radio-resistant tumors
b. **Spine stabilization** – Bone cement (Vertebroplasty) or balloon (kyphoplasty).
c. **Radiotherapy** – most commonly used mode of therapy.
d. **Systemic therapy** – use of chemotherapy or hormone therapy to manage the systemic disease in combination with bone targeted therapies including bisphosphonates, RANK ligand antibodies or targeted radiotherapy agents (Radium 223)

3.11 Supportive and Rehabilitative Care for Brain Tumors

Palliative care is an often overlooked and underutilized part of comprehensive care for brain tumor patients that should be integrated early after diagnosis for both primary and secondary brain cancers. Despite their differences in management, both primary and secondary brain tumor patients experience similar symptom morbidity that impacts their quality of life.

- Rehabilitation provides patients with a chance to achieve optimal functional capacity within the limits of the disease. By setting realistic goals, they can have a better sense of control and reduce their dependency on others.
- Rehabilitation teams can consist of a physical therapist, occupational therapist, speech therapist, social worker, nurse, dietitian, and psychologist. Cancer rehabilitation goals are not universal but should be set according to the prognosis of each patient.
- Supportive rehabilitation attempts to help patients adapt to permanent functional deficits caused by cancer and to maximize their autonomy.

After brain tumor resection, for instance, patients may have cognitive deficits for which they can be taught to compensate with therapy.

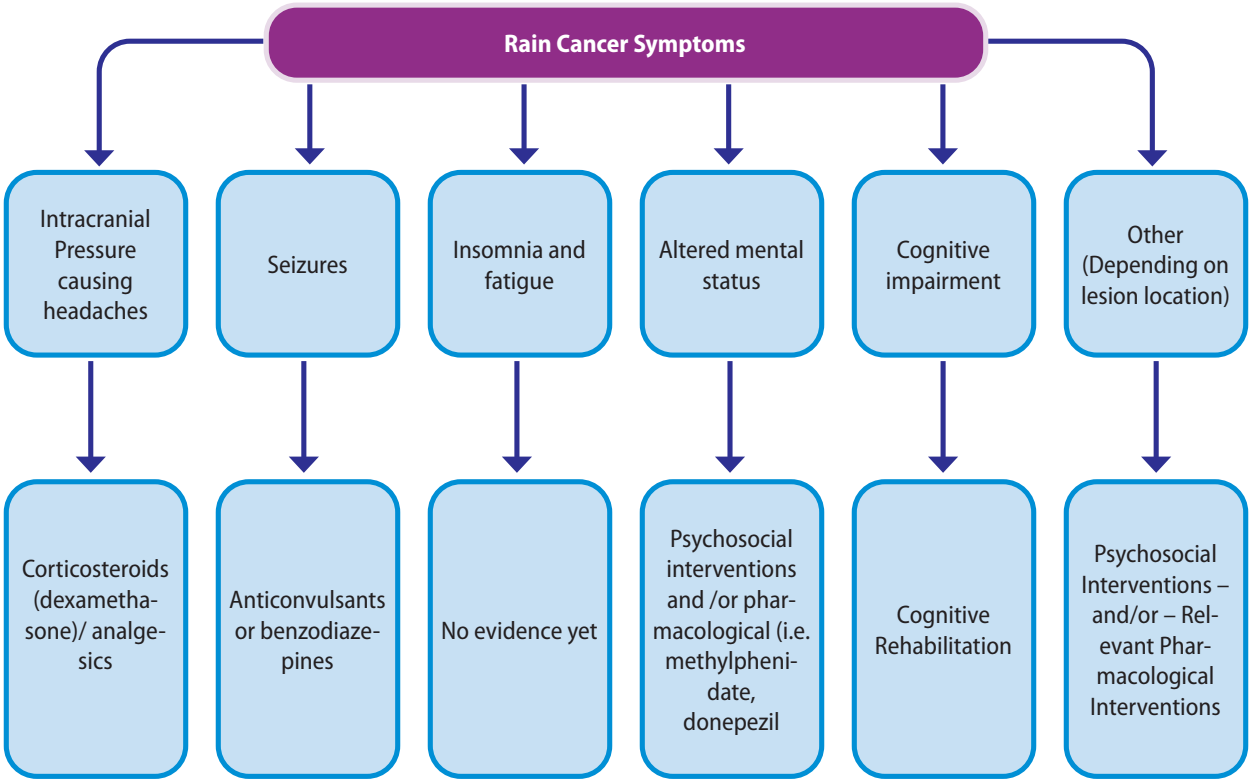


Figure 3.5: Palliative Care Recommendations Concerning the Treatment of Brain Cancer Symptoms
Adopted from the EANO Guidelines for Palliative Care in Adults with Glioma

3.12 Role of facility levels in the care of patients with brain tumors

Table 3.4: Role of facility levels in the care of patients with brain tumors

Level of Health Care	Anticipated Cancer Services
Level 1	Clinical symptoms and signs, refer to County referral hospital, Rehabilitative post-treatment
Level 2,3 and 4	Clinical symptoms and signs, refer to County referral hospital, Rehabilitative post-treatment
Level 5	CT Scan, MRI. Possible tissue biopsy/debulking if Neurosurgical and ICU services available, pathology services available. Refer to Tertiary level for further management
Level 6	All the above plus Comprehensive cancer control services; includes all treatment modalities: radiotherapy, surgery; chemotherapy.

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Chapter 4

Gastrointestinal Cancers

List of Abbreviations

Abbreviation	Definition
BRCA2	Breast Cancer Type 2 Susceptibility Gene
CEA	Carcinoembryonic Antigen
CRT	Chemoradiation
DRE	Digital Rectal Examination
dMMR	Deficient Mismatch Repair
EC	Esophageal Cancer
ECX	Epirubicin / Cisplatin / Capecitabine
EOX	Epirubicin / Oxaliplatin / Capecitabine
ER	Endoscopic Resection
ERCP	Endoscopic Retrograde Cholangio-Pancreatography
FLOT	Fluorouracil / Leucovorin / Oxaliplatin / Docetaxel
FOLFOX	Fluorouracil / Leucovorin / Oxaliplatin
HBV	Hepatitis B Virus

Abbreviation	Definition
HCV	Hepatitis C Virus
MMR	Mismatch Repair
MRCP	Magnetic Resonance Cholangio-Pancreatography
MSI H	Microsatellite Instability High
NACT	Neoadjuvant Chemotherapy
PS	Performance Status
RFA	Radiofrequency Ablation
AC	Adenocarcinoma
SCC	Squamous Cell Carcinoma
SEMS	Self-Expanding Metallic Stents
WHO	World Health Organisation
HER2	Human Epidermal Growth Factor Receptor 2
TNT	Total Neoadjuvant Therapy
TOF	Tracheoesophageal Fistula
GIT	Gastro-Intestinal Tract
CPS	Combined Positive Score

4.1 Esophageal Cancer

4.1.1 Introduction

Kenya has one of the highest incidence rates of esophageal cancer (EC) in Africa. There are approximately 3,340 new cases and 3,132 deaths annually. It is the 3rd leading cause of cancer mortality in Kenya (GLOBOCAN, 2022). There are two main histological subtypes – squamous cell carcinoma (SCC) and adenocarcinoma (AC). Squamous cell carcinoma (SCC) accounts for 90% of EC cases in Kenya.

4.1.2 Risk Factors

- Alcohol intake
- Family history of esophageal cancer
- Age: highest risk is from 45-70years
- Male
- Smoking history
- Overweight and obesity
- Carbon exposure from firewood and other sources
- Barrett's esophagus

4.1.3 Signs and Symptoms

- Dysphagia (difficulty in swallowing)
- Odynophagia (painful swallowing)
- GIT bleeding with haematemesis or melaena stools
- Recurrent aspiration in case TOF
- Post-prandial vomiting
- Weight loss
- Loss of appetite
- Retrosternal pain
- Cough
- Horseness of the voice/ voice changes

4.1.4 Investigations

1. Upper GIT Endoscopy with biopsies for histopathological examination (multiple suspected sites recommended) is the investigation of choice.
   Request for immunohistochemistry (if WHO poorly differentiated and undifferentiated carcinoma on biopsy).
2. **MMR/MSI, PDL-1 (CPS) and for adenocarcinoma HER-2 testing.**
3. Barium swallow (if there is no obstruction, where endoscopy is not available). However endoscopy with biopsy remains the standard of care for diagnosis.
4. ECHO Cardiogram in case of HER 2 positive patients

4.1.5 Investigations for Staging

- CT scan of the chest and abdomen is recommended for staging.
- FDG-PET CT SCAN (if no evidence of M1).
- EUS- (endoscopic ultrasound) for early disease {T1,T2 stage}.
- Diagnostic laparascopy incase of lower esophagus involving GEJ and/or cardia (SEWART III).

Table 4.1: TNM Staging (AJCC 9th edition)

Primary tumor (T)	
TX	Primary tumor cannot be assessed
T0	No evidence of primary tumor
Tis	High-grade dysplasia, * defined as malignant cells confined by the basement membrane
T1	Tumor invades lamina propria, muscularis mucosae, or submucosa
T1a	Tumor invades lamina propria or muscularis mucosae
T1b	Tumor invades submucosa
T2	Tumor invades muscularis propria
T3	Tumor invades adventitia
T4	Tumor invades adjacent structures
T4a	Resectable tumor invading pleura, pericardium, azygos vein, diaphragm, or peritoneum
T4b	Unresectable tumor invading other adjacent structures, such as the aorta, vertebral body, and trachea
*High-grade dysplasia includes all noninvasive neoplastic epithelial lesions formerly called carcinoma in situ; that term is no longer used for columnar mucosae anywhere in the gastrointestinal tract.	
Regional lymph nodes (N)	
NX	Regional lymph node(s) cannot be assessed
N0	No regional lymph node metastasis
N1	Metastasis in 1-2 regional lymph nodes
N2	Metastasis in 3-6 regional lymph nodes
N3	Metastasis in 7 or more regional lymph nodes
Distant metastasis (M)	
M0	No distant metastasis
M1	Distant metastasis

4.1.6 Treatment

Treatment approaches include:
Early-stage disease:

- Endoscopic resection (ER) with/without ablation: is used for very early disease and superficial disease (pTis, pT1a); ER with/without ablation for lesions <2 cm and ablation or esophagectomy for early disease (pT1b) as per institution criteria
- Locally advanced disease (cT1b-T4a, N0/N+).
- Perioperative chemotherapy followed by surgery in adenocarcinoma
- Preoperative chemoradiotherapy followed by surgery
- Definitive chemoradiation - cT4b or for non-surgical candidates
- Preoperative radiotherapy - 41.4Gy - 50.4Gy
- Postoperative radiotherapy - 45Gy - 50.4Gy
- Definitive CCRT - 50Gy - 50.4Gy

Metastatic disease - check performance status

- Chemotherapy- as monotherapy
- Immunotherapy alone or in combination with chemotherapy
- Targeted therapy (e.g. HER2 targeted therapy).
- Palliative radiotherapy
- Best supportive care for poor performance status, stenting and feeding tube where necessary for nutritional support

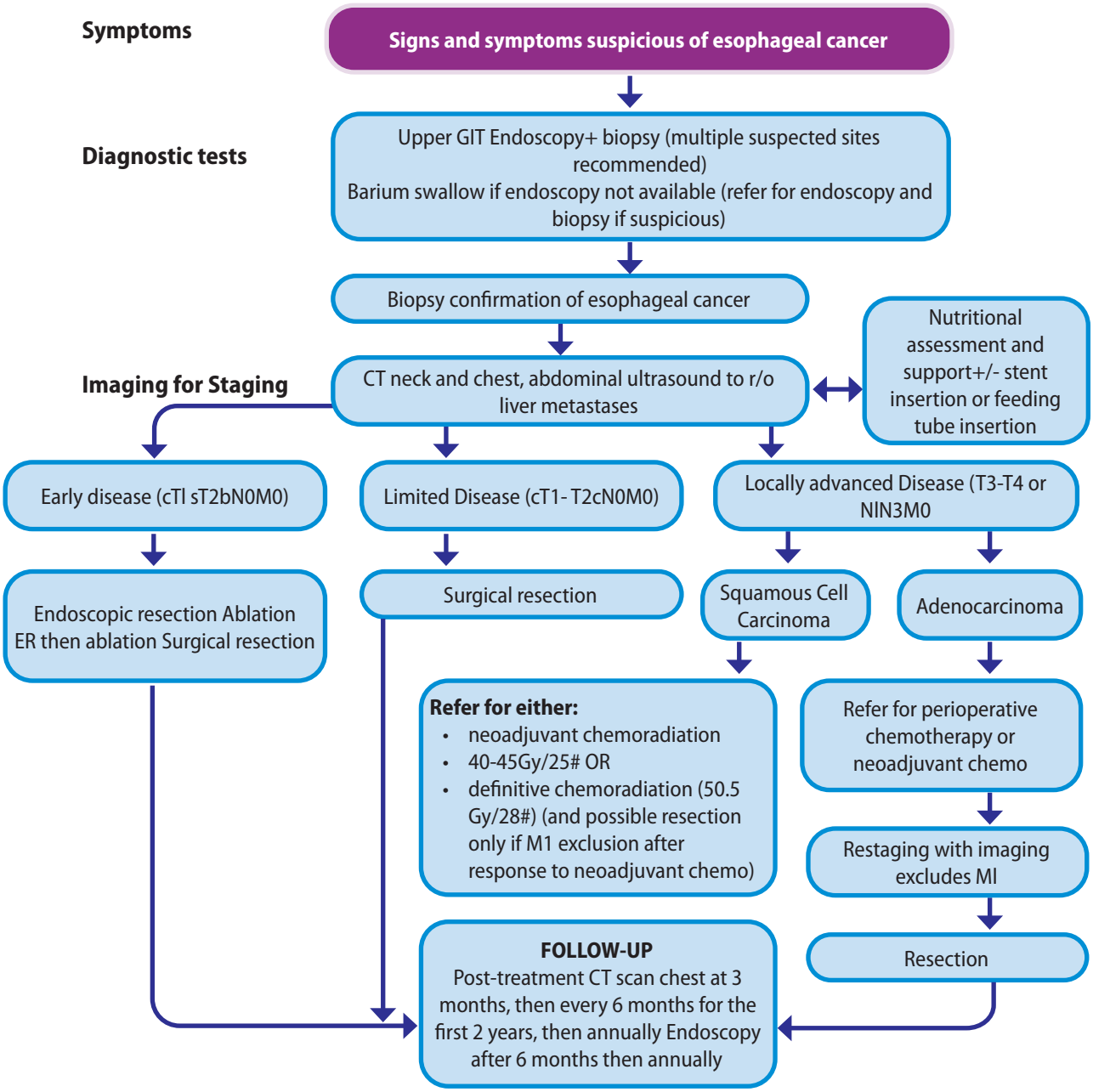


Figure 4.1: Algorithm for Management of Esophageal Cancer

4.1.7 Chemotherapy Protocols

1. **Adenocarcinoma**
   - FLOT regimen: 5-FU/Leucovorin/Oxaliplatin/5-FU/Docetaxel (perioperative chemotherapy)
   - triple chemotherapy reserved for chemotherapy duplet + targeted/immunotherapy
2. **Squamous cell carcinoma**
   - Cisplatin/5-Fluorouracil
   - Carboplatin/Paclitaxel (can be given concurrently with RT as a weekly regimen, or 3-weekly for patients who need systemic therapy)
   - Oxaliplatin, 5-fluorouracil

3. Immunotherapy- Nivolumab/Pembrolizumab for PDL1 positive and CPS >5 and >10 respectively
4. Targeted therapy- Trastuzumab for Her2 positive adenocarcinoma

4.1.8 Supportive Care

- If endoscopic stenting is not possible in patients with unresectable tumor, PEG tubes or feeding gastrostomy or jejunostomy tubes may be required in patients who are obstructed for nutritional support.
- Psychosocial support, counselling and pain management are important.

4.1.9 Principles of Follow up

- All patients should be followed up systemically. However, follow up strategies after successful therapy of esophageal cancer remains institutional based.
- Follow-up visits should be focused on symptoms management, nutrition and psychosocial support as well as investigations guided by clinical judgement.

4.1.10 References

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4.2 Gastric Cancer

4.2.1 Introduction

According to GLOBOCAN 2022, Gastric cancer ranks as the fifth most commonly diagnosed cancer and the fifth leading cause of cancer-related deaths globally for both genders. In Kenya it ranks 6th in incidence and 6th in mortality, with approximately 1899 new cases and 1505 deaths annually (GLOBOCAN, 2022). It is the 3rd most common gastrointestinal malignancy in Kenya after esophageal and colorectal cancers. Over 90% of gastric cancers are adenocarcinomas. (WHO classification of tumors 5th edition, 2019)

4.2.2 Risk Factors

- Male gender (incidence twice as high)
- Infection with Helicobacter pylori
- Lifestyle risk factors: Tobacco use, alcohol use, high-salt diet, processed meat and low fruit and vegetable intake.
- Atrophic gastritis
- Partial gastrectomy
- Pernicious anaemia
- Gastric cardia tumors are associated with obesity, while gastroesophageal junction tumors are associated with reflux and Barrett's esophagus.
- Familial aggregation in ~10% of cases and an inherited genetic predisposition is found in ~1–3%.

4.2.3 Signs and Symptoms

When symptomatic. Gastric Cancer tends to be advanced. Common symptoms include:

- Abdominal pain
- Weight loss
- Dysphagia
- Dyspepsia
- Vomiting
- Early satiety
- Anaemia (usually an iron deficiency anaemia)

Early gastric cancer is often asymptomatic.

4.2.4 Diagnostic Workup

1. Complete history and physical examination
2. Upper GI endoscopy and biopsy: to visualise lesions and obtain tissue for diagnosis. Further testing may be considered to determine MSI/MMR, HER2 and PD-L1 for potential treatment targeting
3. Laboratory tests: FBC, renal function tests and liver function tests, h. pylori testing
4. Contrast enhanced CT chest, abdomen and pelvis.

5. Laparoscopy + peritoneal washings (for patients who are deemed operable and resectable)
6. PET-CT
7. Ideally: Endoscopic ultrasound (where available) for accurate assessment of T and N stage to distinguish between T2/T3 tumors

Table 4.2: TNM Staging of Gastric Cancer as per AJCC, 9th edition

Primary Tumour (T)	Regional lymph nodes (N)	Distant metastasis (M)
TX Primary tumour cannot be assessed	NX Regional lymph nodes cannot be assessed	M0 No distant metastasis
T0 No evidence of primary tumour	N0 No regional lymph node metastasis	Ml Distant metastasis or positive peritoneal cytology
Tis Carcinoma in situ: intraepithelial tumour without invasion of the lamina propria	N1 Metastasis in 1-2 regional lymph nodes	
T1a Tumour invades the lamina propria or the muscularis mucosae	N2 Metastasis in 3-6 regional lymph nodes	
T1b Tumour invades the submucosa	N3 Metastasis in 7 or more regional lymph nodes	
T2 Tumour invades the muscularis propria	N3a Metastasis in 7-15 regional lymph nodes	
T3 Tumour penetrates the subserosal connective tissue without invasion of the visceral peritoneum or adjacent structures	N3b Metastasis in 16 or more regional lymph nodes	
T4 Tumour invades the serosa (visceral peritoneum) or adjacent structures T4a Tumour invades the serosa (visceral peritoneum) T4b Tumour invades adjacent structures		

4.2.5 Treatment

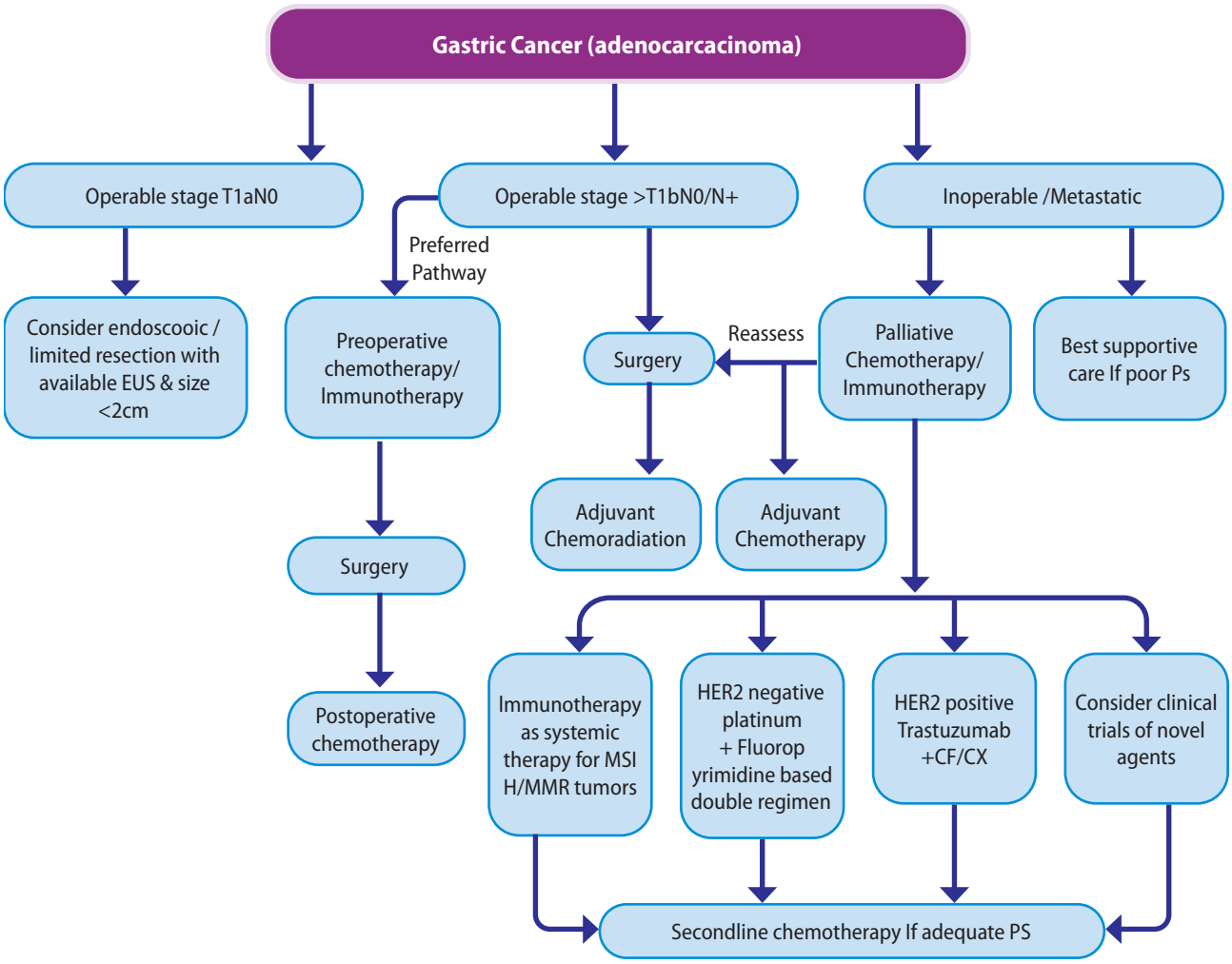


Figure 4.2: Gastric Cancer Treatment Algorithm (Adapted from ESMO clinical practice guidelines)

4.2.6 Treatment Approach

The management of gastric tumors requires a multidisciplinary team (MDT) approach to ensure comprehensive and individualized care, integrating expertise from surgery, oncology, gastroenterology, radiology, pathology, and nutrition.

4.2.7 Early disease

- CTis/cT1a: Endoscopic resection is appropriate for selected medically fit patients.

4.2.8 Locoregional Disease (Stage Ib - III)

- Radical gastrectomy (D2 resection with ≥16 nodes) is indicated in medically fit patients with operable tumors
- Consideration may be given for neoadjuvant or peri-operative systemic therapy before proceeding with surgery based on risk assessment then surgical evaluation within 4-6 weeks (FLOT4 protocol)

- Peri-operative systemic therapy is recommended for patients with ≥cT2 or Node positive tumors.
- The type of surgery is dependent on the location and extent of tumor (distal/subtotal or total gastrectomy)

4.2.9 Perioperative Drug Regimes

- Perioperative (pre- and postoperative) chemotherapy regimens include FLOT, FOLFOX or 5FU/Cisplatin based on performance status
- Others can be considered as per the situation - EOX, ECX
- Immunotherapy - for MSI-H/dMMR tumors - e.g pembrolizumab

4.2.10 Adjuvant Therapy (after surgery)

- Completion of perioperative chemotherapy
- Adjuvant radiotherapy (5FU- based chemoradiation) - patients with R1/R2 resection, combine chemo + Chemoradiation for those pT3/T4 or N+ with less than D2 resection or some select tumors with high-risk features such as higher grade, LVI/PNI+ or <50yr

4.2.11 Advanced and Metastatic Disease

- Features characteristic of unresectable tumors include - invasion of the root of the mesentery, para-aortic lymphadenopathy, invasion/encasement of major vessels, peritoneal seeding or distant metastasis
- Comorbidities, organ function, performance status and patient preference must always be taken into consideration.

4.2.12 Options for Treatment Include

- **Systemic therapy** -
- Chemotherapy (5 FU-based combinations, taxanes with platinum agents or single agent modality),
- Targeted therapy (e.g Trastuzumab for HER2+ tumors),
- Immunotherapy (e.g pembrolizumab based on MSI/MMR or PDL1)

- **Chemoradiation -** for locally unresectable tumors; radiation (45-50.4Gy at 1.8Gy/day) given concurrently with 5FU/Cisplatin or 5FU/Oxaliplatin (Capecitabine can be substituted for 5FU)`
- **Palliative radiotherapy** - bleeding, intractable pain
- **Palliative surgery** - to relieve obstruction, venting gastrostomy, feeding tube insertion, gastric bypass for gastric outlet obstruction
- **Palliative care/best supportive care.**

4.2.13 Follow Up/Survival

This should be tailored to the individual patient and the stage of the disease. Follow up is indicated 3-6 monthly for 3 years then annually. At initial follow-up, restaging is preferred.

- Monitor Vitamin B12 levels and mineral deficiencies in surgically resected patients and provide dietary support as indicated.

- In the advanced disease setting, identification of patients for second-line chemotherapy and clinical trials requires regular follow-up to detect symptoms of disease progression before significant clinical deterioration.
- If relapse/disease progression is suspected, then a clinical history, physical examination and directed blood tests should be carried out.
- Radiological investigations should be carried out in patients who are candidates for further chemotherapy or radiotherapy.

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4.3 Colorectal Cancer

4.3.1 Introduction

Globally, it is the 3rd most common type of cancer with 1,1926425 new cases and 904019 deaths annually. In Kenya it is the 5th most common cancer with an estimated 3091 new cases and 2116 deaths annually (GLOBOCAN, 2022). Majority of colorectal cancers arise from benign polyps and most are adenocarcinomas.

4.3.2 Risk Factors

Non- modifiable

- Older age - > 45 years old
- Inflammatory bowel disease - Crohn's disease or ulcerative colitis
- Family history of colorectal cancer or polyps in a blood relative
- Personal history of colorectal cancer or polyps
- Presence of genetic syndromes, such as familial adenomatous polyposis (FAP) or hereditary non-polyposis colorectal cancer (Lynch syndrome).

Modifiable /Lifestyle Associated

- Intake of red and processed meats
- Physical inactivity
- Low fruit and vegetable intake
- A low-fiber and high-fat diet
- Obesity
- Alcohol intake
- Tobacco use

4.3.3 Clinical Evaluation

History taking

Signs and symptoms depend on the stage at presentation and the location of the tumor in the bowel and the presence of metastasis.

Signs and symptoms of colon cancer:	Signs and symptoms of rectal cancer:
• Altered bowel habits (commonly constipation, diarrhea) • Blood in stool • Anemia or fatigue • Anorexia • Weight loss • Nausea/Vomiting • Abdominal pain • Intestinal obstruction	• Haematochezia • Altered bowel habits • Feeling of incomplete bowel emptying • Cramping or Abdominal pain • Weight loss • Intestinal Obstruction • Narrow stool • Anaemia or fatigue

About 50% of patients are asymptomatic. Inquire and document any risk factors including past medical history, family history and lifestyle risk factors among others.

4.3.3 Physical Examination

- Perform a thorough physical examination including abdominal and digital rectal examination (DRE).
- Vaginal examination should be done in women with rectal cancer.
- examine the supraclavicular and axillary region

4.3.4 Investigations

- Complete colonoscopy with biopsy (for histological diagnosis) is the investigation of choice.
- Molecular investigations including MSI or MMR for all, and expanded IHC for advanced/ metastatic disease such as RAS and BRAF mutations, HER-2 amplifications
- Other routine laboratory tests include Complete Blood Count, Urea Electrolytes & Creatinine, Liver function tests, Carcinoembryonic antigen (CEA)
- DPD (DIHYDROPYRIMIDINE DEHYDROGENASE) for patients to be treated with capecitabine

4.3.5 Imaging (for clinical staging after diagnosis)

1. CT chest, abdomen and pelvis.
2. MRI pelvis is crucial for local staging of rectal disease to identify circumferential resection margin (CRM) involvement which would influence pre-operative treatment.
3. FDG-PET CT scan

Table 4.3: AJCC 9th edition TNM Staging

Primary tumour T	Regional Nodes N	Distant metastasis M
cTX Primary tumour cannot be assessed	cNX Regional lymph nodes cannot be assessed	cM0 No distant metastasis
cTis Carcinoma in situ: invasion of lamina propria	cN0 No regional lymph node metastasis	• cM1 Distant metastasis • cM1a Metastasis confined to one organ (liver, lung, ovary, non-regional • lymph node(s)) without peritoneal metastases • cM1b Metastasis in more than one organ • cM1c Metastasis to the peritoneum with or without other organ • involvement

Primary tumour T	Regional Nodes N	Distant metastasis M
cT1 Tumour invades submucosa	• cN1 Metastasis in 1–3 regional lymph nodes • cN1a Metastasis in 1 regional lymph node • cN1b Metastasis in 2–3 regional lymph nodes • cN1c Tumour deposit(s), i.e. satellites,* in the subserosa, or in non-peritonealised pericolic or peri-rectal soft tissue • without regional lymph • node metastasis	
cT2 Tumour invades muscularis propria	• cN2 Metastasis in 4 or more regional lymph nodes • cN2a Metastasis in 4–6 regional lymph nodes • cN2b Metastasis in 7 or more regional lymph nodes	
cT3 Tumour invades subserosa or into non-peritonealised pericolic or peri-rectal tissues		
cT4 Tumour directly invades other organs or structures and/or perforates visceral peritoneum. cT4a Tumour perforates visceral peritoneum. cT4b Tumour directly invades other organs or structures.		

4.3.6 Treatment of Colorectal Cancers

Surgery is the mainstay of treatment for colon cancer which removes the lesion with wide margins and loco-regional nodes. Early colorectal cancer is potentially curable if treated appropriately.

Table 4.4: Treatment of Early/Non-Metastatic Colon Cancer

Stage	Type of surgery	Chemotherapy
Malignant polyp, no risk of invasion	polypectomy	No benefit
Malignant polyp with risk of invasion	Resection and lymph node dissection	No benefit
Stage 0-(Tis, N0, M0 and T1, N0, M0)	Local excision OR Segmental resection/wide. Colectomy and lymph node dissection >12 nodes.	No benefit
Stage I- T2, N0, M0.		No benefit, discuss case. In MDT (patients. With. Intestinal. Obstruction to be considered for. Defunctioning. Colostomy/stenting and definitive removal. Of tumour)
Stage II-T3-T4, N0, Mo	Colectomy and lymph node dissection	Consider adjuvant treatment if high risk features/MSI low. Or. MMR abundant on IHC. (FOLFOX or CAPEOX)
Stage III-any T, N1-N2, M0	Colectomy and lymph node dissection	Adjuvant chemotherapy indicated-start within 2-4. Weeks. Of. Surgery-(FOLFOX/CAPEOX)

High risk histological features include lymphatic or venous invasion, poorly differentiated (grade 3) tumor, level 4 invasion (invades the submucosa of the bowel wall), serosal involvement, perforated tumours, extramural vascular invasion, less than 12 lymph nodes examined and involved margins of excision, close, indeterminate or positive resection margins (CRM).

4.3.7 Treatment of Early/Non-Metastatic Rectal Cancer

Stage I

1. Total meso-rectal excision (TME) with abdominal-perineal resection (APR – for low lesions) or low anterior resection (LAR- for mid-upper lesions).
2. No adjuvant treatment if pathologically T1-2NO without high-risk features.

Stage II & III (Locoregional disease)

4.3.8 Total Neoadjuvant Therapy

Definition

- Combines systemic chemotherapy and radiation before surgery for locally advanced rectal cancer (LARC - typically stage II/III: cT3–4, node-positive, or involving meso-rectal fascia or with extramural venous invasion), aiming to improve systemic disease control and treatment adherence.
- Differs from traditional neoadjuvant chemoradiotherapy by administering full-dose chemotherapy (not just radio-sensitizing doses).

Indications:

- Clinical stage II–III, especially highrisk features (cT3–4, nodepositive, positive extramural venous invasion on MRI (mrEMVI+).

Rationale

- Addresses distant metastases (a major cause of mortality) by delivering systemic therapy earlier.
- Improves pathologic complete response (pCR) rates, disease-free survival (DFS), and chemotherapy completion rates compared to standard neoadjuvant chemoradiotherapy.
- May enable organ preservation (non-operative "watch-and-wait" strategies) in patients with complete clinical response. (Where feasible.)

Treatment Regimens

- Neoadjuvant chemotherapy, followed by long course chemoradiotherapy (CRT) -50 Gy over 5 weeks- then followed by total meso-rectal excision (TME) -5-12 weeks after CRT, and adjuvant chemotherapy completing a total of 12-16 weeks. Of systemic therapy.
- Short-course radiotherapy (5 × 5 Gy) over a maximum of 8 days) followed by six cycles of CAPOX chemotherapy or nine cycles of FOLFOX4 followed by total meso-rectal excision.**(short course RT not recommended for T4 or multiple clinical LNs, if obstruction is noted, diverting colostomy required prior to neoadjuvant therapy.**
- Chemotherapy options: Doublet (FOLFOX/CAPOX) or triplet (FOLFIRINOX-for Fit -ECOG 0-1- High Risk patients) regimens;

Surgical Resection

Two options exist:

Lower Anterior Resection (LAR)

- This is a sphincter-sparing surgery removing the rectal tumor and surrounding tissue while preserving the anal sphincter.
- Indications include upper/mid rectal cancers (typically ≥5 cm from the anal verge) with adequate distal clearance to retain bowel continuity.

4.3.9 Abdominoperineal Resection (APR)

- This is the removal of the rectum, anus, and distal sigmoid colon, requiring a permanent end colostomy.
- It is indicated for Low rectal cancers involving the sphincter complex or <5 cm from the anal verge (insufficient distal margin for LAR).

Restaging is recommended prior to surgery. Surgery is recommended within 6-8 weeks of treatment.

4.3.10 Metastatic Colon & Rectal Cancers

Available treatment options depend on the patient's predominant symptoms, disease burden, performance status, wishes and prior therapy. For management of metastatic rectal and colon cancers, patients are divided into three groups:

1. Resectable: Surgery of the primary disease + metastasectomy recommended for carefully selected colorectal cancer patients with resectable liver metastasis.
2. Potentially resectable: Surgery for the primary disease with metastasectomy is recommended after neoadjuvant therapy or as a staged procedure.
3. Unresectable: Palliative options such as surgery in cases of obstruction (colostomy), palliative chemotherapy, radiotherapy, with supportive care, pain management and colonic stenting for obstruction.

- Palliative radiotherapy is useful in rectal tumors and fixed caecal tumors for symptoms of bleeding, discharge and local pain.
- Palliative transanal debulking of symptomatic tumors has been shown to improve quality of life.
- Multiple palliative chemotherapy regimens exist, usually given till progression or toxicity.

4.3.11 Chemotherapy Regimens include:

1. Oral Capecitabine monotherapy
2. CAPEOX regimen: Capecitabine plus Oxaliplatin
3. FOLFOX : Fluorouracil/Leucovorin/Oxaliplatin
4. FOLFIRI : Fluorouracil/Leucovorin/Irinotecan
5. FOLFOXIRI / FOLFIRINOX :Fluorouracil/Leucovorin/Oxaliplatin/Irinotecan (respective doses differ in the two regimes)

4.3.12 Targeted/immunotherapy Agents

- Bevacizumab can be used in combination with FOLFIRI, CAPEOX or FOLFOX.
- Pembrolizumab, nivolumab
- Cetuximab & panitumumab can be used for RAS-wild type
- Encorafenib for BRAF V600E positive
- Pembrolizumab
- Aflibercept –with FOLFIRI
- Regorafenib
- Trastuzumab (IV)

4.3.13 Supportive Care

- Psychosocial support
- Stoma care
- Pain management
- Nutrition

4.3.14 Follow Up

- Clinical assessment plus CEA every 3 months for the first 2 years.
- If high risk of recurrence, consider Abdominopelvic CT scan every 4-6 months.
- Colonoscopy in year 1 then every 2 years if negative.

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4.4 Pancreatic Cancer

4.4.1 Introduction

Pancreatic cancer ranks 12th in incidence and 6th in cancer mortality globally. Approximately 95% of pancreatic cancers occur within the exocrine portion. The most common type is ductal adenocarcinoma which accounts for about 80% of all pancreatic cancers. In Kenya, there are approximately 889 new cases and 872 deaths annually(GLOBOCAN, 2022). Approximately 60-70% of pancreatic cancer arises in the head of the pancreas, 20-25% in the body and tail, and the remaining 10-20% diffusely involves the pancreas. The vast majority (>80%) of pancreatic carcinomas are due to sporadically occurring mutations.

4.4.2 Risk Factors

- Age > 65 years
- Cigarette smoking
- Alcohol consumption
- Obesity
- Chronic pancreatitis
- Dietary factors-high consumption of processed food, saturated fat.
- Familial pancreatic cancers (at least two first-degree relatives with pancreatic cancer) account for 5-10% of all cases, mutation in BRCA2 is common.
- Genetic predisposition.

4.4.3 Signs and Symptoms

Early disease is asymptomatic. Tumors at the head of pancreas are likely to present earlier than those at the body or tail.

Patients typically report the gradual onset of non-specific signs and symptoms. Typical symptoms could include:

- Jaundice.
- Abdominal pain (upper abdominal, mid-epigastric)
- Back pain
- Weight loss
- Pruritus
- Others - anorexia, malaise, nausea, fatigue, dark urine, diarrhea and steatorrhea (fatty stools).

Pancreatic tumors may be preceded by newly diagnosed type 2 diabetes.

4.4.4 Diagnostic Work-Up

Imaging

- Contrast enhanced CT (preferred) or MRI abdomen (pancreatic protocol) is the investigation of choice
- Other imaging - CT chest and pelvis, PET-CT
- Magnetic Resonance Cholangiopancreatography (MRCP) for staging

- Endoscopic Retrograde Cholangiopancreatography (ERCP) (plus biopsy) with stenting to relieve obstructive jaundice.
- Pndoscopic ultrasound (EUS) and biopsy

Laboratory Investigations

- Total blood count
- Coagulation profile, PTI/INR
- Biochemistry: U/E/C, LFTs, Blood sugar
- Tumor markers – CA 19-9 and Carcinoembryonic antigen (CEA) can be used **with other tests to aid diagnosis, measure disease burden and to monitor treatment.**
- Consider serum IgG4 (can be helpful in distinguishing autoimmune pancreatitis from pancreatic cancer).
- Biopsy is not indicated if surgery is planned but can be done for patients requiring differential diagnosis and/or initiating chemotherapy;
- Molecular testing e.g KRAS, MSI/MMR/TMB, BRCA1/2

Staging

Stage the tumour (TNM) and categorize it as resectable, borderline resectable, locally advanced or metastatic disease.

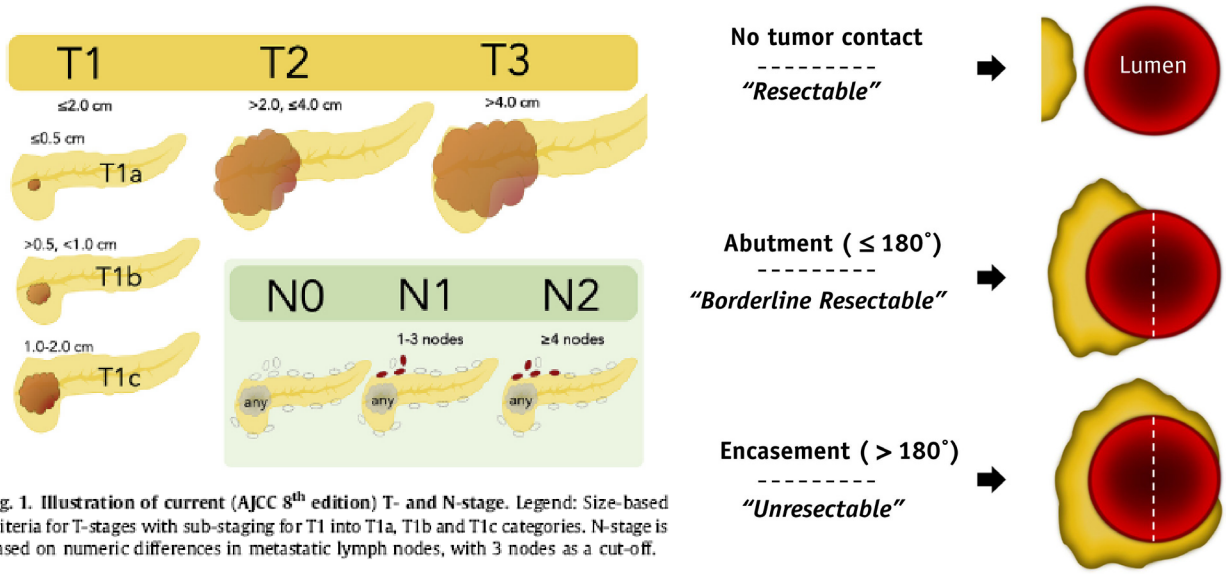


Fig. 1. Illustration of current (AJCC 8th edition) T- and N-stage. Legend: Size-based criteria for T-stages with sub-staging for T1 into T1a, T1b and T1c categories. N-stage is based on numeric differences in metastatic lymph nodes, with 3 nodes as a cut-off.

Figure 4.3: Resectability status at diagnosis

Table 4.5: Criteria Defining Resectability Status At Diagnosis

Resectability Status	Arterial	Venous
Resectable	No arterial tumor contact (celiac axis(CA), superior mesenteric artery (SMA), common hepatic artery (CHA))	No tumor contact with superior mesenteric vein (SMV), portal vein (PV), or =/< 180^0 contact without vein contour irregularity.

Resectability Status	Arterial	Venous
Borderline resectable	Pancreatic head/uncinate process Solid tumor contact with CHA, without extension to CA, hepatic artery bifurcation allowing for safe and complete resection and reconstruction. Solid tumor contact with SMA of =/< 180^0. Solid tumor contact with variant arterial anatomy (Ex: accessory right hepatic artery, replaced right hepatic artery, replaced CHA, and the origin of replaced or accessory artery) and the presence and degree of tumor contact should be noted if present, as it may affect surgical planning. Pancreatic body/tail : Solid tumor contact with CA =/< 180^0	Solid tumor contact with SMV/PV of >180^0, contact of =/<180^0 with contour irregularity of the vein, or thrombosis of the vein but with suitable vessel proximal and distal to the site of involvement, allowing for safe and complete resection and reconstruction. Contact with inferior vena cava.
Locally advanced	Head/uncinate process: Solid tumor contact of > 180^0 with the SMA or CA. Pancreatic body /tail : Solid tumor contact of > 180^0 with the SMA or CA. Solid tumor contact with the CA and Aortic involvement.	Unreconstructible SMP/PV due to tumor involvement or occlusion (can be due to tumor or bland thrombus.)

Adopted from NCCN V1.2022

4.4.5 Treatment

Overview

- Defining the treatment strategy for patients suffering from pancreatic carcinoma requires a specialized multidisciplinary team.
- Surgical resection is the only potentially curative treatment of pancreatic adenocarcinoma.
- Systemic therapy is used in all stages of pancreatic cancer (neoadjuvant, adjuvant, 1st line or subsequent line for advanced or metastatic disease).
- The Choice of treatment depends on patient's performance status and goals of therapy
- Squamous and adenosquamous carcinomas are treated the same as adenocarcinomas

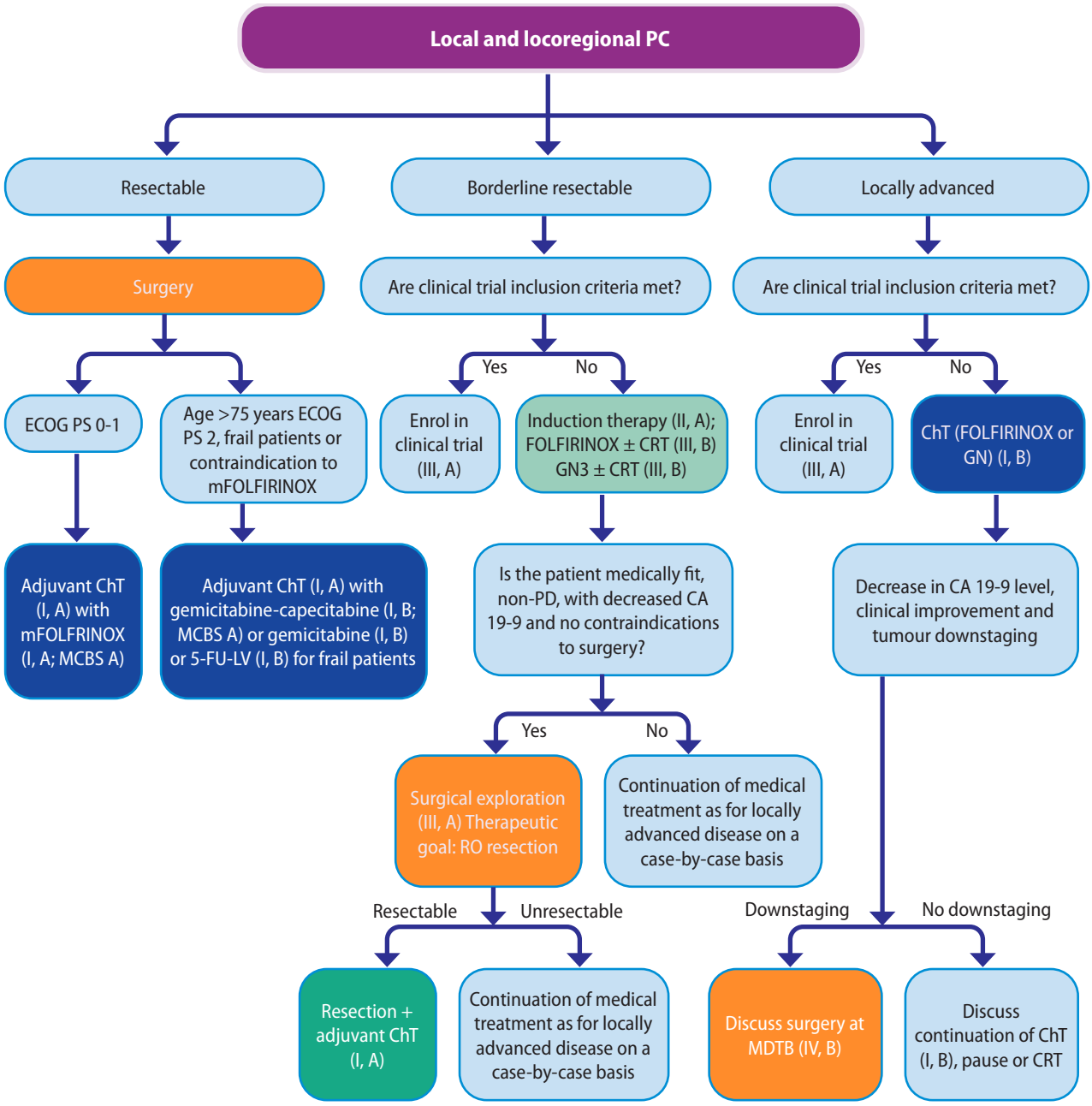


Figure 4.4 : Treatment Algorithm for Pancreatic cancer
Source: ESMO Guidelines

4.4.6 Systemic Regimens

1. mFOLFIRINOX (or FOLFIRINOX) - as neoadjuvant, adjuvant or 1st line for advanced disease
2. Gemcitabine + nab-paclitaxel
3. Gemcitabine + cisplatin (especially if BRCA mutated)
4. Other fluorouracil-based combinations
5. Single agents - capecitabine, gemcitabine, continuous 5FU
6. PARP inhibitors (where applicable) - olaparib, rucaparib
7. Immunotherapy (where applicable)

4.4.7 Role of Radiotherapy

- Radiotherapy (3D CRT/IMRT/ SBRT) can be considered in the neoadjuvant or adjuvant setting: the goal of delivering RT is to sterilize vessel margins, enhance the likelihood of a margin-negative resection, and/or provide adequate local control to prevent or delay progression or prevent local disease recurrence while minimizing the risk of RT exposure to surrounding organs at risk (OARs).
- 3DCRT/IMRT - 45 - 54Gy or SBRT - 3 or 5 fractions
- Radiation can also be used to palliate pain and bleeding or relieve obstructive symptoms in patients who have progressed or recurred locally.

4.4.8 Palliation and Best Supportive Care

Manage pain, obstruction, nutrition, counseling and mental health support, patient and family support, end of life care and thromboembolism risk assessment and prophylaxis

4.4.9 Follow Up Care

Follow up visits should focus on symptoms management, labs and imaging (where appropriate), nutrition and psycho-social support.

4.4.10 References

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4.5 Hepatocellular Carcinoma

4.5.1 Introduction

Hepatocellular carcinoma (HCC) is a primary tumour of the liver that usually develops in the setting of chronic liver disease. It ranks as the 6th in incidence cancer globally and the 3rd most frequent cause of cancer-related deaths. It is more common in males with a male-to-female ratio of 2:1. Over 80% of the cases are diagnosed in developing countries. Highest prevalence is in Asia followed by Sub-Saharan Africa where the annual mortality is 87,890 deaths with 80% of HCC due to HBV infection. GLOBOCAN 2022 data estimates show liver cancer as the 13th most common cancer in Kenya with an estimated 883 new cases and 819 deaths annually.

4.5.2 Risk Factors

- Major risk factors are cirrhosis and chronic liver disease, regardless of etiology
- Viral hepatitis - hepatitis B infection, hepatitis C infection
- Co-infection of HBV/HCV and HIV
- Chronic alcohol intake
- Aflatoxin B1 (mycotoxin)
- Metabolic syndrome (obesity, diabetes & dyslipidemia leading to fatty liver)
- Non-alcoholic steatohepatitis/non-alcoholic fatty liver disease
- Autoimmune hepatitis
- Primary biliary cirrhosis
- Hereditary hemochromatosis (Fe2+ overload)
- Alpha 1 antitrypsin deficiency

4.5.3 Signs and Symptoms

- Common signs and symptoms include: jaundice, ascites, encephalopathy, bleeding, pain and splenomegaly.

4.5.4 Diagnostic Work-up

Imaging

1. CT abdomen (multiphasic - hepatic protocol)
2. Other imaging - CT chest, pelvis
3. FDG-PET-CT scan has limited sensitivity, and thus its use is limited in the workup. it may however be considered when there are equivocal findings on other imaging modalities.
4. Liver biopsy- Liver biopsy can be omitted if AFP and imaging findings are definite for HCC and tumor is resectable. Liver biopsy is indicated if the laboratory, imaging and clinical findings are incongruent or indeterminate for HCC, suspicion of metastatic disease. Additional immunohistochemical staining may also be helpful in diagnosis.

4.5.5 Laboratory Tests

1. Hepatitis panel (HBV - HBsAg, HBeAg/antibody, quantitative HBV DNA, hepatitis B core antibody IgG, Hepatitis C antibody +/- HCV RNA)
2. AFP
3. LFTS, UEC, FBC
4. Prothrombin time/INR/APTT

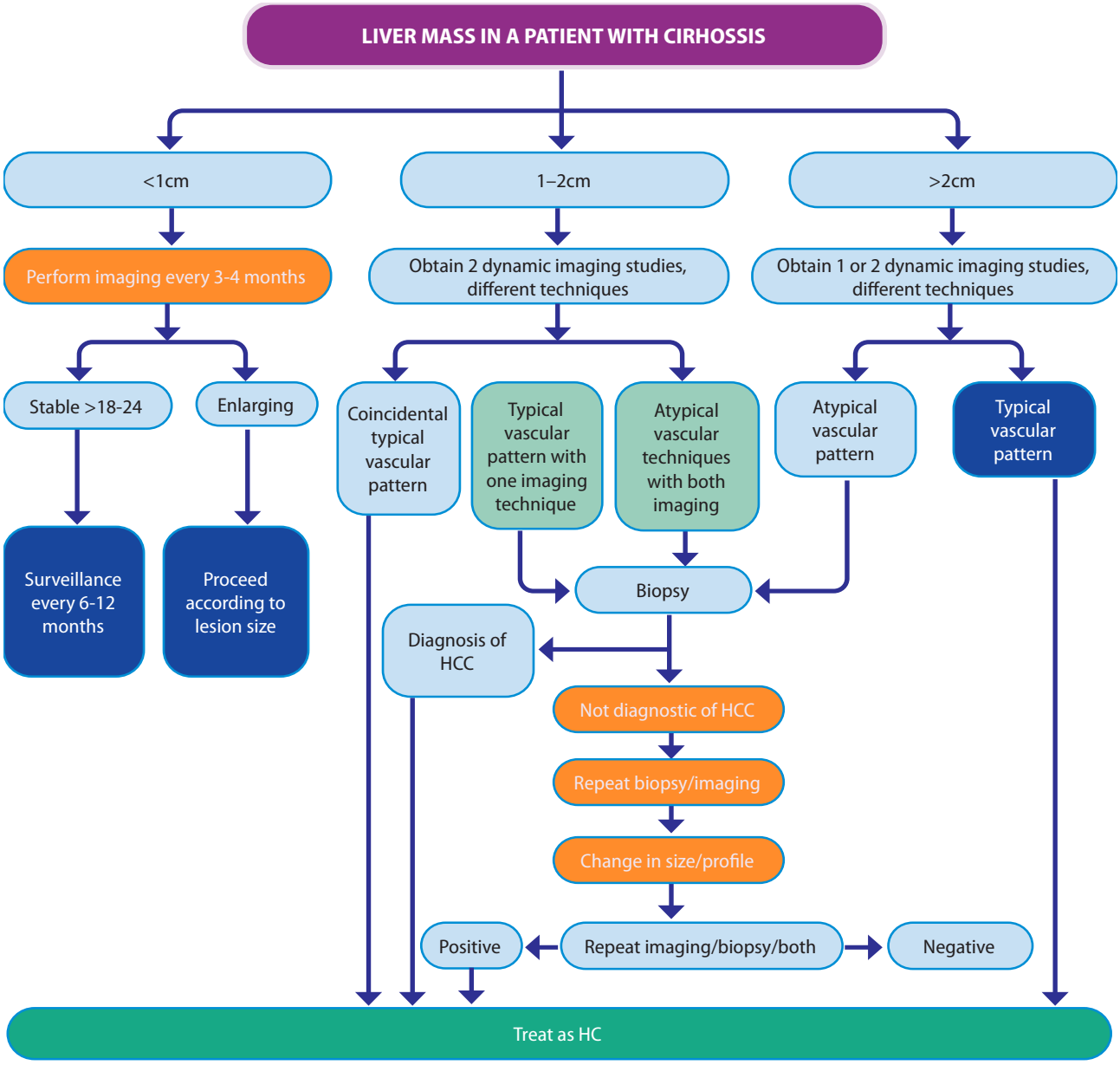


Figure 4.5: Diagnostic Workup Algorithm for Work-Up for a Patient with Liver Mass and Cirrhosis

4.5.6 Staging

Accurate staging helps determine key management decisions, including tumour resectability, transplant eligibility, and the overall operability of the patient.

Several classification systems are relevant to the management of HCC.

- TNM/AJCC staging system
- Child-Turcotte-Pugh score to determine liver functional reserve.
- Transplant eligibility - e.g United Network for Organ Sharing (UNOS) criteria, MELD/PELD scores.
- Barcelona Clinic Liver Cancer (BCLC) staging system - relevant for choice of treatment decision-making

Table 4.6: AJCC 8th edition staging for hepatocellular carcinoma

Primary tumor (T)		Regional lymph nodes (N)		Distant metastases (M)	
T1a	Solitary tumor <2 cm with/ without vascular invasion	Nx	Regional lymph nodes cannot be assessed	M0	No distant metastasis
T1b	Solitary tumor >2 cm without vascular invasion	N0	No regional lymph node metastasis	M1	Distant metastasis
T2	Solitary tumor >2 cm with vascular invasion or multifocal tumors, none >5 cm	N1	Regional lymph node metastasis		
T3	Multifocal tumors at least one of which is >5 cm				
T4	Single tumor or multifocal tumors of any size involving a major branch of the portal vein or hepatic vein or tumor(s) with direct invasion of adjacent organs other than the gallbladder or with perforation of visceral peritoneum				
Stage					
Stage IA	T1a	N0	M0		
Stage IB	T1b	N0	M0		
Stage II	T2	N0	M0		
Stage IMA	T3	N0	M0		
Stage NIB	T4	N0	M0		
Stage IVA	Any T	N1	M0		
Stage IVB	Any T	Any N	MI		

Table 4.7: Child-Turcotte-Pugh Score

Measure	1 point	2 points	3 points	units
Bilirubin (total)	< 22	-3	>3	mg/dL
Serum albumin	>35	28-35	<28	mg/L
INR	<1.7	1.71-2.20	>2.20	no unit
Ascites	None	Suppressed with medication	Refractory	no unit
Hepatic encephalopathy	None	Grade I-II (or suppressed with medication)	Grade III-IV (or refractory)	no unit

Interpretation

The score employs five clinical measures of liver disease. Each measure is scored 1-3, with 3 indicating most severe derangement. Chronic liver disease is classified into Child- Pugh Turcotte class A(5-6), B(7-9) and C (10-15),5 employing the added score from above.

4.5.7 Barcelona Clinic Liver Cancer (BCLC) Staging and Treatment Algorithm

Table 4.8: Treatment Options for Hepatocellular Carcinoma

BCLC stage	Diagnostic description	Standard of care	Alternative Treatment
O	PS O, Child-Pugh A, single HCC <2CM	Surgery or Ablation (RFA)	
A	PS O, Child-Pugh A-B single HCC or 3 nodules, less than 3 cm	Resectable - Surgery Non-Resectable – HDR BT, SBRT, RFA, MWA, TARE Fit for OLT - Transplant	
B	PS O. Child Pugh A-B, Multinodular HCC	Fit for OLT - Transplant Resectable - Surgery TACE, TARE.	Unsuitable for locoregional therapy – systemic therapy
C	PS 1-2, Child Pugh A-B, portal vein invasion, Nodal metastasis, Distant metastasis	Suitable ICI Atezolizumab-Bevacizumab Nivolumab-ipillilumab Durvalumab	Non-ICI suitable Lenvatinib. Sorafenib
D	PS >2, Child-Pugh C	Best supportive care/Palliative care	

4.5.8 Evaluation of the Response to Treatment and Follow-up

Assessment of response should be based on dynamic CT or MRI studies, and the modified Response Evaluation Criteria in Solid Tumours (mRECIST) criteria. Viable tumour should be defined as uptake of contrast agent in the arterial phase on imaging.

4.5.9 Follow Up & Supportive Care

Serum tumour markers especially AFP levels may be helpful particularly in the case of disease that is not easily measurable. They should, however, not be used as the only determinants for treatment decisions. Patients who underwent radical treatments (resection or RFA) should undergo clinical evaluation for liver decompensation and dynamic CT or MRI studies every 3 months for the first 2 years to detect any recurrences and every 6 months thereafter.

Patients with disease recurrence after radical therapies may still be candidates for curative therapies. Patients treated with TACE or systemic agents e.g. sorafenib should be evaluated clinically for signs of liver decompensation and for tumour progression by dynamic CT or MRI every 3 months to guide therapy decisions.

4.5.10 Prevention and Surveillance of HCC

90% of HBV chronicity results from perinatal infection, 20- 50% chronicity from infection before 5 years and only 5% from infection in adulthood. 15-40% of HBV infection leads to cirrhosis, liver failure and eventually HCC. Preventive strategies include:

- HBV vaccination and preventive programs to stop transmission of hepatitis B and C.
- Early antiviral (HBV with Tenofovir & HCV with direct acting antivirals (DAAs) such as sofosbuvir and Ledipasvir/Sofosbuvir for infection eradication.
- Surveillance programs for persons at risk of HCC (patients with cirrhosis, non-cirrhotic HBV carriers with a high viral load, patients with chronic HCV with advanced cirrhosis).
- An abdominal ultrasound (and serum AFP), done every 6 months is the imaging of choice for surveillance of persons at risk of HCC.

Key Recommendations

- Screen all antenatal mothers for HBsAg
- Treat all HBsAg positive mothers with Tenofovir during the 3rd trimester.
- Every child should receive a birth dose of Hep B vaccine.
- For children born of HBsAg positive mothers, they should receive HB-immunoglobulin.
- Hep B vaccination for all children as per immunisation schedule.
- Patients with chronic hepatitis irrespective of the aetiology should be screened for HCC with alpha feto-protein and liver ultrasound every 6 months
- Those found with liver tumours should be referred to national referral hospitals for further management.

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4.6 Anal Carcinoma

4.6.1 Introduction

Anal cancer is considered a relatively rare cancer globally. According to GLOBOCAN 2022, there are an estimated 54,000 new cases every year and about 22,000 deaths annually associated with anal cancer. Approximately 88% of cases are associated with human papillomavirus (HPV) infection worldwide. Anal cancers are predominantly squamous cell carcinomas (95%); the rest are adenocarcinomas, basaloid and cloacogenic among others.

4.6.2 Risk Factors

- Human papillomavirus (HPV) infection
- Human immunodeficiency virus (HIV) infection
- Anal sex
- Older age >50 years
- Multiple sexual partners
- Tobacco use
- History of cervical, vulvar or vaginal cancer
- Immunosuppressive drugs (for example after organ transplantation) or conditions

4.6.3 Risk Reduction Strategies

- Screen high risk populations using cytology, HPV testing, digital anorectal examination and biopsy (The International Anal Neoplasia Society (IANS) and U.S. guidelines recommend initiating anal cancer screening at age 35 for men who have sex with men (MSM) and transgender women living with HIV. For other high-risk groups, screening may begin at age 45).
- HPV vaccinations.
- Treatment of HIV.
- Smoking cessation.

4.6.4 Signs and Symptoms

- Anal bleeding
- Perianal itching
- Anal discharge
- Fecal incontinence and fistula
- Perianal or low abdominal pain
- Non-healing ulcer
- Anal mass
- Altered bowel habits
- Weight loss
- Vomiting of feeds

4.6.5 Patient assessment

- Conduct a complete physical examination
- Digital rectal examination (DRE)
- Inguinal nodal examination
- Vulval/vaginal examination is mandatory in females.

4.6.6 Investigations

- FBC, UEC, LFTs, HIV testing (and if HIV positive, do CD4 counts).
- Rigid proctoscopy/ high resolution anoscopy with biopsies (direct examination under anesthesia).
- Biopsy any visible anal growth or masses are noted. FNA/Biopsy of any palpable inguinal nodes.

4.6.7 Imaging for Staging

- CT chest and abdomen
- Pelvic MRI
- Consider PET-CT (depends on clinical scenario)

4.6.8 Treatment

All patients who are newly diagnosed HIV positive should be started on ARVs. Diverting colostomy should be considered for all patients before starting the definitive treatment.

4.6.9 Role of Surgery

- Anal margin tumors: For stage I, well differentiated lesions, local excision (re-excision or chemo-radiation if involved/close margins on histopathological evaluation).
- Anal canal tumors: Surgery (radical or local excision) is generally contraindicated as primary treatment option; chemoradiation is the treatment of choice.
- Surgery is also considered for persistent or recurrent anal canal tumors after definitive chemoradiation.

4.6.10 Locoregional disease (Any T, Any N, M0)

- Definitive therapy entails chemoradiation:
  - First line: 5-Fluorouracil + Mitomycin + concurrent radiotherapy
  - Alternative: Capecitabine + Mitomycin + concurrent radiotherapy
  - Rarely - cisplatin + 5-FU concurrent with RT
  - Radiotherapy: IMRT preferred; RT dose 50.4Gy - 54Gy

- ***Persistent disease after CCRT may continue to regress even 26 weeks after treatment. After completion of RT patients who have achieved a complete clinical response should be followed up to 6 months with no intervention as long as there is no evidence of progressive disease during this period***

4.6.11 Metastatic Anal Cancer (Any T, Any N, Any M)

Systemic therapy then re-asses for local chemoradiation for local control

Systemic therapy options include:

- Carboplatin + paclitaxel
- others: FOLFCIS, mFOLFOX6, Cisplatin/5FU, modified DCF (docetaxel, cisplatin, 5-FU)
- Subsequent chemoradiation after systemic therapy - 5-FU or capecitabine + RT
- 2nd line and subsequent therapy is customised based on the site(s) of disease, performance status, cost and availability of systemic therapy agents

4.6.12 Follow Up

Re-evaluate patients starting 8-12 weeks after chemo-radiation using pelvic MRI.

i. **Complete Remission**

- DRE and inguinal node examination every 3-6 months for 5 years
- Proctoscopy 6 monthly for 3 years
- CT chest, abdomen and pelvis annually for 3 years

ii. **Persistent disease**

Re-evaluate after 4 weeks:

- If tumor regresses, or no progression, then continue observation every 3 months
- If tumor progresses, re-biopsy and re-stage and consider for AP resection and groin dissection

iii. Progressive disease
Re-biopsy, re-stage and consider for A-P resection and groin dissection

iv. **Metastatic disease**
Palliative treatment

Table 4.9: Roles of Health Care Facilities for Gastrointestinal Cancers

Level	Roles
Level 1,11,111	• Detection of symptoms, signs and risk factors for gastrointestinal cancers. Early and appropriate referral of patients.
Level IV,V	• Detection of symptoms, signs and risk factors for gastrointestinal cancers. Early and appropriate referral of patients for further evaluation, staging and further management. • Targeted investigations for diagnosis to include a total blood count, and liver function tests. • Key diagnostic imaging tests including X-rays, Barium swallow, Abdominal ultrasound, CT- Scan

Level	Roles
Level VI	• Detection of symptoms, signs and risk factors for gastrointestinal cancers. • Full range of diagnostic laboratory investigations • Full range of key diagnostic Imaging tests including a CT- Scan chest/abdomen/ MRI abdomen for diagnosis/ and staging, endoscopy and colonoscopy and FED-PET where applicable. • A bone scans for advanced disease and liver/ pancreatic biopsy for diagnosis. Appropriate TNM and BCLC stage allocation. • Appropriate multidisciplinary team management of these patients. • Develop treatment protocols for various disease stages to include those for liver transplant. • Be able to offer surveillance and follow-up. • Offer best supportive and palliative therapy for advanced disease based on evidence. • Conduct research and training to better improve patient management.

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Chapter 5

Gynaecological Cancers

List of abbreviations

AFP	Alpha Fetoprotein
BSO	Bilateral salpingo-oophorectomy
CA-125	Cancer Antigen 125
EBRT	External Beam Radiation Therapy
EUA	Examination under Anaesthesia
GTD	Gestational Trophoblastic Disease
GTN	Gestational Trophoblastic Neoplasia
HCG	Human Chorionic Gonadotrophin
HIV	Human Immunodeficiency Virus
HPV	Human Papillomavirus
LDH	Lactate Dehydrogenase
PLND	Pelvic lymph node dissection
RMI	Risk Malignancy Index
TAH	Total Abdominal Hysterectomy
TVS	Trans Vaginal Ultrasound

5.1 Cervical Cancer

5.1.1 Introduction

Globally Cervical cancer is the fourth most common cancer in terms of both incidence and mortality in women, with an estimated 660,000 new cases and 350,000 deaths worldwide (Globocan 2022). Most of these cases occur in sub-Saharan Africa as well as South America and South-Eastern Asia. In Kenya it is the leading cause of death contributing to over 10% of all cancer deaths. It accounts for 5845 (13.1%) of new cancer cases and 3,591 (12.2%) of all cancer deaths annually.

5.1.2 Risk Factors

- HPV infection
- HIV infection
- Sexually transmitted infections
- Tobacco use
- Sexual history: Having multiple sexual partners, early onset of sexual activity.
- Multiparity
- Immunosuppression

5.1.3 Clinical Evaluation

Signs and Symptoms

Early
• Vaginal discharge, sometimes foul smelling • Irregular vaginal bleeding • Post coital bleeding in women of any age • Post menopausal bleeding (especially that which does not respond to appropriate treatment)
Late
• Urinary frequency and urgency • Backache • Lower abdominal pain • Severe back pain • Weight loss • Oliguria (due to ureteric obstruction or renal failure) • Urinary/fecal incontinence • Edema of lower limbs

Any woman presenting with any of the above symptoms should have a pelvic examination to visualize the cervix, and any visible lesions should be biopsied +/- colposcopy for diagnosis of cervical cancer.

During pelvic exam, attention should be paid to the size of the tumor and possible involvement of the vaginal fornices, the parametria (transverse cervical and uterosacral ligaments), the pelvic walls, the bladder and the rectum. Examination under anesthesia is recommended in case of doubtful diagnosis or if the patient is too tense or in pain.

Stage I:
The carcinoma is strictly confined to the cervix uteri (extension to the corpus should be disregarded)

- **IA** Invasive carcinoma that can be diagnosed only by microscopy, with maximum depth of invasion <5 mm
  - **IA1** Measured stromal invasion <3 mm in depth
  - **IA2** Measured stromal invasion >3 mm and <5 mm in depth
- **IB** Invasive carcinoma with measured deepest invasion >5 mm (greater than stage IA), lesion limited to the cervix uterib
  - **IB1** Invasive carcinoma >5 mm depth of stromal invasion and <2 cm in greatest dimension
  - **IB2** Invasive carcinoma >2 cm and <4 cm in greatest dimension o IB3 Invasive carcinoma >4 cm in greatest dimension

Stage II:
The carcinoma invades beyond the uterus, but has not extended onto the lower third of the vagina or to the pelvic wall

- **IIA** Involvement limited to the upper two-thirds of the vagina without parametrial involvement
  - **IIA1** Invasive carcinoma <4 cm in greatest dimension o IIA2 Invasive carcinoma >4 cm in greatest dimension
- **MB** With parametrial involvement but not up to the pelvic wall

Stage III:
The carcinoma involves the lower third of the vagina and/or extends to the pelvic wall and/or causes hydronephrosis or non-functioning kidney and/or involves pelvic and/or paraaortic lymph nodes

- **IIIA** Carcinoma involves the lower third of the vagina, with no extension to the pelvic wall
- **NIB** Extension to the pelvic wall and/or hydronephrosis or non-functioning kidney (unless known to be due to another cause)
- **MIC** Involvement of pelvic and/or paraaortic lymph nodes, irrespective of tumor size and extent (with r and p notations)c
  - **IIIC1** Pelvic lymph node metastasis only
  - **IIIC2** Para-aortic lymph node metastasis

Stage IV:
The carcinoma has extended beyond the true pelvis or has involved (biopsy proven) the mucosa of the bladder or rectum. A bullous edema, as such, does not permit a case to be allotted to stage IV

- **IVA** Spread of the growth to adjacent organs
- **IVB** Spread to distant organ

Figure 5.1: Staging for Cervical Cancer (FIGO Staging 2018)

5.1.4 Management of Cervical Cancer

Laboratory test

- Complete blood count
- Renal function tests
- Liver function tests,
- HIV test

Imaging

- Xray chest or CT chest to r/o chest metastasis
- MRI pelvis to assess tumor extent. Transvaginal/transrectal ultrasound is an option if performed by a properly trained personnel in cervical cancer assessment
- CT abdomen/pelvis for evaluation of metastatic disease
- Whole body FDG PET CT (preferred) for evaluation of metastatic disease where available
- Cystoscopy or rectoscopy are not routinely recommended.

5.1.5 Management of IA1 Disease with no LVSI

Diagnosis of IA1 cancer should be based on a conization (or LEEP) specimen examined by an expert pathologist with accurate measurement of depth of invasion, margin status, coexisting pathology, and reliable assessment of LVSI.

Care should be taken to provide an intact (unfragmented) specimen with minimal thermal artifact.

The cone/LEEP specimen should be oriented for the pathologist.

Surgical margins of the cone specimen should be clear of both invasive and preinvasive disease (except for low-grade intraepithelial lesion).

LN staging is not indicated in IAI LVSI-negative patients but can be considered in IAI LVSI-positive patients.

Conization or simple hysterectomy can be considered a definitive treatment

For those with adenocarcinoma, simple hysterectomy should be offered for those who have completed childbearing

5.1.6 Management of IA1 with LVSI, IA2-IB1(Based on Cone Biopsy)

IAI with LVSI

- Those with positive margins – repeat cone, if margins still positive then for radical hysterectomy with PLND
- Those with negative margins – simple hysterectomy + PLND

IA2- IBI meeting the low-risk criteria;

- Conservative surgery including simple hysterectomy with pelvic lymph node dissection (PLND)

All conservative surgery criteria MUST be met,

- No LVSI
- Negative cone margins
- Squamous cell (any grade) or usual type adenocarcinoma (grade 1 or 2 only)
- Tumor size ≤2 cm
- Depth of invasion ≤10 mm on LEEP/cone
- Negative imaging for local-regional disease (MRI mandatory)

For those desiring fertility

- Cone or simple trachelectomy + PLND

IA2-IB1 NOT meeting the conservative surgery criteria, stage IB2 and stage IIA1

- Radical hysterectomy (Type C1) + PLND, +/- para-aortic lymph node dissection (PALN)

For those desiring fertility stage IA2-IB2

- Radical trachelectomy + PLND, +/- para-aortic lymph node dissection (PALN)

5.1.7 Adjuvant Treatment

Patients with high or intermediate risk factors for recurrence should be offered adjuvant treatment

High risk factors (Peters criteria), if any present should be offered concurrent chemoradiation. For patients with high-risk factors, the recurrence risk is approximately 40 percent and the risk of death is up to 50 percent following surgery alone.

- Positive parametria
- Positive margin
- Positive pelvic lymph

Intermediate risk factors (Sedlis criteria), if present should be offered radiation. Risk of death and recurrence is 30% with surgery alone

- Presence of lymphovascular space invasion (LVSI) plus deep one-third cervical stromal invasion and tumor of any size
- Presence of LVSI plus middle one-third stromal invasion and tumor size >2 cm
- Presence of LVSI plus superficial one-third stromal invasion and tumor size >5 cm
- No LVSI but deep or middle one-third stromal invasion and tumor size >4 cm

5.1.8 Management of Stage IB3-IVA

- Concurrent chemoradiation (CCRT): EBRT concurrent platinum-containing chemotherapy+ brachytherapy
- For bulky tumors of more than 4cm neoadjuvant chemotherapy followed by CCRT can be considered
- Metastatic disease (IVB) or recurrent disease
- Systemic therapy – single agent/doublet
- Bevacizumab can be considered
- Pembrolizumab for patients with PDL-1 positive patients

5.1.9 Chemotherapy Regimens

Table 5.1: Cervical Cancer Chemotherapy Regimes

Indication	Chemotherapy agent
Concurrent chemoradiation	Weekly single agent cisplatin/carboplatin
Neoadjuvant chemotherapy	Cisplatin/carboplatin + paclitaxel every 21d x 3 cycles
Metastatic disease	Cisplatin/carboplatin + paclitaxel/docetaxel Bevacizumab (until ds progression or intolerable toxicity) Pembrolizumab (for PDL-1 positive disease only)

5.1.10 Follow up and Surveillance

Patients who have been treated with curative intent should undergo complete history and physical exam with the following evaluation every 3 months for the first 3 years and every 6 months for the next 2 years.

Imaging can be done at follow-up as required based on stage and symptomology
HPV DNA testing for follow-up.

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5.2 Endometrial Cancer

5.2.1 Introduction

Endometrial cancer is a heterogenous group of diseases. Globally, it is the sixth most commonly occurring cancer in women with 420, 242 new cases in 2022 (GLOBOCAN 2022). In Kenya a total of 829 new cases were reported in 2022 (GLOBOCAN 2022). It most commonly occurs in postmenopausal women, with post-menopausal bleeding being the most common presenting symptom.

5.2.2 Classification of Endometrial Cancer

Type I (Non aggressive type) - 80% of Endometrial Cancer

- Endometrioid adenocarcinoma - grade 1 or 2 with less than 50% myometrial invasion

Type II (Aggressive types)

- endometrioid adenocarcinoma, grade 3 with >50% myometrial invasion
- serous adenocarcinoma
- clear cell adenocarcinoma
- mesonephric-like carcinomas
- gastrointestinal-type mucinous endometrial carcinoma,
- undifferentiated carcinomas
- mixed carcinoma
- carcinosarcoma

5.2.3 Risk Factors

- Chronically increased estrogen exposure (exogenous or endogenous)

i. Exogenous

- oral, transdermal patch, vaginal ring) without concurrent administration of a progestin
- Tamoxifen

ii. Endogenous

- Functional ovarian tumors such as granulosa and theca cell tumors
- Obesity
- Chronic anovulation such as PCOS
- Low parity/ nulliparity
- Early menarche (onset below 12 years) and Late menopause (above 52 years)
- Diabetes mellitus

iii. Hereditary Conditions

- Lynch syndrome (hereditary nonpolyposis colorectal cancer)
- Cowden disease or multiple hamartoma syndrome
- First-degree relative with EC

5.2.4 Protective Factors

Breastfeeding reduces the risk of endometrial cancer.
Use of estrogen-progestin oral contraceptives

5.2.5 Clinical Presentation

Abnormal vaginal bleeding including postmenopausal bleeding/spotting occurs in 90% of patients. Though most patients are postmenopausal, about a quarter of them are pre- menopausal with 5% below 40 years of age. The majority of young patients are obese or have high levels of unopposed endogenous estrogen from chronic anovulation due to polycystic ovary disease.

5.2.6 Physical Examination

Initial evaluation should include full physical examination, including a pelvic examination noting the size, position and contour of the uterus.

5.2.7 Diagnosis

Diagnosis requires histopathological evaluation of endometrial tissue. Devices for use in the office such as endometrial pipelle allow endometrial sampling and accurate diagnosis. Hysteroscopic evaluation and biopsy may be performed where facilities are available or dilatation and curettage.

Pelvic ultrasonography may be suggestive of disease. Endometrial thickening above 5 mm in a postmenopausal woman should make the clinician suspicious.

5.2.8 Preoperative Evaluation and Preparation

- Laboratory investigations: total blood count, renal function tests, liver function tests, blood sugar, others as necessary.
- Imaging: chest X-ray, pelvic ultrasonography (transvaginal), pelvic and abdominal CT scan, MRI scan (contrast-enhanced dynamic MRI) for uterine and pelvic spread and nodal assessment
- Whole body PET CT scan in selected patients

Table 5.2: FIGO 2023 Staging for Endometrial Carcinoma

Stage	Description
Stage 1	Confined to the uterine corpus and ovary[c]
IA	• Disease limited to the endometrium OR non-aggressive histological type, i.e. low-grade endometroid, with invasion of less than half of myometrium with no or focal lymphovascular space involvement (LVSI) OR good prognosis disease • IA1 Non-aggressive histological type limited to an endometrial polyp OR confined to the endometrium • IA2 Non-aggressive histological types involving less than half of the myometrium with no or focal LVSI
IB	• IA3 Low-grade endometrioid carcinomas limited to the uterus and ovary[c] • Non-aggressive histological types with invasion of half or more of the myometrium, and with no or focal LVSI[d]
1C	• Aggressive histological types[e] limited to a polyp or confined to the endometrium
Stage II	• Invasion of cervical stroma without extrauterine extension OR with substantial LVSI OR aggressive histological types with myometrial invasion
IIA	• Invasion of the cervical stroma of non-aggressive histological types
MB	• Substantial LVSI[d] of non-aggressive histological types
IIC	• Aggressive histological types[e] with any myometrial involvement
Stage III	Local and/or regional spread of the tumor of any histological subtype
IMA	• Invasion of uterine serosa, adnexa, or both by direct extension or metastasis • IIIA1 Spread to ovary or fallopian tube (except when meeting stage IA3 criteria)[c] • IIIA2 Involvement of uterine subserosa or spread through the uterine serosa
NIB	• Metastasis or direct spread to the vagina and/or to the parametria or pelvic peritoneum • IIIB1 Metastasis or direct spread to the vagina and/or the parametria • IIIB2 Metastasis to the pelvic peritoneum
MIC	• Metastasis to the pelvic or para-aortic lymph nodes or both[f] • IIICl Metastasis to the pelvic lymph nodes • IIICli Micrometastasis • IIIClii Macrometastasis • IIIC2 Metastasis to para-aortic lymph nodes up to the renal vessels, with or without metastasis to the pelvic lymph nodes • IIIC2i Micro metastasis • IIIC2ii Macrometastasis
Stage IV	• Spread to the bladder mucosa and/or intestinal mucosa and/or distance metastasis
IVA	• Invasion of the bladder mucosa and/or the intestinal/bowel mucosa
IVB	• Abdominal peritoneal metastasis beyond the pelvis
IVC	• Distant metastasis, including metastasis to any extra- or intra-abdominal lymph nodes above the renal vessels, lungs, liver, brain, or bone

5.2.9 Management

Surgery followed by adjuvant chemotherapy and or radiotherapy are the mainstay of treatment.

For apparently uterine confined tumors - Total abdominal hysterectomy (TAH), bilateral salpingo-oophorectomy (BSO), pelvic and para-aortic node dissection and omental biopsy (serous carcinoma, clear cell carcinoma, or carcinosarcoma)

For advanced tumor that is amenable to surgery - cytoreductive surgery which involves TAH/BSO/removal of enlarged pelvic and para-aortic lymph nodes and all macroscopic disease.

Adjuvant treatment is given based on a risk stratification (or molecular profile where available) for recurrence.

Molecular analysis where available - POLE mutations, microsatellite instability-high (MSI-H) or mismatch repair deficient (dMMR), no specific molecular profile (NSMP), and p53 aberrant.

The decision to use molecular testing/classification depends on the availability of resources and the multidisciplinary team of each center.

Table 5.3: Treatment algorithm for Endometrial Cancer

Risk Group	Molecular Classification Unknown	Molecular Classification Known[4],*
Low	• Stage IA endometrioid + low-grade** + LVS1 negative or focal	• Stage I-II ***POLE*timiX.** endometrial carcinoma, no residual disease • Stage IA **MMRd/NSMP** endometrioid carcinoma + low-grade** + LVSI negative or focal
Intermediate	• Stage IB endometrioid + low-grade** + LVS1 negative or focal • Stage 1A endometrioid + high-grade** + LVS1 negative or focal • Stage 1A non-endometrioid (serous, clear cell, undifferentia red carcinoma, carcinosarcoma, mixed) without myometrial invasion	• Stage IB **MMRd/NSMP** endometrioid carcinoma + low-grade** + LVSI negative or focal • Stage IA **MMRd/NSMP** endometrioid carcinoma + high-grade** + LVSI negative or focal • Stage IA **p53abn** and/or non-endometrioid (serous, clear cell, undifferentiated carcinoma, carcinosarcoma, mixed) without myometrial invasion
High-intermediate	• Stage 1 endometrioid + substantial LVSI, regardless of grade and depth of invasion • Stage IB endometrioid high-grade**, regardless of LVSI status • Stage 11	• Stage 1 **MMRd/NSMP** endometrioid carcinoma + substantial LVSI, regardless of grade and depth of invasion • Stage IB **MMRd/NSMP** endometrioid carcinoma high-grade**, regardless of LVSI status • Stage 11 **MMRd/NSMP** endometrioid carcinoma

Risk Group	Molecular Classification Unknown	Molecular Classification Known[4],*
High	• Stage I1I-1VA with no residual disease • Stage 1-IVA non-endometrioid Cserous, clear cell, undifferentiated carcinoma, carcinosarcoma, mixed) with myometrial invasion, and with no residual disease	• Stage II1-IVA **MMRd/NSMP** endometrioid carcinoma with no residual disease • Stage I-1VA **p53abn** endometrial carcinoma with myometrial invasion, with no residual disease • Stage I-IVA **NSMP/MMRd** serous, undifferentiated carcinoma, carcinosarcoma with myometrial invasion, with no residual disease
Advanced Metastatic	• Stage I1I-1VA with residual disease • Stage IVB	• Stage 1I1-1VA with residual disease of any molecular type • Stage IVB of any molecular type

Table 5.4: Risk Stratified Adjuvant Treatment For Endometrial Cancer

Risk of recurrence	Adjuvant treatment
Low risk	Observation
Intermediate risk	Observation
High intermediate risk	Adjuvant VBT/EBRT in patients with substantial LVSI
High risk	EBRT with concurrent and adjuvant chemotherapy or alternatively sequential chemotherapy and radiotherapy is recommended Chemotherapy alone

Incorporation of immune checkpoint inhibitors with chemotherapy for selected cancers
For patients with stage III or stage IV mismatch repair deficient (dMMR)/microsatellite instability-high (MSI-H) tumors, the addition of immunotherapy to chemotherapy, followed by immunotherapy maintenance up to 14 cycles.

5.2.10 Metastatic Endometrial Cancer

Those amenable to surgery - cytoreductive surgery, followed by chemotherapy (carboplatin/paclitaxel) with immunotherapy maintenance for MMR deficient tumors with pembrolizumab, dostarlimab or durvalumab until disease progression or intolerable toxicity.

Nonsurgical candidates – Patients who are not surgical candidates should be offered medical therapy (chemotherapy, with or without immunotherapy), provided that they are candidates for treatment.

For those with metastatic, serous EC overexpressing human epidermal growth factor receptor 2 (HER2) - addition of trastuzumab to frontline chemotherapy, with continuation as maintenance until progression rather than immunotherapy. ECHO for follow up of cardiotoxicity.

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5.3 Ovarian Cancer

5.3.1 Introduction

Ovarian cancer is the 8th most commonly occurring cancer and the 8th most common cause of cancer death in women worldwide. Globally the new cases were approximately 324,398 in 2022 and mortality of 324,398, (GLOBOCAN, 2022). In Kenya, it is the 5th leading cause of cancer deaths among women after cervical, breast, esophageal and colorectal cancers. (GLOBOCAN, 2022). There are three main types of ovarian cancer: epithelial, germ cell and sex cord/stromal. Epithelial ovarian cancer accounts for 90% of all cases of malignant tumors of the ovaries. Fallopian tube cancer and primary peritoneal cancer are much rarer, but share histologic, prognostic, and treatment response features with epithelial ovarian cancer.

5.3.2 Risk Factors

- Older age (40-69 years);
- Family history or previous personal history of ovarian, breast or endometrial cancer;
- Hormonal: Early menarche and late menopause, nulliparity or first birth after age 35 years;
- Obesity.

5.3.3 Signs and Symptoms

The disease is often easily missed since it presents with non-specific persistent gastrointestinal signs and symptoms (such as early satiety, indigestion, bloating, mild abdominal or pelvic pain), leading to late diagnosis when the patient has already developed an abdominal mass or ascites.

Other symptoms include vague abdominal discomfort, constipation, poor appetite, nausea, vomiting, frequent micturition, pelvic pressure, abdominal distension, irregular vaginal bleeding, low back pain, fatigue, dyspareunia and weight loss.

5.3.4 Clinical Examination

Any woman presenting with any of the above symptoms should have pelvic examination to rule out ovarian cancer. The presence of a solid, irregular, fixed pelvic mass with/ without an upper abdominal mass and/or ascites is highly suggestive of ovarian malignancy.

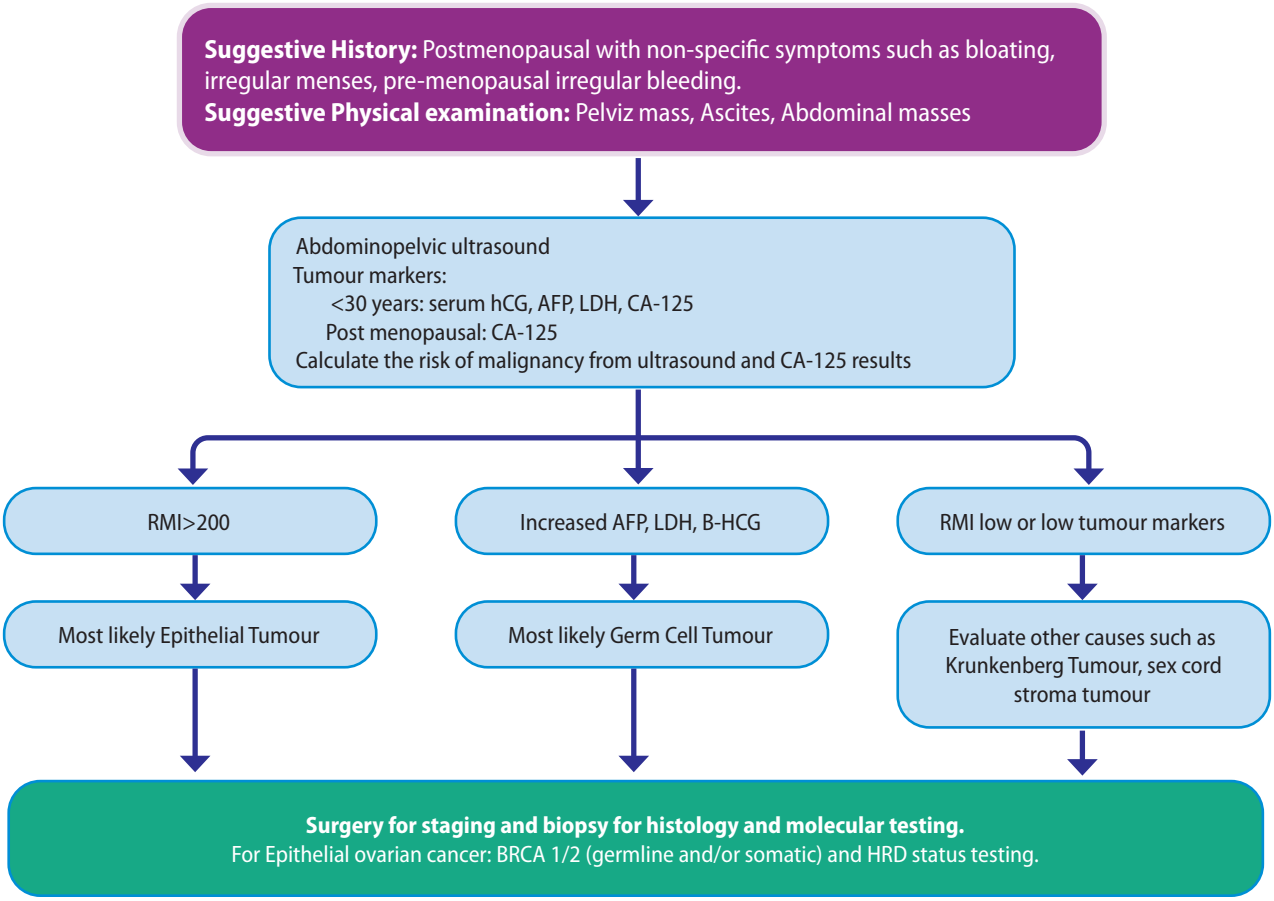


Figure 5.2: Diagnostic Algorithm for Ovarian Cancer

5.3.5 Diagnostic Evaluation

- Transvaginal pelvic ultrasound, (TVS) for characterisation or high resolution (at least 1.5Tesla) for pelvic MRI for mass characterisation.
- Tumor markers CA-125, AFP, LDH, HCG, CEA, CA-19-9
- Abdominopelvic CT scan, Chest CT scan

5.3.6 Epithelial Ovarian Carcinoma

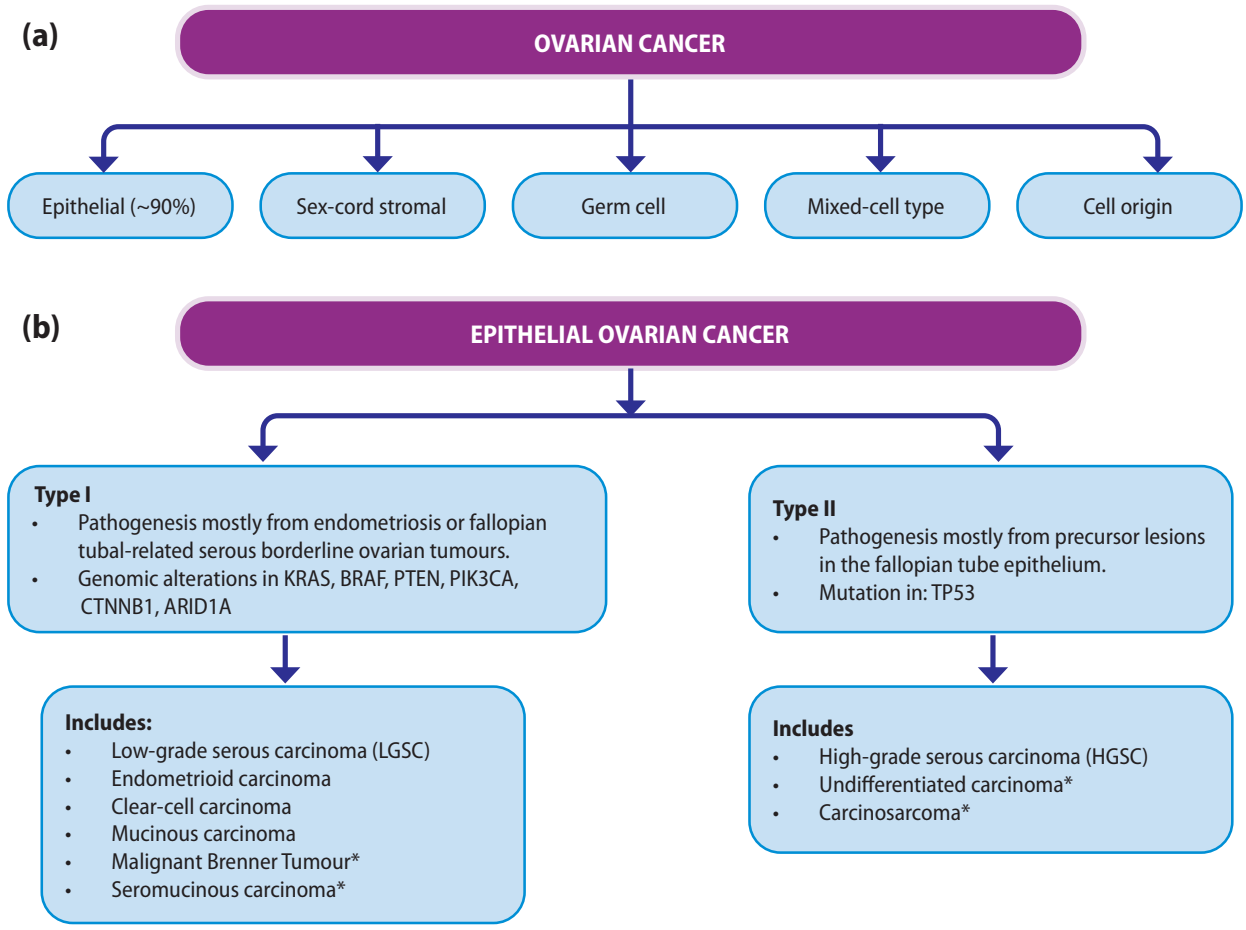


Figure 5.3: Treatment Algorithm for Epithelial Ovarian Cancers

Risk of Malignancy Index (RMI) is used to predict whether an adnexal mass is likely to be malignant, since CA-125 is raised in benign conditions like endometriosis, pelvic inflammatory disease and other abdominal conditions. It combines three pre-surgical features: serum CA-125 (CA-125), menopausal status (M) and ultrasound score (U). The RMI is a product of the ultrasound scan score, the menopausal status and the serum CA-125 level (IU/ml).

RMI = U x M x CA-125.

5.3.7 Fluid Cytology or Omental/pleural Biopsy

Some patients (e.g., **poor performance status, advanced disease that clinically appears unresectable**) require neoadjuvant chemotherapy (NACT) rather than primary surgery. For these patients, it is imperative a diagnosis be made prior to the start of medical treatment which can be accomplished in several ways;

1. **Fluid cytology** – In patients with ascites or pleural effusion, paracentesis or thoracocentesis may be performed, respectively. Cytologic evaluation (**Cell block**) can establish the diagnosis of malignancy,
2. **Omental/pleural biopsies** - In patients with suspected omental or pleural metastases, biopsies may help guide diagnosis as well as management decisions

Image-guided biopsy of the ovary is **not recommended**; incising or rupturing the mass can lead to spillage of malignant cells, which may result in upstaging of the disease and adversely affect prognosis.

Table 5.5: FIGO Classification for Ovarian Cancer for Staging

STAGE 1: Tumor confined to ovaries		
IA		Tumor limited to 1 ovary, capsule intact, no tumor on surface, negative washings
IB		Tumor involves both ovaries otherwise like IA.
1C-Tumor limited to 7 or both ovaries	IC1	Surgical spill
	IC2	Capsule rupture before surgery or tumor on ovarian surface.
	IC3	Malignant cells in the ascites or peritoneal washings
STAGE II: Tumor involves 7 or both ovaries with pelvic extension (below the pelvic brim)		
IIA		Extension and/or implant on uterus and/or Fallopian tubes
IIB		Extension to other pelvic intraperitoneal tissues
STAGE III: Tumor involves 1 or both ovaries with cytologically or histologically		
III A		(Positive retroperitoneal lymph nodes and /or microscopic metastasis beyond the pelvis)
NIB		Macroscopic, extrapelvic, peritoneal metastasis £ 2 cm ± positive retroperitoneal lymph nodes. Includes extension to capsule of liver/spleen
NIC		Macroscopic, extrapelvic, peritoneal metastasis > 2 cm ± positive retroperitoneal lymph nodes. Includes extension to capsule of liver/spleen.
STAGE IV: Distant metastasis excluding peritoneal metastasis		
IVA		Pleural effusion with positive cytology
IVB		Hepatic and/or splenic parenchymal metastasis, metastasis to extraabdominal organs (including inguinal lymph nodes and lymph nodes outside of the abdominal cavity)

5.3.8 Surgical Staging

Epithelial, germ cell, and stromal cancers of the ovary, fallopian tube, and peritoneum are staged surgically

5.3.9 Early-stage Epithelial Ovarian Cancer

- Peritoneal cytology (washings or ascites)
- Visual assessment of the upper abdomen, peritoneal surfaces, and large and small bowel mesentery and other abdominal organs, with biopsies of abnormal findings
- Hysterectomy with bilateral salpingo-oophorectomy
- Pelvic and paraaortic lymph node dissection
- Infracolic or infragastric omentectomy.

5.3.10 For Those who Desire Fertility

Fertility preservation is an acceptable option in young patients with ovarian tumors of low malignant potential or non-epithelial ovarian cancers and patients with stage IA EOC, which includes unilateral salpingo-oophorectomy, peritoneal washings, omentectomy, pelvic and paraaortic lymphadenectomy, and peritoneal biopsies, thorough abdominal exploration and biopsy of any abnormal areas. Endometrial biopsy to exclude synchronous endometrial cancer.

5.3.11 For Advanced Tumors stage II-IV

Cytoreductive surgery - aim of surgery should be removal of all visible tumors. Extensive surgery should be utilized only in a setting where optimal cytoreduction is determined feasible.

- **Role of neoadjuvant chemotherapy followed by interval debulking for stage III-IV -** confirmed poor surgical candidate or low likelihood of optimal cytoreduction
- Initial three cycles of intravenous carboplatin plus paclitaxel, with the addition of bevacizumab for those with high-risk disease (e.g., pleural effusion, ascites)
- Reassess after 3 cycles of chemo, if tumor is resectable, then proceed with interval debulking, followed by adjuvant chemotherapy.

5.3.12 Chemotherapy Regimen for Epithelial Ovarian Cancer

Neoadjuvant therapy:

- Carboplatin and paclitaxel with or without Bevacizumab
- Standard frequency is 3 weekly x 6 cycles.
- Bevacizumab-containing regimens before surgery may be considered in those with high-risk disease (pleural effusion, ascites etc.) should be done with caution due to potential interference with postoperative healing

Before neoadjuvant chemotherapy is delivered, all patients should have confirmation of an invasive ovarian, fallopian tube, or peritoneal cancer

Adjuvant Chemotherapy

- First line - carboplatin and paclitaxel
- Patients with germline or somatic mutation in **BRCA1** or **BRCA2** and a partial or complete response to front-line, platinum-based therapy or other homologous recombination repair-deficient (HRD) cancers may benefit from maintenance therapy with PARPi (Olaparib) until disease progression or intolerable toxicity
- Niraparib can be given for patients without BRCA 1/2 mutation (wild-type).
- Bevacizumab can also be given in the adjuvant setting.

5.3.13 Recurrent/Metastatic Disease:

- Platinum-resistant disease: recurs within 6 months of administration of the 1st line chemotherapy. Pegylated doxorubicin is preferred as the 2nd line in this case.

- Platinum-sensitive disease: recurs after more than 6 months post chemotherapy. The same 1st line regimen can be repeated.
- Bevacizumab in combination with chemotherapy is indicated in metastatic disease.
- PARPi (Olaparib) can be used as maintenance therapy until disease progression, or toxicity.

5.3.14 Low Grade Serous Ovarian Cancer

Stage IA-IB - observation
Stage IC - observation or carboplatin plus paclitaxel or endocrine therapy alone
Stage II to IV

- Paclitaxel/carboplatin/bevacuximab + maintenance bevazuximab.
- Paclitaxel/carboplatin chemotherapy for six cycles followed by endocrine (aromatase inhibitors) maintenance therapy .
- Endocrine/hormonal therapy alone (Aromatase inhibitor: anastrozole, letrozole, exemestane.

5.3.15 Follow up & Surveillance

Three monthly reviews for 2 years, then 6 monthly for 3 years is recommended and annual follow up thereafter.

At follow-up: Take history, conduct physical examination and tests should be done if the patient is symptomatic, including CA-125, CT scan and PET scan. PET CT scan may be useful for assessing recurrence. Palliative care follow-up

5.3.16 Treatment Algorithm for Ovarian Germ Cell Tumors

Germ cell tumors typically respond well to chemotherapy. Most patients are young and therefore extensive surgery is rarely indicated. Fertility-sparing surgery is acceptable even in advanced disease.

5.3.17 Chemotherapy Regimens

- BEP (Bleomycin/ Etoposide /Cisplatin) regimen;
- EP (Etoposide /Cisplatin) regimen is also acceptable.

5.3.18 Follow up & Surveillance

- 3 monthly for two years then 6 monthly for 3 years. At follow-up, take history, perform a physical examination and repeat tumor markers (using the marker that was initially elevated at diagnosis).
- Imaging is necessary if recurrence is suspected;
- The role of CA-125 in follow-up is not fully established.

5.3.19 Palliative Care Management

Treatment Algorithm for Sex-cord Stromal Tumors

- BEP (Bleomycin/ Etoposide /Cisplatin) regimen

5.3.20 References

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5.4 Vulva Cancer

Vulva cancer accounts for 0.8%of all gynecological cancers worldwide, with an estimated 47,342 new cases and 18,579 deaths. (GLOBOCAN 2022). In Kenya it comprises less than 1% of all cancer cases, with 163 new cases and 82 deaths annually (GLOBOCAN, 2022). It is among cancers related to human papillomavirus (HPV) infection and can be detected early through screening. About 90% are squamous cell carcinomas.

5.4.1 Risk Factors

- HPV infection
- Immunodeficiency including HIV
- Tobacco use
- Increasing age
- Inflammatory conditions of the vulva, among others.

5.4.2 Signs and Symptoms

- Some are asymptomatic.
- Elderly women could present with vulvar pain, burning, pruritus and soreness. They could also present with vulvar lump or mass, groin mass, vulva bleeding or discharge.
- Any vulvar symptom should prompt an examination of the genital tract.

5.4.3 Physical Examination

The physical exam requires careful inspection in good light. One may observe signs of vulvar pruritus which occurs several years before actual cancer. Colposcopy may be useful after application of 2% acetic acid if no lesion is visible grossly. Warts are uncommon in elderly women and should be regarded with suspicion. Vulvar lesions may be raised, fleshy, ulcerated, leukoplakia or warty mostly on labia majora but also on labia minora, clitoris. The lesion may be multifocal. There may also be change in colour of the skin to brown or black, change in surface from smooth to scaly or ulcerated among others.

5.4.4 Investigations

Tissue biopsy (punch biopsy using keyes) is the preferred method for diagnosis. Include underlying dermis and connective tissue for evaluation of depth & nature of stromal invasion. Preferably leave primary lesion in situ to allow the treating surgeon to fashion adequate surgical margins.

Fine needle aspiration cytology of suspicious inguinal lymph nodes may be performed in the background of suggestive clinical scenarios.

Imaging of the pelvis, abdomen, or chest may complement physical examination. MRI pelvis may assist in defining the local extent of disease if surgery is planned. Chest X-ray or CT or MRI scan /PET CT of pelvis, abdomen and chest to evaluate local-regional and distant metastases when necessary.

Laboratory work up should include full blood counts, renal function tests, liver function tests, VDRL, HIV test (CD4 counts and VL) and random blood sugar

Table 5.6 : New (2021) FIGO staging for carcinoma of the vulva

Stage	Description	
I	Tumor confined to the vulva [a]	
	IA	Tumor size ^2 cm and stromal invasion <s1 mm [a]
	IB	Tumor size >2 cm or stromal invasion >1 mm
II	Tumor of any size with extension to lower one-third of the urethra, lower one-third of the vagina, lower one-third of the anus with negative nodes	
III	Tumor of any size with extension to upper part of adjacent perineal structures, or with any number of nonfixed, nonulcerated lymph node	
	IIIA	Tumor of any size with disease extension to upper two-thirds of the urethra, upper two-thirds of the vagina, bladder mucosa, rectal mucosa, or regional lymph node metastases ≤5 mm
	IIIB	Regional [b] lymph node metastases >5 mm
	IIIC	Regional [b] lymph node metastases with extracapsular spread
IV	Tumor of any size fixed to bone, or fixed, ulcerated lymph node metastases, or distant metastases	
	IVA	Disease fixed to pelvic bone, or fixed or ulcerated regional [b] lymph node metastases
	IVB	Distant metastases

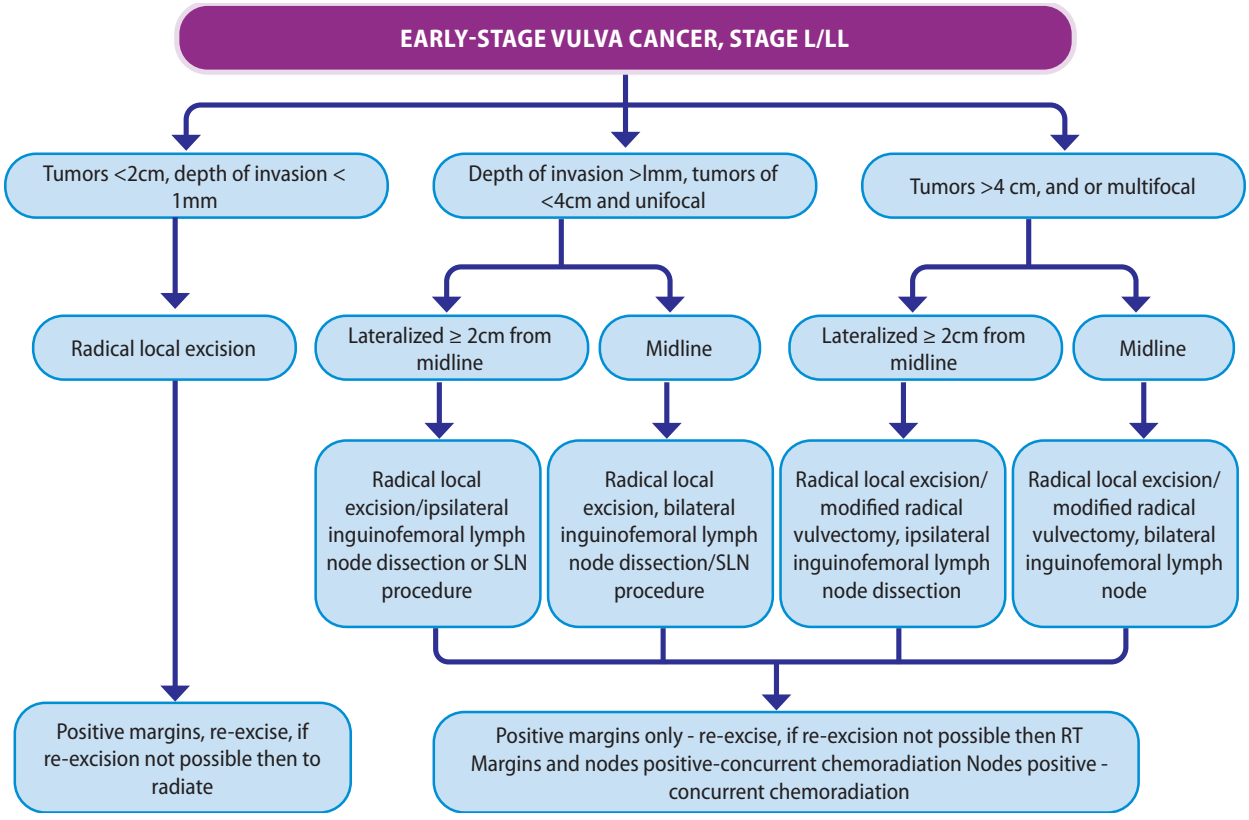


Figure 5.4: Treatment Algorithm for Early Vulvar Cancer

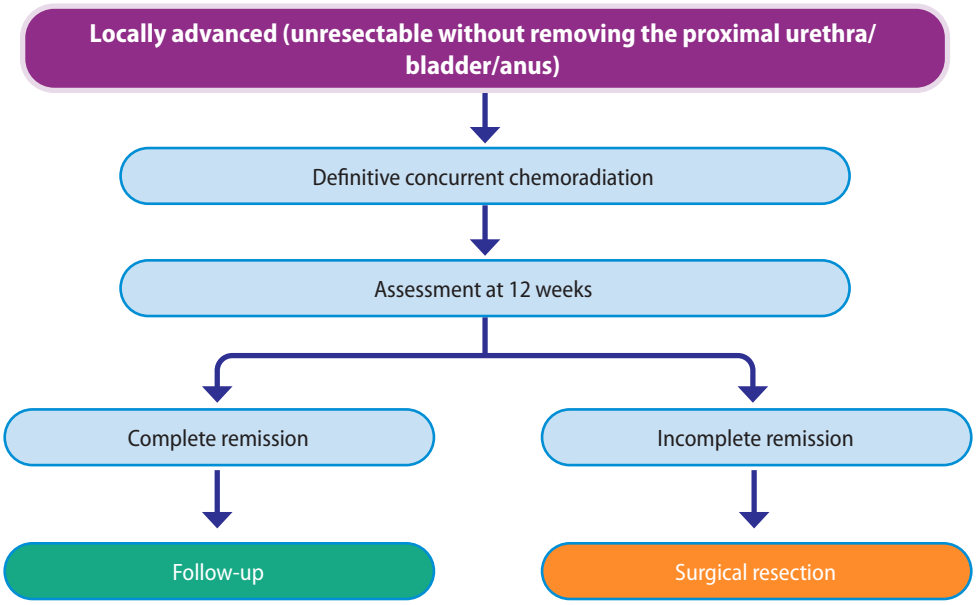


Figure 5.5 : Treatment Algorithm for locally advanced Vulvar Cancer

5.4.5 Metastatic Vulvar Cancer

- EBRT for local-regional control/symptom palliation and/or
- Systemic therapy, or
- supportive care alone

5.4.6 Follow up and Surveillance

- A three monthly for the first 2 years then six monthly for up to 5 years then yearly thereafter.
- Follow-up with, history and physical exam and imaging if recurrence is suspected is recommended.
- Palliative care follow-up

5.4.7 References

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5.5 Gestational Trophoblastic Neoplasm

5.5.1 Introduction

Gestational trophoblastic neoplasia (GTN) - A group of malignant neoplasms that consist of abnormal proliferation of trophoblastic tissue, and may follow a hydatidiform mole or a non-molar pregnancy.

GTN is comprised of the following histologic types:

- Invasive mole
- Choriocarcinoma
- Placental site trophoblastic tumor (PSTT)
- Epithelioid trophoblastic tumor (ETT)

Approximately 50% of cases of GTN arise from molar pregnancy, 25% from pregnancy loss or tubal pregnancy, and 25% from term or preterm pregnancy.

Invasive mole and choriocarcinoma are characterized by high levels of human chorionic gonadotropin (hCG), while PSTT and ETT typically have low hCG levels.

Gestational trophoblastic disease (GTD) – Lesions characterized by abnormal proliferation of trophoblast of the placenta. This category comprises benign, nonneoplastic lesions, including placental site nodule (both typical and atypical), exaggerated placental site, and hydatidiform mole.

5.5.2 Risk Factors

- Maternal age (younger than 20 or older than 35; higher in >40 years);
- Previous history of molar pregnancy- hydatidiform mole
- Nulliparity
- Gestational trophoblastic disease
- Nutritional deficiencies -decreasing levels of consumption of dietary carotene (vitamin A precursor) and animal fat

5.5.3 Clinical Presentation

- Abnormal and excessive vaginal bleeding is the commonest mode of presentation and may or may not follow a period of amenorrhea.
- Pelvic pain or sensation of pressure
- Bilateral adnexal masses
- Enlarged uterus
- In addition to the above symptoms, GTN may present symptoms and signs depending on the site of metastasis e.g. lungs, liver and brain.
- Gestational Trophoblastic Neoplasia should also be suspected in any woman presenting with per vaginal bleeding after any gestation term pregnancy, abortion or ectopic.

5.5.4 Physical Examination

Examination includes speculum pelvic examination which may show vaginal deposits.

5.5.5 Investigations

- Pregnancy Detection Test: A positive pregnancy test should prompt further diagnostic workup.
- Quantitative hCG
- UECs/LFTS/TFTs, TBCs
- Imaging (usually for evaluation of metastatic disease): Pelvic ultrasound and Chest X-ray.
- Additional imaging is indicated only when there is evidence of metastatic disease;
- MRI brain
- CT abdomen/pelvis

5.5.6 Diagnostic Criteria for GTN

GTN is a **clinical diagnosis** made based upon elevation of serum human chorionic gonadotropin.

5.5.7 Following a Molar Pregnancy

Following evacuation of a complete or partial molar pregnancy, serial serum hCG levels are measured, typically on a weekly schedule, until an undetectable level is reached. If hCG levels rise or remain elevated over several weeks, the diagnosis of post molar GTN based on International Federation of Gynecology and Obstetrics (FIGO) criteria

- Weekly hCG levels plateau (remain within ±10 percent of the previous result) over a three-week period
- hCG level increases >10 percent across three values recorded over a two-week duration
- Histologic diagnosis of choriocarcinoma or invasive mole and/or the identification of clinical or radiological evidence of metastases

5.5.8 Following a Non-molar Pregnancy

The diagnosis is made based upon an elevated hCG, with the exclusion of any other explanation than GTN.

Biopsies should **NOT** be performed because GTN lesions are highly vascular and may cause vigorous bleeding.

5.5.9 Staging and Risk Stratification

Each patient with a gestational trophoblastic neoplasia is assigned both a FIGO stage and a WHO prognostic score. The patient's diagnosis is assigned to a FIGO stage represented by a Roman number I, II, III, IV separated by a colon and the WHO risk score. For example, II: 7 to mean stage II with a WHO score of 7.

Table 5.7: FIGO Staging of Gestational Trophoblastic Neoplasia (GTN) and modified WHO Prognostic Scoring System as adapted by FIGO

Stage I	Disease confined to the uterus
Stage II	GTN extends outside of the uterus, but is limited to the genital structures (adnexa, vagina)
Stage III	GTN extends to the lungs, with or without genital tract involvement
Stage IV	All other metastatic sites
The stage should be followed by the sum of the risk factors (eg, 111:5)	

Risk factor	Score			
	0	1	2	3
Age (years)	<40	>40	-	-
Antecedent pregnancy	Mole	Abortion	Term	-
Interval (months)*	4	4 to 6	7 to 12	>12
Pretreatment serum hCG (mIU/mL)	<103	103 to 104	104 to 105	>105
Largest tumor (including uterus)	<3 cm	3 to 4 cm	>5 cm	-
Site of metastases	Lung1	Spleen, kidney	GI tract	Brain, liver
Number of metastases	-	1 to 4	5 to 8	>8
Prior failed chemotherapy	-	-	Single drug	>2 drugs

The use of the WHO Prognostic Scoring System is more predictive of clinical outcome than the use of individual risk factors

- A score of 0 to 6 suggests **low risk** of resistance to single-agent chemotherapy.
- A score of ≥7 predicts a **high risk** of resistance to monotherapy and requires combination chemotherapy.
- A score ≥12 is considered as **ultra-high risk** and particularly associated with non-molar antecedent pregnancy, brain metastases, and failed prior multiagent chemotherapy.

5.5.10 Management of Low Risk GTN

FIGO stage I GTN or Stage II or III GTN with a WHO risk score 0-6.

Low risk GTN is a highly curable disease.

Usually treated with single-agent chemotherapy, using either methotrexate (MTX) or dactinomycin (ActD) Select patients with WHO score of 5 or 6 may require combination chemotherapy in whom other high-risk features are **also** present:

- No metastases and no histopathologic choriocarcinoma (CCA), but hCG >410,000
- Either metastases from non-CCA or histopathologic CCA, and hCG >150,000
- Metastatic CCA

5.5.11 Management of High Risk GTN

Defined as Stage IV disease or Stages II and III with risk score >6

Patients with high-risk GTN are likely to develop drug resistance if single-agent therapy is administered. Therefore, these patients are commonly treated with multi-agent regimens.

The most commonly used is EMA-CO (etoposide, methotrexate, actinomycin D, cyclophosphamide, vincristine).

EMACO is usually administered every 2 to3 weekly

Induction Therapy in Patients with High Tumor Burden

Induction with EP in patients with high tumor burden can eliminate early deaths (first 4 weeks) due to complications arising as a result of tumor lysis syndrome.

Induction therapy with low dose etoposide and cisplatin on days 1 and 2, repeated weekly for 1 to 3 weeks, before starting the normal therapy in patients with high tumor burden.

Criteria for induction with low dose EP:

- Extensive disease in chest, brain or liver
- FIGO> 12 or histology of choriocarcinoma
- Major Bleeding

5.5.12 Treatment of Patients with Brain Mets

Higher MTX dose than what is routinely administered.
The higher dose of parenteral MTX allows for adequate levels of MTX within the cerebrospinal fluid (CSF).
Dexamethasone to decrease cerebral edema and antiseizure treatment

5.5.13 Management of Ultra-High Risk GTN

WHO risk score of ≥12
Commonly used regimens – EMA/EP or TP/TE (less toxic)

5.5.14 Consolidation Therapy

After complete remission is attained, consolidation therapy should be administered to prevent relapse. This generally consists of three courses of the last effective regimen

5.5.15 PSTT and ETT management

Both PSTT and ETT are less chemosensitive than choriocarcinoma.
Hysterectomy is the primary mode of treatment in most cases
In the advanced stage, EP- EMA or TE/TP should be considered
Interval from antecedent pregnancy of more than 48 months is the most significant adverse prognostic factor.

5.5.16 Indications of Surgery in GTN

Role of surgery in GTN is secondary, chemotherapy is effective in vast majority

5.5.17 Hysterectomy Indications

1. For primary treatment of PSTT, ETT, or other chemotherapy-resistant diseases.
2. For treatment of severe uncontrollable vaginal or intra-abdominal bleeding as an emergency procedure.
3. Adjuvant hysterectomy decreases the total dose of chemotherapy needed to achieve clinical remission in low-risk GTN.

5.5.18 Monitoring During Treatment

Serial measurements of serum human chorionic gonadotropin (hCG) at the start of treatment and then at weekly intervals during therapy, until 3 consecutive weekly negative levels

Remission is achieved when the quantitative hCG level becomes undetectable for three consecutive weeks. Given the sensitivity of this tumor marker, no imaging is required if levels are consistent with remission because abnormalities on imaging can persist despite the attainment of undetectable hCG levels, representing fibrosis rather than active tumor.

5.5.19 Follow up & Surveillance

Monitor serum β-HCG levels monthly. In low-risk disease, this is carried on for a year. In high-risk patients, this should be performed for a period of 2 years. During this period of monitoring, it is important to prescribe effective contraception e.g. combined oral contraceptive pills to avoid pregnancy.

Fertility, psychosocial support and support groups should be emphasized.

Prognosis is good even with metastatic disease especially when only lungs are involved.

5.5.20 References

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Chapter 6

Head And Neck Cancers

List of Abbreviations

Ab	Antibody
CD4	Cluster of differentiation 4 type T-cells
CT	Computed tomography scan
CXR	Chest X-Ray
EBRT	External Beam radiotherapy
EBV	Epstein Barr Virus
ENT	Ear nose and throat
FNA	Fine needle aspirate
HIV	Human Immunodeficiency virus
HPV	Human papilloma virus
IMRT	Intensity modulated radiotherapy
LFT	Liver function tests
MDT	Multidisciplinary team
MRI	Magnetic resonance imaging
PET	Positron emission tomography
PNS	Post nasal space
RFT	Renal function test
SCC	Squamous cell carcinoma
SLT	Speech and language therapist
TBC	Tuberculin test
TFT	Thyroid function test
TNM	Tumour node metastasis
WCC	White cell count
WHO	World Health Organization

6.1 Introduction

These are tumors affecting the upper aerodigestive tract above the level of the clavicle (excluding those of the central nervous system and skin). They constitute about 6.1% of the cancer burden in Kenya. (Globocan 2022) Anatomically, these include oral cavity; oropharynx; hypopharynx; larynx; nasopharynx; sinonasal regions; thyroid and salivary glands tumors. Majority are squamous cell carcinomas.

6.2 Risk Factors

A high index of suspicion is recommended for patients who are:

- Heavy smokers or other forms of tobacco use and/or chewing betel (areca nut)
- History of harmful use of alcohol
- Aged over 45 years
- Male gender

Other risk factors include:

- exposure to carcinogenic human papilloma virus (HPV) strains, Epstein Barr virus EBV, Human Immunodeficiency Virus (HIV).
- environmental/occupational inhalants (textile, glue, leather, wood dust) Agrochemicals, smoke.
- Previous radiotherapy.

Risk factors for thyroid malignancies. include:

- Age >65Y,
- Female gender,
- Previous radiotherapy,
- positive family history of thyroid ca

6.3 Signs & Symptoms – this Depends on the Affected Subsite

6.3.1 Urgent Referral Considerations

Head & Neck Cancers

- Hoarseness persisting for > 2weeks
- Ulceration of oral mucosa persisting for > 2weeks
- Oral or facial swellings persisting for > 2 weeks
- All red or red and white patches of the oral mucosa
- Dysphagia and odynophagia persisting for > 2 weeks
- Unilateral/bilateral nasal obstruction, particularly when associated with purulent discharge
- Unexplained tooth mobility and /or pain not associated with periodontal disease

- Unresolving neck masses persisting for > 2 weeks
- Cranial neuropathies
- Orbital masses, strabismus, proptosis (unilateral or bilateral) and visual impairment visual impairment.
- Ear pain without evidence of local ear abnormalities
- Unilateral middle ear effusion or hearing loss especially in adults,
- unexplained Epistaxis.
- Hyponasal speech.

6.4 Thyroid Cancer

Patients with thyroid lump/swelling/enlargement AND:

- Stridor
- Cervical lymphadenopathy
- Voice change
- Thyroid swelling/ enlargement,
- dysphagia & odynophagia,
- symptoms of thyrotoxicosis,
- bone pains (advanced disease)

6.5 Clinical Evaluation

All patients with the above symptoms should be referred IMMEDIATELY to the ENT or maxillofacial clinic for further evaluation and management.

6.6 History and Physical Examination

- Take a detailed history
- Perform a complete physical examination
- **Refer the patient for examination and biopsy by an ENT specialist.** This may include indirect laryngoscopy, direct laryngoscopy with hypopharyngoscopy under general anesthesia, or rigid nasal endoscopy under local or general anesthesia.
- For patients presenting with metastatic neck disease and an occult primary tumor, tissue biopsies should be obtained from the nasopharynx, oropharynx, hypopharynx, and tonsillar region during examination under anesthesia.

6.7 Imaging

- CT scan with contrast can be used for primary staging of head and neck cancers. Where available, PET CT-Scan is the gold standard for staging locally advanced disease T3-T4 primary or ≥ N1.

However, FDG-PET cannot rule out brain metastases and where this is a concern in cases such as mucosal melanoma or high-grade neuroendocrine carcinoma or adenocarcinoma, contrast enhanced brain MRI should be performed.

- CT scan*, MRI**, abdominal ultrasound and chest x-ray (CXR) are used for assessment of the primary and staging (Appendix 6- IMAGING PROTOCOLS).

**Features of mastoiditis on CT should prompt evaluation of the PNS.*
***MRI is used where CT is inconclusive or contraindicated and also for imaging of oral cavity tumours.*

6.8 Pathology

- Tissue biopsy*** – for all patients for histological diagnosis
- Immunohistochemistry - an important adjunct for confirmation of diagnosis.
- Fine needle aspirate (FNA) cytology of suspected metastatic neck nodes should also be done. FNAC is also essential in diagnostic work-up of thyroid carcinoma.
- Molecular diagnostics and genomics

Open biopsy of metastatic neck disease is not recommended.
*****Patients with metastatic neck disease and occult primary should have tissue biopsy obtained from the nasopharynx, oropharynx and hypopharynx during examination under anaesthesia.**

6.9 Laboratory Tests

Total Blood Count, Liver Function Tests, Renal Function Tests, HIV test + CD4 levels, Eb- stein Barr Virus and HPV Antibody titers, Thyroid Function Tests, Parathyroid hormone and Bone Marrow Aspirate and ultrasound.

6.10 Hearing Evaluation

Audiometry should be done prior to treatment.

6.11 Dental Evaluation

Dental. Evaluation and preparation (which may involve extraction of loose and infected teeth) should be done for those receiving irradiation.

6.12 Nutritional Evaluation

For possible feeding gastrotomy or PEG tube insertion before start of any radiotherapy to head & neck. In addition, nutritional planning and advise for high calorie and protein diet should be done.

6.13 Airway Evaluation

For possible tracheostomy insertion before treatment to prevent airway obstruction as an emergency.

6.14 Staging

Table 6.1: Stage Grouping of Head and Neck Cancers

Primary Tumour (T)	
TX P	primary tumour cannot be assessed
TO N	o evidence of primary tumour
Tis	Carcinoma in situ
TI T	tumour < 2 cm in greatest dimension
T2 T	tumour < 2 cm but 4 cm
T3 T	tumour < 4 cm
T4	Tumour invades adjacent structures (defined under specific site)
T4a	Tumour invades adjacent structures, but is potentially resectable
T4b	Tumour invades adjacent structures, but is unresectable
Regional Lymph nodes (N)	
Nx R	regional lymph nodes cannot be assessed
NO N	o regional node metastasis
N1 M	metastasis in a single ipsilateral lymph node, < 3 cm
N2a	Metastasis in single ipsilateral lymph < 3 cm but <6 cm
N2b	Metastasis in multiple ipsilateral lymph nodes, none > 6 cm
N2c	Metastasis in bilateral or contralateral lymph nodes/ none 6 cm
N3 M	metastasis in a lymph node > 6 cm in greatest dimension
Distance Metastasis (M)	
Mx	Distant metastasis cannot be assessed
M0No	Distant metastasis
M1	Distant metastasis

Table 6.2: Stage Grouping of Head and Neck Cancers

0 I II	Tis T1 T2	N0 N0 N0	M0 M0 M0		
III	T1-3	N1	M0	T3 N0	M0
IVA	T4	N0-2	M0	T1-3 N2	M0 (advanced, resectable disease)
IVB	Any T	N3	M0	T4 Any N	M0 (Advanced disease, unresectable)
IVC AnyT AnyN M1 (advanced distant, metastatic disease)					

Above staging excludes nasopharyngeal carcinoma
T staging of hypopharyngeal and salivary gland carcinomas is similar to others in tumour size but also depends on local tumour extension.
Carcinomas of the larynx and paranasal sinuses have specific definitions for all T stages that depends on tumour location rather than size
Adapted from AJCC 8th Edition 2018

Table 6. 3: Nodal Assessment of Thyroid Cancers

Nx N0	Nodal disease cannot be assessed One or more cytological or histologically confirmed benign node
N1a	Metastases to level VI and VII nodes (pre-tracheal, para-tracheal, pre-laryngeal/Delphian or upper mediastinal
N1b	Metastases to unilateral, bilateral or contralateral lateral neck nodes (level I, II, III, IV, V) or retropharyngeal nodes

Adopted from AJCC 9th edition 2022

Table 6.4: Nasopharynx Cancer staging (Adopted form AJCC 9th Edition)

Staging		
T	cTX	Primary tumour cannot be assessed
	C T0	No evidence of primary tumour, but EBV-positive (EBV-associated) cervical node(s) metastasis present
	cTis	Carcinoma in situ
	cT1	Tumour confined to nasopharynx or Tumour extends to oropharynx and/or nasal cavity without parapharyngeal involvement
	cT2	Tumour with extension to parapharyngeal space or Tumour infiltration of the medial pterygoid, lateral pterygoid and/or prevertebral muscles
	Ct3	Tumour invades bony structures of skull base, cervical vertebrae, pterygoid structures and/or paranasal sinuses

Staging		
	CT4	Tumour with any of the following: ● Intracranial extension ● Unequivocal clinical and/or radiological involvement of cranial nerves ● Involvement of hypopharynx ● Invading orbit (including inferior orbital fissure) ● Involvement of parotid gland ● Infiltration beyond the anterolateral surface of the lateral pterygoid muscle
N	cNX	cNX Regional lymph nodes cannot be assessed cN1 Unilateral metastasis in cervical lymph node(s), and/or unilateral or bilateral metastasis in retropharyngeal lymph nodes, and 6 cm or less in greatest dimension, and above the caudal border of cricoid cartilage, and without advanced clinical/radiological extranodal extension*
	cNO	cN0 No regional lymph node metastasis
	cN1	Unilateral metastasis in cervical lymph node(s), and/or unilateral or bilateral metastasis in retropharyngeal lymph nodes, and 6 cm or less in greatest dimension, and above the caudal border of cricoid cartilage, and without advanced clinical/radiological extranodal extension*
	cN2	Bilateral metastasis in cervical lymph nodes, and 6 cm or less in greatest dimension, and above the caudal border of cricoid cartilage, and without advanced clinical/radiological extranodal extension*
	cN3	Metastasis in cervical lymph node(s) greater than 6 cm in greatest dimension Or Extension below the caudal border of cricoid cartilage or Advanced clinical/radiological extranodal extension*
M	cM0	No distant metastasis
	M1	Distant metastasis. M1a Distant metastasis. Three or fewer lesion(s) in one or more organs M1b Distant metastasis of more than three lesions in one or more organs

Stage-Nasopharynx			
Stage 0	Tis	N0	M0
Stage IA	T1,T2	N0	M0
Stage IB	T0,T1,T2	N1	M0
Stage II	T0,T1,T2	N2	M0
	T3	N0,N1,N2	M0
Stage III	T4	Any N	M0
	Any T	N3	M0
Stage IVA	Any T	Any N	M1a
Stage IVB	Any T	Any N	M1b

Figure 6.1 Nasopharynx Cancer Prognostic staging (Adopted from AJCC 9th edition)

6.15 Management

Challenges in the management of head and neck cancers include the preservation of functional integrity of the upper aerodigestive tract, minimizing the adverse effects of the different treatment modalities, and aesthetics. The core multidisciplinary team (MDT) needed for their management includes a maxillofacial and reconstructive surgeon, pathologist, radiologist, radiation oncologist, clinical pharmacist, nutritionist, oncology nurse, physiotherapist and counsellor.

Patients with the below features must be booked with the multidisciplinary team for a decision on the appropriate modality of treatment:

- Extensive primary and/or nodal that is medically inoperable
- Distant metastases
- Recurrent or persistent disease not amenable to surgery
- Performance status >2
- Requiring radiotherapy for symptom control

6.17 Head and Neck Cancers

Surgery and concurrent chemo-radiotherapy are the key modalities in the treatment of head and neck cancer.

6.18 Surgery

- Surgery is the mainstay of treatment in majority of early head and neck cancers and is guided by clinical staging and performance status.
- Early tumors of the larynx/pharynx will be assessed for open or laser excision or radical radiotherapy.
- Tumors that are not of mucosal origin are surgically treated where amenable. Post-surgical Histological grade may also determine the need for post-operative radiation with or without chemotherapy.

6.19 Radiotherapy

Radical radiotherapy (given concurrently with chemotherapy) requires 1-2 weeks for planning and 6-7 weeks to complete the actual treatment. These timelines need to be well communicated to the patient.
Indications for post-operative radiotherapy include:

- Incompletely excised lesions (margins <5mm)
- Perineural invasion
- T3/T4 tumors
- 2 or more positive lymph nodes
- Extra-capsular extension of nodes
- Vascular invasion
- High-grade thyroid tumors

Palliative radiotherapy can be given in metastatic disease e.g. fractures, brain metastasis, bleeding or ulceration for symptom control.

Note that:

- Nasopharyngeal tumors are treated with radiotherapy with or without chemo therapy, depending on stage.
- Where the primary lesion is occult, treat the neck disease appropriately as the search to identify the primary continues.

6.20 Systemic Therapy Protocols

This consist of targeted therapy and chemotherapy. Systemic therapy can be used as induction/neo-adjuvant, adjuvant or concurrent with radiotherapy in a curative/radical setting. It can also be used in a palliative setting in case of advanced, recurrent or metastatic disease.

Table 6.5: Systemic Therapy Regimens For Head & Neck Tumors

Regime	Indications
Cisplatin single agent	• For definitive concurrent/chemoradiotherapy • Post-operative concurrent /chemoradiotherapy for fit patients with high risk features. • Palliative chemotherapy for unfit or unresected recurrence
Carboplatin AUC 5 single agent	• As above, where cisplatin is not tolerated or is contraindicated.
Cetuximab Single agent	• Concurrent/Chemoradiation for definitive and postoperative therapy. • Palliative therapy
Single agent: Methotrexate, Docetaxel, Ifosfamide, Bleomycin OR 5FU	• Metastatic or unresectable recurrent with unfit or poor performance status
Cisplatin and 5FU doublet Or Carboplatin and 5FU doublet	• Induction chemotherapy • Metastatic or unresected recurrent very fit/ performance status 1-2
Cisplatin, 5FU and Docetaxel	• Induction chemotherapy
Cisplatin/ carboplatin andTaxane OR Cisplatin/ cetuximab	• Metastatic fit patients • Unresectable recurrent and fit.

6.21 Chemotherapy Regimens for Nasopharyngeal Carcinoma

- All patients without contraindications should receive concurrent cisplatin with radiotherapy. Weekly (≈ 40 mg/m^2) or tri-weekly (≈ 100 mg/m^2, or at least 80 mg/m^2) administrations are acceptable.
- The cumulative dose of cisplatin in the concurrent phase should reach at least ~200 mg/m^2 (e.g. 3 cycles triweekly, or ~7 weekly doses) to get full benefit.
- For patients with contraindications to cisplatin: alternatives include nedaplatin, carboplatin (AUC 5-6 in triweekly schedule), or oxaliplatin weekly.
- Induction chemotherapy is recommended for stage III-IVA (except "T3N0" in AJCC 8th) before concurrent chemoradiotherapy. Regimens include:
    - GP (gemcitabine + cisplatin)
    - TPF (docetaxel + cisplatin + 5-FU)
    - PF (cisplatin + 5-FU)

- Adjuvant chemotherapy (PF: cisplatin + 5-FU) may be used after definitive chemoradiotherapy in some patients.

Note:

Renal function should be monitored initially and a full blood count performed weekly. In patients with renal compromise, reduce Methotrexate dose or discontinue. Do not give if the patient has pleural effusion or ascites, because of the 'reservoir effect'. Mild stomatitis signals low white cell count; wait until WCC recovers before commencing

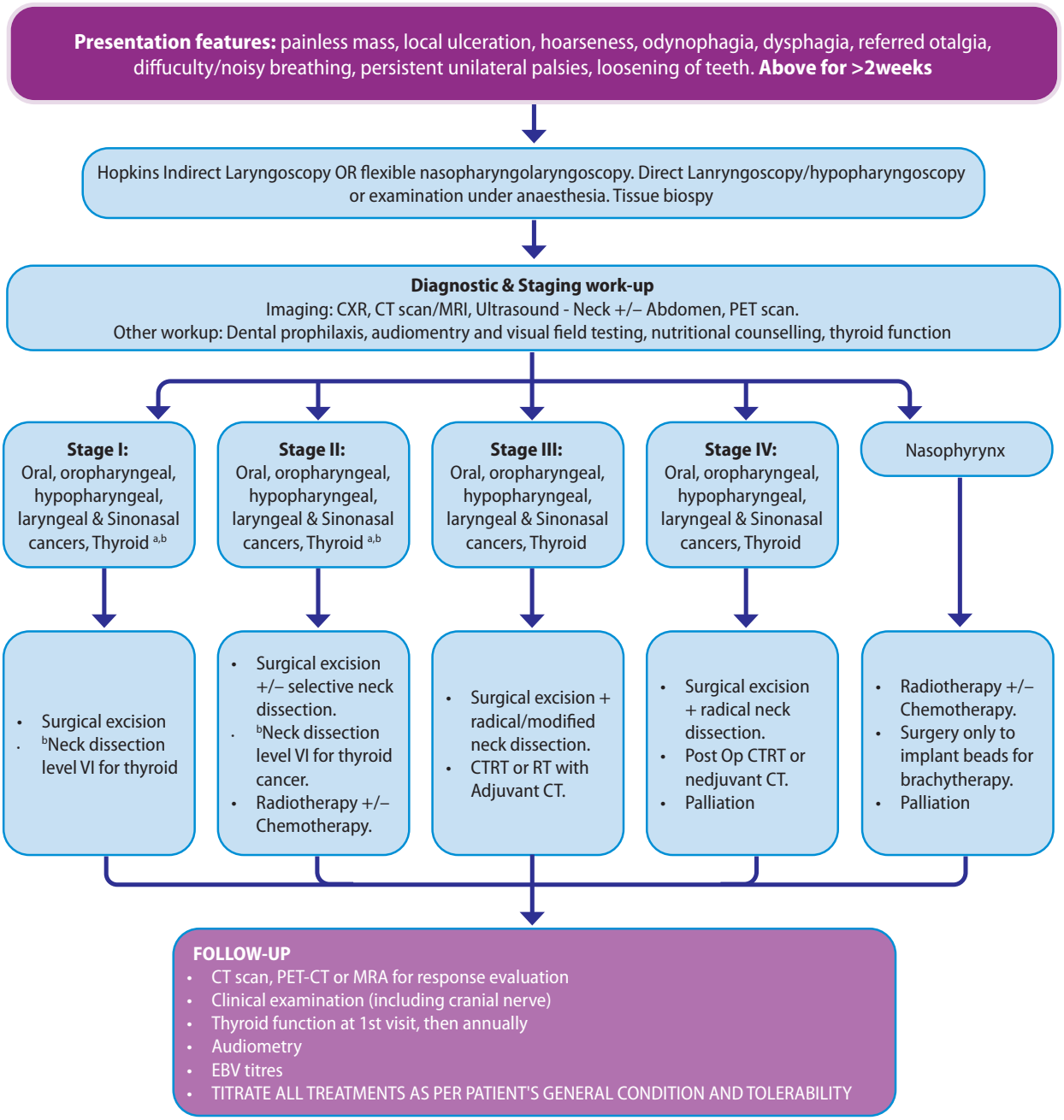


Figure 6.2: Management Algorithm for Head & Neck SCC and Thyroid Cancers

6.22 Thyroid Carcinomas

They Include differentiated subtypes (follicular, Hurthle cell, medullary and papillary) and un-differentiated subtypes (anaplastic).

Surgery remains the mainstay treatment modality for thyroid cancers amenable to surgery.
Iodine131 ablation should be carried out at designated centers for: follicular and papillary subtypes, tumor diameter > 1 cm; tumor with gross extra-thyroid extension and distant metastases. A post-ablation scan is indicated 3–10 days after the procedure.

For Hurthle, medullary and undifferentiated subtypes, chemotherapy, radiotherapy and Tyrosine Kinase Inhibitors can be given either in adjuvant or palliative setting.

6.23 Palliative Care

Palliative care is required for all patients.

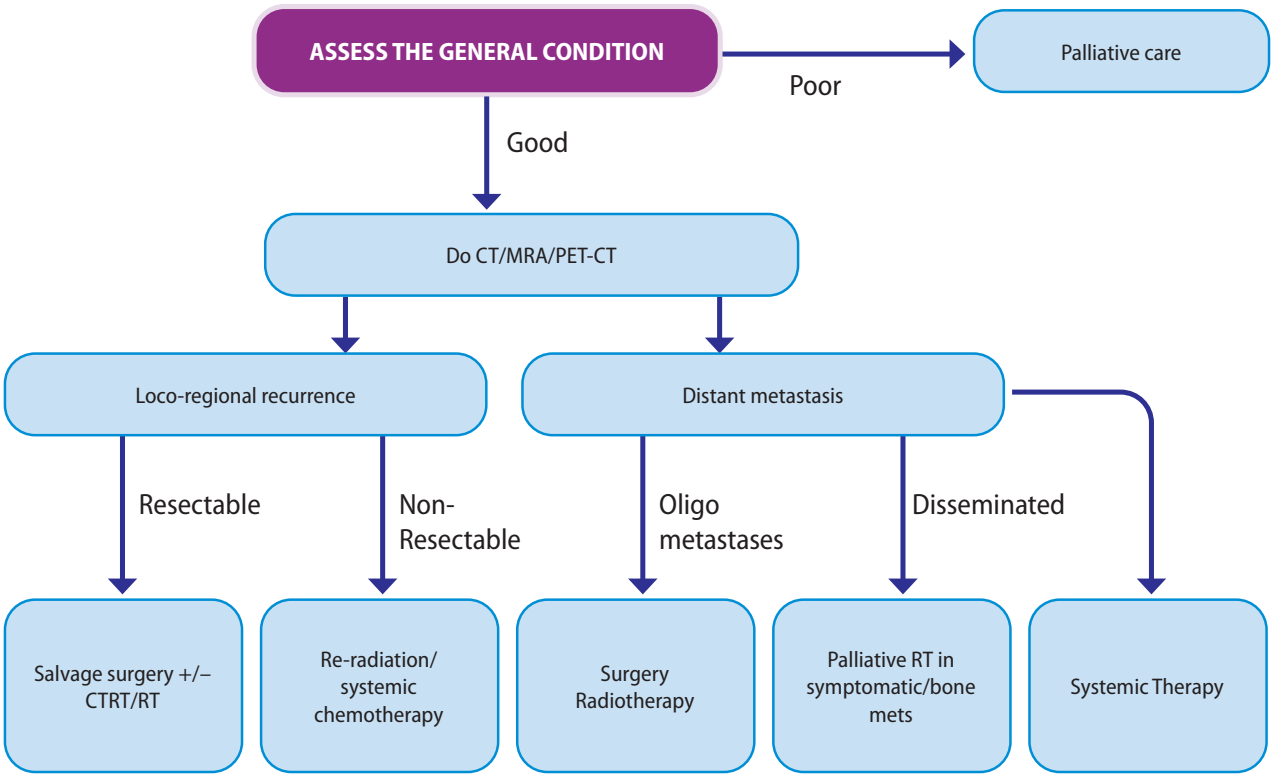


Figure 6.3: Management Algorithm for Recurrent/Metastatic Head & Neck Cancers
NB: palliative care should be offered for all patients regardless of stage or prognosis

Table 6.6: Supportive Management for Head and Neck Cancer

Symptom/sign	Referrals/assessments/interventions to be considered
Shoulder/accessory nerve Problem	• Referral to physiotherapy • Referral to specialist shoulder unit • Referral back to MDT
Problems with speech	• Referral to speech and language therapist (SLT) • Laryngectomy patients should maintain lifelong links with local head and neck SLT services
Dysphagia	• Referral to SLT/referral to dietitian Assessment by SLT Assess weight loss • Dietary modification and dietary advice Referral back to MDT

Symptom/sign	Referrals/assessments/interventions to be considered
Trismus **Dental problems/** **ORN Dry mouth**	• Referral to SLT, Dietary advice, • Referral Back to MDT • Referral back to specialist MDT Artificial saliva • Carry water bottle
Excessive fatigue	• Referral to occupational therapy/physiotherapy for fatigue-management strategies • Support for family and care givers • Training in management strategies
Weight changes	• Consider if steroid related and reduce/increase dose if clinically indicated • Referral to dietitian • If weight loss related to nausea and vomiting, consider anti-emetics • If weight gain, consider reducing dietary intake
Raised blood sugars	• Referral to diabetic nurse specialist/GP • Start medication if appropriate • Monitor blood sugars regularly • Encourage low sugar diet Referral to dietitian for dietary advice
Constipation	Regular laxatives, ENCOURAGE ORAL fluid intake, referral to dietitian
Tobacco use	Cessation services
Anxiety/depression	• Non-pharmacological techniques: counselling, cognitive behaviour therapy, • Psychologist • Antidepressant • Consider psychiatric referral if depression unresponsive to treatment or if suicide risk identified

6.24 Patient Follow-Up

Any patients with a new primary or a suspected recurrence after radical treatment should be referred to ENT/ maxillofacial clinic for re-assessment and further treatment. Recommended post-treatment follow-up intervals

- Monthly for the initial 3 months, then 3-monthly for 1 year
- 6-monthly in the second year
- Annually until the fifth year

Follow up will include clinical examination, radiological studies and lab investigations.

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Chapter 7

Adult Haematological Cancer Treatment Guidelines

List of abbreviations

Abbreviation	Definition
AML	Acute Myeloid Leukemia
ALL	Acute Lymphoid Leukemia
ATRA	All-trans Retinoic Acid
LDH	Lactate Dehydrogenase
CNS	Central Nervous System
FISH	Fluorescence In Situ Hybridization
HIV	Human Immunodeficiency Virus
HLA	Human Leukocyte Antigen Typing
HSCT	Hematopoietic Stem Cell Transplant
APML	Acute Promyelocytic Leukaemia
FLAG-Ida	Fludarabine, Cytarabine, G-CSF and Idarubicin
HiDAC	High Dose Cytarabine
NGS	Next Generation Sequencing
PCR	Polymerase Chain Reaction
TKI	Tyrosine Kinase Inhibitor
FDG-PET CT	Fluorodeoxyglucose Positron Emission Tomography–Computed Tomography
IFRT	Involved Field Radiotherapy
ISRT	Involved Site Radiotherapy
MGUS	Monoclonal Gammopathy of Undetermined Significance
SFLC	Serum Free Light Chain
SPEP	Serum Protein Electrophoresis
UPEP	Urine Protein Electrophoresis

7.1 Acute Leukaemias

7.1.1 Introduction

Acute leukaemia is characterized by the presence in the bone marrow and/or peripheral blood circulation of immature cells (blasts) with lymphoblasts in acute lymphoblastic leukaemia (ALL) and myeloblasts in acute myeloid leukaemia (AML). Both have a slight male preponderance with an aggressive biological behavior and rapid progression to fatality if not properly treated. They are the 7th most common prevalent cancers in Kenya with about 1729 new cases annually.

Symptoms include:	**Examine for:**
• Persistent or recurrent infections • Persistent fatigue • Generalized body weakness • Shortness of breath/Dyspnea on exertion • Bleeding or easy bruisability • Epistaxis, Gum bleeding or Menorrhagia • Unexplained weight loss • Drenching night sweats • Persistent fever • Anorexia • Bone pains • Nodal swellings • Abdominal swelling	• Fever • Pallor • Petechiae • Bruising (purpura) • Gum hypertrophy • Splenomegaly • Hepatomegaly • Lymphadenopathy (Assess all nodal groups) • Abnormal swelling (to suggest leukemic infiltration) • Features of hyperleukocytosis in high count leukemia

Figure 7.1: Signs and Symptoms

Do a full blood count in patients with the above signs and symptoms and refer for further care.

7.1.2 Diagnostic Workup

1. Full Blood Count with differential count (cytopenia's or leukocytosis) and peripheral blood film examination (blasts).
2. Bone Marrow Aspirate examination and trephine biopsy {>20% blasts is diagnostic**).
3. Immunophenotyping; Flow cytometry on peripheral blood or bone marrow aspirate or immunohistochemistry on trephine biopsy
4. **Cytogenetics and molecular studies for establishing subtype, risk stratification and prognostication**

   ***Both stained and unstained slides should be prepared at site and sent with the whole blood to the pathology laboratory for review*

   *** According to WHO 2022, the 20% blast threshold is no longer required for the diagnosis of AML in those with defining genetic abnormalities—except in cases such as AML with BCR:ABL1 fusion, CEBPA mutations, and certain other rare genetic alterations, which still require ≥20% blasts for diagnosis.*

7.1.3 Additional Tests

1. Liver and renal function tests, lactate dehydrogenase (LDH) and uric acid
2. Serology for HIV (mandatory), Hepatitis B and Hepatitis C (strongly recommended).
3. Pregnancy testing for childbearing women
4. Coagulation profile if indicated
5. Lumbar puncture for cytology is mandatory for all patients with ALL.
6. Others depending on clinical presentation: i.e. septic screen,
7. ECHO for planned anthracycline use etc.
8. HLA typing of patients requiring allogeneic transplant must be done at or soon after.

7.1.4 Treatment

- **Urgently refer all patients suspected to have acute leukemia for hemato-oncologist review and management in tertiary facilities with adequate supportive care services.**

- Definitive treatment includes induction and consolidation for AML for fit patients. Less intense treatment is available for those not fit to undergo aggressive chemotherapy.

- The definitive treatment for ALL includes induction, consolidation and maintenance therapy along with CNS prophylaxis or treatment.

- Management within a Multi-disciplinary Team setting is encouraged.

- The role of hematopoietic stem cell transplant (HSCT) should be discussed before the initiation of treatment. Allogeneic HSCT is an effective post remission strategy for patients with high risk acute leukemias in first complete remission post induction treatment.

7.1.5 Supportive Care

- **Transfusion** of blood components (packed cells and platelet concentrates) as needed. Leukocyte-depleted products should be used for transfusion where available.

- **Cytoreductive Therapy:** *R*isk factors for leukostasis include a markedly elevated WBC count, especially > 100,000/mm^3. Symptoms include dyspnea due to pulmonary infiltration and altered mental status due to CNS effects. Patients should receive emergency cytoreductive therapy, steroids (ALL) and Hydroxyurea (AML). Leukapheresis can be considered if available to rapidly reduce the WBC count where available.

- **Prophylaxis and treatment of tumor lysis syndrome:** *R*isk factors include elevated WBC count, especially > 50,000/mm^3; marked elevation of LDH, especially > 1000 U/L and baseline hyperuricemia. Patients should receive aggressive hydration @ 3L/m^2 and allopurinol; high-risk patients should also receive uricolytic agents such as rasburicase.

- **Prevention and treatment of infections:** Most patients are neutropenic and immunocompromised therefore barrier nursing and isolation should be emphasized.

- Afebrile patients usually receive prophylactic antibiotics, such as ciprofloxacin or levofloxacin, antivirals such as acyclovir, and an azole antifungal such as posaconazole, voriconazole, itraconazole, or fluconazole during induction therapy.

- Febrile neutropenic patients require prompt initiation of broad-spectrum antibiotics regimens which include a third- or fourth-generation cephalosporin or carbapenem with or without vancomycin.

7.1.6 Acute Lymphoblastic Leukaemia

Diagnostic and Treatment

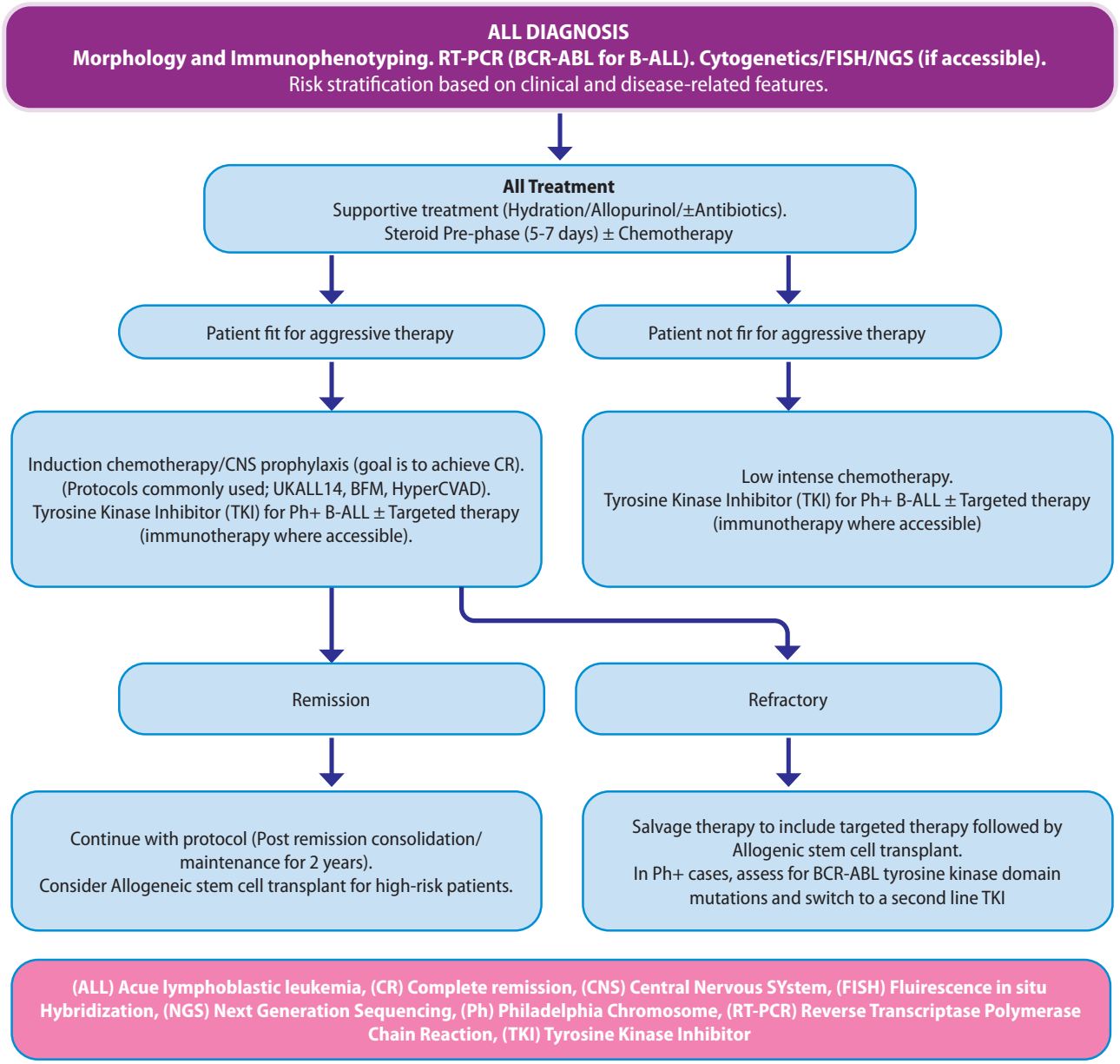


Figure 7.2: Diagnostic and treatment algorithm for Acute Lymphoblastic Leukemia:

The entire treatment course will run over 2½ years with the first 6 months requiring admission for intravenous chemotherapy from time to time.

Some of the chemotherapy protocols include:

- Hyper-CVAD
- UKALL 14
- BFM protocol
- GMALL
- GRALL
- CALGB

For frail and elderly patients and in those with major comorbidities, less intensive chemotherapy protocols can be used

7.1.7 CNS Prophylaxis

Patients with ALL have an increased risk of CNS involvement; therefore, intrathecal therapy (methotrexate, and corticosteroid) for CNS prophylaxis either after consolidation or during maintenance.

7.1.8 Role of Stem Cell Transplantation in High-risk Patients

Allogeneic transplant is used in high risk ALL patients as a consolidative therapy in first complete remission or in the relapse setting.

If HLA-matched siblings are available: consider Allogeneic Transplantation post induction chemotherapy for high-risk patients. Those who are not able/ not willing to undergo a stem cell transplant proceed with consolidation therapy followed by maintenance therapy for at least 2 years.

- **Stem cell transplantation is the most intensive post remission therapy and potentially increases a patient's chance for cure.**

NOTE: Allogeneic stem cell transplantation patients frequently develop graft versus host disease that can affect long-term survival as well as quality of life.

7.1.9 Treatment of Relapse

Though the prognosis after relapse is generally poor, the best outcome is obtained if patients achieve a second remission and then proceed to allogeneic stem cell transplantation. Most of the earlier regimens used for first line therapy may be used for salvage therapy though other protocols for consideration may include: FLAG-IDA, Nelarabine based therapies for T cell ALL, Immunotherapy for B ALL, second or third generation TKI for Ph+ B cell ALL and CAR-T therapies where available.

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7.2 Acute Myeloid Leukaemia

This is a heterogeneous group of diseases which, though infrequent, are highly malignant yet curable in a sizable proportion of cases. AML shows 2 peaks in occurrence, one in early childhood with the majority occurring later in adulthood. The 2022 WHO classification (WHO-HAEM5) categorizes AML into four categories as:

- AML with defining recurrent genetic abnormalities e.g. PML-RARA fusion, RUNX1::RUNX1T1 fusion, CBFB::MYHH11 fusion, DEK::NUP214 fusion, among others.
- AML, defined by differentiation e.g. AML with maturation, without maturation, with minimal differentiation, basophilic, myelomonocytic among others.
- AML myelodysplasia related.
- Myeloid sarcoma

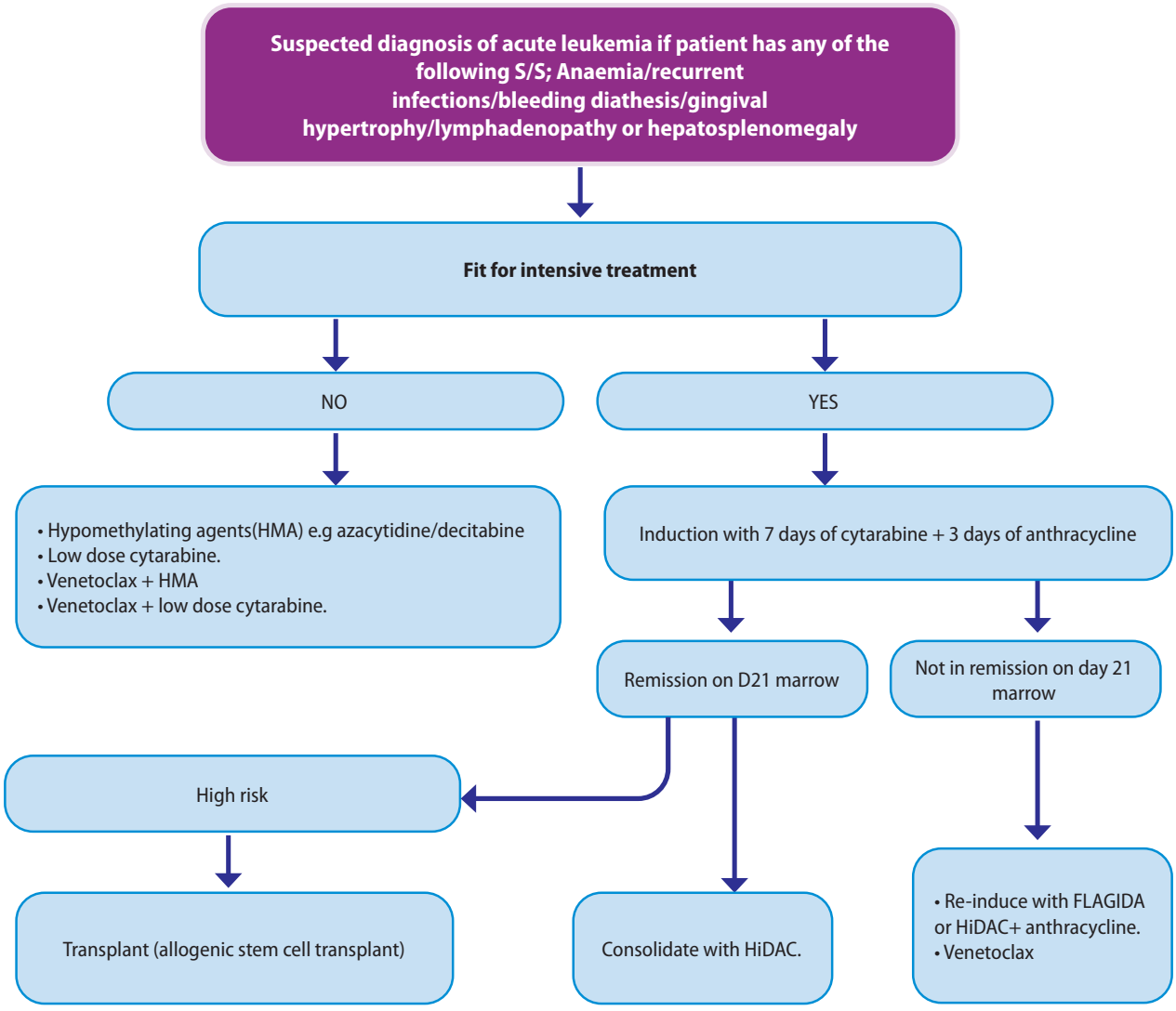


Figure 7.3 : Acute Myeloid Leukaemia Treatment Algorithm

Treatment recommendations for FIT patients defined as < 60y OR select patients ≤ 75y (good performance status, with minimal/no comorbidities);

Induction therapy: with a combination of cytarabine and anthracycline. Assess for remission status by repeating a bone marrow aspirate/immunophenotype reassessment on Day 21.

Consolidation therapy: Assess the risk stratification and remission status to decide on consolidative management. In those in remission:

Table 7.1: Risk Stratified Management Of Acute Myeloid Leukemi

Good risk patient	High dose cytarabine every 12 hours on days 1,3 and 5 for 2-3 cycles.
Standard or intermediate risk	High dose cytarabine IV on days 1,3 and 5 for 4 cycles OR Allogenic stem cell transplant
Adverse risk	Allogenic stem cell transplant OR Clinical trial OR High dose cytarabine IV every 12 hours on day 1, 3 and 5; if clinical trial or transplant are not available

*** Patients receiving HiDAC have a high risk of cerebellar toxicity. Stop if you notice any of these signs: nystagmus, slurred speech etc.*
*** Steroid eye drops should be administered to both eyes four times daily until 24 hours after cytarabine.*

Risk stratification has a bearing on management of the patient and time of referral for transplant.

- Low risk patients can be managed with chemotherapy unless they do not achieve remission in first induction.
- High risk patients are preferentially referred for transplant on achieving remission post first induction (CR1).

Treatment recommendations for patients not considered fit for intensive therapy ≥ 60yrs:

Azacitidine 75 mg/m2 SC or IV for 7 days every 28 days and venetoclax PO once daily (100 mg day 1, 200 mg day 2, and 400 mg days 3 and beyond) Decitabine 20 mg/m2 IV for 10 days every 28 days and venetoclax PO once daily (100mg Day 1, 200mg Day 2, and 400 mg day 3 and beyond. LDAC 20 mg/m2 SC for 10 days every 28 days and venetoclax PO once daily (100 mg day 1, 200 mg day 2, 400 mg day 3 and 600 mg day 4 and beyond
Azacitidine 75 mg/m2 SC or IV days 1–7 of each 28-day cycle Decitabine 20mg/m2 IV days 1-10 of each 28-day cycle LDAC 20 mg/m2/day SC days 1–10 of each 28-day cycle
:supportive care with transfusion and infection management

**It is important to note that despite these being low intensity treatments, best supportive care is still crucial.*

Treatment Recommendations for Relapsed or Refractory Disease;

- Response rates depend on duration of first remission
- Patients in complete remission (CR) longer than 2years have a 60% chance of responding to front-line regimens
- Patients in CR 1-2y have a 40% chance of responding to front-line regimens; clinical trials are preferred
- Patients in CR less than 1y are unlikely to respond to front-line regimens and should be referred for clinical trials
- The prognosis for patients beyond first salvage is very poor.

Table 7.2: treatment recommendation for relapsed or refractory disease

Discuss with family about need for transplant. In addition, HLA typing to identify potential donors should be discussed and process started.
FLAG-IDA (Fludarabine, Cytarabine, Idarubicin and filgrastim)
HiDAC with anthracyclines
Venetoclax based combinations
Other targeted therapies if accessible i.e. FLT3 inhibitors, IDH mutation inhibitors.
Refer for transplant once patient goes into remission.

7.2.1 Role of Radiotherapy

Patients who present with isolated extramedullary disease (myeloid sarcoma) should be treated with systemic therapy and radiation therapy used for residual disease. In a small group of patients where the extramedullary disease is causing nerve compression, radiotherapy may be considered to reduce the disease burden.

7.2.3 Acute Promyelocytic Leukaemia

Acute Promyelocytic Leukemia (APML) is relatively rare and comprises about 7% to 8% of adult AML. It represents a medical emergency with a high rate of early mortality, often due to hemorrhage from a characteristic coagulopathy. Clinicians need to have a high index of suspicion.

It is classified as low risk (WBC <10,000 per uL)) and high risk (WBC > 10,000 per uL) based on the white cell count. Patients with low-risk disease undergo induction with all-trans retinoic acid (ATRA) and anthracycline and continue with consolidation as per protocol. In the event Arsenic trioxide is available, ATRA and Arsenic trioxide can be used.

High risk disease is treated with induction using a combination of all-trans retinoic acid (ATRA), arsenic trioxide (where available) and anthracyclines (such as daunorubicin or idarubicin followed by consolidation therapy. Maintenance therapy is part of standard of care and may go up to 24 months.

7.2.4 Treatment

- Consolidation regimen is dependent on risk with high risk patients having Cytarabine as part of their regimen.
- In those with coagulopathies, monitor coagulation parameters daily until coagulopathy is corrected by administration of fresh frozen plasma and platelets.
- Arsenic trioxide: monitor electrolytes (maintain K >4. mmol/l and Mg >1.8mmol/l) and monitor for QTc prolongation.
- Patients should have reassessment marrow on day 21 which should include testing for PML RARA gene.

7.2.4 Response criteria for Acute myeloid Leukemias

- Complete remission (CR): bone marrow blasts < 5% in aspirate with spicules, no evidence of extramedullary disease, transfusion independence, PCR negative, absolute neutrophil count (ANC) > 1000/µL with platelet count > 100,000/µL
- CR with incomplete blood count recovery (CRi) or incomplete platelet count recovery (CRp): CR plus persistence of cytopenia (e.g., underlying myelodysplastic syndrome [MDS] in the elderly)
- Treatment failure: 5-25% blasts in bone marrow or 50% decrease in blasts and normalization of blood counts.

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7.3 Chronic Leukemias

Chronic Lymphocytic Leukemia (CLL) /Small Lymphocytic Lymphoma (SLL)

7.3.1 Introduction

A monoclonal disorder characterized by a progressive accumulation of functionally incompetent lymphocytes. CLL is the most common leukemia globally especially in the western world with an incidence of 4.2/100 000/year, seen in elderly patients with a mean age at diagnosis of 72 years.

Symptoms to look out for:	Clinical signs to look out for in a full clinical examination:
• Recurrent infections • Bleeding or easy bruisability • Unexplained weight loss • Drenching night sweats • Persistent fever • Waxing and waning l lymph nodes	• Pallor (anaemia) • Lymphadenopathy • Splenomegaly • Hepatomegaly • Bruising (purpura)

7.3.2 Laboratory Evaluation

1. Full blood count (FBC), with differential count and peripheral blood film examination
2. The diagnosis of CLL is established by the presence in the peripheral blood of
3. **>5000 monoclonal lymphocytes/l for the duration of at least 3 months.** Patients with cytopenias i.e. haemoglobin < 11g/dl, platelets <100,000 and progressive splenomegaly have more aggressive disease and will require urgent referral for treatment.
4. Immunophenotyping is recommended-On peripheral blood (flow cytometry- to confirm clonality of circulating B lymphocytes and distinguish it from other lymphoid proliferations that can present with leukemic phase such as mantle cell lymphoma).
5. Ancillary tests: Cytogenetics and FISH where available for risk stratification.
6. Pre-treatment investigations: LFTS, U/E/Cr, Uric acid, LDH, ALP, HBV, HCV, HIV.
7. Bone Marrow Aspirate and trephine biopsy only in refractory cytopenias

7.3.3 Imaging

Imaging using CT scans or FDG/PET is recommended if Ritcher's transformation is suspected.

7.3.4 Treatment

This differs based on stage, age and cytogenetics. Start prophylaxis for tumor lysis syndrome in patients with high tumor burden, infectious disease prophylaxis and blood transfusions as necessary. Treatment is only indicated in advanced and symptomatic disease. The only curative treatment is allogeneic stem cell transplant.

Table 7.3: Modified RAI Staging & Classification

Risk Classification	Rai Stage	Features	Consideration
Low risk	0	Peripheral blood lymphocyte count > 5.0 x 10^9 /L. Bone marrow, if done, contains > 30% lymphocytes*	Observe
Intermediate risk	1	Stage 0 + lymphadenopathy	Observe if no indications, start treatment if there are indications
	2	Stage 0 + hepatomegaly and splenomegaly	
High risk	3	Stage 0 + anemia (Hgb <110 g/L)	Therapies started based on the individual patient's age and performance status.
	4	Stage 0 + thrombocytopenia (Platelets < 100 xl 09 /L).	

The other staging criteria is the Binet system.

7.3.5 Special Considerations for Treatment

- For relapse of disease that occurs more than 3 years after initial treatment, re-treat with first-line therapy.

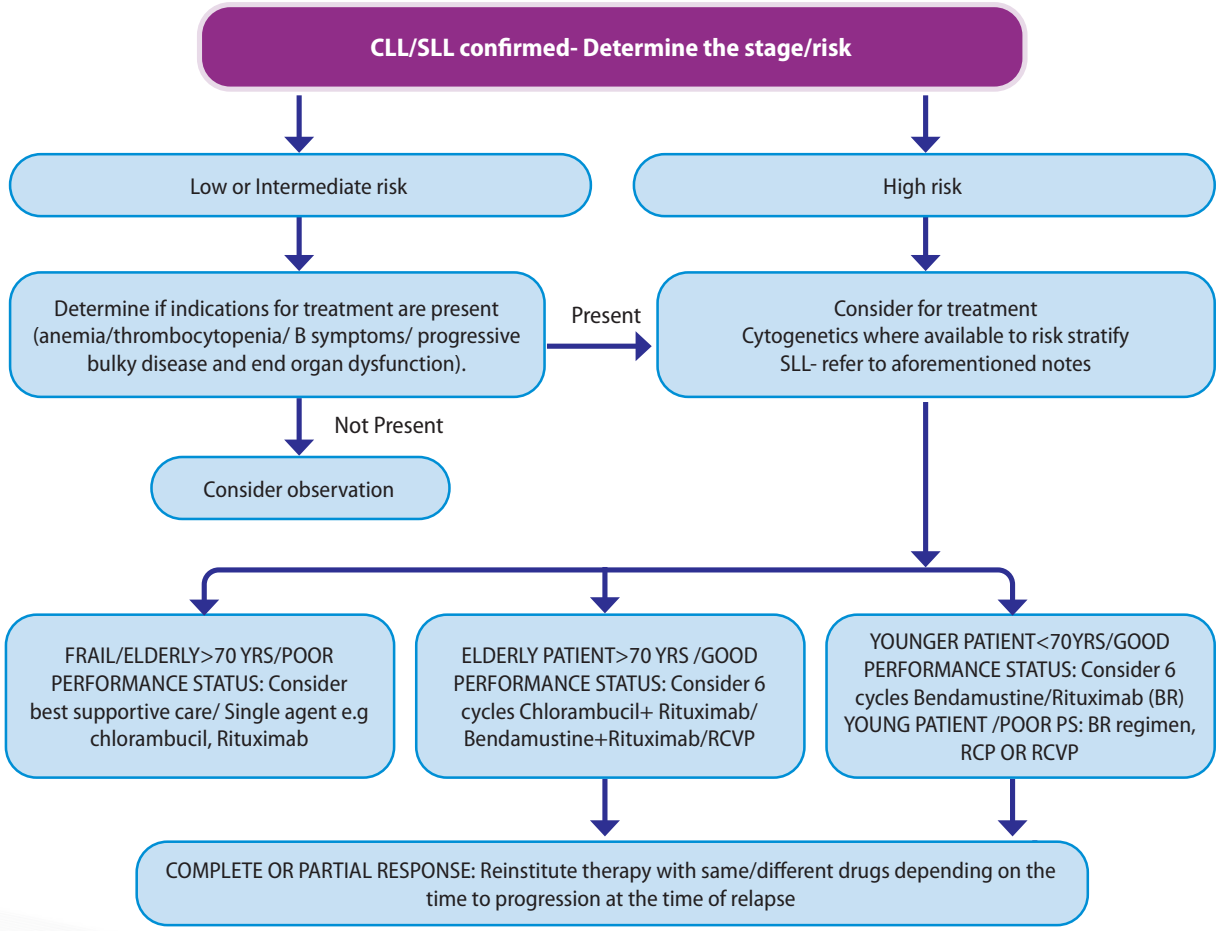


Figure 7.4: Treatment Algorithm for CLL/SLL

Chemoimmunotherapy has been the standard of care for treatment naive CLL/SLL patients; However, highly active and efficacious targeted agents are currently being used as first line therapy where available and accessible. Example of these agents include:

1. Bruton tyrosine kinase inhibitors (BTKis) i.e. Ibrutinib, Acalabrutinib, Zanubrutinib.
2. Venetoclax (BCL-2 inhibitor) plus obinutuzumab (anti-CD20 monoclonal antibody).

7.3.6 Examples of Chemo-immunotherapy Regimens include:

1. Bendamustine-based therapy: BR regimen (bendamustine and rituximab). The regimen is given every 28 d for 6 cycles.
2. Fludarabine-based therapy for those with low-risk cytogenetic abnormalities: FCR (fludarabine, cyclophosphamide, and rituximab) and FR (fludarabine and rituximab). Both regimens are given every 28 d, along with infectious disease (ID) prophylaxis.
3. Chlorambucil based therapy: (Rituximab and Chlorambucil with or without prednisone).
4. Cyclophosphamide based therapy: RCVP (Rituximab with Cyclophosphamide, vincristine and prednisone.

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7.4 Chronic Myeloid Leukemia (CML)

7.4.1 Introduction

A myeloproliferative neoplasm arising from a primitive hematopoietic stem cell in the bone marrow that involves all three myeloid lineages: granulocytes, erythroid cells, and megakaryocytes. It is defined by the presence of the BCR/ABL1 fusion gene, which results from the reciprocal translocation t (9; 22) (q34; q11.2). The Philadelphia chromo- some refers to the abnormally short chromosome 22 in CML.

CML occurs in three phases: Chronic/Stable phase (most patients are first diagnosed here), accelerated phase and blast transformation/crisis (acute myeloid or lymphoid leukemia).

CML has an annual global incidence of 1:1.5 per 100000 and comprises 10-15% of all the leukemias with most cases occurring sporadically. The M: F ratio is 1.4:1 and median age of occurrence is about 50 years. In Kenya, Abinya et al found an M: F ratio of 1.26:1 with median age of 44 years.

7.4.2 Clinical Evaluation

The most common presenting symptoms include:

- Fatigue (usually related to mild to moderate anemia)
- Weight loss
- Left upper quadrant fullness/pain
- Early satiety from splenomegaly.

7.4.3 Other Presentations that are Less Common include:

- Priapism caused by marked leukocytosis +/- thrombocytosis
- Gouty arthritis due to increased uric acid levels
- Bleeding caused by platelet dysfunction or low platelets
- Deafness
- Upper GIT ulceration and bleeding due to histamine release associated with basophilia.
- Hyperviscosity symptoms can occur due to sludging of cells in pulmonary or cerebral vessels resulting in confusion, drowsiness or dyspnea.
- In accelerated phase, constitutional symptoms of fever, sweating and weight loss are present.
- Bleeding can occur especially in accelerated and blast phases of the disease.

7.4.4 Physical Examination

1. Pallor.
2. Splenomegaly: record spleen size by palpation (cm below costal margin).
3. Hepatomegaly.
4. Lymphadenopathy and skin infiltration are rare.

7.4.5 Laboratory Evaluation

1. Full blood count with differential count.
2. Peripheral blood smear: demonstrate full maturation spectrum of myeloid series.
3. Molecular genetic test: Quantitative reverse transcriptase –PCR for BCR-ABL1
4. Bone marrow aspirate for characterization of phase of CML.
5. Molecular genetic test: Quantitative reverse transcriptase –PCR for BCR-ABL1 or FISH to detect translocation t (9, 22).
6. UEC and LFTs.
7. Uric acid level.

Other investigations based on clinical presentation or co-morbidities.

7.4.6 Treatment

- Imatinib Mesylate is the treatment of choice, in all newly diagnosed patients.
- Switch to second generation line TKI if suboptimal molecular response.

Use second generation line TKI based on existing co-morbidities (pre-existing vascular disease-avoid Nilotinib; lung disease/pleural effusion-avoid Dasatinib) and mutational analysis. If positive for T315I mutation switch to ponatinib. If negative, options are dasatinib, nilotinib or bosutinib.

NB: There may be mutations that affect the other drugs therefore where possible a full mutation panel is recommended however this is more expensive.

Blood counts need to be monitored during treatment to avoid cytopenias and loss of hematological response.

Regular molecular monitoring once a year is acceptable or earlier in cases of loss of hema- tological or clinical response.

Role of hydroxyurea: has no effect in treating the underlying molecular abnormality in CML but is useful in controlling high white cell counts and most patients are indicated on this and allopurinol as they await transition to a Tyrosine Kinase Inhibitor.

Table 7.4: Assessment of response criteria

Definition	Response criteria
Complete Hematologic Response (CHR)	• Complete normalization of peripheral blood white count (<10x1 09), and platelet count (< 450 x 109) sustained for at least 4 weeks
Major Cytogenetic Response (MCR)	• < 35% Philadelphia positive bone marrow metaphases by conventional cytogenetic analysis. • This is equivalent to a 1 log reduction in BCR/ABL transcripts from baseline as measured by quantitative PCR (<1 0%)
Complete Cytogenetic Response (CCR)	• No Philadelphia positive bone marrow metaphases by conventional cytogenetic analysis. • This is equivalent to a 2 log reduction in BCR/ABL transcripts from baseline as measured by quantitative PCR (1%)
Major Molecular Response (MMR)	• Greater than or equal to a 3 log reduction in BCR/ABL fusion transcripts compared to baseline by PCR analysis (0.1 *%)*
Deep Molecular Response (CMR)	• 4 log reduction in BCR-ABL (<0.01 *%)*

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7.5 Lymphomas

These are cancers of lymphocytes that represent clonal proliferation derived from various lymphocytic cell lines, mainly B cells and also T-cells or NK cells at different stages of differentiation and/or activation. They are broadly classified as Hodgkin Lymphoma (HL) and Non-Hodgkin Lymphoma (NHL) with the latter occurring more commonly. Both have a male preponderance.

7.5.1 Clinical Evaluation

History: Symptoms to look out for:

- Painless progressive lymph node
- Enlargement
- B- symptoms:
  - Unexplained Weight loss >10% in months
  - Drenching night sweats
  - Persistent fever >38˚C
- Others: HL- pruritus, alcohol-induced pain
- on involved nodes

Examination: Clinical signs to assess for in a full physical examination:

- Pallor (anaemia)
- Lymphadenopathy (Map all groups of lymph nodes including the Waldeyers ring), size
- Hepatosplenomegaly
- Ascites
- Extranodal sites- testis, CNS

Refer all patients presenting with unexplained lymphadenopathy or splenomegaly or associated symptoms particularly; Fever, Night sweats, Shortness of breath, Puritan Weight loss

7.5.2 Diagnosis

1. Biopsy: a lymph node excisional biopsy is preferred.
   It is recommended that a surgical excision biopsy of a significantly enlarged lymph node (>2cm) is carried out in patients clinically suspected to have lymphoma, and referred immediately for histopathological evaluation in order to reduce diagnostic delay. Fine needle aspirations (FNA) are inappropriate for a reliable diagnosis. Core needle biopsies can be performed in cases of deep-seated nodes e.g. retroperitoneal or intra-abdominal sites.
2. Immunohistochemistry to subtype the lymphoma.
3. Molecular and cytogenetic studies are optional but may be helpful where morphology and immunohistochemistry are inconclusive.

7.5.3 Staging and Risk Assessment

1. Full blood count/ESR
2. Bone marrow aspirate and trephine biopsy is strongly recommended for staging.
3. Biochemistry: LFTs, RFTs, LDH, uric acid, and liver enzymes.
4. Viral serology for HIV, Hepatitis B and Hepatitis C (strongly recommended).
5. Serum Protein Electrophoresis should be considered in some cases.
6. Lumbar puncture is indicated if there is clinical suspicion of CNS involvement, or for high-risk lymphoma that have propensity for CNS involvement involving CNS.
7. Pregnancy test for childbearing women.
8. Fertility preservation for those at risk.

7.5.4 Imaging

1. CT-Scan of neck, chest, abdomen and pelvis is strongly recommended.
2. If CT-scan is unavailable, then CXR and abdomino-pelvic ultrasound should be done.
3. MRI is the investigation of choice for suspected CNS disease.
4. FDG-PET/CT scan where available for high-grade lymphomas.
5. Echocardiogram for those who anthracycline based chemotherapy will be used.

7.6 Hodgkin Lymphoma

7.6.1 Introduction

It comprises about 1.7% of all cancers in Kenya with about 420 new cases and approximately 182 deaths annually. It has a worldwide bi-modal age distribution with a peak of 15-30 and 50-70 years. However, in Kenya a single peak at about 17 years has been observed. The male to female ratio is 4:3 worldwide although in Kenya, it is about 3:1.

7.6.2 Risk Factors

There are no clear risk factors for HL, but several associations have been observed such as:

- EBV infection associated mainly with mixed cellularity type (30%)
- HIV infection
- Woodworking
- Chemical exposure mostly from agricultural chemicals
- Ionizing radiation
- Genetic predisposition: First degree relatives have 5-fold increased risk for HL
- High socio-economic status
- Men gender
- Race: Caucasians have a higher preponderance.

Table 7.5: Diagnosis and Work-up

Diagnosis	Confirmatory workup: excisional biopsy/core biopsy of the lymph node. Diagnosis rests on identification of RS cell or its variants with IHC demonstrating positivity for CD15,30 and possibly EBV markers for classical HL and CD20, CD45, EMA for nodular lymphocyte predominant HL
Staging and risk stratification	Medical history and physical examination X-ray of the chest ***Contrast-enhanced CT scan of the neck, chest abdomen and pelvis*** Whole body FDG PET CT scan TBC, blood chemistry, ESR, LDH, ***BMA &Trephine not necessary with PET*** HBV, HCV and HIV screening
Pre-treatment examinations	ECG, Echocardiography Pulmonary function test, Reproductive counselling (in patients of reproductive age) Serum pregnancy test (female of reproductive age)

Table 7.6: Ann Arbor Staging for Hodgkin's Lymphoma

Ann Arbor Staging (1971)	
STAGE	**DESCRIPTION**
1	Involvement of a single lymph node region or a single extralymphatic organ or site (IE)
II	Involvement of two or more lymph node regions on the same side of the diaphragm (II) or localized involvement of an extralymphatic organ or site (HE)
III	Involvement of lymph node regions on both sides of the diaphragm (III) or localized involvement of an extralymphatic organ or site (IIIE) or spleen (INS) or both (IIISE)
IV	Diffuse or disseminated involvement of one or more extralymphatic organs with or without associated lymph node involvement. Bone marrow and liver involvement are always stage IV.
• Identification of the presence or absence of symptoms should be noted with each stage designation: A, assymptomatic; B, fever, sweats, weight loss more than 10% of body weight.	

Metabolic response reflected by FDG-PE/CT has emerged as one of the most important prognostic factors for survival, and it forms the rationale for response-adapted therapy as shown in the treatment algorithm.

7.6.3 Early-Stage Hodgkin Lymphoma (Stage I/II)

Early-stage Hodgkin lymphoma is further divided into:

- IA/IIA: Favorable (Non-Bulky)
- IA/IIA: Unfavorable (B symptoms, bulky mediastinal disease, adenopathy of > 10cm)

Treatment consists of chemotherapy and involved site radiation therapy (ISRT). In certain cases, chemotherapy alone can be used.

- ABVD (Adriamycin, bleomycin, vinblastine, and dacarbazine) is the commonest chemotherapy regimen utilized
- Response assessment with available imaging (FDG/PET is ideal) is done after 2 cycles of ABVD.
- Patients with positive FDG-PET scan after 2 cycles of ABVD need individualized treatment.
- Multidisciplinary team (MDT) should be used to guide further treatment.

7.6.4 Stages III-IV Hodgkin Lymphoma

Systemic chemotherapy with ABVD is the preferred (first line) regimen.
Consider ISRT following chemotherapy to initial bulky sites of disease or to sites that are still FDG PET positive post treatment.

7.6.5 Relapsed/Refractory Hodgkin Lymphoma

The treatment approach is influenced by the time to relapse and the nature of the initial treatment, but generally consists of a multistep process of salvage chemotherapy, followed by autologous stem cell rescue (HDC-ASCT). This can result in cure for approximately 50% of the patients.

The following regimens can be utilized in the relapsed/refractory settings:

- **BV:** Brentuximab Vedotin
- **BV** + Bendamustine
- **ICE** regimen (ifosfamide, carboplatin, and etoposide)
- **ICE** + Brentuximab Vedotin
- **ICE** + Nivolumab
- **ICE** + Pembrolizumab
- **DHAP** regimen (high-dose cytarabine [Ara-C], cisplatin, and dexamethasone
- **ESHAP** regimen (etoposide, methylprednisolone, Ara-C, and cisplatin)
- **GDP** (Gemcitabine, carboplatin, dexamethasone):
- **GVD** (gemcitabine, vinorelbine, liposomal doxorubicin):
- **IGEV** (ifosfamide, gemcitabine, vinorelbine
- Immunotherapy based combinations
- Immunotherapy single agents
- **GEMOX** Gemcitabine and Oxaliplatin
- Bendamustine
- Bendamustine, Carboplatin, Etoposide
- Gemcitabine
- Lenalidomide

7.6.6 Treatment of Nodular Lymphocyte Predominant Hodgkin's Lymphoma

Early-stage disease:

- Either radiotherapy alone or abbreviated chemotherapy plus radiotherapy is utilized.
- Patients with stage IA disease with full excision of a solitary lymph node may be observed.
- The choice of chemotherapy is controversial; most groups recommend ABVD for two to four cycles plus radiation, but retrospective data suggest that alkylating agents are more effective, and CHOP or CVP can be used.
- Survival for early-stage disease is excellent and approximates 90% with at least 10 years of follow-up.

Advanced-stage disease:

- Six cycles of ABVD, CHOP, or CVbP are recommended.
- The addition of rituximab to CD20 expressing NLPHL is recommended as this has shown to improve response rates.
- The major risk in NLPHL survivors appears to be transformation to DLBCL T-cell rich diffuse large B-cell lymphoma,), which should be treated as any transformed aggressive B-cell lymphomas.

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7.7 Non- Hodgkin Lymphoma

7.7.1 High Grade B Cell Lymphomas

Diffuse Large B Cell Lymphoma

Diffuse large B-cell lymphoma (DLBCL) constitutes 30%–60% of lymphoma series worldwide including Kenya. Significant risk factors for the development of the disease include underlying immune deficiency, such as HIV or in the post-solid organ transplant setting.

Clinical Features

Rapidly enlarging tumor mass at single or multiple nodal or extranodal sites. Constitutional symptoms which are often dictated by the organ or anatomical site involvement.

Treatment

- The standard therapy is Rituximab plus CHOP 6 cycles.
- Other recommended regimens: Dose-adjusted EPOCH (with rituximab) for specific cases.
- There is a role for radiotherapy in DLBCL, as adjuvant therapy post chemotherapy to initial bulky disease sites or persistent FDG-PET avid areas.
- CNS disease is best treated with high doses of CNS penetrating chemotherapy

Refractory/Relapsed Disease

The prognosis of patients with primary refractory disease or relapsed disease after R-CHOP chemo-immunotherapy is poor, especially relapses observed within 1 year of treatment.

The management of refractory and relapse disease involves use of second line chemo-immunotherapies, stem cell transplant and CAR-T therapies.

Salvage regimens include:

- **R-GDP:** Rituximab, Gemcitabine, Dexamethasone, Cisplatin
- **R-DHAP:** Rituximab, Dexamethasone, Cytarabine, Cisplatin
- **R-ESHAP**: Rituximab, Etoposide, Cytarabine, Cisplatin, Methylprednisolone
- **R-ICE:** Rituximab, Ifosfamide, Epirubicin, Etoposide
- **R-IVE:** Rituximab, Ifosfamide, Epirubicin, Etoposide
- **R-GemP**: Rituximab, Gemcitabine, Cisplatin, Methylprednisolone
- **R-Gem-Ox:** Rituximab, Gemcitabine, Oxaliplatin
- **R- MINE:** (mesna, ifosfamide, mitoxantrone, etoposide) ± rituximab
- Ibrutinib (non-GCB DLBCL)
- R- Lenalidomide + rituximab (non-GCB DLBCL)
- **Elderly patients:** consider Bendamustine plus Rituximab.

CAR-T cell therapy, Bispecific antibody agents, antibody drug conjugates are increasingly being used in the relapsed/refractory settings.

Primary CNS Lymphoma

- The usual histology of PCNSL is DLBCL (90% of cases).
- The treatment of primary DLBCL of the CNS includes remission induction with regimens that contain high-dose based methotrexate.
- The addition of cytarabine and thiotepa improve remission rate and outcome.
- Whole brain radiotherapy (WBRT) in those not fit for systemic therapy
- Chemotherapy treatment should be given in conjunction with rituximab as the combination has been shown to further improve response rates and survival.

7.7.2 Low Grade B cell Lymphomas

Follicular Lymphoma

Follicular lymphoma (FL) is the most common form of indolent lymphoma in the Western World and second most common form of Non-Hodgkin's lymphoma presenting in 5/100,000 population/year with a median age of onset of around 60 years.

Treatment

Follicular lymphoma (FL), grades 1, 2 and 3a

a) Limited stage disease (Stage 1A +/-2A non-bulky)

- FL is highly radiosensitive. Limited stage 1 or 2 disease and no adverse factors may be considered for local radiotherapy utilizing (ISRT) with curative potential.
- ISRT maybe combined with rituximab with or without chemotherapy
- A lower radiation dose may be appropriate in cases where a palliative approach is pursued or radiation toxicity requires particular consideration.

b) Advanced stage disease

There is no overall survival benefit to initiation of anti-lymphoma therapy in advance stage disease based solely on the presence of FL in the absence of other adverse factors. Therefore, one must determine whether treatment is indicated where indications for treatment include:

- Candidate for a clinical trial
- Rapidly progressive disease
- B symptoms
- Pruritus
- Vital organ compression/compromise
- Significant bone marrow infiltration/haematopoietic suppression
- Bone lesions
- Renal infiltration
- Ascites
- Pleural effusion
- Splenomegaly
- Rapid disease progression
- Bulky disease including multiple (>3LNs each >3cm in diameter or single sites >7cm in diameter).

Be aware that patients with bulky disease are likely to have a short duration to progression.

Systemic Treatment Regimens

Combination immuno-chemotherapy (rituximab + chemotherapy).

Front-line Therapy:

- R-CHOP regimen should be considered for FL that demonstrates clinically aggressive behaviour
- Rituximab, cyclophosphamide, vincristine, prednisolone (R-CVP) in non-aggressive disease
- Bendamustine + Rituximab
- Rituximab and Lenalidomide
- Rituximab-Chlorambucil for frail patients

Consolidation of Response

Maintenance: Rituximab given once every 8-12 weeks for 2 years is recommended if a complete or partial response is received following first-line systemic therapy.

Second line Management

At progression or relapse, biopsy of index lesions is strongly recommended particularly to assess the possibility of disease transformation. The patient's preference, initial treatment response duration, evidence of aggressive disease transformation, previous treatment regimens, toxicities, patient fitness, co-morbidities and the stage of disease help guide management.

- Less fit patients with limited stage disease can have a palliative approach to treatment with local radiotherapy.
- In advanced disease, particularly if low bulk and in absence of other indications for treatment, selected patients may again be appropriate for watch and wait with close disease monitoring.
- responding
- Fit patients who demonstrate a sufficient response to second line therapy should be considered for auto-HSCT
- Rarely, particularly in young, otherwise fit patients with rapidly progressive/early refractory disease then allo-HSCT should be considered.

Systemic Second-line Treatment Regimens

- An alternative regimen to the initial one used should be considered, especially if remission duration was less than 12-24 months.
- Re-treatment with a previously used regimen may be appropriate in selected cases who experienced a long treatment response duration (>24 months)

Consolidation of Response

Maintenance rituximab given once every 8-12 weeks for 2 years if a complete or partial response is received following second-line systemic therapy.

7.7.3 Follicular Lymphoma, Grade 3b

This should be managed as per diffuse large B cell lymphoma (DLBCL) guidelines.

Transformation

FL has a risk of undergoing histological high-grade disease transformation to aggressive lymphoma (typically DLBCL) at a rate of around 3% of cases per year. If transformation is suspected, biopsy is recommended. Clinical indicators suggestive of possible transformation include:

- Sudden rise in LDH to ≥twice upper limit of normal
- Rapid discordant localised nodal growth (detected clinically or by imaging)
- New involvement of unusual extranodal sites (e.g. liver, bone, muscle, brain)
- New B symptoms
- New hypercalcaemia.

Transformation to DLBCL should be managed as per de novo DLBCL with the anthracycline containing R-CHOP regimen and consideration of consolidation of response by AutoHSCT in patients fit enough for this approach.

Alternative regimens to R-CHOP (such as rituximab, dexamethasone, high-dose cytarabine, cisplatin; R-DHAP) can be considered in patients fit enough for this approach

Peripheral T-Cell Lymphomas

Peripheral T-cell lymphoma (PTCL) is a heterogeneous group of rare malignancies accounting for approximately 10–12% of all lymphoid neoplasms. Most have an aggressive clinical behaviour and, apart from anaplastic large cell lymphoma (ALCL), a poor response to conventional chemotherapy with only 30% survival at 3 years.

Treatment

- The standard therapy is with multi-agent chemotherapy-based regimens: 6 cycles of either, DA-EPOCH, CHOEP, CHOP is employed.
- Consider consolidation with HSCT in those responding to first line chemotherapy.
- There is a role for ISRT for stage I/II disease or, as adjuvant therapy post chemotherapy to initial bulky disease sites or persistent FDG-PET avid areas.
- Response assessment with Imaging (FDG/PET is ideal) post 4 cycles of chemotherapy.
- Completion of chemotherapy upto 6 cycles is recommended for those whose imaging shows complete/partial response
- Those with no response or progression of disease on imaging, should be treated as refractory disease.

Refractory/Relapsed Disease

- The prognosis of patients with primary refractory disease or relapsed disease after multi-agent chemotherapy is poor, especially relapses observed within 1 year of treatment.
- The management of refractory and relapse disease involves use of second line chemotherapy agents (either single agent/combined therapies) or targeted therapies.

- Salvage chemotherapy regimens include:
  - Bendamustine
  - ICE (Ifosfamide, carboplatin, etoposide)
  - DHAP (Dexamethasone, Cytarabine and Platinum
  - GDP (Gemcitabine dexamethasone, platinum)
  - ESHAP (Etoposide, methylprednisolone, Cytarabine and Platinum)
  - GEMOX (Gemcitabine and Oxaliplatin)
  - Lenalidomide
  - Gemcitabine

7.8 Plasma Cell Dyscrasias

Encompass a spectrum of diseases where a clone of plasma cells produces a homogeneous monoclonal immunoglobulin or its fragments. This monoclonal protein can be detected in the blood or urine and is indicative of an underlying plasma cell disorder.

7.8.1 Classification and Diagnostic Criteria

The International Myeloma Working Group (IMWG) classifies plasma cell dyscrasias into several categories, each with specific diagnostic criteria:

1. Monoclonal Gammopathy of Undetermined Significance (MGUS)

- **Serum M protein**: < 3 g/dL
- **Bone marrow plasma cells**: < 10%
- **Absence of**:
  - End-organ damage (CRAB features: hypercalcemia, Renal insufficiency, Anemia, Bone lesions)
  - Myeloma-defining events
- **Clinical course**: Asymptomatic; progresses to multiple myeloma at a rate of approximately 1% per year.

2. Smoldering (Asymptomatic) Multiple Myeloma (SMM)

- **Serum M protein**: ≥ 3 g/dL and/or
- **Bone marrow plasma cells**: 10–60%
- **Absence of**:
  - End-organ damage (CRAB features)
  - Myeloma-defining events
- **Clinical course**: Asymptomatic; higher risk of progression to symptomatic multiple myeloma.

3. Multiple Myeloma (MM)

- **Bone marrow plasma cells**: ≥ 10% or biopsy-proven plasmacytoma
- **Presence of**:
- End-organ damage (CRAB features) and/or
- Myeloma-defining events (SLiM criteria):
  - **S**: Sixty percent or more clonal plasma cells in bone marrow
  - **Li**: Involved/uninvolved serum free light chain ratio ≥ 100
  - **M**: MRI with more than one focal lesion (>5 mm)

4. Other Plasma Cell Dyscrasias

- **Solitary Plasmacytoma**: A single mass of clonal plasma cells in bone or soft tissue without systemic involvement.
- **Primary Amyloidosis (AL)**: Deposition of light chain-derived amyloid fibrils in organs, leading to dysfunction.
- **Plasma Cell Leukemia**: A rare and aggressive form characterized by high levels of plasma cells in the peripheral blood.

7.9 Multiple Myeloma

7.9.1 Introduction

A plasma cell malignancy characterized by abnormal proliferation of clonal plasma cells. It can either be secretory (majority, secretes an abnormal serum or urine immunoglobulin referred to as M protein or paraprotein in majority of cases) or non-secretory (2-5%, fails to produce abnormal immunoglobulin). Patients can also present with solitary or multiple masses (plasmacytomas).

7.9.2 Epidemiology

It is one of the common blood malignancies in Kenya with 800 new cases and 677 deaths reported in 2022. It occurs almost exclusively in the middle-aged to elderly population all over the world with a male preponderance. Local studies have shown a lower median age of 53 to 59 years with a slight male predominance.

7.9.3 Clinical Presentation

Multiple myeloma is often manifested by complications of enhanced bone loss associated with diffuse osteopenia or lytic bone lesions, renal failure, hypercalcemia, immuno- suppression or anemia.

7.9.4 Signs and Symptoms

• Fatigue and generalized body weakness • Bone pains or back pain • Recurrent infections • Weight loss • Pathological fracture • Paraplegia due to cord compression. • Paraesthesias. • Mass (more commonly in head)	The following clinical signs should be looked in a full physical examination: • Pallor (anemia) • Bone tenderness • Vertebral collapse or spinal cord compression

7.9.5 Diagnostic Workup

1. Total blood count with differential and peripheral blood smear.
2. Bone marrow aspirate (mandatory) and trephine (recommended) OR tissue biopsy for histology and immunohistochemistry in case of suspected plasmacytoma.
3. Serum protein electrophoresis (SPEP) and urine protein electrophoresis to establish clonality. Serum Immunofixation to identify immunoglobulin sub- type (IgG, A, D, M). Serum free light chains (SFLC) and SFLC ratio.
4. Investigations to look for end organ disease include: UECs, Total protein, albumin and Serum calcium
5. Myeloma FISH panel ,B2 microglobulin and serum LDH for prognostication
6. Viral serology for HIV is mandatory; hepatitis B and hepatitis C is strongly recommended.

7.9.6 Imaging

a) Whole body low dose CT imaging without contrast.
b) Magnetic resonance imaging (MRI) covering the axial skeleton or the whole-body
c) Skeletal survey (has a low sensitivity and misses early bone disease can be performed if above are not available). Chest Radiograph, anteroposterior and lateral views of cervical spine and thoraco-lumbar spine, pelvis and skull. Radiographs of other bones depending on clinical presentation.
d) Whole body PET CT scan (where available).

***If skeletal survey is negative, Whole-body MRI or PET CT scan should be done to differentiate active from smoldering disease. Recommendations are MRI with contrast*

Table 7.7: Diagnostic Criteria as Defined by the International Myeloma Working Group (2014)

Monoclonal Gammopathy of undetermined significance	Smoldering Myeloma	Multiple Myeloma
Serum paraprotein <30g/l BMA No myeloma defining events plasmacytosis of <10% **No evidence of B cell lymphoproliferativ e disease or other plasma cell dyscrasias.**	Serum paraprotein >30g/l Urine monoclonal protein >500mg in 24hours BMA plasmacytosis 10-60% No myeloma defining events * No amyloidosis	BMA plasmacytosis >10% or extramedullary plasmacytoma And Myeloma Defining Events* or Biomarkers of malignancy**

**Myeloma defining events: hypercalcemia (corrected serum calcium >2.75mmol/l, renal insufficiency (serum creatinine >177umol/l, anemia Hb<10gm/dl, lytic bone lesions*
***Biomarkers of malignancy: clonal BMA plasmacytosis >60%, involved to uninvolved serum free light chain ratio >100 and >1 focal lesion on MRI (>5mm).*

Table 7.8: International Staging System Risk Stratification Model

Stage	B-2 microglobulin	Albumin
I	<3.5mg/dl	>3.5mg/dl
II	Not I or III *B2M <3.5 and low albumin <35g/l *B2M 3.5- 5.5 irrespective of albumin	
III	>5.5mg/dl	Normal or low

**Revised ISS (R2-ISS) risk stratification model incorporates serum lactate dehydrogenase and chromosomal abnormalities detected by interphase Fluorescent in-Situ Hybridization (FISH) or conventional cytogenetics.7*

Table 7.9: Second Revision: R2-ISS International Staging System (ISS) Risk Stratification for Multiple Myeloma (2022)

R2-ISS Risk Group / Score	Criteria and Point Allocation
Risk Factors (Scoring Components)	• ISS II = 1.0 point • ISS III = 1.5 points • del(17p) = 1.0 point • t(4;14) = 1.0 point • 1q gain/amplification = 0.5 point • Elevated LDH = 1.0 point
R2-ISS I (Low risk, 0 points)	No adverse factors present. Median OS ≈ 95 months
R2-ISS II (Low–intermediate, 0.5–1.0 points)	One minor risk factor (e.g., ISS II, 1q+, or LDH ↑) Median OS ≈ 80 months

R2-ISS Risk Group / Score	Criteria and Point Allocation
R2-ISS III (Intermediate–high, 1.5–2.5 points)	Two cumulative risk factors Median OS ≈ 55 months
R2-ISS IV (High, ≥3 points)	Three or more risk factors including del(17p) or t(4;14) Median OS ≈ 30 months

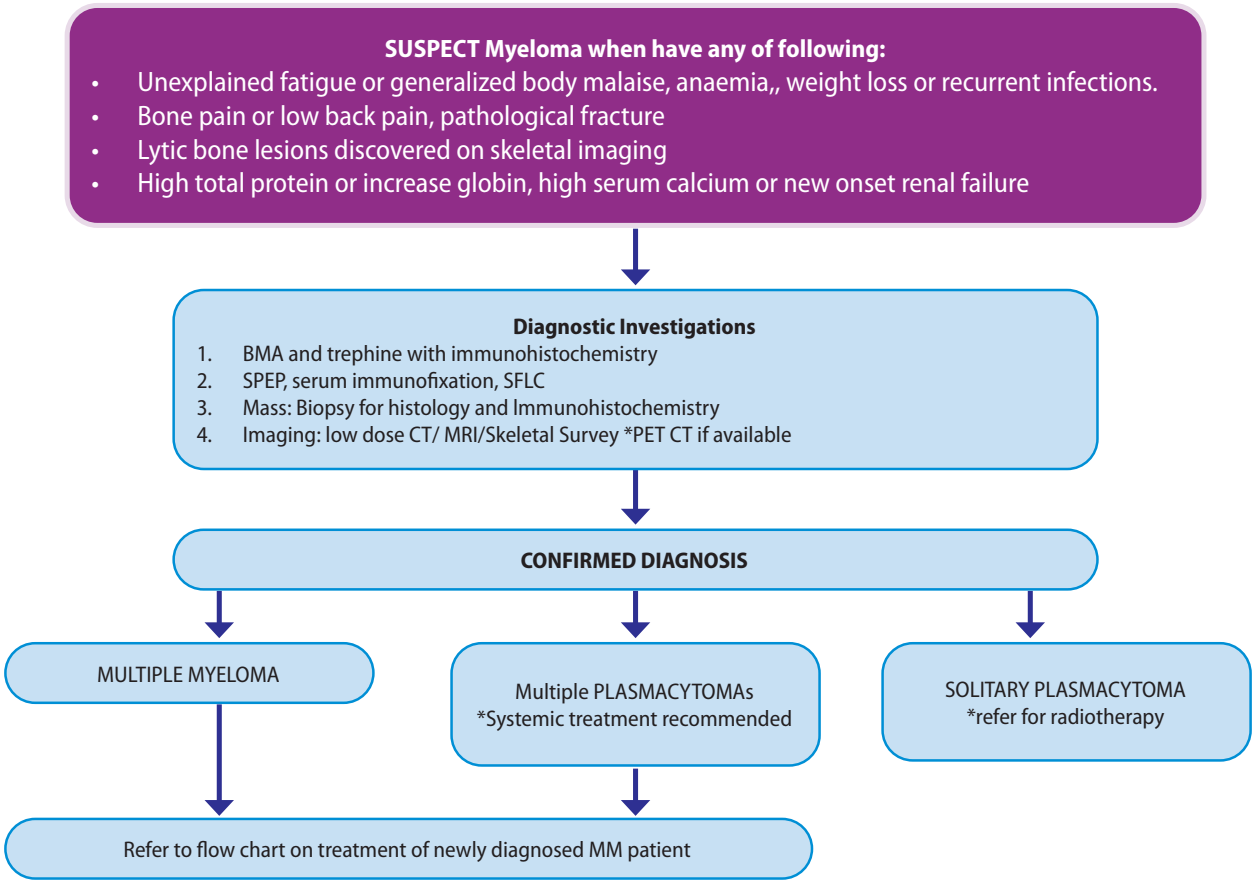


Figure 7.5: Diagnostic Algorithm for Multiple Myeloma

Urgently refer all patients suspected to have myeloma or with confirmed diagnosis for specialist management.

7.9.7 Treatment of Multiple Myeloma

The of options of treatment in MM include a variety of drug combinations, including quadruplet and triplet regimens; autologous stem cell transplantation; and chimeric antigen receptor (CAR) T-cell therapies for relapsed or refractory MM; along with adjunctive radiation and surgical care as necessary.

The approach to newly diagnosed MM is as follows:

- Induction
- Consolidation
- Maintenance
- Management and prevention of myeloma bone disease

Table 7.10: Multiple myeloma treatment options

Classes	Drug
Chemotherapy	Cyclophosphamide, Melphalan
Steroids	Dexamethasone and Prednisolone
Proteasome Inhibitors	Bortezomib, Carfilzomib, and ixazomib
Immunomodulators	Thalidomide, lenalidomide, and pomalidomide
Immunotherapies	daratumumab and elotuzumab
Bispecific Agents	Elranatab Teclistamab

1. Initial Assessment

- Eligibility for Autologous Stem Cell Transplant (ASCT): Patients are evaluated for transplant eligibility based on age, performance status, and comorbidities.
- Risk Stratification: Cytogenetic and molecular analyses help determine disease risk, guiding treatment intensity.
- Often a combination of drugs is used, such as triplet regimens (three drugs) or quadruplet regimens (four drugs) and this is guided by the performance status of the patient, affordability and availability of the drugs and systemic issues like renal insufficiency

2. Induction Therapy

The goal of induction is to reduce the myeloma cells and achieve a deep response rate.

For Transplant-Eligible Patients:

- **Standard Regimen:** A combination of a proteasome inhibitor (e.g., bortezomib), an immunomodulatory drug (e.g., lenalidomide), and dexamethasone (VRd) is commonly used.
- **Alternative Regimens:** Depending on patient factors, other combinations may include carfilzomib, daratumumab, isatuximab-based therapies.

For Transplant-Ineligible Patients:

- **Standard Regimen:** Similar to transplant-eligible patients, VRd is often used, with dose adjustments as needed.
- **Considerations:** Patient frailty and comorbidities may necessitate modified dosing or alternative regimens. Daratumumab, carfilzomib, isatuximab, cyclophosphamide based combinations maybe used

3. Stem Cell transplantation

- **Autologous Stem Cell Transplant (ASCT):** Typically performed after 3–4 cycles of induction therapy in eligible patients to deepen response.
- **Timing:** ASCT can be conducted upfront or delayed until first relapse, based on patient preference and disease characteristics.

4. Maintenance Therapy

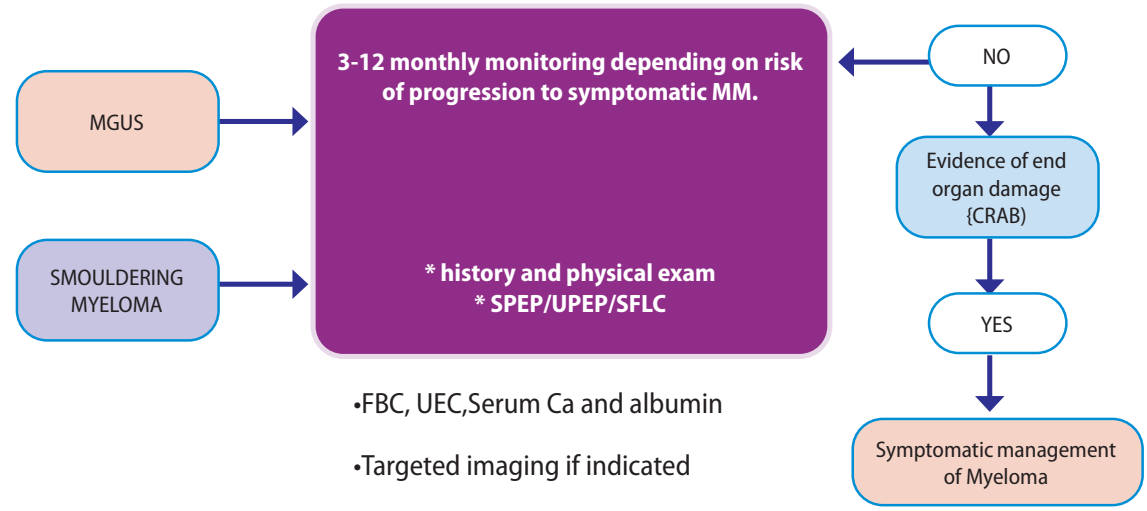
- **Lenalidomide:** Commonly used as maintenance therapy to prolong progression-free survival.
- Duration: Maintenance therapy is continued until disease progression or unacceptable toxicity.
- Other recommended regimens for maintenance include bortezomib or carfilzomib with or without lenalidomide.

5. Treatment of Relapsed/Refractory Multiple Myeloma

- **Therapy Selection:** Depends on prior treatments, duration of response, and patient-specific factors.
- Options Include:
  - **Proteasome Inhibitors:** Carfilzomib, ixazomib.
  - **Immunomodulatory Drugs:** Pomalidomide.
  - **Monoclonal Antibodies:** Daratumumab, isatuximab.
  - CAR T-Cell Therapy
  - **Bispecific Antibodies:** Emerging therapies targeting BCMA. (Elranatanab, Teclistamab?)

6. Supportive Care

- **Bone Health:** Bisphosphonates or denosumab to prevent skeletal-related events.
- **Hypercalcemia:** hydration, bisphosphonates, steroids
- **Anemia:** Transfusion, EPO
- **Infection Prophylaxis:** Vaccinations and antiviral prophylaxis as appropriate.
- **Thromboprophylaxis:** Aspirin or anticoagulation for patients on immunomodulatory agents.
- **Monitoring:** Regular assessments for treatment response and toxicity management.



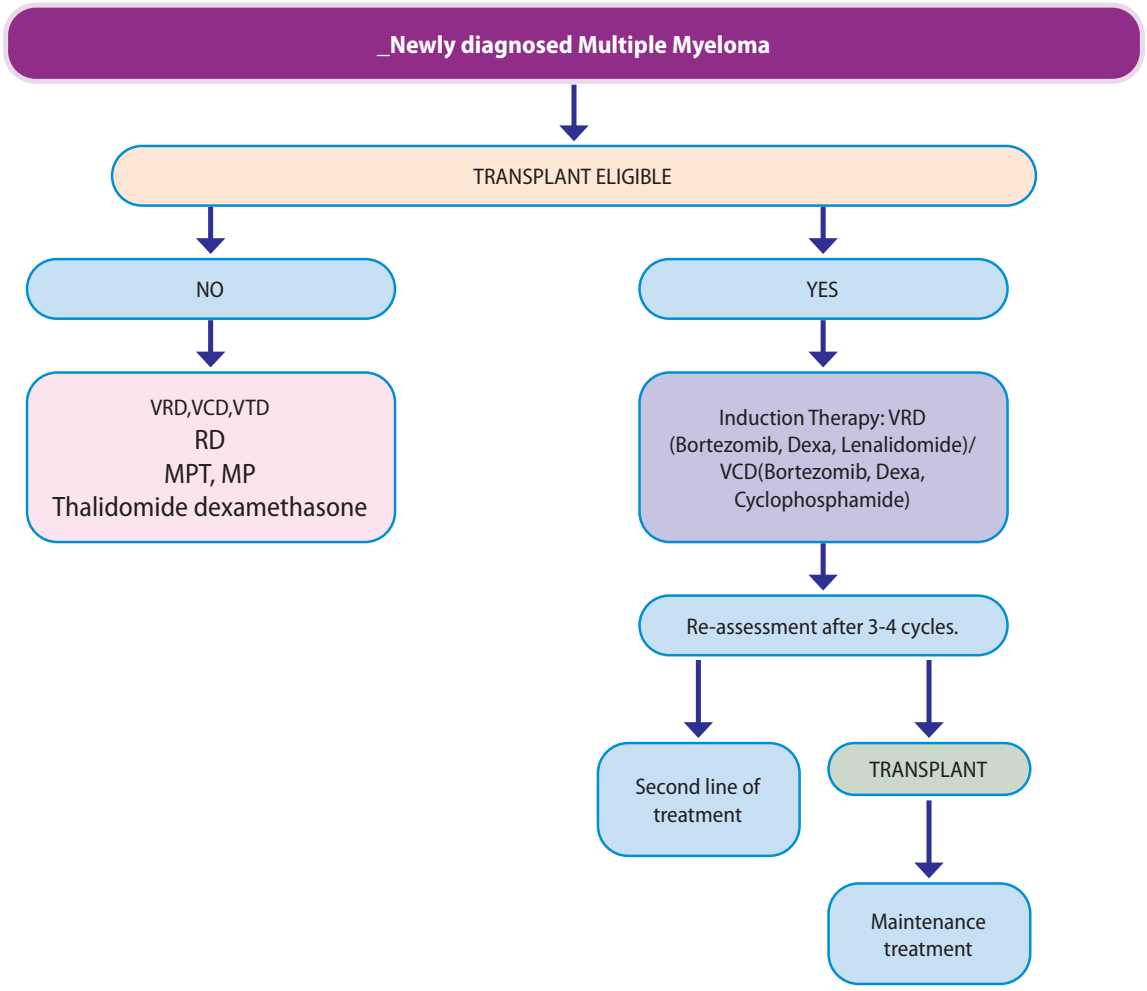


Figure 7.6: Multiple Myeloma Treatment Flow Chart

***Assessment needs to be individualized to the patient based on the baseline measurable disease*

Transplant Eligible Regimens

Prefered therapy regimens:
Bortezomib/Lenalidomide/Dexamethasone
Bortezomib/Cyclophosphamide/Dexamethasone

Other regimens:
Bortezomib/Thalidomide/Dexamethasone
Cyclophosphamide/Thalidomide/Dexamethason
Bortezomib/Dexamethasone in some circumstances.

Transplant Ineligible

Bortezomib/Lenalidomide/Dexamethasone
Bortezomib/Cyclophosphamide/Dexamethasone
Bortezomib/Thalidomide/Dexamethasone
Bortezomib/Melphalan/Prednisone
Lenalidomide/Dexamethasone
Thalidomide-Dexamethasone

Maintenance Therapy

Lenalidomide
Thalidomide

Relapse/ Refractory Myeloma

Consider previous treatment and duration of response.
>6months consider primary treatment.

Radiotherapy may be indicated for plasmacytomas, fractures, pain control and spinal cord compression.

7.9.8 Staging Response Criteria

Table 7.11: International Myeloma Working Group Response Criteria

Criteria	Definition
CR	Disappearance of any soft tissue plasmacytomas and <5% plasmacytosis in BMA.
VGPR	>90% reduction in measurable disease
PR	50% reduction in measurable disease
Progressive disease	Increase in 20% from lowest confirmed response

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Chapter 8

Kaposi's Sarcoma

List of Abbreviations

Abbreviation	Definition
cART	Combined Antiretroviral Therapy
CXR	Chest X-ray
ECG	Electrocardiography
GIT	Gastrointestinal Tract
HIV	Human Immunodeficiency Virus
HHV8	Human Herpes Virus-8
KS	Kaposi's Sarcoma
KSHV	Kaposi's Sarcoma Herpes Virus
MCD	Multicentric Castleman's Disease
MTCT	Maternal to Child Transmission
OI	Opportunistic Infections
PJP	Pneumocystic Jiroveci Pneumonia (PJP), (MCD)
PLWHIV	People Living with Human Immunodeficiency Virus

8.1 Introduction

Kaposi's Sarcoma (KS) is a multi-centric angio-proliferative malignancy, originating from the vascular endothelium with simultaneous eruptions in several parts of the body, more commonly on the skin and oral cavity. It is the most common opportunistic malignancy associated with HIV infection. Non-compliance to antiretroviral medication is related to a greater incidence of KS. All types of KS are associated with Human Herpes Virus-8 (Kaposi's sarcoma Herpes Virus, KSHV). Epidemiological forms of KS include classic KS, endemic KS, immunosuppression-associated KS and epidemic KS. Epidemic KS occurs in advanced HIV disease, is more aggressive and affects muco-cutaneous and visceral sites.



Figure 8.1: Physical Signs Of Kaposi Sarcoma On The Skin And Mucosa

It has a global incidence of 35 813 with a mortality of 16 169 deaths annually in 2022. In Kenya, it ranked as the 18th most common cancer with an estimated 658 cases and the 17th most common cause of cancer related deaths with 383 deaths annually in 2022 (Globocan 2022 Kenya). A study conducted at Kenyatta National Hospital, reported a decreasing male to female ratio among patients with muco-cutaneous Kaposi's sarcoma.

Epidemic KS has been reported to affect young patients with a median age of 34 years. Transmission of KSHV is postulated to occur through sexual intercourse, saliva, mother to child transmission (MTCT), transfusion of contaminated blood and consumption of infected breast milk.

Clinical Evaluation History Taking Inquire about:

1. Constitutional symptoms: Fever, night sweats, weight loss
2. KS specific signs and symptoms- muco-cutaneous lesions (presents as papules, macules, nodules, ulcers, plagues, and fungating lesions) -enquire about onset of lesions, location, appearance, distribution; Melena stools or oral lesions (GIT involvement); cough, chest pain, haemoptysis (Pulmonary involvement)
3. Comprehensive HIV and opportunistic infection (OI) history, excluding other OIs such as tuberculosis, Pneumocystic Jiroveci Pneumonia(PJP), multicentric Castleman's disease (MCD) and lymphoma.
4. Medication history: If HIV positive ask for first line medication, second line medication, dates started, dates changed and reasons for change; history of additional immunosuppressive disease or therapy such as transplant, local or systemic corticosteroid

8.2 Physical Examination

- Conduct a comprehensive skin, oral and lymph node examination and document type of lesion, site, size, number, morphology, and limb circumference.
- Evaluate edema comprehensively
- Photography of oral, conjunctival, and cutaneous lesions should be done for documentation of extent of disease.

Look out for the signs and symptoms tabulated below:

Table 8.1: Signs And Symptoms Of Kaposi Sarcoma

Site	Clinical presentation
Mucocutaneous	Skin lesions in the limbs, trunk, groin, face, eyes, and external genitalia increase as disease progresses. They are often raised, violaceous/black, brown, plaques, macules, papules & nodules along tension lines of the skin
Extremities, groin, face, peri-orbital tissues	Lymphoedema occurs due to vascular obstruction by tumor or effect of cytokines released by KSHV.
Oral cavity	Bleeding painful lesions in the hard palate, gingiva, tongue, uvula, tonsils, pharynx, trachea
Cervical and inguinal areas	Lymphadenopathy- moderate or massive painful nodes
Pulmonary	Lesions in the pulmonary parenchyma, pleura, airways, and intra- thoracic nodes. Pleural effusions are common and may be hemorrhagic.
Gastro-Intestinal system	Lesions may involve the entire GIT resulting in bleeding, ulceration and intestinal obstruction

8.3 Diagnosis Laboratory Tests

- Human immunodeficiency virus test.
- HIV viral load and CD4 to evaluate viral control and immune function.
- If chemotherapy or radiotherapy is planned: Do tests to evaluate organ function such as Total blood count and biochemistry (Liver Function Tests, Renal Function Tests) and Pregnancy test (for women in child bearing age), transthoracic ECHO if treatment with doxorubicin planned. These tests should be done for everyone at diagnosis of KS.

8.4 Diagnostic Pathology

- Tissue biopsy (punch biopsy preferred or if not, excisional biopsy, incisional biopsy).
- Immunohistochemistry for HHV8 and LANA-1 where the microscopy is not clear. Silver Stains to exclude Bacillary Angiomatosis where the diagnosis is not clear-cut on morphology and immunohistochemistry.

Biopsy of other sites if co-existing disorders suspected (such as infection, multicentric, Castleman's Disease, lymphoma).

8.5 Imaging & Other Procedures

- CT Abdomen/ Pelvis or Abdominal ultrasound (if CT not available).
- Contrast enhanced CT Chest or Chest X-ray (if CT not available).
- Bronchoscopy if pulmonary symptoms present
- OGD and colonoscopy if gastrointestinal sites are symptomatic by history and physical examinations

FDG-PET if MCD or lymphoma suspected

Differential diagnoses include: *Purpura, hematomas,angiomas, dermatofibromas, nevi, bacillary angiomatosis, extrapulmonary PCP, disseminated cryptococcosis, lymphedema, keratoacanthoma, pyogenic granuloma and fibrosarcoma.*

8.6 Staging and Prognostication of Kaposi's Sarcoma

The AIDS Clinical Trial Group (ACTG) staging classifies patients with KS into good-risk or poor-risk groups based on:

- Tumor extent (T)
- Immune profile status (I) as measured by CD4 cell count
- Evidence of HIV-associated systemic illness (S)

Table 8.2: Revised AIDS Clinical Trial Group (ACTG) staging classification for KS

	Good Risk (0) (all of the following)	Poor Risk(1) (any of the following)
Tumor (T)	Confined to skin and/or lymph nodes and/or minimal oral disease	Tumor associated edema or ulceration. Extensive oral KS or KS in other non-nodal viscera. Mucous lesion located on the conjunctiva
Immune system (1)	CD4 cells >200/pl_	CD4 cells < 200/pL
Systemic illness (S)	No history of OI or thrush No 'B' symptoms Kanorfsky performance status > 70%	History of OI and/or thrush 'B' symptoms present Performance status < 70%

The ACTG staging predicts the prognosis of Kaposi's sarcoma; survival is significantly shorter in the poor risk category.

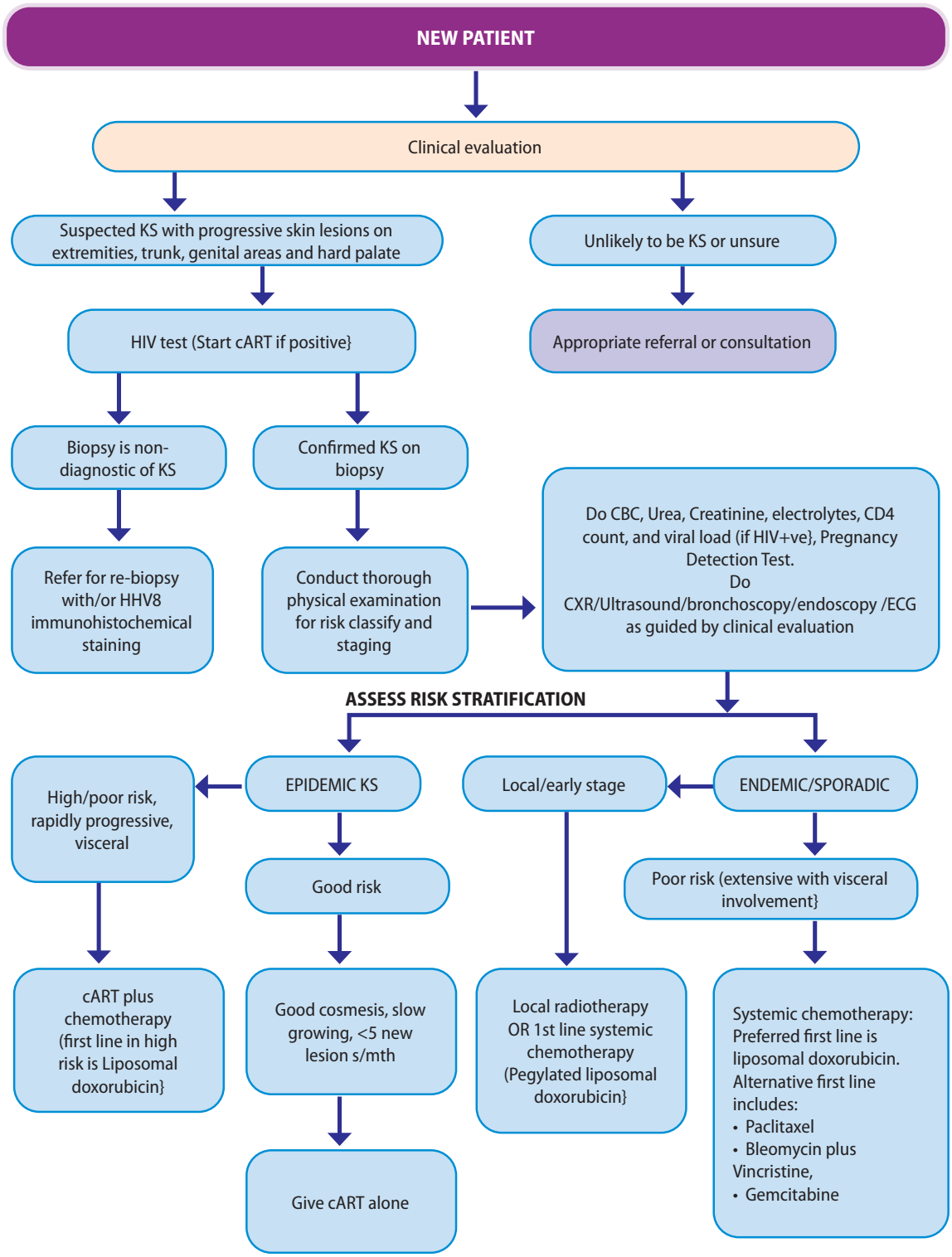


Figure 8.2: Treatment Algorithm for Kaposi's sarcoma

- **For PLWHIV, optimize HIV control and immune function**
- **Collaborate with a HIV specialist in patient management**
- **Treatment with systemic therapy is generally continued till toxicity or a plateau in response is reached**
- **Avoid corticosteroids should the patient develop IRIS**

Table 8.3: Systemic Chemotherapeutic Agents for KS: Include bortezomib, vinorelbine, etoposide, Imatinib, Nivolumab, pembrolizumab as other options for relapsed/refractory

Treatment	Drugs
First line	**Preferred:** Pegylated liposomal doxorubicin or liposomal daunorubicin (anthracyclines)- every 3 weeks **Alternate:** Paclitaxel - every two weeks **Alternate:** Bleomycin/Vincristine with sequential addition of Adriamycin in patients with poor response by cycle 3 Adriamycin/Bleomycin/Vincristine - may be considered as first line in early stage endemic/ sporadic KS
Second line	Gemcitabine Pomalidomide Thalidomide Lenalidomide

8.7 Radiotherapy

Patients with epidemic KS are managed with radiotherapy for ulcerated lesions, head and neck mucocutaneous disease, or residual disease post chemotherapy.

- Radiotherapy schedule is based on site and volume of disease with fractionation ranging from 30Gy in 10 fractions to 20Gy in 5 fractions and single shot of 8Gy.
- In Endemic KS radiotherapy doses will range from 30-45 Gy given in 1.8Gy-3G fractions.

8.8 Treatment Response

Table 8.4: Evaluation of Treatment Response used in ACTG (Aids Clinical Trial Group)

Response	Definition
Complete Response (CR)	Absence of detectable lesion, including tumour associated oedema, for more than 4 consecutive weeks
Partial Response (PR)	• Diminution of 50% of the total number and/or the size of previously existing lesions (skin, oral, measurable visceral disease) for more than 4 consecutive weeks and/or flattening of 50% of the nodular lesions without occurrence of new muco-cutaneous lesions (oral or skin or new visceral sites) • Complete response but with persistence of residual lymphoedema
Stable disease (S)	Any response not meeting the criteria for progression or PR
Progression	• > 25% increase in the number and/or size of previously existing lesions • Appearance of new muco-cutaneous lesions (nodules, plaque, macula, edema, ulceration or infiltration) or new localizations

8.9 Supportive Care

This includes psycho-social support, pain management, spirituality counselling and sexuality counselling. Systemic involvement with KS, particularly visceral and lymph node involvement can carry a poor prognosis. Supportive care may be required for patient's nutritional, mobility, social, and psychological wellbeing, to control pain, and to monitor and treat disease complications if and when they occur.

8.10 Anticipated Services per Level of Care

Table 8.5: Anticipated Services per Level of Care

Level of Care	Activities
Level 1	Awareness creation, appropriate referral
Level 2 & 3	Awareness creation, history taking and examination, appropriate referral, palliative care
Level 4	History taking and examination, laboratory & histologic diagnosis, appropriate referral, palliative care
Level 5 & 6	History taking and examination, laboratory & histologic diagnosis, endoscopy, staging, advanced diagnosis, treatment, palliative care

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Chapter 9

Lung Cancers

List of Abbreviations

Abbreviation	Definition
ALK	Anaplastic Lymphoma Kinase
CNS	Central Nervous System
COPD	Chronic Obstructive Pulmonary Disease
CRT	Chemo-radiotherapy
CT	Computerized Tomography
ECHO	Echocardiogram
ECOG	Eastern Cooperative Oncology Group Performance status
EGFR	Epidermal Growth Factor Receptor
FEV	Forced Expiratory Volume
FVC	Forced Vital Capacity
KRAS	Kirsten Rat Sarcoma viral oncogene
MRI	Magnetic Resonance Imaging
NSCLC	Non-Small Cell Lung Cancer
PET	Positron Emission Tomography
PCI	Prophylactic Cranial Irradiation
ROS1	A type of receptor tyrosine kinase
SIADH	Syndrome of Inappropriate Anti-Diuretic Hormone
SCLC	Small Cell Lung Cancer
SQLC	Squamous cell lung cancer
SVC	Superior Vena Cava
TB	Tuberculosis

9.1 Introduction

Lung cancer is the most common cancer and leading cause of cancer related deaths globally. Incidence in Africa however, remains generally low. In Kenya there are an estimated 903 new cases with 822 deaths annually (Globocan 2022).

There are two main types of lung cancer:

- Non-Small Cell Lung Cancer (NSCLC) constitutes 80-85% of cases. The most common subtypes are adenocarcinoma, Squamous cell lung carcinoma (SQCLC) and Large cell,
- Small Cell Lung Cancer (SCLC) constitutes 15% of all lung cancers, is associated with smoking.

Majority (70%) have advanced disease at diagnosis. A molecular study done in Kenya shows EGFR as the most prevalent biomarker (35%) followed by ALK (19%), ROS (6%) and PDL1(1%) (Mwanzi S, et al, 2025)

9.2 Risk Factors

- Tobacco smoking, both active and passive (associated with 85% of all cases),
- Environmental exposure (asbestos exposure, radon exposure, air pollution, and occupational exposure),
- Family history of lung cancer
- Chronic lung diseases such as COPD and TB
- Prior history of lung cancer
- **Prior exposure to radiation (head & neck cancer, breast cancer)**
- **Immunosuppressive status (HIV, Immuno-modulators,)**
- Age 55-74 years

Patients aged 55-74 years with ≥ 30 pack years of smoking are at the highest risk of developing lung cancer.

9.3 Clinical Evaluation

History Taking

Suspect lung cancer in patients with persistent pulmonary symptoms getting TB treatment.

- Inquire about onset and duration of:
  - **Constitutional symptom**s such as anorexia, unexplained weight loss, and sweating;
  - **Symptoms of primary lung lesion** such as cough, dyspnea, chest pain, hemoptysis, wheezing;
  - **Symptoms of regional invasion** such as dysphagia, Superior Venacava (SVC) obstruction, Horner's syndrome and Brachial plexus obstruction.

- **Metastatic stage**; patients may present with bone pain and tenderness, CNS symptoms such as headache, seizures, motor sensory deficits, and symptoms of liver metastasis such as abdominal pain and hepatomegaly.
- Paraneoplastic symptoms are also common in lung cancer, especially in SCLC and include Syndrome of Inappropriate ADH secretion (SIADH), Cushing's syndrome, ectopic parathyroid hormone secretion, hypercalcemia and Eaton-Lambert syndrome.

- Assess for risk factors especially smoking history, family history, and occupational exposure.
- Document the past medical history including history of chronic lung diseases, medication history among others.

9.4 Physical Examination

Conduct a thorough physical examination and assess for pleural effusion and lung mass among others.

Diagnosis

Initial Imaging

- Chest X-ray (can be AI assisted)
- CT Scan chest and upper abdomen (to include liver and adrenal glands): further defines abnormalities seen on Chest X-ray, detects very small nodules in the lung and is useful for staging.

Urgently refer for Lung Cancer or Mesothelioma management in patients with chest x-ray or CT chest findings that suggest Lung Cancer or Mesothelioma

Further Imaging (Staging and Metastatic work up)

This is usually guided by CT chest and upper abdomen findings when a confirmed lung cancer diagnosis on tissue biopsy.

- PET CT scan
- MRI brain: in SCLC whether symptomatic or asymptomatic; in NSCLC from Stage II and above.
- EBUS- endoscopic bronchial ultrasound
- Radionuclide bone scan should be done for all SCLC at diagnosis or during follow-up if bone symptoms develop.

Biopsy/ Specimen Collection

- Cell block preparation of effusion fluid specimen or fine needle aspiration including endobronchial ultrasound guided FNA provide excellent diagnostic specimens that are suitable for biomarkers testing and is highly encouraged.

Biopsy the most accessible site using minimal invasive techniques first for example cervical/ supraclavicular/ axillary lymph nodes.

- Flexible bronchoscopy or CT-guided biopsy for tissue diagnosis and assessing local extent of disease.
- Thoracocentesis, Pleural fluid for evaluation (cell block for histology analysis)

9.5 Pathology

- Biopsy testing should include HE and Immunohistochemistry to ensure diagnosis of primary lung Cancer.

For -adenocarcinoma (TTF1+ and P63/Napsin+), SQCLC (p63+ and p40+) and SCLC (TTF1+, CD56+, chromogranin+, synaptophysin+).

- Pretreatment evaluation; EGFR, PDL-1, HER2, ALK/ROS & Kras mutation tests to guide in the choice of targeted therapy
- Pleural fluid evaluation (cell block)

9.6 Baseline Evaluation Tests

- UECs, CBC, LFTs, HIV test
- Pulmonary & cardiac function tests such as ECHO, FEV, FVC will determine eligibility for surgery for those with early stage NSCLC only.

9.7 Staging

Stage the patient using the TNM and/or AJCC staging system.

9.8 Treatment

The management of lung cancer is complex and requires a multidisciplinary team approach, whose core team includes a thoracic surgeon, radio-oncologist, medical oncologist and palliative care specialist, among others. Prognosis should be explained to the patients, relatives and medical caretakers. Ideally, surgery should be done by thoracic surgeons who have the requisite skills and experience.

Table 9.1: Treatment Summary for Lung Cancer

Stage		Treatment summary
Stage 1	T1 -2N0	Surgery followed by adjuvant chemo except for stage IA OR Stereotactic RT OR Concurrent chemo-radiotherapy (CCRT) if inoperable
Stage IIA	T2N0	
Stage IIB	T1 -2N1 orT3N0	
Stage IIIA (N1)	orT3N1 orT4N0-1	
Stage IIIA (N2)	T1 -2N2	Neoadjuvant chemo then resection and radiotherapy OR Concurrent chemo-radiotherapy (CCRT) if unresectable
Stage IIIB	T1-2N3 orT3-4N2	
Stage IIIC	T3-4N3	Concurrent chemo-radiotherapy (CCRT)
Stage IV	Contralateral lung nodule, pleural or pericardial effusion (Mia), single extrathoracic metastasis (M1b), multiple extrathoracic metastasis (Mic)	Systemic chemotherapy or targeted therapy WBRT if indicated Bisphosphonate if indicated

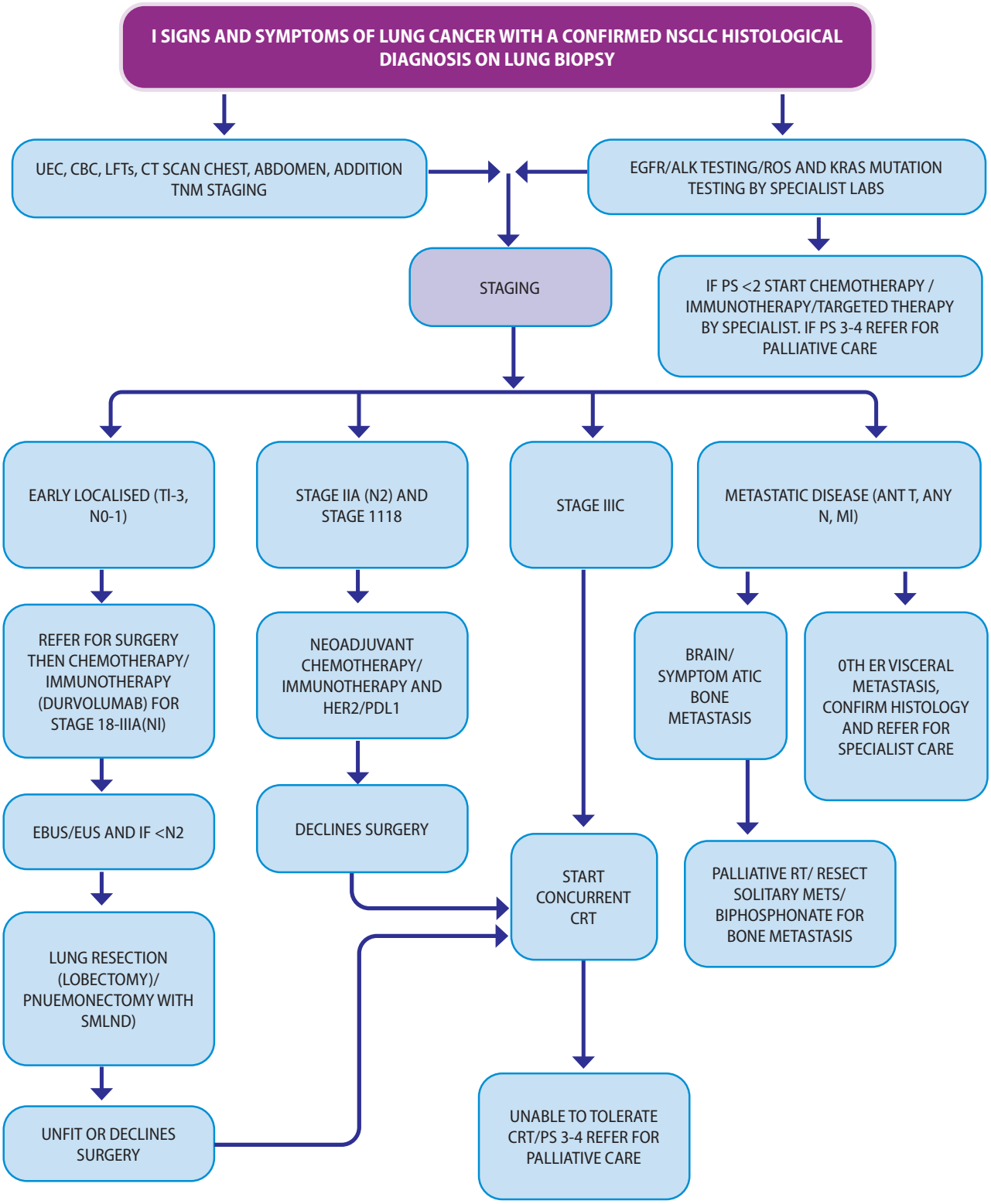


Figure 9.1: Algorithm for Non-Small Cell Lung Cancer- PDL-1. Her2

1. In some cases, surgery may precede histological diagnosis.
2. Adjuvant chemotherapy is not recommended for stage IA, but maybe considered for stage IB, should be given to stage IIA-IIIA.
3. Re-resect tumor if margins positive.

4. PET/CT should be considered for suspected stage I/II disease ONLY; other diagnostic procedures include bronchoscopy and biopsy; transbronchial needle aspiration, transthoracic needle aspiration, video assisted thoracoscopy, endoscopic bronchial ultrasound and thoracentesis, and brain MRI for asymptomatic stage IIIA/B and T4.
5. Resectability should be determined by cardiothoracic surgeons. Pulmonary and cardiac reserves should be determined prior to surgery. Anatomic resections are preferred.
6. Concurrent chemoradiation (CRT) is preferred, at doses of at least 60Gy, but sequential may be given to unfit patients. CCRT may be given pre-operatively for resectable stage Pancoast tumors
7. Targeted therapy is 1st line for eligible stage IV patients in developed countries.
8. Avoid chemotherapy in Performance score (ECOG 3-4).

Table 9.2: Systemic Chemotherapy Protocols

	Adenocarcinoma	Squamous cell lung cancer	Small cell lung cancer
1st line neoadjuvant, adjuvant & palliative chemo	4-6 cycles of Pemetrexed/platinum Paditaxel /carbo/bevacizumab Paclitaxel/platinum Docetaxel/ platinum Gemcitabine/platinum	Gemcitabine/ Cisplatin	Carboplatin/ etoposide Cisplatin/etoposide
Maintenance chemo	Pemetrexed		
Chemotherapy used concurrently with RT	Paclitaxel /platinum Docetaxel/ platinum Etoposide/cisplatin Pemetrexed/cisplatin		Cisplatin /Etoposide Carboplatin/ Etoposide
2nd line chemo	Docetaxel Gemcitabine/ Vinorelbine		Topotecan Irinotecan Paclitaxel Docetaxel Temozolomide Vinorelbine Etoposide Gemcitabine
First line Immunotherapy PDL1 >50%	Pembrolizumab		
Metastatic adenoCa with EGFR mutational antibodies	Tyrosine kinase inhibitors (where available) Gefitinib Afatinib Erlotinib		
ALK/ROS1 mutation present	ALK inhibitors (where available) Crizotinib		

9.9 Small Cell Lung Cancer (SCLC) Staging and Prognosis

Use both TNM system and the Veterans Administration Lung Study Group (VALSG) system as below to classify the tumor stage.

Table 9.3: The Veterans Administration Lung Study Group System

Stage	Treatment summary
Limited disease (to ipsilateral hemithorax)	Surgery Or Neoadjuvant chemoradiotherapy (NACT) then surgery Or Concurrent chemoradiotherapy if node positive Palliative care
Extensive disease	Systemic chemotherapy (thoracic radiotherapy for residual disease) Or Concurrent chemoradiotherapy Palliative care

** Immunotherapy to be considered where available for extensive disease*
***In SCLC, Prophylactic Cranial Irradiation (PCI) is recommended for both limited and extensive disease.*

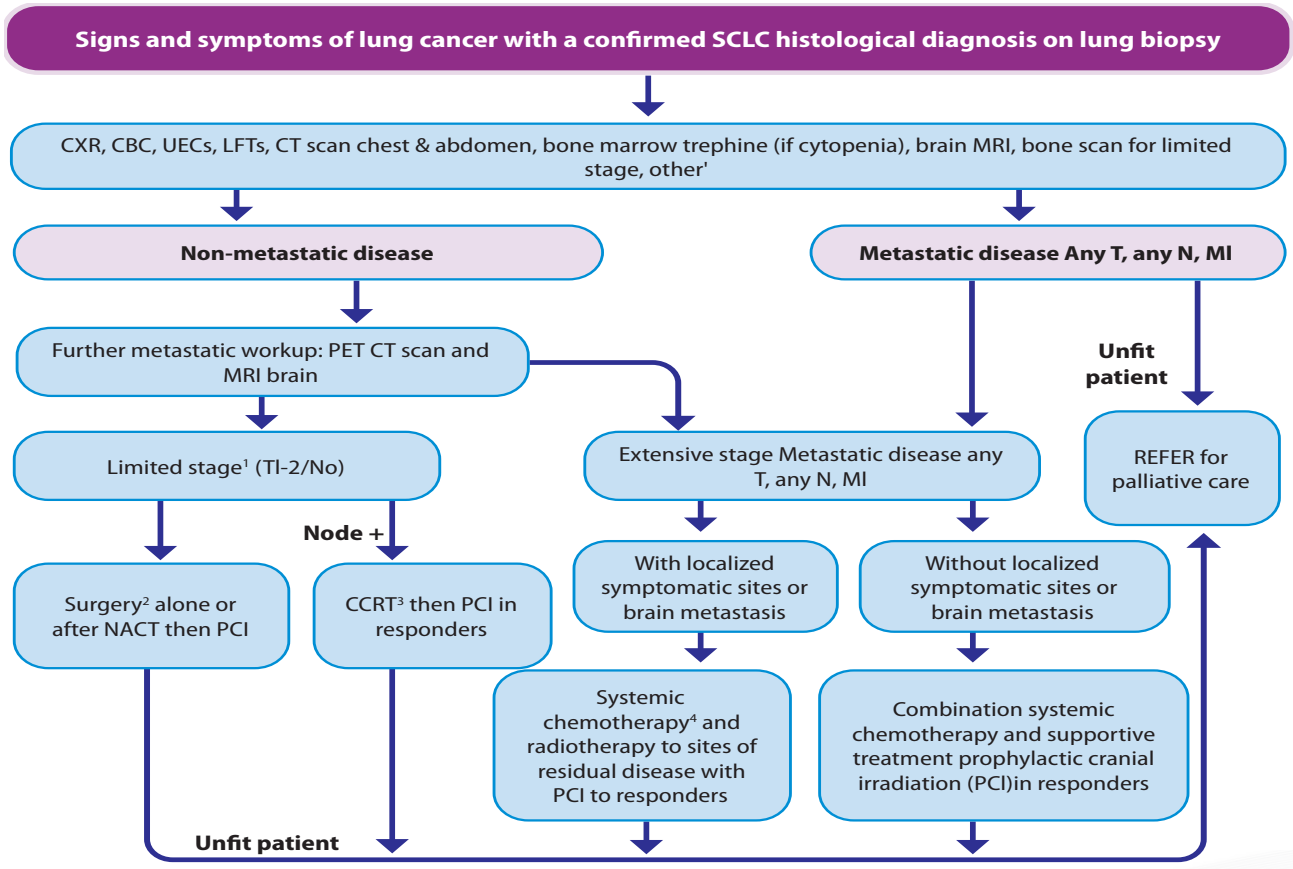


Figure 9.2: Algorithm for Small Cell Lung Cancer

1. Mediastinoscopy or other staging should be done to rule out nodal disease
2. Pulmonary function tests to evaluate for surgery. Patients in excess of T1-2N0 do not benefit from surgery.

Concurrent chemoradiation preferred to sequential typically 45Gy in 3 weeks twice daily. Concurrent chemoradiation should be used only in patients with a suitable performance status. Otherwise, give sequential as tolerated. **Palliative and Supportive Care**

- Support groups with smoking cessation services.
- Psycho-social support and spirituality support in advanced cases.
- Counselling on healthy lifestyle and sexuality should be available.
- **Palliative Radiotherapy** can alleviate symptoms in 41–95% of cases. Indications of palliative radiotherapy include:
  - Brachytherapy (in case of disturbance of ventilation due to airway obstruction),
  - pain and hemoptysis,
  - symptomatic metastases to the central nervous system or bone.

- **Palliative Chemotherapy**
  - This can relieve pain, cough and other ailments.
  - If there is shortness of breath following the toxic effects of chemotherapy on pulmonary parenchyma, discontinuation of administration of cytotoxic drugs and incorporation of steroids is necessary.
  - When chemotherapy starts to increase side effects then it should be discontinued.

- **Symptom management** e.g. dyspnea, cough, hemoptysis, pleural effusion, pain management, superior vena cava syndrome and fatigue among others.

Table 9.4: Recommended Treatment Regimes

	Adenocarcinoma	Squamous Cell lung cancer	Small cell lung cancer
1st line neoadjuvant, adjuvant & palliative chemo 4-6	4-6 cycles of Pemetrexed/platinum Paclitaxel /carbo/bevacizumab Paclitaxel/platinum Docetaxel/ platinum Gemcitabine/platinum Gemcitabine/	Gemcitabine/ Cisplatin	Carboplatin/ etoposide Cisplatin / etoposide
Maintenance chemo	Pemetrexed		
Chemotherapy used concurrently with RT	Paclitaxel /platinum Docetaxel/ platinum Etoposide/cisplatin Pemetrexed/ cisplatin		Cisplatin / etoposide Carboplatin/ etoposide
2nd line chemo	Docetaxel Gemcitabine Vinorelbine		

	Adenocarcinoma	Squamous Cell lung cancer	Small cell lung cancer
First line metastatic chemotherapy PDL1>/= 50%	Pembrolizumab ATEZOLIZUMAB		
Metastatic adenoCa with EGFR Mutational antibodies	Tyrosine kinase inhibitors (were available) Osimertinib Gefitinib Afatinib Erlotinib		
ALK/ROS 1 Mutation present	ALK inhibitors (where available) Crizotinib		

9.10 Follow Up

Stage I and II Lung Cancer

- 3 to 6 months after initial treatment.
- Then every 6 months for 5 years.

During follow-up visits: Clinical evaluation includes history of chest symptoms, and for smokers, whether they have stopped, examination for effusion or mass in chest or abdomen, CT scan chest, ALP for liver and bone metastasis.

Stage IIIA

- Surveillance every 6 months for 2 years with history, physical examination and contrast-enhanced chest CT at least at 12 and 24 months, thereafter every 12 months is recommended.

Stage IIIB

- 6-monthly CT chest scans for 3 years are recommended for patients who are suitable for salvage treatment.

Table 9.5: Anticipated lung cancer services per level of health care

Level of Health Care	Anticipated Cancer Services
Community Health Care Level	**Awareness creation** **Patient identification** Intergrate Lung cancer assessment in TB and cough assesment Signs and symptoms of lung cancer **Referral to level 4** **Palliative care**

Level of Health Care	Anticipated Cancer Services
Primary Health Care Level (Dispensary and Health centres)	**As above** Intergrate Lung cancer assessment into TB and cough clinics
Secondary Health Care Level (County Referral Hospitals)- previously level 5	**As above plus;** Intergrate Lung cancer assessment into TB and cough clinics **CXR, CT scan chest and upper abdomen** **Bronchoscopy, mediastinoscopy, CT guided biopsy, thoracentesis** **Lung biopsy and review by specialist** **Metastatic work-up:** v. Refer for PET/CT scan ONLY in early disease. vi. Brain MRI may be utilized Pulmonary & cardiac function tests e.g. ECHO, FEV, FVC Therapy: i. Surgical management by thoracic surgeons ii. Can manage patients with metastatic disease with chemotherapy under guidance of specialist/oncologists iii. Palliative care iv. Supportive services v. Patient follow-up
Tertiary Hospitals (previously level 6)	**All above plus:** • **Immunohistochemistry &targeted therapy** • PET CT scan • **Radiotherapy**

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Chapter 10

Urological Cancers

List of abbreviations

Abbreviation	Definition
ALP	Alkaline Phosphatase
ADT	Androgen Deprivation Therapy
CT	Computer Tomography Scan
BCG	Bacille Calmette Guerin
CXR	Chest X ray
DRE	Digital Rectal Examination
ECOG	Eastern Cooperative Oncology Group (performance status)
EBRT	External Beam Radiotherapy
HPC1	Hereditary Prostate Cancer 1
HIV	Human Immunodeficiency Virus
IB	Interstitial Brachytherapy
ISUP	International Society of Urological Pathology
KUB	Kidney Ureters & Bladder Ultrasound
MRI	Magnetic Resonance Imaging
PO BID	Per Oral Twice a Day
PCA3	Prostate Cancer Antigen 3 Gene
PSA	Prostate-Specific Antigen
RP	Radical Prostatectomy
RT	Radiotherapy
SC	Subcutaneous
TRUS	Transrectal Ultrasound
TURBT	Transurethral Resection of Bladder Tumor

10.1 Prostate Cancer

10.1.1 Introduction

Globally, prostate cancer is the second leading cause of cancer in males and the 8th common cause of overall cancer related mortality. In Kenya, prostate cancer is the leading cause of cancer and cancer related death in males with 3,582 cases (8%) and 2029 deaths (6.9%) as per GLOBOCAN 2022 data.

10.1.2 Clinical Evaluation

Risk Factors

- Advancing age
- African ancestry
- Heredity and genetic predisposition
- Male gender

History Taking

Prostate cancer usually progresses slowly and produces no symptoms in its initial stages.
Symptoms in both stages may include:

- Lower urinary tract symptoms such as:
- Difficult or painful urination;
- Hesitancy (delay in starting) and dribbling at the end of urine flow;
- Frequent urination, especially at night;
- Urgency in urinating;
- A sense of incomplete emptying of the bladder;
- Loss of bladder control;
- Decreased flow of urine stream; and
- Blood in the urine (hematuria) or semen.

These symptoms, however, may have other overlapping causes, such as infection or benign prostate enlargement (BPE).

- Unexplained bone pain in the axial and/or appendicular skeleton may signal metastases.

Other important aspects of history include

- Risk factor assessment
- Past medical history.
- Co-morbidities such as diabetes and hypertension.
- Modifiable lifestyle risk factors such as unhealthy diet, alcohol and tobacco use.

Physical Exam

Conduct a complete physical examination including:

- Digital rectal exam – this assesses the size, nodularity, shape, firmness, tenderness of the prostate and mobility of the rectal mucosa.
- Systemic examination

10.1.3 Diagnosis

Laboratory

- Prostate-specific antigen (PSA) test – raised PSA level does not necessarily indicate the presence of cancer.
- Urgently refer men whose PSA level is above the age-specific reference range to a urologist.

Table 10.1: PSA reference range

Age range	PSA reference range(ng/ml)
40-49	≤2.5
50-59	≤3.5
60-69	≤4.5
70-79	≤6.5

Prostatic Core Biopsy- Confirmatory Diagnosis

A minimum of 10-12 cores should be taken targeting the peripheral zones of the prostate. Image-guided biopsies yield better sampling of the prostate.

The Indications for Prostate Biopsy

- PSA > 10ng/mL – generally should lead to a biopsy.
- PSA 4 – 10ng/mL – this requires further interrogation with adjunctive studies which include multi-parametric MRI (1.5 tesla) of the prostate.

In addition, DRE, PSA derivatives (PSA density and age specific reference ranges) and PSA kinetics (velocity and doubling time), PSA molecular forms (percent free PSA and pro-PSA), newer urinary markers (PCA3), and prostate imaging should be considered secondary tests (not primary screening tests) that can help to guide on the justification for a prostate biopsy.

NB:

- **The clinician should first consider other factors that lead to an increased PSA e.g. recent history of infection, inflammation, instrumentation etc. before proceeding to take a prostate biopsy. Other factors to also consider are PIRADS scoring on imaging.**
- **Do not subject men to PSA testing without first counselling them on the potential risks and benefits.**

Other laboratory tests may include: Full blood count, urinalysis and urine microscopy, renal function test, Liver function test, Lactate dehydrogenase test, Serum calcium.

PI-RADS
PI-RADS 1 = Very low - clinically significant cancer highly unlikely **PI-RADS 2** = Low - clinically significant cancer unlikely **PI-RADS 3** = Intermediate - clinically significant cancer equivocal **PI-RADS 4** = High - clinically significant cancer likely **PI-RADS 5** = Very high - clinically significant cancer highly likely

Figure 10.1: PIRAD Criteria

Imaging

Imaging is useful for diagnosis and staging

Diagnostic Imaging

- Multi-parametric MRI (1.5 tesla) is done upon suspicion of prostate cancer in an asymptomatic man with a normal DRE and a PSA of 3-20ng/ml. This should be done before any biopsy is obtained.
- Trans-rectal ultrasound (TRUS) and multi-parametric MRI are useful for image guided biopsy.

Staging Imaging

Metastatic work up is done using

- Pet PSMA is recommended for diagnosing metastases
- Pre biopsy multiparametric MRI (1.5 tesla) is recommended for diagnosing and staging local disease.

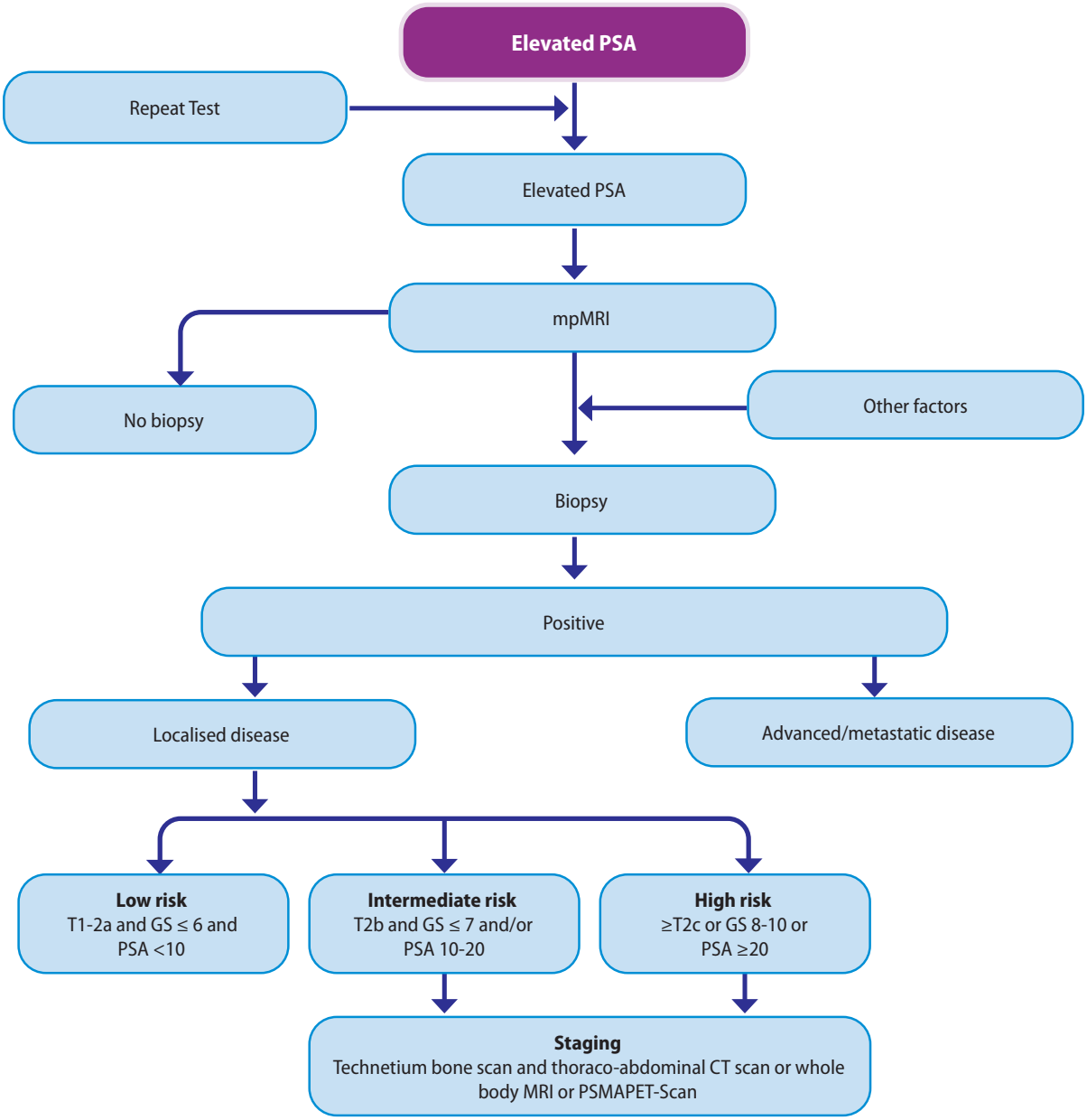


Figure 10.2: Prostate Cancer Diagnostic Approach Algorithm

10.1.4 Metastatic Disease Workup

For suspected skeletal metastases:

- Bone scan: Indications include PSA above 10, T3-T4 disease, Gleason 8-10/ISUP Group 4 and 5; bone pain, raised alkaline phosphatase.

NB: A bone scan is recommended for all patients before any definitive curative treatment.

- PSMA PET should be considered for specific cases if available due to its increased diagnostic accuracy.

Table 10.2 :TNM staging classification (AJCC) 8th edition

Primary Tumor	
Tx	Primary tumor cannot be assessed
TO	No evidence of primary tumor
T1	Clinically inapparent tumor not palpable not visible by imaging
Tla	Incidental tumor in < 5% of TUR tissue
Tib	Incidental tumor in > 5% of TUR tissue
Tic	Needle biopsy prompted by elevated PSA
T2	Organ confined
T3	Tumor extends beyond the prostatic capsule
T3a	Extracapsular, unilateral and bilateral or microscopic invasion of bladder neck
T3b	Tumor invades seminal vesicles (s)
T4	Tumor invades external sphincter, rectum, pelvic side wall
Lymph Nodes	
Nx	Regional nodes were not assessed
NO	No regional (below level of bifurcation of common iliac arteries) nodes
N1	Regional node metastases - including pelvic, hypogastric, obturator, iliac, sacral
Distant Metastases	
Mx	Regional nodes not assessed
MO	No Metastases
Ml	No distant
Mia	Non-regional lymph nodes (outside true pelvis)
M1b	Bone(s)
Mlc	Other site(s) with or without bone disease

Table 10.3: International Society of Urological Pathology 2014 grade group system

Gleason score	ISUP grade
2-6	1
7 (3+4)	2
7 (4+3)	3
8(4+4 or 3+5 or 5+3)	4
9-10 (4+5 or 5+4 or 5+5)	5

Table 10.4: EAU risk groups for biochemical recurrence of localized and locally-advanced prostate cancer based on systematic biopsy

Definition				
Low-risk	**Intermediate-risk**		**High-risk**	
	Favourable	**Unfavourable**		
ISUP grade 1 and PSA < 10 ng/mL and cT1-2a*	ISUP grade 2 and PSA < 10 ng/mL and cT1-2b* Or ISUP grade 1 and PSA 10 -20 ng/mL and cT1-2b* Or ISUP grade 1 and PSA < 10 ng/mL and cT2b*	ISUP grade 2 and PSA 10- 20 ng/mL and cT1-2b* Or ISUP grade 3 and cT1-2b*	ISUP grade 4/5 Or PSA > 20 ng/mL Or cT2c*	cT3-4* and/ orcN+** any ISUP grade* any PSA
Localised				**Locally advanced**

ISUP = International Society of Urological Pathology;
PSA = prostate-specific antigen.
** Based on digital rectal examination.*
*** Based on CT/bone scan.*

Treatment

The treatment of prostate cancer generally depends on the stage of the cancer, risk stratification and the life expectancy of the patient. Decision making during MDT is preferred. Treatment modalities include:

1. Active surveillance (actively monitoring the course of the disease with the expectation to intervene with potentially curative therapy if the cancer progresses);
2. Watchful waiting (actively monitoring the course of disease with the plan to deliver stage specific therapy in the event of development of symptoms or a change in exam of PSA that suggests symptoms/progression is imminent);
3. Surgery- radical prostatectomy, preferably laparoscopic, or robotic.
4. Radiation therapy - External Beam Radiation Therapy, preferably Stereotactic body radiotherapy; or Brachytherapy, depending on risk stratification and primary presentation of the patient.
5. Hormonal Therapy.
6. Chemotherapy.
7. Targeted therapy - PARP inhibitors, etc.
8. Primary Palliative care.

Table 10.5: Stage-matched therapeutic strategies

Type of disease	Risk profile	Treatment
Localized disease	Low risk	• Active surveillance • Brachytherapy • Radical Prostatectomy (RP) • Radical Radiotherapy (Radical RT)
	Intermediate risk	• RP • Radical RT± neoadjuvant ADT • Brachytherapy • Active surveillance
	High risk	• Long-term ADT + radical RT • ±neoadjuvant docetaxel • RP + pelvic lymphadenectomy
Localized advanced disease		• Neoadjuvant ADT + radical RT + • adjuvant ADT • ± neoadjuvant docetaxel • RP + pelvic lymphadenectomy
	High risk	• ADT + apalutamide • ADT + darolutamide • ADT + enzalutamide
	Hormone-naive	• ADT + abiraterone • ADT + docetaxel • ADT + enzalutamide • ADT + apalutamide • RT for low volume • ADT alone for frail patients who cannot tolerate the above treatments • Bone health agent
	Castration-resistant (first line)	• Abiraterone • Docetaxel • Enzalutamide • Radiofrequency Ablation is preferred for patients unfit for above treatments (and bone-only metastases)
	Second line or post docetaxel	• Abiraterone • Cabazitaxel • Enzalutamide • Radiofrequency Ablation

10.1.5 Clinically Localized Prostate Cancer

Patients with localized prostate cancer should be categorized according to their risk based on PSA levels, Gleason Score and clinical staging. The risk stratification below is based on ESMO guidelines 2015 thus for;

1. Low risk disease (PSA ≤10ng/ml and Gleason Score 6 and T1-T2a)
    a. If the expected survival is < ten years: watchful waiting.

b. If the life expectancy is > ten years then treatment options include:
   i. active surveillance,
   ii. definitive external beam radiotherapy,
   iii. interstitial brachytherapy, and
   iv. Radical prostatectomy with a lymph node dissection.

2. Intermediate (moderate) risk disease (PSA 10-20 ng/ml or Gleason Score 7 or T2b /T2c)
   a. If the expected survival is < ten years: watchful waiting.
   b. If the life expectancy is > ten years then treatment options include:
      i. active surveillance,
      ii. radical prostatectomy,
      iii. definitive RT* + short term hormonal therapy 4-6 months, or
      iv. Adjuvant RT after RP (if +ve Surgical margins or Lymph nodes).

3. High risk disease (PSA >20ng/ml or Gleason Score 8-10 or T2c-T3).
   a. Neoadjuvant Androgen Deprivation Therapy (ADT) followed by Radical Radiotherapy (RT) and Long-term ADT for 2-3 years.
   b. Radical prostatectomy can be offered to a select group of patients but with the understanding that there is a 50% chance of requiring additional therapy.

10.1.6 Metastatic Disease

Castration sensitive metastatic disease

- High risk/High Volume:
  - ADT plus Docetaxel and Predisolone up to 6 cycles OR
  - ADT plus Abiraterone and Prednisolone OR
  - ADT alone (especially if there is visceral crisis and patient cannot tolerate systemic therapy)

- Low risk/Low volume metastatic disease: ADT alone.
- Palliative radiotherapy ± chemotherapy: radiation may be used for palliation in patients with painful bone metastases or impending spinal cord compression.
- Surgical intervention may be necessary for weight-bearing bones affected by pathological fracture.

Castrate-resistant metastatic disease

- Maintain Androgen Deprivation Therapy to keep the testosterone below · castrate levels (less than 50ng/dl).
- Docetaxel; repeat cycle every 21 days for up to a total of 10 cycles. (especially if there is visceral crisis)
- Abiraterone 1000 mg PO daily plus prednisone 5 mg PO BID until disease progression.
- Enzalutamide or other androgen receptor pathway inhibitors may be administered until disease progression.
- Cabazitaxel with prednisone can be used for patients who have hormone-refractory metastatic prostate cancer that was previously treated with a docetaxel-containing treatment regimen.
- Symptomatic men who are not candidates for docetaxel-based regimens may be candidates for Mitoxantrone IV on day 1 plus prednisone 5 mg PO BID daily; repeat cycle every 21 days.

- Sipuleucel-T immunotherapy (also known as Provenge) is useful in low risk, low-burden disease.
- Olaparib can also be used in patients who have HRR(BRCA) mutations
- For metastatic prostate cancer in patients with impending spinal cord compression, urgent radiotherapy should be administered within 48hrs.

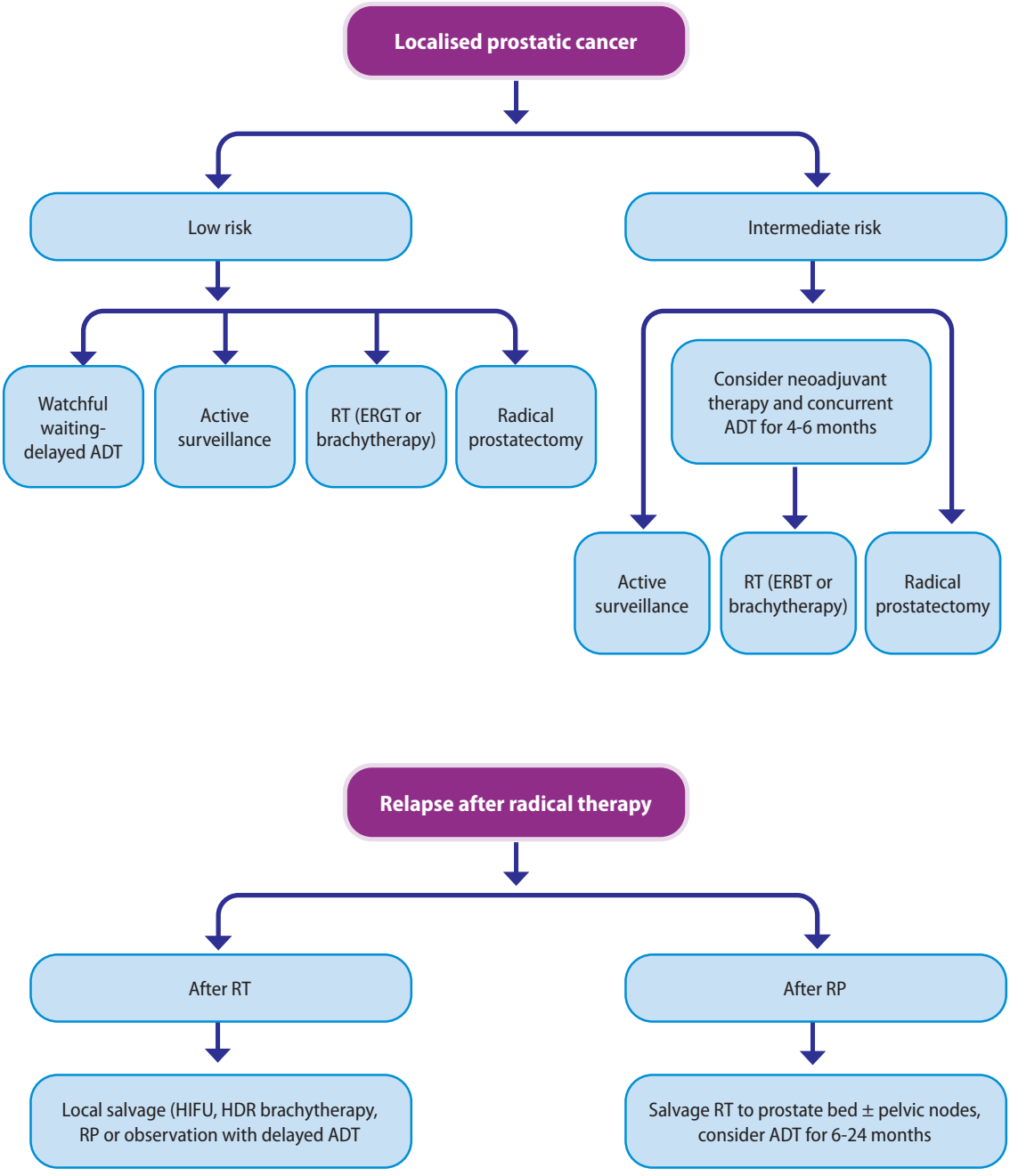


Figure 10.3: Treatment Algorithm for Prostate Cancer localized prostate cancer

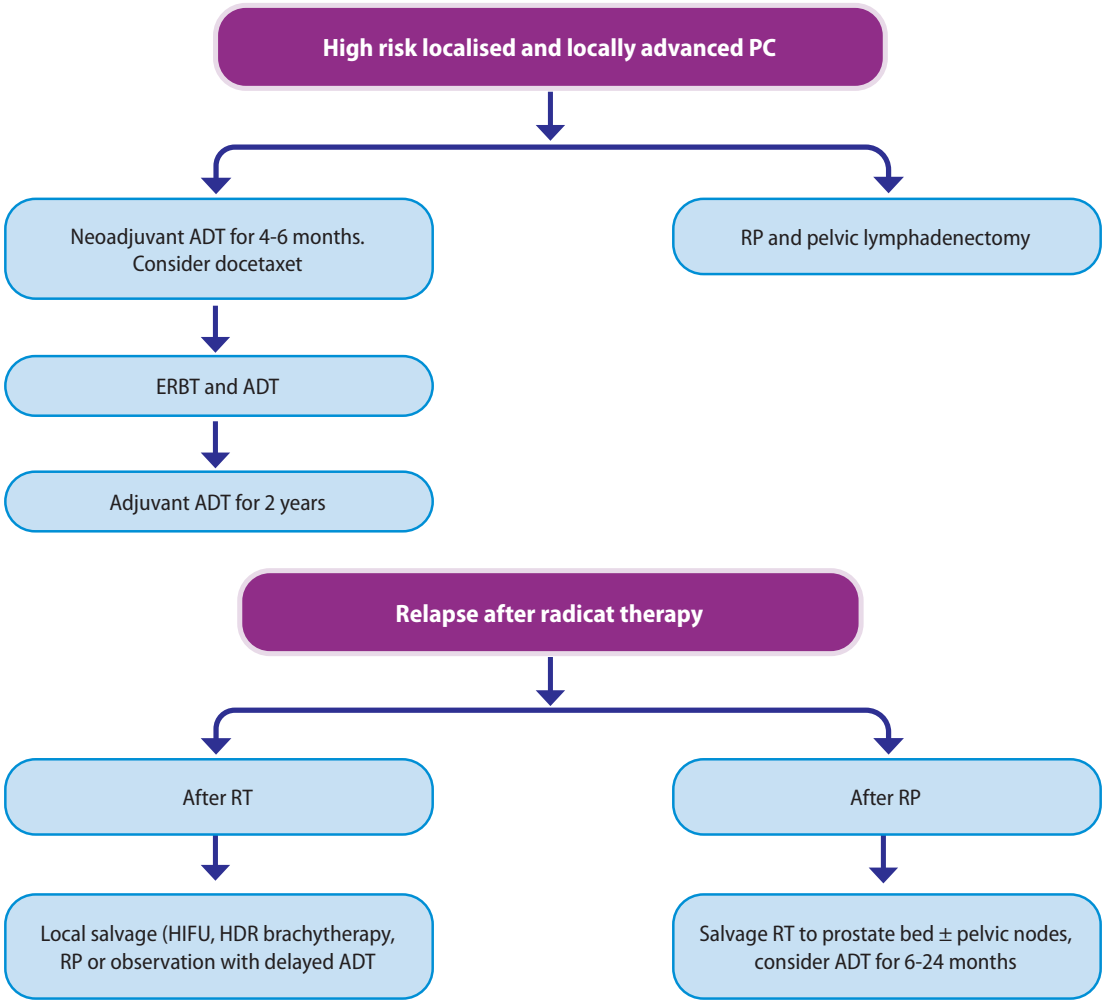


Figure 10.4: Treatment Algorithm for Prostate Cancer high risk localized and locally advanced prostate cancer

Castration-Resistant Prostate Cancer and Bone Metastases

Bisphosphonates are recommended for all men with hormone-refractory prostate cancer and symptomatic bone metastases as appropriate. They reduce skeletal-related events.

- Zoledronic acid 4 mg IV 3 monthly over no less than 15 min (check renal functions and adjust accordingly) or
- Denosumab 120 mg SC every 4 week.
- The radiopharmaceutical radium-223 dichloride (XoFIGO) is approved for men with castration-resistant prostate cancer with symptomatic bone metastases and no known visceral metastatic disease.

Biochemical Failure

- Approximately 20-30% of patients treated with intent for cure will have biochemical recurrence, which most commonly manifests as a rising PSA level (within 2-3 years of radical treatment).
- Biopsy of the prostatic bed is usually not recommended unless the patient is a candidate for salvage therapy.
- Monitor PSA, alkaline phosphatase, and calcium every 3 months and determine whether the doubling time is less than 1 year.

Post–Radical Prostatectomy Recurrence

- Defined as a detectable PSA that increases on two subsequent measurements or a
- PSA that fails to fall to undetectable levels.
- Best PSA threshold unknown but probably between 0.2 and 0.4 ng/ml.
- PSA that fails to fall to undetectable levels after prostatectomy may indicate residual prostate cancer or prostate cancer recurrence.
- Treatment options include salvage radiation therapy, androgen deprivation, and surveillance.
- Adjuvant radiation therapy may be more beneficial than salvage radiation therapy in men with poor pathologic features

Post–Radiation Therapy Recurrence

There are two recognized definitions of recurrence after radiotherapy

- The 1996 ASTRO definition of PSA failure following EBRT is three consecutive PSA rises, with the time of failure backdated to the midpoint between the PSA nadir and the first rising PSA, or any rise great enough to provoke initiation of salvage therapy; a minimum follow-up of 2 years was recommended for presentation or publication of data.
- The "Phoenix Definition" (current ASTRO/RTOG definition) of PSA failure after EBRT, with or without short-term HT;
  - Defined as a rise in PSA of 2 ng/ml or more above the nadir (defined as the lowest PSA achieved after radiotherapy).
- Note that the PSA can bounce up and down after radiation.

Treatment options include:

- salvage prostatectomy, ADT, surveillance, high-intensity focused ultrasound (clinical trials), cryotherapy.

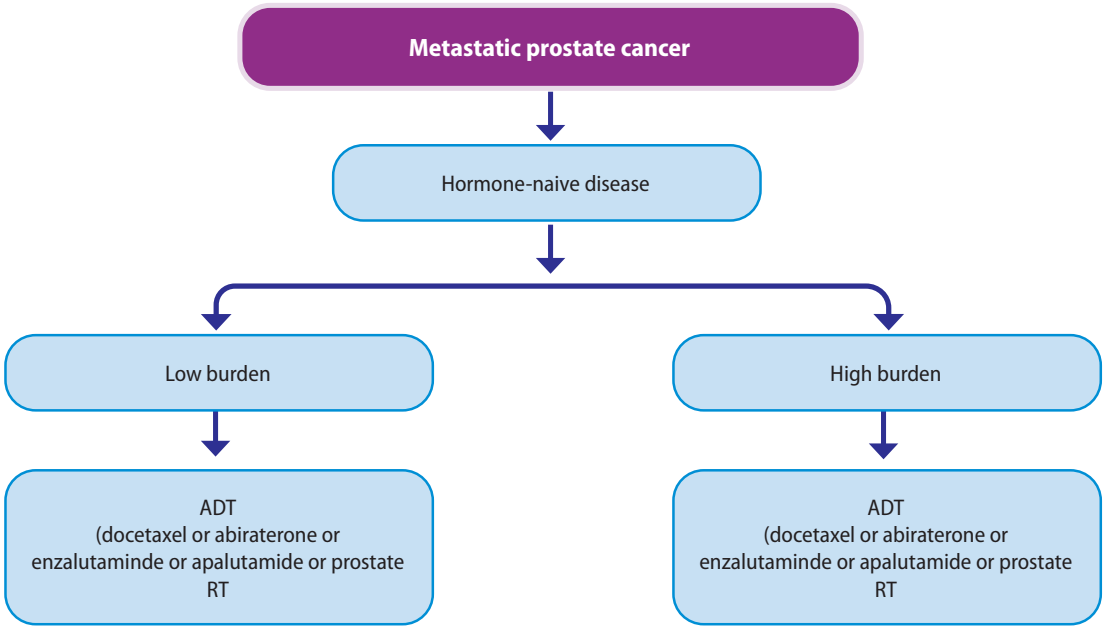


Figure 10.5: Treatment Algorithm for Metastatic Prostate Cancer

Follow-Up

- After initial definitive therapy, perform PSA every 6 t o12 months for 5 years and then annually.
- DRE every year but may be omitted if PSA is undetectable.
- For metastatic disease –physical exam and PSA every 3 to 6 months or earlier if clinically
- indicated; bone imaging for symptoms and as often as every 6-12 months.

Supportive Care for Prostate Cancer

- Incontinence management
- Pain and symptom management
- Sexuality counseling and management
- Rehabilitative therapy
- Anticipatory approach to symptom management
- Oncologic emergencies e.g. bone pain, spinal cord compression

10.1.7 References

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10.2 Testicular Cancer

10.2.1 Introduction

Testicular cancer is a highly curable cancer that usually develops in young and middle-aged men mostly in the third and fourth decade of life. As per Globocan data 2022 testicular cancer is not very common, with an incidence rate of 0.08%. Despite its high curability the mortality still remains high in our setting at 0.04%, this translates to half of the cases leading to death. This can be attributed to either, poor health seeking behavior, low index of diagnostic suspicion and poor health referral system.

They are clinically divided into seminomatous and nonseminomatous. The risk associated with testicular cancer include undescended testis, family history, previous history of contralateral testicular cancer. People diagnosed with seminomatous testicular cancer have a 25% risk of contralateral testicular cancer with the first two years after treatment. Additionally, men aged between 15-45 years are at a high risk of getting testicular cancer. White race is at a higher risk of developing testicular cancer compared to black race. The clinical presentation of testicular cancer includes testicular painless mass. In advanced diseases one can experience abdominal/lower back pain and scrotal pain.

Diagnosis:

Physical Examination
Presence of a palpable testicular mass unilaterally or bilaterally
NB: small intratesticular lesions in a minority may not be palpable
Testicular tenderness

10.2.2 Investigation

Laboratory

- Basic investigations- Hemogram, LFT's, UEC's
- Testicular tumour markers - BetaHCG, Alpha-fetoprotein, LDH
- Seminalysis and possible sperm banking are important consideration if fertility is desired especially in bilateral disease

Radiology

- High frequency Scrotal ultrasound - is vital in diagnosis and confirmation of physical findings and assesses size of mass, extra or intra testicular position and multiplicity of those masses.
- Contrast enhanced CT of the abdomen and chest is the gold standard used in staging
- Scrotal MRI provides higher specificity and sensitivity but is limited as a screening tool and

routine investigation due to cost and availability. It should however be considered when ultrasound findings are inconclusive.

- There is not enough evidence to support routine use of PET CT as a preferred staging tool/ modality in testicular cancer.

Staging

The 2016 Tumour, Node, Metastasis (TNM) classification of the International Union Against Cancer (UICC) is recommended to assess the anatomical extent of the disease.

Table 10.6: TNM classification for testicular cancer (adapted from UICC, 2016, 8th edition.)

pT - Primary Tumour1	
pTX	Primary tumour cannot be assessed (see note1)
pTO	No evidence of primary tumour (e.g., histological scar in testis)
pTis	Intratubular germ cell neoplasia (carcinoma in situ)*
pTI	Tumour limited to testis (including rete testis) and epididymis without vascular/lymphatic invasion and without invasion of the epididymis**
pT2	Tumour limited to testis with vascular/lymphatic invasion, or invading hilar soft tissue or the epididymis or tumour extending through tunica albuginea with involvement of visceral tunica vaginalis***
pT3	Tumour invades spermatic cord with or without vascular/lymphatic invasion**
pT4	Tumour invades scrotum with or without vascular/lymphatic invasion
N - Regional Lymph Nodes - Clinical	
NX	Regional lymph nodes cannot be assessed
NO	No regional lymph node metastasis
N1	Metastasis with a lymph node mass 2 cm or less in greatest dimension or multiple lymph nodes, none more than 2 cm in greatest dimension
N2	Metastasis with a lymph node mass more than 2 cm but not more than 5 cm in greatest dimension; or more than 5 nodes positive, none more than 5 cm; or evidence of extranodal extension of tumour
N3	Metastasis with a lymph node mass more than 5 cm in greatest dimension
Pn - Regional Lymph Nodes - Pathological	
pNX	Regional lymph nodes cannot be assessed
pNO	No regional lymph node metastasis
pNI	Metastasis with a lymph node mass 2 cm or less in greatest dimension and 5 or fewer positive nodes, none more than 2 cm in greatest dimension
pN2	Metastasis with a lymph node mass more than 2 cm but not more than 5 cm in greatest dimension; or more than 5 nodes positive, none more than 5 cm; or evidence of extranodal extension of tumour
pN3	Metastasis with a lymph node mass more than 5 cm in greatest dimension

M-Distant Metastasis	
	MX Distant metastasis cannot be assessed **M0** No distant metastasis **M1** Distant metastasis ** **M1a** Non-regional lymph node{s) or lung metastasis **M1b** Distant metastasis other than non-regional lymph nodes and lung

S - Serum Tumour Markers (Pre-chemotherapy)			
sx	Serum marker studies not available or not performed		
so	Serum marker study levels within normal limits		
	LDH (U/l)	hCG (mIU/mL)	AFP (ng/mL)
SI	< 1.5 x N and	< 5,000 and	<1,000
S2	1.5-10 x Nor	5,000-50,000 or	1,000-10,000
S3	> 10 x N or	> 50,000 or	>10,000

N indicates the upper limit of norma/.
LDH = lactate dehydrogenase; foCG = human chorionic gonadotrophin; AFP - alpha-fetoprotein.
1 Except for pTis and pT4, where radical orchidectomy is not always necessary for classification purposes, the extent of the primary tumour is assessed in the radical orchidectomy specimen; see pT. In other circumstances, TX is used if no radical orchidectomy has been performed.
** The current "Carcinoma in situ"nomenclature is replaced by GCNIS.*
** AJCC eighth edition subdivides T1 Pure Seminoma by Tla and Tib depending on size no greater than 3 cm or greater than 3 cm in greatest dimension [29].*
w AJCC eighth edition considers hilar soft tissue invasion and epididymal invasion as pT2, while the discontinuous involvement of the spermatic cord is considered as pMI [29].
** As updated in the U/CC 8th Edition Errata [28bJ.*

Table 10.7: Prognostic groups for testicular cancer (UICC, 2016, 8th edition.)

Stage grouping	T	N	M	S
Stage 0	pTis	N0	M0	S0
Stage 1	pT1-T4	N0	M0	sx
Stage IA	PT1	N0	M0	S0
Stage IB	pT2 - pT4	N0	M0	S0
Stage IS	Any pT/TX	N0	M0	S1-3
Stage II	Any pT/TX	N1-N3	M0	SX
Stage IIA	Any pT/TX	N1	M0	S0
	Any pT/TX	N1	M0	S1
Stage MB	Any pT/TX	N2	M0	S0
	Any pT/TX	N2	M0	S1
Stage IIC	Any pT/TX	N3	M0	S0
	Any pT/TX	N3	M0	S1

Stage grouping	T	N	M	S
Stage III	Any pT/TX	Any N	M1a	SX
Stage IIIA	Any pT/TX	Any N	M1a	S0
	Any pT/TX	Any N	M1a	S1
Stage II IB	Any pT/TX	N1-N3	M0	S2
	Any pT/TX	Any N	M1a	S2
Stage IIIC	Any pT/TX	N1-N3	M0	S3
	Any pT/TX	Any N	M1a	S3
	Any pT/TX	Any N	M1b	Any S

Management

- Based on staging and histological type
- Modalities utilized are surgery, chemotherapy and radiotherapy.
- Outcomes are based on selective criteria that determine prognostic grouping

Table 10.8: Prognostic-based staging system for metastatic germ cell cancer (IGCCCG)

Good-prognosis group	
NSGCT 5-year PFS 90% 5-year survival 96%	All of the following criteria: ● Testis/retro-peritoneal primary ● No non-pulmonary visceral metastases ● AFP < 1,000 ng/mL ● p-hCG < 5,000 IU/L (1,000 ng/mL) ● LDH < 1.5 x ULN
SGTC 5-year PFS 89% 5-year survival 95%	All of the following criteria: ● Any primary site ● No non-pulmonary visceral metastases ● Normal AFP ● Any (3-hCG ● Any LDH
Intermediate-prognosis group	
NSGCT 5-year PFS 78% 5-year survival 89%	Any of the following criteria: ● Testis/retro-peritoneal primary ● No non-pulmonary visceral metastases ● AFP 1,000-10,000 ng/mL or ● p-hCG 5,000 - 50,000 IU/L or ● LDH 1.5-10 x ULN
SGCT 5-year PFS 79% 5-year survival 88%	All of the following criteria: ● Any primary site ● Non-pulmonary visceral metastases ● Normal AFP ● Any p-hCG ● Any LDH

Poor-prognosis group	
NSGCT 5-year PFS 54% 5-year survival 67%	Any of the following criteria: ● Mediastinal primary ● Non-pulmonary visceral metastases ● AFP > 10,000 ng/mL or ● p-hCG > 50,000 IU/L (10,000 ng/mL) or ● LDH > 10 x ULN
SGCT	No patients classified as poor-prognosis

** Pre-chemotherapy serum tumour markers should he assessed immediately prior to the administration of chemotherapy (same day).*
AFP = alpha-fetoprotein; fl-hCG = human chorionic gonadotrophin; LDH = lactate dehydrogenase;
PFS = progression-free survival.

10.23 Disease Management by Stage

Modalities utilized in management of testicular cancer include:

1. **Surgery-** Surgery is the mainstay of treatment in early /localized disease. (Radical inguinal Orchidectomy with high ligation of spermatic cord when the contralateral testis is normal is the gold standard surgical procedure)
2. **Radiotherapy**- can be given in early disease as an alternative to surgery. It can also be given in the adjuvant setting if there are any histological adverse features e.g. LVI, PNI, close/positive margins, lymph node and RETE testis involvement,
3. **Chemotherapy**- Can be given in all stages of seminomatous disease, if surgery and radiotherapy are not a viable/accessible option. Adjuvant therapy may be required if there are any adverse histological features post orchidectomy.
4. **Active surveillance -** used for early disease which is low risk and for patients who prefer to preserve their fertility. It is done using serial tumor markers, physical exams and cross-sectional imaging.

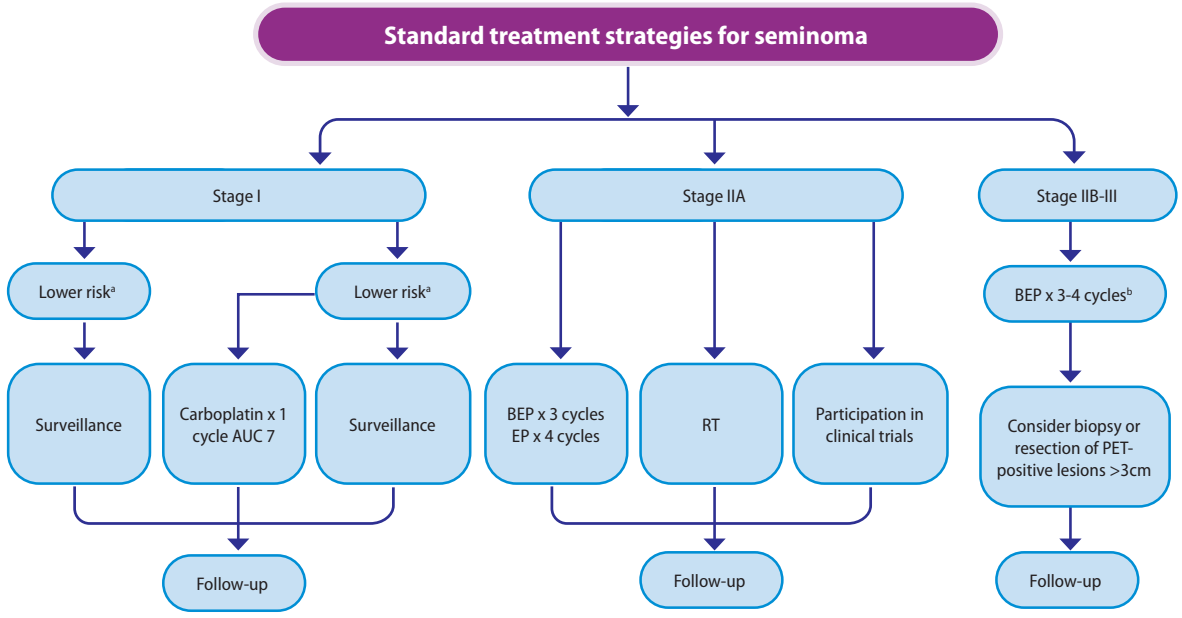


Figure 10.6: Treatment algorithm for seminomatous testicular cancer
(Oldenburg et al., 2022)

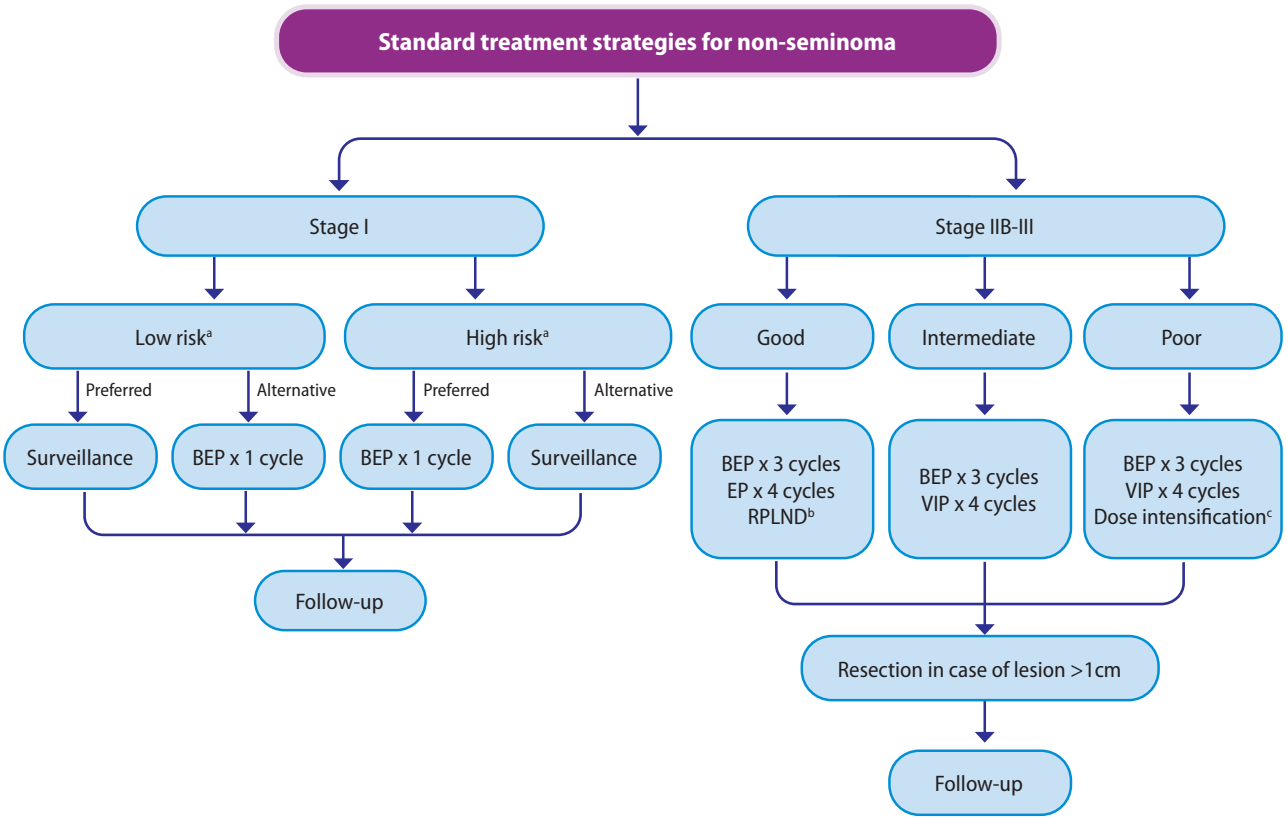


Figure 10.7: Treatment algorithm for non-seminomatous testicular cancer
(Oldenburg et al., 2022)

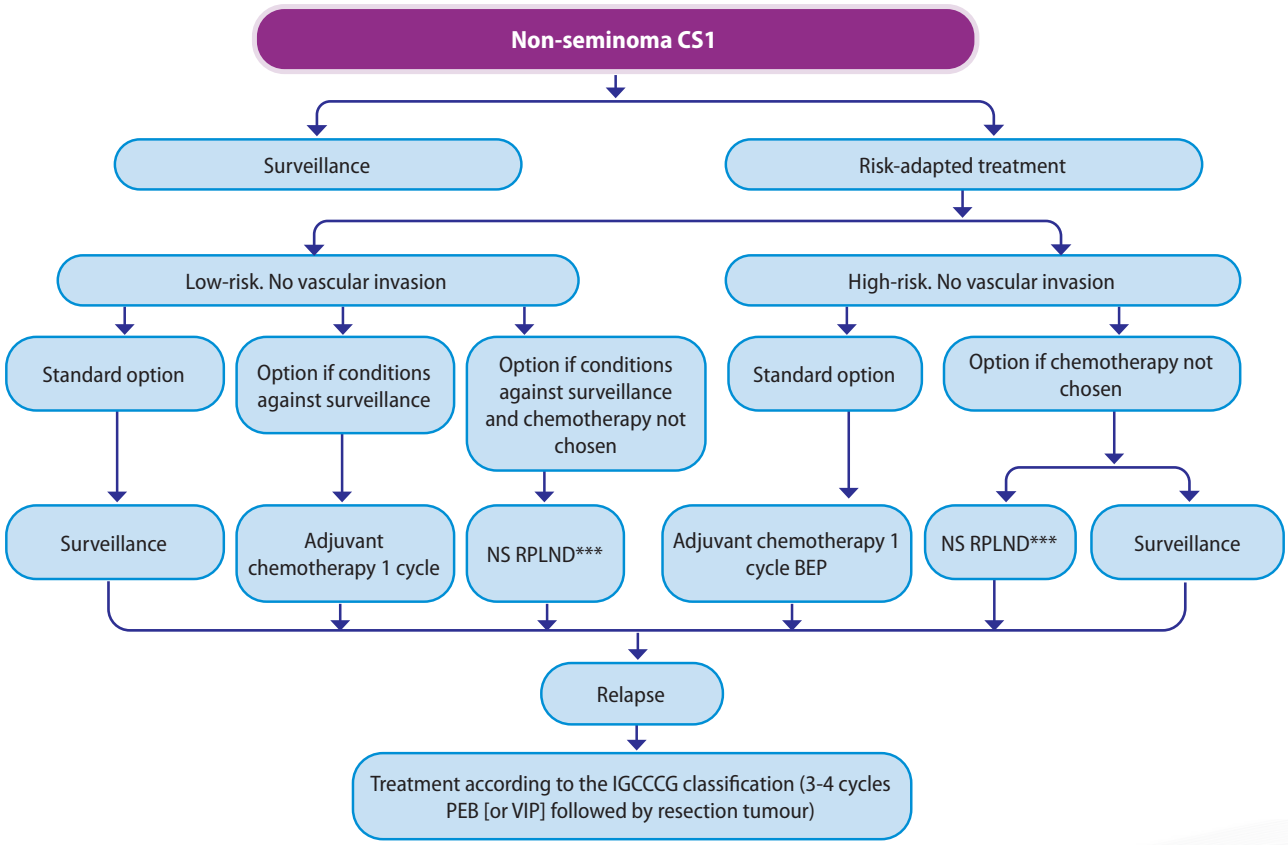


Figure 10.7: Risk-adapted treatment in patients with clinical stage I non-seminoma NSGCT

Follow Up

Table 10.9: Clinical Stage I Seminoma: Surveillance After Orchiectomy (NCCN 2025)

	Year				
	1	2	3	4	5
H&P[a,b]	Every 4-6 mo	Every 6 mo	Every 6-12 mo	Annually	Annually
Abdomen ± Pelvis CT[c] or MRI[d]	Every 4-6 mo	Every 6 mo	Every 6-12 mo	Every 12-24 mo	
Chest X-Ray	As clinically indicated, consider chest CT with contrast in symptomatic patients.				

Table 10.10: Clinical Stage I Seminoma: Surveillance After Adjuvant Treatment (Chemotherapy or Radiation)

	Year				
	1	2	3	4	5[e]
H&P[a,b]	Every 6-12 mo	Every 6-12 mo	Annually	Annually	Annually
Abdomen ± Pelvis CT[c] or MRI[d]	Annually	Annually	Annually	As clinically indicated	
Chest X-Ray	As clinically indicated, consider chest CT with contrast in symptomatic patients.				

Table 10.11: Clinical Stage IIA and Non-Bulky IIB Seminoma: Surveillance After Radiotherapy or Chemotherapy or Both

	Year				
	1	2	3	4	5[e]
H&P[a,b]	Every 3 mo	Every 6 mo	Every 6 mo	Every 6 mo	Every 6 mo
CT[g] or MRI[d] of Abdomen + Pelvis	Annually	Annually	Annually	As clinically indicated	
Chest X-Ray[h]	Every 6 mo	Every 6 mo	As clinically indicated		

Table 10.12: Surveillance After Primary RPLND +/- Adjuvant Chemotherapy for stages II bulky & non-bulky disease

	Year				
	1	2	3	4	5[e]
H&P[a,b]	Every 4 mo	Every 6 mo	Every 6 mo	Every 6 mo	Every 6 mo
CT[g] or MRI[d] of Abdomen + Pelvis	Every 4 mo	Every 6 mo	Annually	As clinically indicated	
Chest X-Ray[h]	Every 4 mo	Every 6 mo	Annually	As clinically indicated	

Non-Seminoma

Table 10.13: Clinical Stage 1 ,II & III NSGCT after RPLND +\- Adjuvant chemotherapy

	Year				
	1[h]	2[h]	3	4	5
H&P and Markers[b]	Every 2 mo	Every 3 mo	Every 6 mo	Every 6 mo	Every 6 mo[k]
CT[c,i] or MRI[d] of Abdomen ± Pelvis	Every 4-6 mo	Every 6-12 mo	Annually	As clinically indicated[l]	
Chest X-Ray[e,j]	Every 4-6 mo	Every 6-12 mo	As clinically indicated[m]		

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10.3 Bladder Cancer

10.3.1 Introduction

Cancer of the urinary bladder is the 9th most common cancer worldwide. It is the 20th most commonly occurring cancer in Kenya (GLOBOCAN 2022). It occurs more in males than females and is the 7th most commonly diagnosed cancer in males. Incidence and mortality rates do vary across countries due to different risk factors but the disease is seen to occur more with Caucasians than Africans. The most common histological diagnosis seen is transitional cell carcinoma. Other variants include primary squamous cell carcinoma, adenocarcinoma and carcinosarcomas.

10.3.2 Risk Factors

- Tobacco use;
- Age >65 years;
- Occupational Exposure to chemicals used in rubber, dye, textile and print industries, aromatic hydrocarbons and arsenic compounds;
- Exposure to radiation and chemotherapy agents such as cyclophosphamide;
- Chronic bladder problems such as bladder stones, long term catheterization and chronic cystitis such as infection with Schistosomiasis.

10.3.3 Clinical Presentation

Symptoms:

Haematuria, lower urinary tract symptoms consisting of hesitancy, urgency, dribbling, dysuria and pelvic pain. Symptoms associated with anaemia are common. In addition, advanced disease would present with lower limb swelling, bone pains and systemic

Signs:

There will be frank haematuria, and pelvic tenderness.

Diagnostic Testing:

Urinalysis and urine microscopy, with description of lower urinary tract haemorrhage, where red cell casts are not identified.

Recommendation on Referral and Care:

Patients presenting with haematuria, where there is suspicion of bladder cancer, should be urgently referred to a qualified urologist for assessment and care, including diagnostic cystoscopy and biopsies.

Diagnosis

Diagnostic confirmation is performed on cystoscopy biopsies; these are examined by qualified pathologists. The histologic classification, grading of the tumor staging through identification of smooth muscle involvement is essential for guiding therapy. Open bladder biopsy or excision is contraindicated.

Imaging

- Kidney, Ureter and bladder ultrasound – if negative, further imaging and cystoscopy still needed.
- CT Urogram is essential in assessing the entire urinary tract for tumor
- CT scan of the chest, abdomen and pelvis (Staging)
- MRI of the pelvis; if utilized should be done before resection of bladder tumor (TURBT)
- Cystoscopy and transurethral resection of bladder tumor (TURBT) – for local
- staging of disease.
- Consider bone scan (if symptomatic or elevated alkaline phosphatase).

Other Tests

- Urinalysis and urine microscopy
- Urine Cytology (25ml of a fresh urine sample with adequate fixation)
- FBC, liver function tests and renal function tests.

Staging

Use the TNM Staging (AJCC version 8). Once diagnosis is done, risk stratification is used to guide in treatment. Bladder cancer risk stratification is divided into low, intermediate, high and very high risk.

Treatment

These regimens can be used in the neoadjuvant, adjuvant and metastatic setting. Decision making is based on MDT discussion.

Non muscle invasive disease (CIS, Ta, T1 disease)

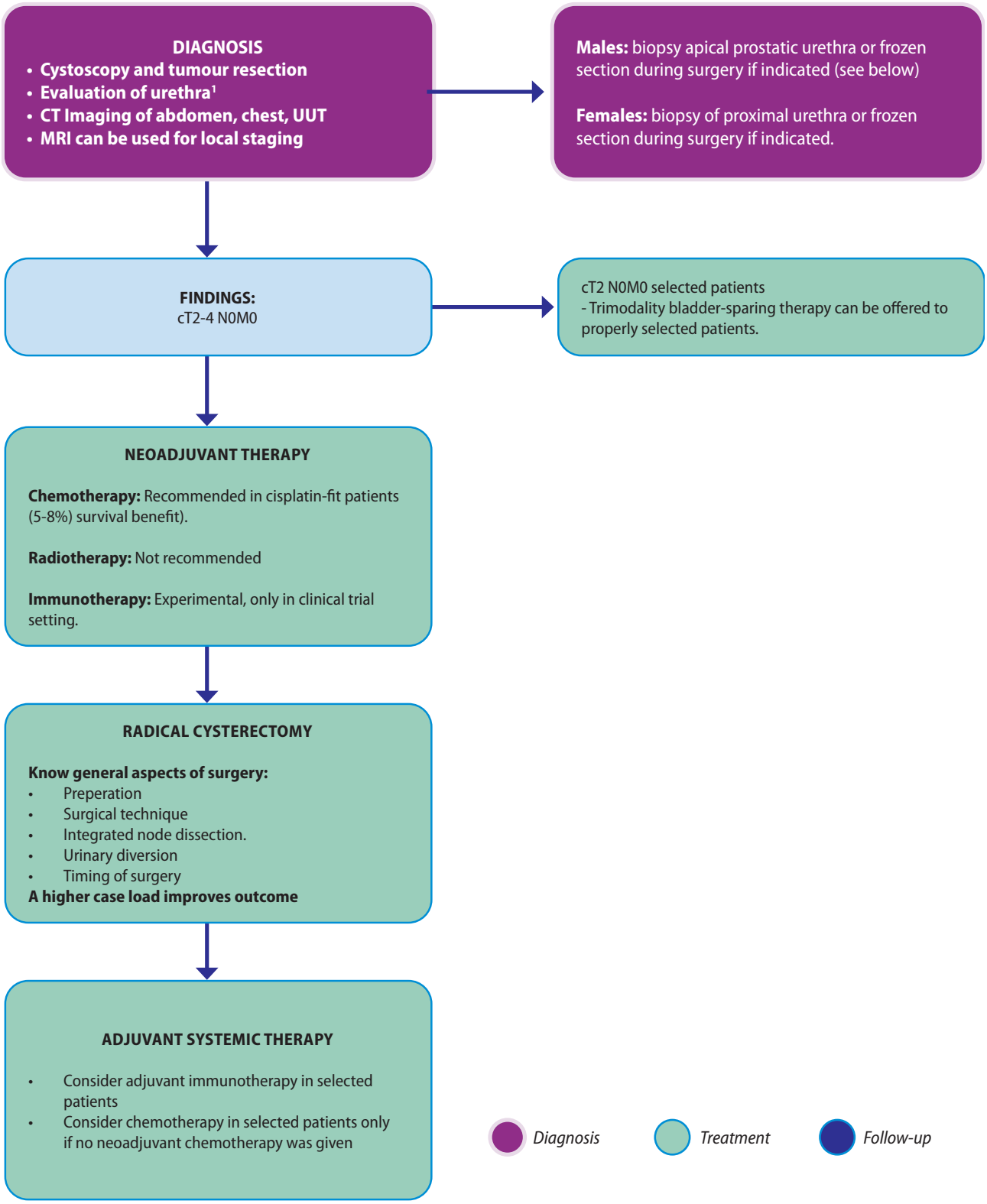
- Complete TURBT, followed by Intravesical treatment options, which include intravesical chemotherapy, BCG.

Muscle invasive disease (T2 - T4)

- Radical cystectomy with lymph node dissection, and urine diversion.
- Neoadjuvant and adjuvant chemotherapy may be considered in high volume disease.
- Trimodal therapy consisting of radiation, chemotherapy and TURBT may be considered where organ preservation is preferred.

Metastatic Disease

- Systemic therapy consisting of chemotherapy or immunotherapy.
- First line agents are Enfortumab vedotin and Pembrolimumab.



CT = computed tomography; MRI = magnetic resource imaging; UUT = upper urinary tract

Figure 10.8: Options for Muscle Invasive Disease

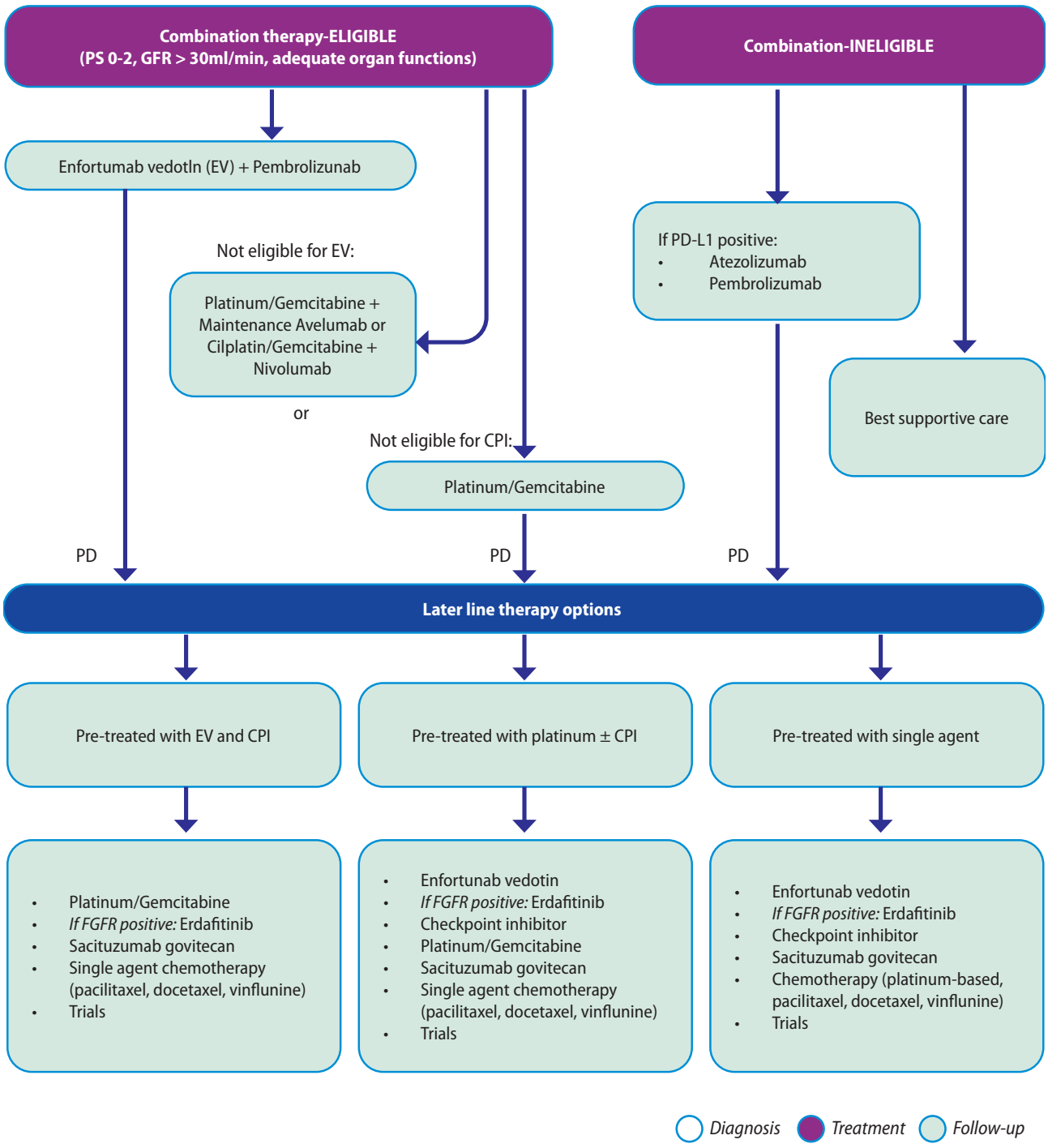


Figure 10.9: Systemic therapy - Flow Chart for management of Metastatic Disease.

Follow Up & Surveillance

Non-muscle-invasive bladder cancer (NMIBC) patients will experience an early recurrence; an initial follow-up cystoscopy is necessary for all patients. Thus a 3-4 monthly followup is recommended after initial treatment. This allows doctors to check for cancer return and confirm whether the initial tumor was fully removed.

10.3.4 References

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Chapter 11

Childhood Cancers

List of abbreviations

Abbreviation	Definition
ALL	Acute Lymphoblastic Leukaemia
AML	Acute Myeloid Leukaemia
ATRA	All Trans-retinoic Acid
BEP	Bleomycin Etoposide Cisplatin
BMA	Bone Marrow Aspirate
BMI	Body Mass Index
CCNU	Lomustine
CCRT	Concurrent Chemotherapy and Radiation Therapy
CNS	Central Nervous System
CRT	Chemoradiation
CSI	Cranio-spinal Irradiation
CXR	Chest X-ray
DNA	Deoxyribonucleic Acid
EBV	Epstein-Barr Virus
EUA	Examination Under Anaesthesia

Abbreviation	Definition
FAB	French-American-British Classification
GCT	Germ Cell Tumors
IQ	Intelligence Quotient
JPA	Juvenile Pilocytic Astrocytoma
LDH	Lactate Dehydrogenase
LFTs	Liver Function Tests
MDS	Myelodysplastic Syndrome
MSR	Maximal Safe Resection
PNET	Primitive Neuroectodermal Tumors
RB	Retinoblastoma
RMS	Rhabdomyosarcoma
UECr	Urea Electrolyte Creatinine
VShunt	Ventriculoperitoneal Shunt
WBC	White Blood Cell
WBRT	Whole Brain Radiotherapy



11.1 Introduction

Childhood cancer represents cancers diagnosed in children between 0 and 19 years of age. The leading childhood cancers globally include, acute leukaemias, brain tumors and lymphomas. The Global Cancer Observatory (GLOBOCAN) provides an estimate incidence of childhood cancers in all countries, and estimates Kenya to have had 2294 childhood cancer cases in 2022.

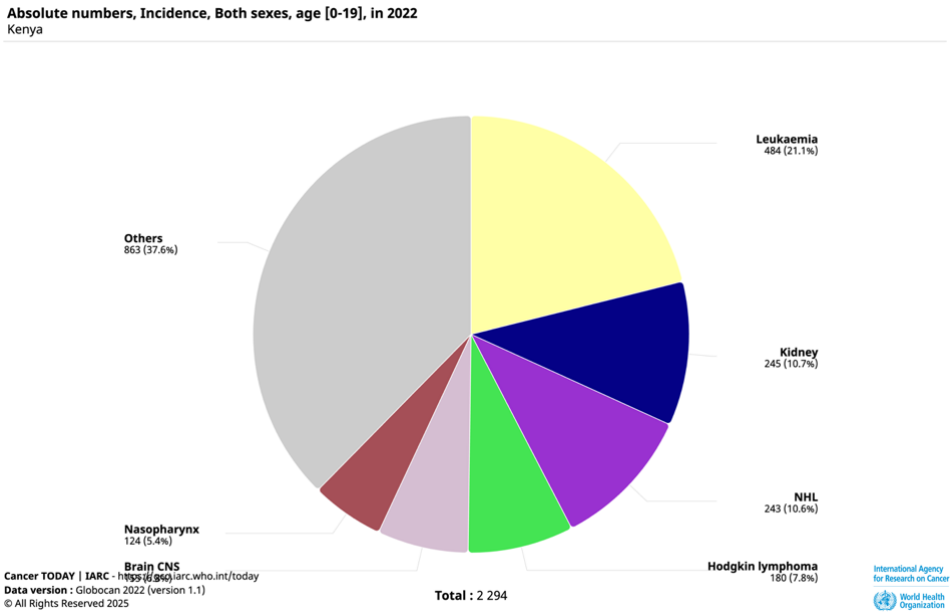


Figure 11.1: Estimate Incidence Of Childhood Cancers In Kenya

Data from one of the main treatment centres in Kenya (Moi Teaching and Referral Hospital) has shown the commonest diagnosed cancers to be NHL, ALL and Wilm's tumor. In the same Centre, survival rate was 32%, in comparison to an average survival rate of >80% in high income countries.

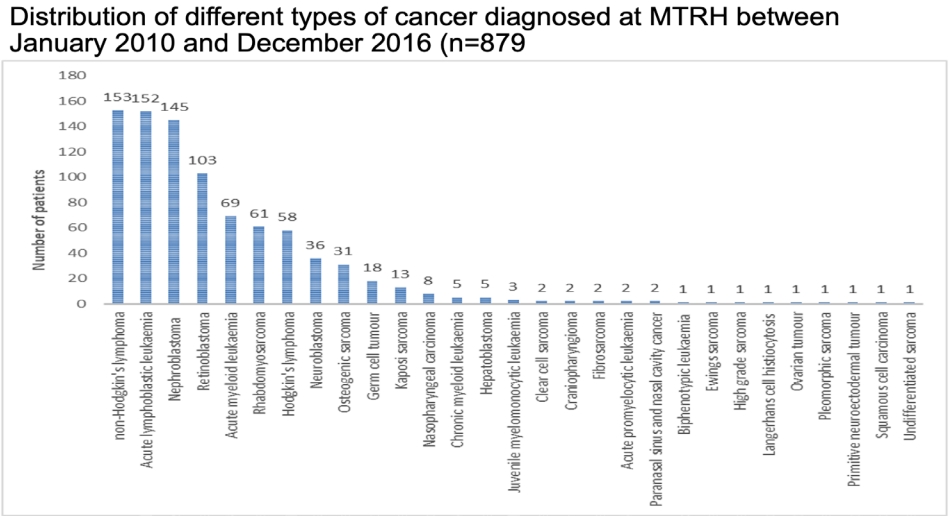


Figure 11.2: distribution of different types of cancer diagnosed at MTRH between January 2010 and December 2016.(n=879)

11.2 Leukemias

11.2.1 Acute Lymphoblastic Leukaemia (ALL)

Introduction

Globally and in Kenya, it is the most common cancer in children, with peak age at diagnosis at 2 to 5 years, and 85% of patients diagnosed between ages 2 and 10 years.

Risk factors

For most cases, no risk factor can be identified and the cause remains unknown. Where risk factors are identified, they include:

- Exposure to ionizing radiation
- Genetic conditions:
  - Identical twins (20% increased risk in 2nd twin if the first twin develops leukemia in the first 5 years of life)
  - Chromosomal abnormalities such as Trisomy 21(Down syndrome)
  - Genetic conditions such as ataxia telangiectasia, Li-Fraumeni syndrome, neurofibromatosis, diamond-Blackfan anemia, Bloom syndrome

Most cases of leukemia are associated with somatic genetic alterations and not inherited genetic predisposition.

Signs and Symptoms

- Fever (61%)
- Recurrent infections
- Arthralgia
- Anemia and lethargy (50%)
- Bleeding tendencies (petechiae, purpura, ecchymoses) (48%)

Clinical features of extramedullary involvement include:

- Bone (and joint pains) (23%)
- Lymphadenopathy (50%)
- Splenomegaly (63%)
- Hepatosplenomegaly (68%)
- Central Nervous System: headache, cranial nerve palsies, convulsions, vomiting and hyperphagia
- Testicular and ovarian involvement: painless enlargement of the testis
- Renal involvement: occasionally hematuria and hypertension.

Laboratory tests and features

1. Total Blood Count: combinations of anemia (usually normocytic normochromic), thrombocytopenia and leucopaenia, leukocytosis or normal leucocyte count.
2. Peripheral blood smear: Lymphoblasts are often present, may be absent especially in leucopaenia

3. Bone marrow aspirate/ biopsy is confirmatory:
   a) Morphological identification of >20% blast cells.
   b) Flow cytometry on the aspirate/ peripheral blood, is important in differentiation of precursor B-cell ALL from, T-cell ALL or AML
   c) Immunohistochemistry on the bone marrow biopsy
   d) Cytogenetics to identify genetic abnormalities

4. Cerebrospinal fluid: Important in all leukemia, to diagnose CNS disease when there are >5lymphoblasts/microliter and/or presence of CNS signs and symptoms.
5. Blood chemistry: Uric acid, LFTs, blood urea, electrolytes, creatinine, calcium, phosphorus prior to starting chemotherapy.
6. Infectious disease profile: HIV screening.
7. Grouping and crossmatch

Imaging

1. Chest X-ray : may show a mediastinal mass especially in T-cell ALL
2. Echocardiogram is required due to use of anthracyclines for treatment.

Treatment

Urgently refer all suspected cases of ALL to childhood cancer treatment facilities for further investigation and treatment.
Selection of treatment protocol depends on prognostic features as shown below:

Table 11.1: Risk stratification (ALL)

Factor	Favourable (Standard risk)	Less favourable (High risk)
Age in years	1-9	<1 or >9
WBC count(x109/L)	<50	>50
CNS involvement	absent	present
Immunophenotype	B-precursor cell	T-cell
Genetics	Hyperdiploidy >50 chromosomes or DNA index>1.16	Hypodiploidy <44 chromosomes or DNA index <0.81
	Trisomy 4,10 and 17	MLL rearrangement
	t(12;21)/ ETV6-RUNX1	Philadelphia chromosome = t(9;22)/ BCR-ABL1 iAMP21

The most important prognostic factor is response to treatment and minimal residual disease at the end of induction. Patients who are not in remission at the end of induction therapy and those who relapse early in maintenance phase have a poor prognosis and together with other High-risk group should be fast tracked for bone marrow transplant when feasible.

The phases of treatment of Acute Lymphoblastic Leukemia include:

Remission induction: This may include a steroid pre-phase from Day 1 to 7. Do a peripheral blood blast count on Day 8. Patients with standard risk disease receive a 3-drug induction that includes Prednisone, Vincristine and L-asparaginase while patients with high-risk disease will have *anthracycline added to standard therapy. An end of induction BMA and assessment for Minimal Residual disease (MRD) should be done at end of induction. Children in remission should proceed to consolidation. For those not in remission, it is important to counsel the family and consider supportive care only, relapse protocol or BMT where feasible.

Central Nervous System directed prophylaxis/treatment: is administered from induction to maintenance phase

- CNS negative disease at diagnosis: intrathecal methotrexate and hydrocortisone from induction to maintenance phase. These do not require craniospinal irradiation.
- CNS positive disease at diagnosis: weekly intrathecal 3-drug (Methotrexate, Cytosar, Hydrocortisone) until first two negative CSFs then continue with protocol. These should receive craniospinal irradiation.
- CNS relapse during treatment : Give weekly triple intrathecal drugs until first two negative CSFs and continue with protocols and give craniospinal irradiation.

Consolidation: This phase has the intensity increased with different drugs to eradicate disease in the sanctuary sites (brain, testes or ovaries) and obtain more cure with continued CNS prophylaxis / treatment. The drugs used are cyclophosphamide, cytarabine, 6MP

Interim Maintenance phase: This is usually with IV methotrexate and 6-mercaptopurine meant to allow bone marrow recovery pending re-intensification.

Delayed intensification (Reinduction and Reconsolidation): This phase aims to intensify therapy before maintenance phase. It is one phase that has been shown to improve survival in the treatment of acute lymphoblastic leukemia. The drugs used are vincristine, *anthracyclines, dexamethasone, asparaginase

Long-term Maintenance. The optimal duration has been shown to be 2 years. The drugs used are IV vincristine, oral steroids, 6MP and methotrexate.

***For anthracyclines, daunorubicin is preferable to doxorubicin.**

Follow up

Monthly follow up with clinical examination and full blood count for the first 3 months, every 3 months until the end of the first year then twice yearly till 3 years and then annually. Further work up should be based on clinical findings.

Vaccination is recommended after 3 months of treatment completion, for those who did not complete KEPI vaccination schedule.

A comprehensive multidisciplinary follow-up is recommended and should include rehabilitation, nutritional and psychosocial support.

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11.3 Acute Myeloid Leukaemia (AML)

11.3.1 Introduction

AML constitutes 15-20% of all childhood acute leukemia cases. Survival rates have in- creased over the past few decades to about 70% for low-risk AML in high income countries, due to improved supportive care, optimized risk stratification and intensified chemotherapy. In most children, AML presents as a de novo entity, but in a minority, it is a secondary malignancy.

11.3.2 Risk Factors

For most cases, no risk factor can be identified and the cause remains unknown. Where risk factors are identified, they include:

- Genetic factors include Trisomy 21, Fanconi Anaemia, Bloom Syndrome, Diamond-Blackfan anaemia, and Noonans syndrome
- Environmental factors include treatment with alkylating agents, ionizing radiation exposure.

11.3.3 Clinical Features

The common presenting features include fever, anemia, unusual bleeding and bruising, petechiae, ecchymoses, gingival hypertrophy, chloromas, frequent or persistent infections and other symptoms depending on systems affected by infiltrating disease.

11.3.4 Investigations

1. Total Blood Count: combinations of anemia (usually normocytic normochromic), thrombocytopenia and leucopaenia, leukocytosis or normal leucocyte count.
2. Peripheral blood smear: Myeloblasts are often present, may be absent especially in leucopaenia
3. Bone marrow aspirate/ biopsy is confirmatory:
   a) Morphological identification of >20% blast cells.
   b) Flow cytometry on the aspirate/ peripheral blood, is important in differentiation of precursor B-cell ALL from, T-cell ALL or AML
   c) Immunohistochemistry on the bone marrow biopsy
   d) Cytogenetics to identify genetic abnormalities
4. Cerebrospinal fluid: Important in all leukemia, to diagnose CNS disease when there are >5 blasts/microlitre and/or presence of CNS signs and symptoms.
5. Blood chemistry: Uric acid, LFTs, blood urea, electrolytes, creatinine, calcium, phosphorus prior to starting chemotherapy.
6. Infectious disease profile: HIV screening.
7. Grouping and crossmatch
8. Echocardiogram is needed due to use of anthracyclines

11.3.5 Prognostic Factors

The prognostic factors are both patient and disease-related characteristics:

Patient Characteristics

- Patients with Down's syndrome have a better prognosis
- BMI of >95 percentile confers a poor prognosis

Clinical Characteristics

- WBC count: a high count confers a poorer prognosis
- Response to therapy: Early response which is usually assessed after the first course confers a good prognosis
- Cytogenetic/Molecular characteristics: Favorable; t(8:21), inv(16), t(15:17) (the latter only if ATRA and/or ATO is used), Unfavorable (Monosomy 5, 7, MLL gene rearrangements among others).

11.3.6 Treatment

Treatment consists of a combination of intensive anthracycline and cytarabine-containing chemotherapy and stem cell transplantation in selected genetic high-risk cases or poor responders. In general, phases in the treatment of AML include:

1. **Induction:** The induction phase consists of two cycles of chemotherapy. Intrathecal chemotherapy is also administered with each cycle as prophylaxis/treatment. The drugs used are daunorubicin and cytarabine. Remission assessment is done using BMA for morphology and immunophenotyping.

2. **Consolidation:** Patients in remission are given an additional 2 cycles of high dose Cytarabine. (HiDAC)

Those not in remission are referred for best supportive care, palliative chemotherapy or Hematopoietic Stem Cell Transplant (HSCT).

Meticulous supportive care which includes infection control and timely availability of blood components is required for these patients.

Treatment of APML is based on Sanz risk stratification to high risk (WBC >10K) and low risk disease.

Induction

- Low risk- Drugs used are ATRA (all-trans retinoic acid) and ATO (Arsenic trioxide) . In the absence of ATO, use an anthracycline
- High risk-Drugs used are ATRA, ATO and an anthracycline
- Intrathecal chemotherapy for high-risk disease

Consolidation

Drugs used are ATRA, ATO and Cytarabine

11.3.7 Maintenance

Drugs used are ATRA, 6MP and oral methotrexate, for 2 years.

11.3.8 Follow-up Care

Monthly follow up with clinical examination and full blood count for the first 3 months, every 3 months until the end of the first year then twice yearly till 3 years and then annually. Further work up should be based on clinical findings.

Vaccination is recommended after 3 months of treatment completion, for those who did not complete KEPI vaccination schedule.

A comprehensive multidisciplinary follow-up is recommended and should include rehabilitation, nutritional and psychosocial support.

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11.4 Non-Hodgkin Lymphoma

11.4.1 Introduction

NHLs are the 3rd most common cancers in children globally.

They arise in peripheral lymphatic structures, mostly lymph nodes and have a high propensity to disseminate and all organ systems may be involved. If the bone marrow infiltration is 25% or more, the disease by definition will be referred to as leukemia. Childhood NHL are almost exclusively aggressive malignant lymphomas. The table below summarizes the histologic categories of Non-Hodgkin Lymphoma in Children and Adolescents.

Table 11.2: Classification of Non-Hodgkins Lymphoma

WHO classification	Lymphoma immunophenotype	Clinical presentation	Prevalence (Global)
Lymphoblastic Lymphoma*	T cell Pre B cell	Mediastinal Bone marrow, skin, Bone	30%
Burkitt and Burkitt like Lymphomas	Mature B cell	Intra-abdominal (sporadic), jaw (endemic), head and neck (non-jaw, sporadic) bone marrow, CNS	40%
Diffuse Large B Cell Lymphoma	Mature B Cell	Nodal, abdominal, bone, primary CNS, mediastinal	10-15%
Anaplastic large cell-lymphoma, systemic	T cell or Null Cell	Variable but systemic symptoms often prominent	10-15%

**Lymphoblastic Lymphoma should be treated on high risk ALL protocol.*

11.4.2 Burkitt's Lymphoma

Risk Factors

- Infectious agents: HIV infection, coinfections with Epstein–Barr virus (EBV) and Plasmodium falciparum malaria early in life
- Congenital immunodeficiency disorders such as severe combined immunodeficiency, hypogammaglobulinaemia, common variable immunodeficiency, Wiskott-Aldrich syndrome, and ataxia-telangiectasia.
- Environmental exposures

Clinical Presentation

Burkitt's lymphoma often presents as a rapidly-growing painless mass at extra nodal sites commonly on the jaw but can also occur as a progressive abdominal swelling, in the mediastinum and to a lesser extent in the bone. CNS involvement can occur and present as headaches, cranial nerve palsies or paraplegia. It

is a fast-growing tumor and the tumor size can double within a 48-hour period. It thus requires urgent diagnostic and treatment intervention.

Table 11.3: The St. Jude's Staging System

Stage 1	A single tumor (extranodal) or single anatomical area (nodal), not mediastinum or abdomen.
Stage II	Single tumor (extranodal) with regional node involvement, Two or more nodal areas on the same side of the diaphragm, A primary GIT tumor with or without involvement of mesenteric nodes.
Stage III	Two single tumors or nodal areas on opposite sides of the diaphragm, Primary intrathoracic tumors, extensive intra-abdominal disease, and all paraspinal or epidural tumors.
Stage IV	Any of the above with initial CNS and/or bone marrow involvement.

It should be handled as an oncological emergency with prompt confirmation of diagnosis and staging before start of treatment since it is a fast-growing tumor.

Evidence shows no advantage in performing extensive surgery or debulking, since tumor re-growth is fast and surgery delays and may complicate chemotherapy.

Prognostic factors include the stage and LDH level.

Investigations

1. **Diagnosis: Take tissue biopsy for histology, immunophenotyping and cytogenetics.**
2. **Staging:** CXR, abdominal ultrasound, CT of primary site, chest, abdomen and pelvis, CSF cytology, bone marrow aspiration or trephine biopsy for evaluation of marrow involvement,
3. **Laboratory:** HIV status, CBC, LDH, LFTs, uric acid, UECs, calcium, magnesium and phosphorus levels.

Treatment

Multi-drug chemotherapy is the mainstay of treatment. Treatment involves 6-8 cycles of Cyclophosphamide, Vincristine, Prednisone, etoposide, high dose Methotrexate with folinic acid rescue and high dose Cytarabine. CNS directed therapy is also essential with IT Methotrexate and Hydrocortisone (add IT Cytarabine in confirmed CNS disease).

Supportive Care

Supportive care is important in the management of Burkitts lymphoma prior and during treatment.

Tumor Lysis Syndrome Prevention:

- Start allopurinol 10 mg/kg/dose TDS upon diagnosis, at least 1 day before starting chemotherapy.
- Start hyper hydration 12-24 hours before chemotherapy with 3000 ml/m2/day normal saline till 48 hours after chemotherapy and then continue hydration at maintenance level. Initial aim is adequate urine output of >3ml/kg/hr. which must be achieved before start of chemotherapy.

- Check daily weight; and creatinine and electrolytes at 24 and 48 hrs. after commencement of chemotherapy.

Spinal cord compression occurs frequently in Burkitts lymphoma. Presenting features include: back pain, incontinence, urinary retention, paraplegia and other neurological deficits.

Increased intracranial pressure may occur due to pleocytosis and lumbar puncture should be performed with caution.

Other emergencies that may occur with Burkitt's lymphoma include **superior vena cava syndrome, mediastinal mass and intestinal obstruction.**

Second line chemotherapy
This will be determined on a case-by-case basis by the oncologist

Follow Up

After chemotherapy, follow up will be done: Monthly for 3 months; then every 3 months for a year; then every 6 months for 2 years then annually thereafter. At follow-up con- duct a clinical examination and other appropriate investigations.

A multidisciplinary follow up plan should be instituted.

11.5 Hodgkin Lymphoma

11.5.1 Introduction

Hodgkin lymphoma is a lymphoid neoplasm of B cell origin composed of large neoplastic binuclear and multinucleate (Reed-Sternberg cell) and their mononuclear variants (Hodgkin cells) surrounded by a variable mixture of mature non-neoplastic inflammatory cells.

11.5.2 Clinical Presentation:

- Lymphadenopathy
- Fever, weight loss, night sweats (B symptoms)
- Fatigue,
- Chest pain, cough, shortness of breath
- Splenomegaly and/or hepatomegaly

Histopathology:

- Lymph node excision biopsy; AVOID lymph node fine needle aspirates
- Histology for Reed-Sternberg cells and variants
- Immunohistochemistry: CD45, CD20, CD15, CD30, CD3

11.5.3 Laboratory Investigation

- Complete blood count
- Serology for HIV
- Liver function tests
- Renal function test (U/E/C)
- Bone marrow biopsy (

Imaging:

- Plain Radiograph (Chest X-ray -posteroanterior and lateral).
- CT scan (neck, chest, Abdomen, pelvis).
- PET (recommended where available).
- Cardiac Echocardiography

WHO classifies Hodgkin's lymphoma as Classical Hodgkin's Lymphoma (CHL) and Nodular Lymphocyte Predominant Hodgkin's Lymphoma (NLPHL).

Classical Hodgkin Lymphoma (CHL)

Four histological variants are present based on the characteristic of reactive infiltrate and to a certain extent, the morphology of the Hodgkin/Reed Sternberg cells (HRS).

1. Nodular Sclerosis CHL (most common 80% of cases)
2. Lymphocytic rich CHL (rare in children)

3. Mixed cellularity CHL (20% of cases)
4. Lymphocyte depleted CHL (rare in children)

Nodular Lymphocyte Predominant HL

Malignant cells in NHPHL are called LP cells have distinct characteristics that distinguish them from Reed-Sternberg cells

Principles of Management

Management is based on a risk-adapted treatment approach as tabulated below. Combination of Chemotherapy and Radiation therapy is the preferred management modality. Staging of HL is achieved by use of the Ann Arbor staging system while the Childhood Hodgkin's International Prognostic Score (CHIPS) is used in risk stratification.

Stage	Involvement
1	Single lymphatic region (1) or 1 extra lymphatic site (IE)
II	Two or more lymph node regions on same side of diaphragm (II) or 1 localized extralymphatic site plus 1 or more lymph node regions on same side of the diaphragm (HE)
III	Lymph node regions on both sides of diaphragm (III), which may be accompanied by extralymphatic extension (HIE), splenic involvement (IMS), or both (IIIE#S)
IV	Diffuse involvement of 1 or more extralymphatic organs or sites
Suffix	**Features**
A	No B symptoms
B	Presence of at least one of the following: unexplained weight loss >10% of usual body weight in 6 months before diagnosis; unexplained fever >38°C; drenching night sweats
X	Bulky tumor is defined as either a single mass of tumor tissue exceeding 10 cm in largest diameter or a mediastinal mass with size greater than or equal to one-third of the maximum intrathoracic diameter measured on a standard posterior-anterior chest radiograph

Data from Armitage JO. Staging non-Hodgkin lymphoma. CA Cancer J Clin 2005;55(6):368-76; and Connors JM. State-of- the-art therapeutics: Hodgkin's lymphoma. J Clin Oncol 2005;23(26):6400-8.

Figure 11.4 : Ann Arbor Staging for Hodgkin's Lymphoma

Table 11.5: Risk Group stratification for Hodgkin Lymphoma

Risk group	Stage I	Stage II	Stage III	Stage IV
Low	A	A		
Intermediate	Any of B, E OR X	Any of B, E OR X	A	A
High			B	B

11.5.4 Treatment For Hodgkins Lymphoma

Treatment modalities: Chemotherapy ± Radiation therapy
Risk stratification and response-adapted therapy guide intensity of chemotherapy and whether radiation is indicated.

1. Limited Stage (stage IA, IB or IIA and the absence of bulky disease)

- 2 cycles of ABVD followed by CT evaluation
- 2 further cycles of ABVD for limited stage rapid early responders (RER-reduction by 70% then 2 cycles); End of treatment revaluation by CT scan (PET is preferable). No disease-No DXT

If still disease positive (slow early response – (SER-less than 70% reduction) give 4 more cycles (total is 6) of ABVD. End of treatment revaluation by CT scan (PET is preferable). IFRT at end of treatment if imaging remained positive.

2. Advanced Stage (all others)

- 6 cycles of ABVD treatment followed by repeat imaging.
- Followed by IFRT (involved field radiotherapy) to areas of residual disease.
- Interim repeat imaging – after 3 cycles

11.5.5 Relapsed/Refractory Hodgkins Lymphoma

The treatment approach is influenced by the time to relapse and the nature of the initial treatment, but generally with chemotherapy, immunotherapy and stem cell transplant where available.

11.5.6 References

1. Selected Schedules in the Therapy of Childhood Cancers (Gunter Henze and Heinz Wein- Berger, Editors). 2nd Edition 2014; Pages 230- 240. Baxter Oncology GmbH
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11.6 Nephroblastoma (Wilms' Tumour)

11.6.1 Introduction

It is one of the most common abdominal childhood malignancy accountings for 6 - 7% of all childhood malignancies in Kenya. (references)

Peak incidence is between 2 and 6 years and it occurs with an equal frequency in both sexes. About 5% of children present with bilateral disease. Wilms tumour may be associated with hereditary cancer syndromes such as Beckwith-Weiderman syndrome, WAGR syndrome (Wilms tumour, aniridia, malformations of the urinary genital system and mental retardation) Denys-Drash syndrome and neurofibromatosis type 1. The major differential diagnosis for nephroblastoma is neuroblastoma which affects the same age group.

11.6.2 Clinical Presentation

- Abdominal mass
- Abdominal pain
- Haematuria
- Hypertension
- Respiratory symptoms maybe present due to huge abdominal mass or lung metastases.

11.6.3 Diagnostic Workup

Imaging

- Abdomino-pelvic ultrasound
- Abdomino-pelvic CT scan
- CXR or chest CT scan
- Cardiac Echocardiogram

Laboratory Investigations

Complete Blood Count, Liver Function Tests, U/E/Cr, Calcium, Urinalysis, Coagulation profile and Urinary catecholamines.

HIV Screening

Staging

Table 11.6 : The National Wilms Tumor (NWT) study staging system

Stage 1	Tumor within renal capsule - completely resected
Stage II	Tumor extends beyond the kidney but is completely resected
Stage III	Gross or microscopic residual tumor remains post-operatively, spillage of tumor preoperatively or intraoperatively, regional lymph node metastasis
Stage IV	Haematogenous metastasis or lymph node metastasis outside the abdomen (often to lungs and liver)
Stage V	Bilateral tumors

Treatment

Treatment is usually commenced on the basis of clinical presentation and imaging findings of an intrarenal mass on CT scan of the abdomen. **CT scan should be reviewed together by the radiologist and oncologist at the time of initial patient presentation.** Biopsy is not usually required.

A multidisciplinary stage and risk adapted approach is recommended which combines surgery, chemotherapy and radiotherapy.

Initial Treatment Plan

Neo adjuvant chemotherapy

Table 11.7: Neo-adjuvant chemotherapy (Wilms Tumor)

Stage	Chemotherapy
Localized	Vincristine + Actinomycin-D – 4 weeks
Metastatic	Vincristine + Actinomycin-D + Doxorubicin – 6 Weeks
Bilateral	• 6 weeks vincristine/actinomycin-D • If progressive then switch to etoposide/carboplatin

Surgery

Surgery should be done within 1 week of completion of neo-adjuvant chemotherapy. If there is any delay for more than 1 week, give additional dose of vincristine.

Surgical staging should be done intra-operatively, and should include lymph node sampling.

Table 11.8: Pathologic staging of disease (after surgical resection, post-neoadjuvant chemotherapy)

Low-risk	Intermediate-risk	High-risk
No anaplasia	Focal anaplasia	Diffusely anaplastic or blastemal type WT

Post-operative Treatment Plan

This will depend on surgical and pathologic staging and histological risk stratification. Treatment should start by the post-operative day 10.

11.6.4 Adjuvant

Table 11.9: Adjuvant treatment algorithm

Disease	Tumour volume after preoperative chemotherapy	Treatment		
		Stage 1	Stage II	Stage III
Low-risk	All	None	AV (27 weeks)	AV (27 weeks)
Intermediate-risk, all subtypes	<500 ml	AV (4 weeks)	AV (27 weeks)	AV (27 weeks) + flank radiotherapy
Intermediate-risk, stromal or epithelial-type	>500 ml	AV (4 weeks)	AV (27 weeks)	AV (27 weeks) + flank radiotherapy
Intermediate-risk, nonstromal, nonepithelial	>500 ml	AV (4 weeks)	AVD (27 weeks)	AVD (27 weeks) + flank radiotherapy
High-risk blastemal type Wilms tumour	All	AVD (27 weeks)	HR-1 (34 weeks)	HR-1 (34 weeks) + flank radiotherapy
High-risk diffuse anaplasia	All	AVD (27 weeks)	HR-1 (34 weeks) + flank radiotherapy	HR-1 (34 weeks) + flank radiotherapy

AV: Actinomycin-D and vincristine
AVD: Actinomycin-D , vincristine and doxorubicin
HR: alternating cycles of Cyclophosphamide/ doxorubicin and carboplatin etoposide

11.6.5 Radiotherapy

Indications for Radiotherapy

- All stage III & IV including tumour rupture
- Stage II Unfavorable Histology, Anaplastic
- Stage I – CCSK, RTK
- High risk histology (pathology report)

Timing of Radiotherapy

- Referred within 1 week of surgery
- Start RT within 10-14 days of surgery or as soon as possible (tight timeline is more important for Unfavourable Histology)

11.6.6 References

1. Selected Schedules in the Therapy of Childhood Cancers (Gunter Henze and Heinz Wein- Berger, Editors). 2nd Edition 2014; Pages 230- 240. Baxter Oncology GmbH
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11.7 Osteogenic Sarcoma

11.7.1 Introduction

Bone tumours constitute 3-5% of all childhood cancers. Osteosarcoma is a primary malignant mesenchymal bone tumour. It is the most commonly diagnosed primary malignant bone tumour, with incidence peak during the pubescent growth spurt,. While its aetiology is un- known, some factors that correlate with occurrence include: bone growth, genetic factors, and human predisposition syndromes

11.7.2 Clinical Evaluation

- Pain at tumour site
- Local swelling
- Decreased range of motion
- Pathologic fracture

Sites:

- Metaphysis: 90%
- Diaphysis: 8-10%
- Epiphysis: 1%

Occurrence:

- Long Bones: 70%-80%
- Distal Femur (40%)
- Proximal Tibia (20%)
- Proximal Humerus (15%) Axial Skeleton
- Pelvis Jaw

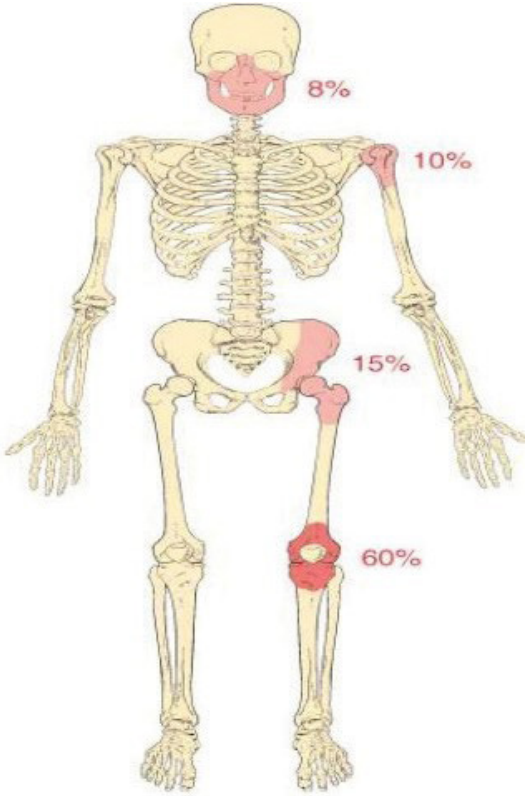


Figure 11.3 : Common sites of involvement of osteosarcoma in children

Diagnosis Initial Imaging

- Plain X-ray of affected limb: characteristic radiological features are sun-burst appearance, periosteal lifting with formation of Codman's triangle, new bone formation in the soft tissues along with permeative pattern of destruction of bone and other features for specific types of osteosarcomas.

Imaging (for diagnosis and staging)

- Chest CT Scan: early detection of small sub-centimetre nodules in the lung.
- MRI: most accurate tool for determining the limits of tumour within and outside the bone. Include the whole of the involved bone with one joint above and below so that skip lesions are not missed in the same bone and across the joint. MRI accurately and precisely delineates (1) extent of the tumour into the soft tissues and the medullary canal, (2) involvement of joint, (3) crossing of the lesion through and/or around the growth plate, (4) any skip lesion in the same bone and across the joint in other bone, (5) proximity/encasement of the neurovascular bundle by the tumour.

Laboratory

Biochemical tests include:

- Serum alkaline phosphatase (ALP): elevated in osteosarcoma due to increased osteoblastic activity. Higher levels are associated with heavy tumour burden and poor prognosis Other tests include: Full Blood count, , BUN, Creatinine, LFT, Pathology:

Biopsy using percutaneous core needle or open biopsy

Other tests will include a baseline audiogram and echocardiogram prior to chemotherapy

Management

Doxorubicin, cisplatin and ifosfamide based protocols - Give cycles of neoadjuvant chemotherapy then perform surgical resection on all sites of disease if possible including an amputation/limb salvage procedures. Adjuvant chemotherapy is also required.

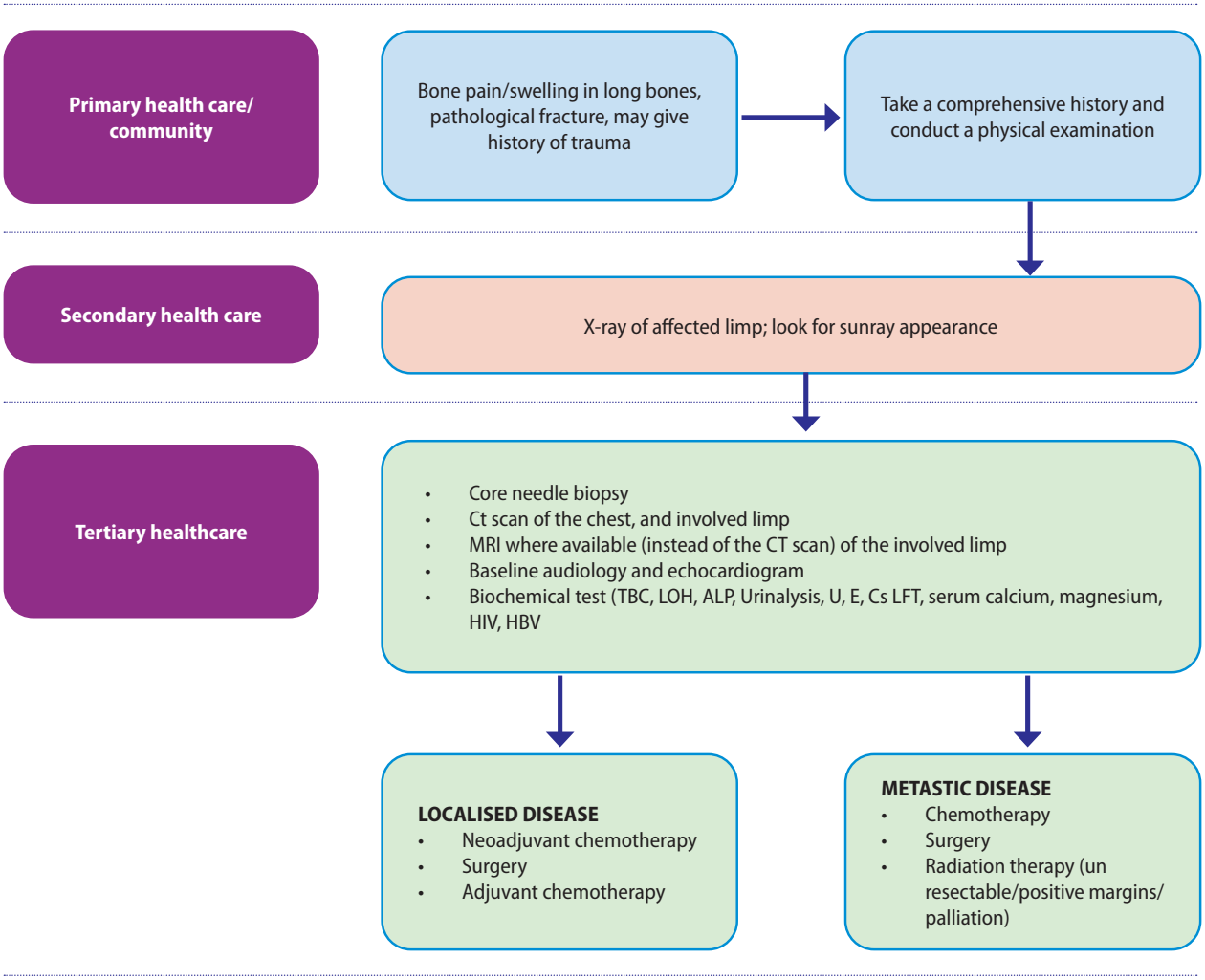


Figure 11.4 : Algorithm for Management of Osteogenic Sarcoma

11.7.3 References

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11.8 Rhabdomyosarcoma

11.8.1 Introduction

It is the most common soft tissue sarcoma in children. It is more common in boys than girls, with bimodal distribution and peak incidence in the 0-4 and 14-18 age ranges. (REF: Ognjanovic S et al Trends in childhood rhabdomyosarcoma)

11.8.2 Clinical Presentation

Head and neck region: signs and symptoms include;

- Orbital- proptosis,
- Mastoid- bloody aural discharge, cranial nerve paralysis and eventually a polypoid mass in the middle ear,
- Nasopharyngeal tumours- airway obstruction, difficulty in mastication and trismus
- Neck tumours- invasion of cervical or brachial plexus with peripheral neurological deficits.

Pelvic region:

- Prostate or bladder area -urinary obstruction
- Scrotum and spermatic cord- scrotal masses.
- In females it commonly presents as the classic sarcoma botryoides (cluster of grapes) of the margins of vagina and uterus. Other sites include gluteal or perianal region, paravertebral and flank muscles, hands, thighs, arms and legs. Metastasis can be seen in the regional nodes, lungs, bone marrow and other bones resulting in pulmonary insufficiency, cachexia, pancytopenia and pneumonia.

11.8.3 Investigations

Tissue biopsy for histology and immunophenotyping. Histological subtypes include embryonal, alveolar, spindle cell, botryoid and pleomorphic.

Imaging with CT scans of the primary site and metastatic sites (Chest, abdomino-pelvic)
Bone marrow aspirate for staging.

Laboratory testing- CBC, Renal function and liver function testing, HIV

Table 11.10: Intergroup Rhabdomyosarcoma Study Group Staging System

STAGE	SITE OF PRIMARY TUMOR	TUMOR SIZE	LYMPH NODES	DISTANT METASTASES
1.	Orbit, non-PM head/neck; GU non-bladder/prostate; biliary tract	Any size	NO, N1	MO
2.	All other sites	<5cm	NO	MO
3.	All other sites	< 5cm	N1	MO
		>5cm	NO orNI	MO
4.	Any site	Any size	NO orNI	MI

Unfavorable sites: Parameningeal areas, kidney, bladder, prostate, extremities
Favorable sites: Orbit, head and neck, genitourinary tract and biliary tract

11.8.4 Clinical Group

The clinical group is based on the extent of the disease and how completely it is re- moved during initial surgery. The groups are defined as follows:

Group I

Localised tumor, completely excised by surgery. Negative margins and no lymph node involvement

Group II

Localised tumor with microscopic positive margins with or without lymph node involvement

Group III

Localised tumor with gross residual disease (post-surgery or biopsy only) Lymph node involvement present but no distant metastases. Group IV
Any tumor size with distant metastases present at diagnosis.

11.8.5 Treatment

Treatment is multi-modal.

Chemotherapy:

Drugs used include Vincristine, Actinomycin D, Adriamycin and Cyclophosphamide (VAC).
4 cycles neoadjuvant followed by local control (Surgery +/- radiotherapy). Adjuvant chemotherapy with 10 cycles of VAC given every 21 days.

Surgery:

Gross total resection, preferably after neoadjuvant chemotherapy.
Upfront mutilating surgery is discouraged.

Radiotherapy:

Patients with microscopic, locoregional or gross residual tumor receive radiation therapy at week 13 of treatment.

Relapse:

Any recurrence whether local, regional or distant has a poor prognosis. It requires a multi-disciplinary approach.

11.8.6 References

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11.9 Retinoblastoma

11.9.1 Introduction

Retinoblastoma (RB) is a hereditary childhood cancer that arises in the immature cells of the retina. It is the most common primary malignant intraocular cancer in children, and it is almost exclusively found in young children less than 5 years. It is curable if detected and treated early and therapeutic approaches need to consider not only the cure of the disease but also the need to preserve vision with minimal long-term side effects.

While in the developed world, most children (over 90%) treated for retinoblastoma survive, in Africa most of them die from advanced disease. RB is associated with the retinoblastoma gene (a tumor suppressor gene), a mutation in the long arm of chromosome 13.

Retinoblastoma presents in two distinct clinical forms:

1. A bilateral or multifocal, heritable form (25% of all cases), characterized by the presence of germ-line mutations of the RB1 gene; and
2. A unilateral or unifocal form (75% of all cases), of which are non-hereditary.

The hereditary disease is diagnosed at an earlier age than the non-hereditary disease.

11.9.2 Clinical Presentation

Patients may present with the following:

- white pupil (leukocoria),
- squint (strabismus),
- proptosis,
- Others: red painful eye orbital cellulitis

Patients with a family history of retinoblastoma should undergo an ophthalmology review from birth.

Physical examination: Conduct a thorough physical examination and examine for white reflex

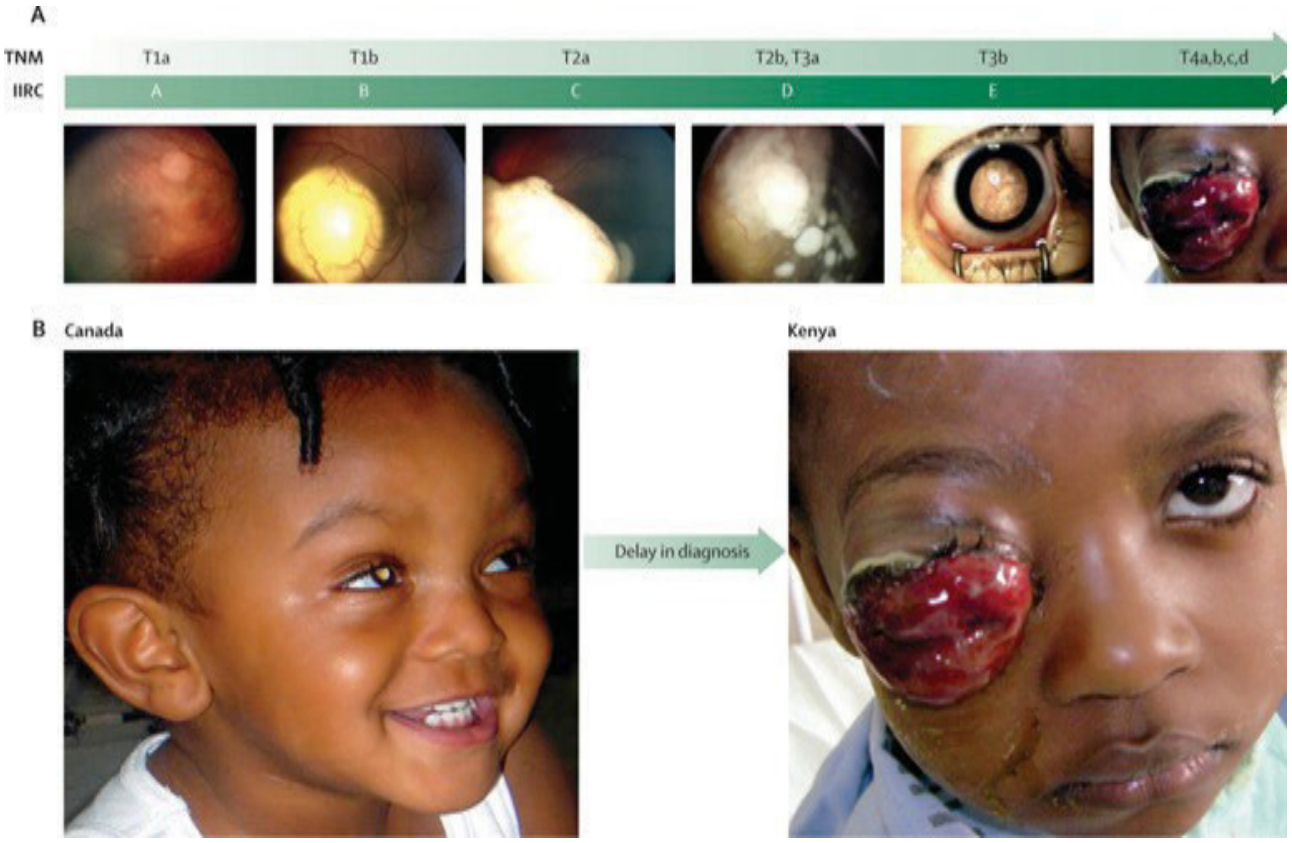


Figure 11.5: Clinical findings for retinoblastoma

Additional test include:

- Imaging
  - Head and orbit MRI
- Laboratory
  - CBC, UECs, LFT, HIV
- Audiogram

11.9.3 Treatment

Should be done at a tertiary center by a multidisciplinary team Couples should undergo genetic counselling. Treatment is individualized to the patient depending on the stage of the disease.

Treatment options include enucleation, cryotherapy, laser therapy, intravitreal chemotherapy. Systemic chemotherapy is given for both initial and recurrent disease utilizing carboplatin, etoposide and vincristine. These are given for a total of 6 cycles every 21 days.

Table 11.11: Treatment of retinoblastoma

Group A	Tumour<3 mm	Local therapy(laser/cryotherapy)
Group B	Tumour >3 mm, clear subretinal fluid	Enucleation and Chemotherapy
Group C	subretinal and vitreous seeds< 3 mm	Enucleation and chemotherapy
Group D	Diffuse seeds >3 mm	Enucleation and chemotherapy and radiotherapy
Group E	Extensive disease involving > 50% of the globe	Enucleation, chemotherapy and radiotherapy

11.9.4 Follow-Up

- Patients who are treated for retinoblastoma should have regular eye exams every 3 months till the age of 5 years to assess for recurrence or development of tumour in the opposite eye.
- All patients should have prostheses put in the enucleated eye
- Visual acuity examination should be done for all school going children and appropriate care given.
- All siblings of children with retinoblastoma should be evaluated by the ophthalmologists from birth till the age of 5 years

11.9.5 References

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11.10 Brain and Central Nervous System Tumours

11.10.1 Introduction

CNS tumours are the 2nd commonest childhood malignancy globally. In Kenya majority remain undiagnosed due to low index of suspicion and lack of diagnostic capabilities . tumours). Most childhood brain tumours occur in the infratentorial region.

11.10.2 Clinical Presentation

- **Intracranial pressure/Obstruction:** headache, vomiting, hydrocephalus -papilledema, sun setting eyes, Cushing's Triad(bradycardia, hypertension, irregular breathing), increased head/ fontanelle size.
- **Focal neurological deficits:** Frontal lobe (personality changes, seizures, headache),Temporal lobe (seizures, speech problems), suprasellar (endocrinopathies, visual problems), thalamus (motor sensory deficits), cerebellar (nystagmus, ataxia, hydrocephalus, resting tremor, incoordination), brainstem (cranial neuropathies, gait, swallowing, incoordination, drooling, breathing dysfunction), posterior fossa (nausea, vomiting).
- **Brain dysfunction:** mental function deterioration, personality changes, seizures.

11.10.3 Diagnosis

- Contrast-Enhanced MRI
- CT scan head and neck (If MRI not available or condition does not allow)

Urgently refer all suspected cases for review to level 5 or 6 facilities with a multidisciplinary team consisting of a paediatrician, peadiatric neurologist, paediatric neurosurgeon, paediatric oncologist, ophthalmologist, radiologist, medical and radiation oncologist among other

Table 11.12: Management Approach For Childhood Brain Tumors

Diagnosis	Grade/Risk Category	Surgery	Chemotherapy	Radiotherapy	Imaging
Cerebellar Astrocytoma	JPA	Maximal safe resection(MSR)	Nil	50.4Gy/28#	MRI Brain
Medulloblas-toma	Average risk	VP shunt +/- MSR	Adjuvant Vincristine/ Cisplatin/ CCNUOR Carboplatin/ Vincristine/ Etoposide	CSI 23.4Gy/13#, Boost 32.4Gy/18#	MR Brain and whole spine, CSF assessment
High Risk	VP shunt +/- MSR	CCRT with Vincristine then adjuvant 6-8 cycles as above	CSI 36Gy/20#, Boost 19.8Gy/11 #	MR Brain and whole spine, CSF assessment	
Ependymoma	WHO Grade 1- 3, Ependymoblast oma(PNET)	MSR	No benefit	54Gy/30# local disease, CSI in disseminated disease	MR Brain and whole spine, CSF assessment
Brainstem Glioma	Diffuse/intrinsic, Low/high grade	Nil	Nil	56Gy/28# on 2D, Consider 54Gy/30# with 3DCRT	Diagnosis on imaging
Intracranial Germ Cell Tumors	Germinoma	Biopsy	CE with extended local RT	WBRT 45Gy/25# or Chemo and 25.2Gy/14#	
Non-Germinomatou s	For residual masses post chemo		BEP/ CE	Focal RT or CSI (metastatic disease) 30.4Gy/19#, Boost 19.8Gy/11 # Post Chemo CSI 24Gy/15#, Boost 30.6Gy/17#	
Hemispheric glioma	MSR	No Gold standard Adjuvant Vinc, CCNU, Prednisolone to be considered	45Gy/25# Large Volume, Boost 14.4Gy/8# (Adjust boost for OAR as needed)	Prognosis depends on resection	
Visual pathway glioma (OPG)	JPA/Fibrillary	Nil (protect visual pathway)	Emerging evidence for Vincristine/ Carbopl atin)	45Gy/30#	Observatio n in NF1 patients
Craniopharyn-gioma	Adamantinoma tous	MSR	Intra Omayo Bleomycin/ interferon alpha	54Gy/30#	Follow up in combined paeds clinic

Key: CCRT: Concurrent Chemotherapy and radiation therapy
CSI : Cranio-spinal radiation
CCNU : Lomustine
MSR : Maximal Safe resection
JPA : Juvenile pilocytic Astrocytoma
Treatment Protocols

11.10.4 Sequelae of Therapy & Follow Up

1. Neurocognitive effects - patients need to be followed up in a combined clinic set up with access to a developmental specialist as they experience a decline in IQ with problems in organization, attention learning and executive functions. Memory impairment and a wide range of fine and visual motor skills can be affected.
2. Neuroendocrine effects- Truncal growth failure should be monitored at every clinic visit (sitting and standing height), hypothyroidism, delayed or precocious puberty and adrenocortical dysfunction should be monitored.
3. Secondary malignancies - mostly occur in the radiotherapy field being meningiomas and high-grade gliomas. Acute lymphocytic leukaemia can occur due to use of alkylating agents.
4. Hearing impairment- occurs due to chemotherapy and radiotherapy given. Should be monitored on follow up.
5. Psychological sequelae – usually relate to post therapy changes including but not limited to short stature, deafness, abnormal gait and learning or coping difficulties. A clinical psychologist needs to be part of the multidisciplinary team looking after these children.
6. Management of intracranial hypertension and spinal cord compression.

11.10.5 References

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Chapter 12

Occult Primary (Metastatic Carcinoma of Unknown Primary)

12.1 Introduction

12.1.1 Definition

Cancer of unknown primary (CUP) is defined as a carcinoma or undifferentiated neoplasm for which a standardised diagnostic work-up fails to identify the primary tumour responsible for metastatic seeding.

12.1.2 Epidemiology

- Cancer of unknown primary (CUP) represents less than 5% of all cancer cases, but due to its high mortality, it contributes disproportionately to cancer-related deaths.
- The overall incidence of CUP has been on the decline, likely because of better success in identifying primary tumor sites. Incidence rates rise with age and are generally higher in men than in women.
- Adenocarcinoma is the most frequently observed histological type. Roughly half of CUP cases are classified as well- to moderately differentiated adenocarcinomas, about 30% as poorly differentiated or undifferentiated carcinomas, approximately 15% as squamous cell carcinomas, and around 5% as undifferentiated neoplasms.
- Sarcomas, melanomas, germ cell tumours, neuroendocrine tumours and haematological malignancies whose exact site of origin is not established are not included in the CUP definition
- Many patients are diagnosed with metastases in multiple organs—most commonly the liver—followed by the lungs, lymph nodes, abdominal cavity, bones, and brain.

12.1.3 Risk Factors

- Lifestyle and Metabolic Factors:
  - tobacco use
  - heavy alcohol consumption.
  - Type 2 diabetes.
  - Obesity, and high waist circumference.

- Chronic Infections such as Human papillomavirus (HPV), Epstein–Barr virus (EBV), Chronic hepatitis B and C infections.
- Occupational Exposures such as long-term exposure to asbestos, aromatic hydrocarbons, or industrial chemicals.
- Immunosuppression.
- Older age >60yrs
- Genetic and Molecular Alterations such as TP53 mutations.

12.1.4 Presentation

CUP typically presents with metastatic disease without an identifiable primary tumor despite a standard diagnostic workup.

Common clinical features include:

1. Non-specific Symptoms
   - Weight loss, fatigue, loss of appetite, and pain at metastatic sites.

2. Site-Specific Symptoms Based on Metastases
   - Liver involvement: right upper quadrant pain, hepatomegaly, abnormal liver function tests.
   - Lung involvement: cough, shortness of breath, chest pain.
   - Bone involvement: localized bone pain or pathological fractures.
   - Lymphadenopathy: painless, firm lymph node enlargement (cervical, axillary, or inguinal).
   - Brain metastases: neurological deficits, headaches, seizures.

3. Paraneoplastic Syndromes
   - Occasionally, CUP may present with paraneoplastic features such as hypercalcemia or syndrome of inappropriate antidiuretic hormone secretion (SIADH).

12.1.5 Diagnostic Approach and Considerations

CUP diagnosis involves a thorough investigation to locate the underlying primary tumor, which is why new primary cancers are rarely identified shortly after a CUP diagnosis. In most cases, the primary tumor remains undetected throughout the disease course. However, some individuals who survive CUP may later develop either a primary tumor that was initially hidden and caused the metastases or an entirely separate second primary cancer. Research has shown an increased risk of developing certain second primary cancers, particularly those of the small intestine, male reproductive organs, and the aerodigestive tract.

12.1.6 Assessment

- Initiate assessment with a minimal basic workup and subsequent testing is informed by findings.
- Intensive and unnecessary testing delays initiation of treatment and also has negative implications on morbidity and finances.

MDT collaboration between clinicians, pathology and radiology departments.

1. Complete history and physical exam – include all relevant areas (nodal, organ systems)
2. Review past medical history – prior biopsies or excisions/surgeries, prior history of masses/lesions that may have spontaneously regressed, prior imaging or other results
3. Labs – UEC, Ca2+, CBC, LDH, LFT, urinalysis, stool occult blood
4. Imaging guided by site of lesion – CT CAP, if no primary detected get PET-CT;
   a. Others – mammography/breast MRI

5. Endoscopy guided by site of lesion
6. Biopsy – core needle biopsy preferred
   a. IHC – best approached in collaboration with the pathologist to maximize the yield
      i. Subclassify as epithelial, lymphoma/hematologic, thyroid, melanoma, sarcoma, GCT, or non-malignant tumor
      ii. Further IHC testing to narrow down to potential site of origin – e.g ER/PR, TTF1, PSA, AFP

b. Molecular test after initial determination of histology
c. Tonsillectomy - for isolated SCC in neck nodes

7. Tumor markers (serum) – relevant to clinical and pathology findings e.g AFP, PSA, CA125, CA15-3, CEA, CA19-9, chromogranin A

Table 12.1: Broad tumor markers

Markers*	Most Likely Cell Lineage
Pan-keratin (AE1/AE3 & CAM5.2)	Carcinoma
CK5/6, p63/p40	Squamous cell carcinoma
S100, SOXIO	Melanoma and sarcoma
LCA± CD20± CD3±	Lymphoma
OCT3/4± SALL4±	Germ cell tumor
WT1, calretinin, mesothelin, D2-40	Mesothelial tumor

These markers are not uniformly specific or sensitive and can be present on other tumors.

Table 12.2: Site specific tumor markers

TUMOR-SPECIFIC MARKERS AND THEIR STAINING PATTERN[1,2]		
Marker	**Tumor**	**Staining Pattern**
Argmase-1	Hepatocellular	Nuclear/cytoplasmic
Calretinin	Mesothelioma, sex cord-stromal, adrenocortical	Nuclear/cytoplasmic
CDX2	Colorectal, other gastrointestinal, pancreaticobiliary tract	Nuclear
D2-40	Mesothelioma, lymphatic endothelial cell marker	Membranous
EBV	Nasopharynx	Nuclear
ER/PR	Breast, ovary, endometrium	Nuclear
GATA3	Breast, urinary bladder, salivary gland	Nuclear
Glypican-3	Hepatocellular	Cytoplasmic
HepPar-1	Hepatocellular	Cytoplasmic
HPV	Cervix, vulva, vagina, penis, anal, oropharynx	Nuclear (DNAISH); nuclear/cytoplasmic (RNA ISHi
Inhibin	Sex cord-stromal, adrenocortical	Cytoplasmic
Melan-A	Adrenocortical, melanoma	Nuclear
Napsin A	Lung	Cytoplasmic
NKX3-1	Prostate	Nuclear

TUMOR-SPECIFIC MARKERS AND THEIR STAINING PATTERN[1,2]		
Marker	**Tumor**	**Staining Pattern**
PAP	Prostate	Membranous
PAX8	Thyroid, renal, ovary, endometrium, cervix, thymus	Nuclear
PSA	Prostate	Cytoplasmic
RCC marker	Renal	Membranous
SF-1	Adrenocortical, sex-cord stromal	Nuclear
SATB2	Colorectal, osteosarcoma, and other gastrointestinal tract	Nuclear
Thyroglobulin	Thyroid	Cytoplasmic
TTF-1	Lung, thyroid	Nuclear
Uroplakin III	Urothelial	Membranous
Villin	Gastrointestinal (epithelia with brush border)	Apical
WT1	Ovarian serous, mesothelioma, Wilms	Nuclear

N/B – collaborate with pathologist and/or MDT for the most efficient approach

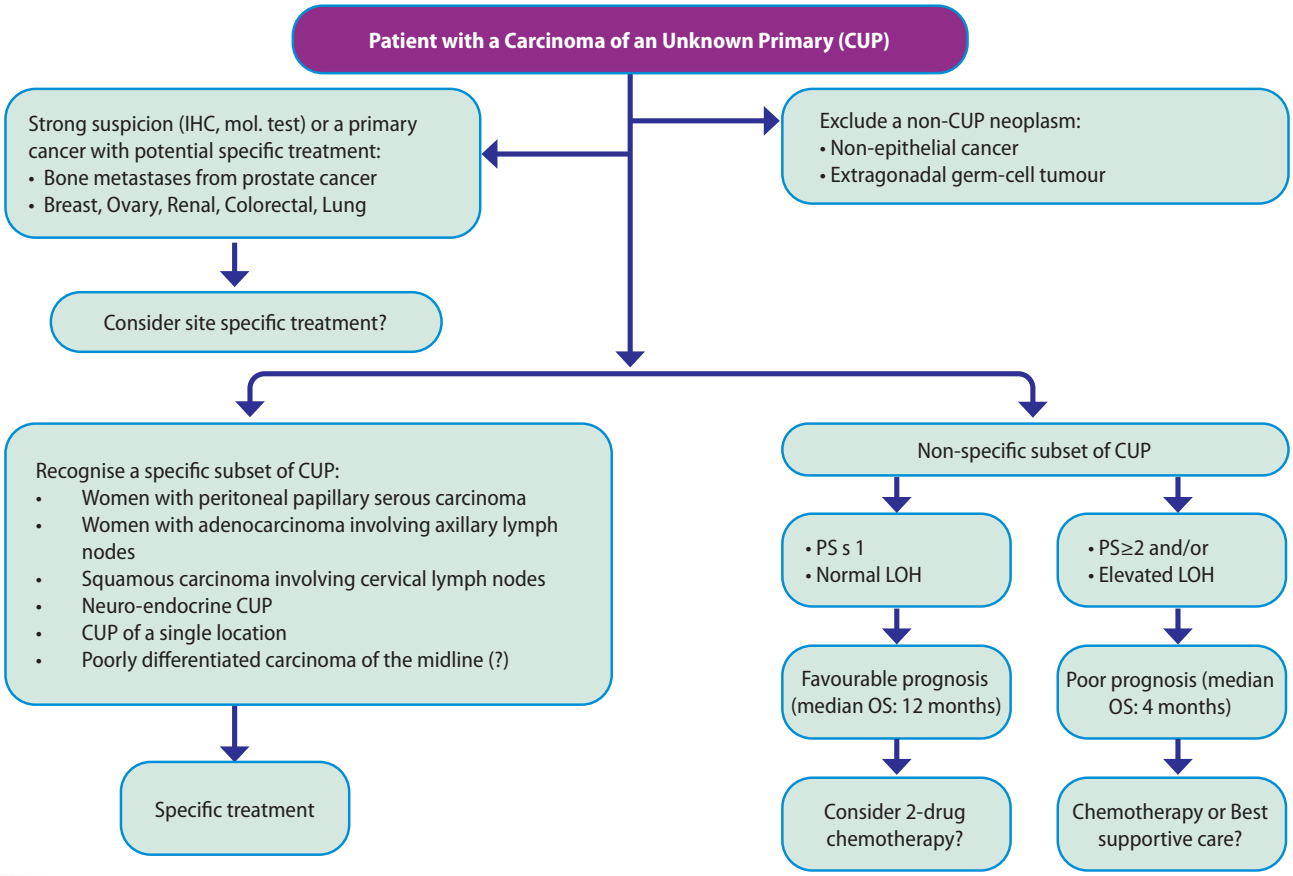


Figure 12.1: Diagnostic Algorithm

12.1.7 Prognosis

Depends on a number of factors such as single vs multiple sites of lesions, patients' performance status
Poor prognostic factors include

1. Male sex
2. Multiple brain metastases
3. Pleural/lung involvement
4. Liver or adrenal involvement
5. Adenocarcinoma histology

Favorable CUP entities have been described and these could include:

- Single metastatic deposit or oligometastatic disease amenable to local ablative treatment
- In women, isolated axillary nodal metastases
- In women, peritoneal adenocarcinomatosis of a serous papillary histologic type
- In men, blastic bone metastases or IHC/serum prostate-specific antigen (PSA) expression
- Squamous cell carcinoma involving non-supraclavicular cervical lymph nodes (head-and-neck)
- Adenocarcinoma with a colorectal immunohistochemical (CK7-negative, CK20-positive, CDX2-positive) or molecular profile
- Carcinoma with a renal-cell histologic and immunohistochemical profile

12.1.8 Staging

Carcinoma of unknown primary is presumed to be stage IV at diagnosis

12.1.9 Approach to Management

Histopathology findings/pathways

1. Epithelial malignancy
    a. Adenocarcinoma/carcinoma, not otherwise specified – approach investigations as per site of presentation
    b. Squamous cell carcinoma

12.1.10 Systemic Therapy

Choice of regimen is guided by pathology (IHC +/- molecular markers) and site of clinical presentation
Occult adenocarcinoma regimens may include:

- Carboplatin and paclitaxel
- Cisplatin and gemcitabine
- CapeOx/ mFOLFOX6/FOLFIRI – presumed GIT
- Other single or combined agents
- Immunotherapy
- Targeted therapy

Occult squamous cell carcinoma regimens may include:

- Carboplatin (or cisplatin) and paclitaxel/docetaxel
- mFOLFOX6
- cisplatin and fluorouracil
- Gemcitabine and cisplatin

- Other single agents or combined agents
- Targeted therapy
- Immunotherapy

12.1.11 Role of Radiotherapy

May be considered as:

- Definitive RT – including conventional RT or SBRT for limited sites (1-3), where appropriate
- Adjuvant RT – if only single nodal site (e.g. head and neck)
- Palliative – pain control, fracture risk, cord compression

12.1.12 References

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Chapter 13

Common Oncologic Emergencies

An acute life-threatening condition that occurs due to the cancer itself, or its treatment leading to morbidity or mortality. Most common oncologic emergencies can be categorized as either:

- Metabolic- Tumor lysis syndrome, hypercalcemia of malignancy, syndrome ofinappropriate ADH secretion
- Hematologic- hemorrhage, hemoptysis, hemorrhagic cystitis, thrombosis, hyperviscosity syndrome, hyper leukocytosis,
- Structural- Superior Vena Cava syndrome, spinal cord compression, urinary tract obstruction, pathological fractures
- Treatment related- tumor lysis syndrome, febrile neutropenia and sepsis, hemorrhagic cystitis, anemia.
- Brain Metastasis

13.1 Febrile Neutropenia and Sepsis

Neutropenia: Absolute neutrophil count = ANC (% neutrophils x total WBC)

- Mild: ANC < 1000/mcL
- Severe: ANC < 500/mcL

Fever: Single axillary temperature > 38.50 C OR temperature of >38.00 C in 2 readings 1 hour apart.

13.1.1 Evaluation of Febrile Neutropenia

History

- Age
- Time since last chemotherapy
- Recent antibiotic therapy / prophylaxis HIV status
- Other co-morbidities Exposure to sick persons
- Exposure to animal excreta/ live plants/ flowers

Laboratory

- Blood slide for malaria parasites CBC including differentials
- CXR if respiratory symptoms Blood culture
- Urine culture if symptoms or catheter
- Or if severe neutropenia Stool culture if diarrhea

Initial Therapy for Fever and Neutropenia

- Haematopoietic growth factors such as filgrastim can be initiated prophylactically as clinically indicated post-chemotherapy in patients at high risk for severe neutropenia.
- Re-emphasize infection prevention measures.

Examination

- Site of branula Skin
- Lungs
- Alimentary canal (mouth, pharynx) Perivaginal / perirectal

- Choice of antibiotics should be tailored to findings on history and physical exam as well as local antibiotic sensitivity patterns.

13.1.2 High Risk Patient:

- Prophylaxis is indicated with ciprofloxacin, fluconazole and Acyclovir.
- IV antibiotics should be used, covering gram +ve and gram –ve organisms
- Monotherapy: Cefepime, Ceftazidime or Meropenem are the best.
- Use reliable ceftriaxone or combination drugs: Aminoglycoside + antipseudomonal penicillin + beta lactamase inhibitor
- If in 48 hours no response: Add AB covering anaerobes (IV Metronidazole)
- If in 72 hours no response: Add antifungal (Amphotericin-B, Fluconazole if ampho-B is unavailable)

13.1.3 Low Risk Patient (and outpatient Rx):

- Oral antibiotic allowed: Ciprofloxacin or Amoxicillin/Clavulanic acid.
- Mouth care with antiseptic.
- Close observation, and review by a clinician within 72 hours.

NOTE:
Assess for **Systemic Inflammatory Response Syndrome (SIRS)** criteria, symptoms or signs, in an afebrile patient currently receiving chemotherapy (some patients cannot mount fever for example in long term use of corticosteroids). If hypotensive, tachycardic, unstable, and neutropenic, institute febrile neutropenia procedures.

Systemic Inflammatory Response Syndrome (SIRS) is recognized by:

- Temperature > 38° C or < 36° C
- Heart rate > 90 beats/minute (tachycardia)
- Respiratory rate > 20 breaths/minute (tachypnea)
- Systolic blood pressure ≤ 90 mm Hg (hypotension)
- Abnormally high or low white blood cells

SIRS criteria (1 or more) predict serious infection and sepsis in a chemotherapy patient at time of presentation / triage even in the absence of fever.

Table 13.1: Infection Prevention during Chemotherapy

Ward	• Patient should be in isolation area (away from acute infections) • Only one patient per bed • In severe neutropenia one patient per cubicle • Strict asepsis for invasive devices • Change nebulizers, humidifiers, water in jugs daily • Strict hand washing / hand sanitizer in between patients
Staff	• In severe neutropenia handling of the patients should be as little as possible fresh • linen should be provided daily • Live flowers and plants should be out of patients room
Food	Only use safe foods (no salads, raw eggs, left over foods)
Visitors	Allow only limited visitation, without respiratory infections Visitors should wash hands before entering patients room
Personal	• Prevention of infection. Regular hand washing • Oral hygiene: Cleaning of mouth with water after every meal (with assistance if needed) Use of soft tooth brush twice daily Betadine mouth wash in severe neutropenia
Drugs	For paediatrics, Pneumocystis carinii prophylaxis in all patients on chemotherapy: • Use Cotrimoxazole 15+3 mg/kg OD • Continue till end of maintenance

Treatment recommendations for Brain Metastasis (Handbook of evidence -based Radiation Oncology)

13.1.4 Respiratory Emergencies

Airway Obstruction:

- Common in lung cancer, head and neck tumors, lymphomas, thymomas and thyroid malignancies.
- Symptoms include cough, dyspnea, stridor
- Management:
  - Surgical intervention (Intubation and tracheostomy) with or without appropriate medication (steroids,
  - bronchodilators, chemo/radiotherapy)
  - Oxygen prn.
  - Definitive Treatment: Chemoradiotherapy, radiotherapy

Hemoptysis:

- bleeding from the lower respiratory tract.
- The most important aspect of managing massive hemoptysis is protecting the airway.
- This may need intubation through bronchoscopy.

Management:

- Any coagulopathy should be corrected, and cough sup- pressed with codeine or other agents.
- Options include bronchial artery embolization or tumor resection, external beam radiation therapy (EBRT), local epinephrine injection, laser treatment, electrocautery, photocoagulation, balloon tamponade, or iced-saline lavage.

Toxic Lung Injury:

- Presents as acute respiratory distress syndrome.
- Management: Support as appropriate.

13.2 Spinal Cord Compression

- 2.5–5.0% of patients have spinal cord compression (SCC) within the last 2 years of their illness.
  - Prostate, breast cancer, lung cancer most common each ~15–20%
  - NHL, multiple myeloma, and renal cancer ~5–10% of patients

- Thoracic spine affected in 60-80% of cases though 50% may present with disease in multiple spinal areas.

13.2.1 Signs and Symptoms

- Presents with new onset back pain (initially localized, typically increasing in intensity, worsens when the patient is lying down, pain with percussion of vertebral bodies), weakness (60-85% of patients present with weakness), about 2/3 are non-ambulatory at presentation.

- Late neurologic signs are associated with permanent deficits such as paraplegia, urinary retention and loss of sensory function.
- Often leads to permanent neurologic dysfunction though 80-90% of patients ambulant at treatment retain function. Ambulatory status is therefore the most important prognostic feature.

13.2.2 Imaging

- Non-contrast MRI of whole spine is recommended
- If MRI not available, can use Myelography/CT
- Biopsy if: metastatic disease not proven/documented, or no previous diagnosis of cancer.

13.2.3 Management

- strict bed rest, start on steroids immediately to reduce edema and further cord compression.
- with fracture and unstable bone fragments: surgical decompression and stabilization.
- with fracture, stable fragments: radiation therapy
- cord compression with no fracture: radiation therapy or chemotherapy.

13.2.4 Tumour Lysis Syndrome

Tumor Lysis Syndrome (TLS) is an oncological emergency caused by rapid destruction of tumor cells, releasing intracellular contents into the bloodstream.

This results in metabolic disturbances that can be life-threatening.

At-Risk Populations

- Hematologic malignancies: Acute leukemias; High- and intermediate-grade lymphomas
- Solid tumors (less common): Small-cell lung cancer; Germ cell tumors
- Risk factors: Bulky or rapidly proliferating tumors; Initiation of cytotoxic therapy; Underlying renal impairment

Biochemical & Clinical Features

- Biochemical abnormalities: Hyperuricemia, Hyperkalemia, Hyperphosphatemia leading to hypocalcemia
- Clinical consequences: Acute kidney injury, Cardiac arrhythmias (ventricular tachycardia, ventricular fibrillation), Seizures, sudden death in severe cases

Prevention

General Principles

- Initiate preventive measures in all high-risk patients before chemotherapy.

Pharmacological Measures

- Allopurinol – for moderate-risk patients
- Rasburicase – preferred in high-risk patients or when allopurinol is contraindicated
- Avoid concomitant allopurinol with drugs that increase risk of skin reactions

Supportive Measures

- Vigorous IV hydration to maintain high urine output
- Consider urine alkalinization if indicated
- Loop diuretics if inadequate diuresis despite hydration

Monitoring

- Baseline: Electrolytes, phosphate, calcium, uric acid, creatinine
- Moderate risk: Check daily
- High risk: Check every 6 hours for first 24–48 h after chemotherapy
- If abnormalities present: Repeat every 6–12 h until stabilization

Management of Established TLS

General Measures

- Aggressive IV hydration
- Continuous cardiac monitoring if hyperkalemia or hypocalcemia present
- Correct metabolic abnormalities promptly

Hyperuricemia

- Treat with rasburicase
- Continue hydration

Hyperphosphatemia

- Oral phosphate binders (e.g., aluminum hydroxide)

Hyperkalemia

- Mild: IV fluids ± diuretics
- Moderate: Add potassium-binding resins
- Severe or with arrhythmia: IV calcium for cardioprotection, Insulin with dextrose, Albuterol as adjunct, Dialysis if refractory Hypocalcemia
- Correct only if symptomatic (tetany, seizures, arrhythmias) with IV calcium

Dialysis Indications

- Refractory hyperkalemia
- Severe hyperphosphatemia or hypocalcemia with symptoms
- Worsening renal failure

Table 13.2: Management of Tumor Lysis Syndrome

Category	Key Recommendations
Prevention	Allopurinol (moderate risk), Rasburicase (high risk), hydration, consider alkalinization, diuretics if needed
Monitoring	Daily labs for moderate risk; every 6 h for high risk
Hyperuricemia	Rasburicase preferred
Hyperphosphatemia	Oral phosphate binders
Hyperkalemia	Fluids → diuretics → resins → insulin/dextrose → calcium (if ECG changes) → dialysis if refractory
Dialysis	Refractory electrolyte imbalance or renal failure

TLS Management Flow (Algorithm)

Risk assessment →

- Moderate risk → Allopurinol + hydration + daily monitoring
- High risk → Rasburicase + aggressive hydration + monitoring q6h

If TLS develops:

1. Hydration + electrolyte correction
2. Hyperuricemia → Rasburicase
3. Hyperphosphatemia → Phosphate binders
4. Hyperkalemia → Stepwise therapy (fluids → diuretics → resin → insulin/dextrose → calcium → dialysis)
5. Continuous cardiac and renal monitoring

13.3 Superior Vena Cava Syndrome

This refers to impaired blood flow from the SVC to the right atrium due to obstruction which may either be: internal (for example, thrombus) or external (malignant/non-malignant). Malignancies associated with superior vena cava syndrome include: Lung Cancer – SCLC, NSCLC, Non- Hodgkins Lymphoma, Thymoma, Germ cell tumors of mediastinum and Metastatic tumors.

13.3.1 Clinical Presentation

- Early symptoms – Dyspnea and non-productive cough, headache, dysphagia, hoarseness, chest pain, facial puffiness.
- Late symptoms -Visual disturbances, dizziness, syncope, lethargy, irritability and mental status changes.
- Physical examination often reveals engorgement of veins and collaterals in the upper extremities.

13.3.2 Management

- Management is based on the cause.
- Bed rest with head elevation, Oxygen administration as necessary, corticosteroids, and restrict intravenous fluids, no IV line in upper limbs.

13.3.3 Role of Radiotherapy and Other Modalities

- **Chemotherapy**: Preferred first-line for highly chemosensitive tumors (SCLC, lymphoma, germ cell).
- **Radiotherapy**: Used when tumor is not highly chemosensitive (e.g., NSCLC) or as adjunct after chemo.
- **Endovascular stenting**: Best for rapid symptomatic relief, often combined with RT/chemo for long-term control.

13.4 Brain Metastasis

- Brain metastasis is the most prevalent kind of cerebral tumor, it occurs in about 20–30% of cancer patients overall.
- Melanoma, lung, and breast cancers are the primary tumors most likely to spread to the brain.
- The most common examples of hemorrhagic metastases include melanoma, choriocarcinoma, and renal cell carcinoma.

13.4.1 Workup and Treatment Approach

- History and physical examination with emphasis on a detailed neurologic exam
- Brain MRI: T1 pre- and post-contrast. T2 FLAIR
- For new cancer diagnosis, staging workup for identification of primary, evaluation of systemic disease burden, and tissue biopsy
- Headaches are among the neurologic symptoms that steroids can alleviate.
- Patients having seizures are advised to use antiepileptic medications.
- Prophylactic antiepileptic medication is not beneficial for patients who have never experienced seizures.
- For the best care, medical oncology, neurosurgery, and radiation oncology must coordinate across several disciplines.
- Clinical situation, tumor histology, intracranial and extracranial disease burden, performance status, life expectancy, cancer-directed therapy options, and patient preferences should all be taken into consideration when customizing a patient's course of treatment.
- When using whole brain radiation therapy for neuroprotection, consider the use of memantine (where available) or hippocampal sparing treatment plans.
- Close follow-up with surveillance MRIs is advised to optimize the chance for salvage treatment(s), as the use of SRS alone is linked to a higher probability of distant brain failure in comparison to whole brain radiation.
- It is important to thoroughly examine and talk about the simultaneous administration of systemic medicines and brain radiation.

Table 13.3: Treatment recommendation for Brain Metastasis

Clinical scenario	Treatment recommendation
Acute decompression required	Surgery plus: • Adjuvant **SRS** (small cavity) or fractionated **SRT** (large cavity) for limited brain metastases • Adjuvant WBRT for leptomeningeal or diffuse brain metastases
Single large (>3 cm) metastasis, surgical candidate	• Surgery + adjuvant **SRS** (small cavity) or adjuvant FSRT (large cavity) • Adjuvant **WBRT** may be considered if SRS or FSRT is not available or for patient preference
Single large metastasis (>3 cm), not surgical candidate	• Fractionated SRT (preferred). Single-session SRS is associated with increased risk of local failure and toxicity • **WBRT** + SRS boost
Single small metastasis, good prognosis	• SRS or surgery + adjuvant SRS. **LC** for small tumors with SRS is excellent
Limited metastases, good prognosis	• SRS and close observation, reserving **WBRT** or additional SRS for salvage
Diffuse metastases, good prognosis	WBRT
Prior brain radiation	• For limited volume brain metastases, SRS • Repeat WBRT for selected cases (good **PS**, absent or limited and controlled extracranial disease, prolonged disease-free interval)
Poor prognosis	• Best supportive care or WBRT • Prognosis is poor and options are limited
Leptomeningeal disease	• WBRT or focal **RT** to bulky disease • Intrathecal chemotherapy • Shunting if **CSF** obstruction suspected (communicating or noncommunicating)

Table from Handbook of Evidence Based Radiation Oncology

13.4.2 Extravasation

Extravasation is an uncommon but high-risk adverse event that occurs when an agent with the potential to cause tissue damage leaks out of the intended administration space into the surrounding area. The frequency of extravasation in cancer care is reported to be between 0.1% and 6% among adult cancer patients.
Risk Factors:

1. Patient Related: small and fragile veins in infants, children or elderly patients, patients with hardened or thickened vessels due to frequent venipuncture, veins that move easily, excised lymph nodes, limb amputation, obesity and highly mobile patients during chemotherapy administration.
2. Procedure related: untrained/ inexperienced staff, multiple attempts of canulation, high flow pressure, inadequate choice of equipment i.e. catheter or steel needle size, inadequate dressing, poor canula fixation
3. Drug related: necrotizing capacity of the drug, amount of extravasated agent (volume and concentration), duration of exposure to extravasation, pH value, presence of excipients such as benzyl alcohol (has local anesthetic effect)

13.4.3 Classification of Cytotoxic Drugs Based on Tissue Damage Potency

1. Vesicants

Causes pain, inflammation and blistering of local skin, underlying structures leading to tissue death and necrosis. Depending on mechanism of action, they are divided into two.

a) DNA binding vesicants. Vascular leakage of these agents can cause combination with the nucleic acid within the cells, resulting in the spreading of toxicity to surrounding tissue, and the degree of damage can be much more severe. The aim in management of the extravasation is therefore to localize and neutralize. Examples of these drugs include dacarbazine, dactinomycin, daunorubicin, doxorubicin, Epirubicin, idarubicin, mechlorethamine and mitomycin

b) Non-DNA binding vesicants. The aim in management of the extravasation of these agents is therefore to disperse and dilute. Examples of these agents include vinblastine, vincristine, vinorelbine and paclitaxel

2. Non Vesicant

Extravasation of these agents does not cause ulceration though they can cause mild to moderate inflammation and flare in local tissues. Extravasation rarely progresses to an acute reaction or necrosis. The management is purely geared towards symptomatic relief.

3. Irritant and Exfoliants

Exfoliants are drugs capable of causing inflammation and shedding of skin but less likely to cause tissue death while irritants may cause inflammation, irritation or pain at site of extravasation but rarely cause tissue breakdown. Symptoms for both irritants and exfoliants resolve within weeks and rarely progress to long term. The aim of management is to localize the extravasation. Cisplatin and oxaliplatin are examples of exfoliants while carboplatin, etoposide and irinotecan are examples of irritants.

13.4.4 Clinical Presentation of Extravasation

The symptoms of extravasation include pain, stinging or burning sensation, edema around the IV site, tissue dysfunction or physical defects.



Figure13.1: Extravasation caused by Epirubicin.

13.4.5 Management of Extravasation

1. Stop the infusion.
2. Disconnect IV tube from canula (Do not flush the line)

3. Aspirate any residual drug from the cannula with 10-29ml cc. Avoid manual pressure on extravasated region.
4. Administer drug specific antidote.
   a. DNA Binding
      - Topical application of DMSO- dimethyl sulfoxide 99% solution (do not apply if there is blistering)
      - Apply hydrocortisone 1% cream after 3 hours from the initial application of DMSO.
      - Dexrazoxane – derivative of EDTA- given if more than 3ml is extravasated. This should never be given with DMSO

   b. Non- DNA binding
      - SC/ intradermal 150IU diluted in 1ml of sterile water given as 5 (0.2ml) injection of Hyaluronidase around extravasated area. This should begin within 1 hour of extravasation for best results.
      - Apply hydrocortisone 1% cream every 6 hours for as long as erythema persists.

5. Elevate the limb with extravasation site.
6. Local thermal treatments to decrease the site reaction and absorption of infiltrate for 15-20 min every 4hrs for 24-48hrs.
   a. Local cooling, i.e. ice packs- aids in vasoconstriction theoretically limiting drug dispersion. for DNA binding vesicants except mechlorethamine (nitrogen mustard), contrast media and hyperosmolar fluids.
   b. Local warming, i.e. dry heat- enhances vasodilation, enhances dispersion of the vesicant and decreasing drug accumulation in local tissue. For non-DNA vesicants.

7. Pain management. Consider opioids for moderate to severe pain especially in the setting of widespread necrosis.
8. Apply sterile dressing on area of extravasation and consult department of cosmetic surgery if necessary.
9. Record the extravasation extent and management therapies instituted.

STEP 1

Stop the infusion.

STEP 2

Disconnect IV tube from canula (Do not flush the line).

STEP 3

Aspirate any residual drug from the cannula with 10-20ml. Avoid manual pressure on extravasated region.

STEP 4

Administer drug specific antidote.

DNA Biding

- Topical application of OMSO- dimethyl sulfoxide 99% solution (do not apply if there is blistering)
- Apply hydrocortisone 1% cream after 3 hours from the initial application of DMSO.
- Dexrazoxane- derivative of EDTA· given if more than 3ml is extravasated. This should never be given with DMSO

Non- DNA binding

- SC/ intradennal lSOIU diluted in 1ml of sterile water given as 5 (0.2ml) injection of Hyaluronidase around extravasated area. This should begin within 1 hour of extravasation for best results.
- Apply hydrocortisone 1% cream every 6 hours for as long as erythema persists.

STEP 5

Elevate the limb with extra11asation site.

STEP 6

- **Local thermal treatments to decrease the site reaction and absorption of infiltrate for 15-20 min every 4hrs for 24-48hrs.**
- **Local cooling, i.e. ice packs- aids in vasoconstriction theoretically limiting drug dispersion, for DNA binding vesicants except mechlorethamine (nitrogen mustard), contrast media and hyperosmolar fluids.**
- **Local warming, i.e. dry heat- enhances 11asodilation, enhances dispersion of the vesicant and decreasing drug accumulation in local tissue. For non-DNA vesicants.**

STEP 7

Pain management. Consider narcotics for severe pain especially in the setting of widespread necrosis.

STEP 8

Apply sterile dressing on area of extravasation and consult department of cosmetic surgery if necessary

Fig 13.2: Extravasation Management Algorithm

13.5 Syndrome of Inappropriate ADH Secretion

A paraneoplastic syndrome in which antidiuretic hormone (ADH) is secreted inappropriately from the posterior pituitary gland, despite lower serum osmolality.

15.5.1 Management:

- Restricting water to 500 to 1000 mL/day from all sources and treating the underlying disorder.
- Demeclocycline (600-1200 mg/day) can be used in divided doses two to four times per day.
- Slow infusion of 3% normal saline; care must be taken not to increase serum sodium by more than 0.5 to 1 mEq/h to prevent central pontine myelinolysis.

13.5.2 Hematological Emergencies

Hyperviscosity Syndrome:

- Serum viscosity >4.0 Ostwald units.
- Treatment; intravenous fluids followed by diuresis, plasma exchange, chemotherapy

Hyperleukocytosis:

- WBC count in the peripheral blood higher than 100,000/uL.
- Management: involves lowering the WBC count, which can be accomplished with leukapheresis or chemotherapy.

Thrombosis:

- Both deep venous thrombosis and pulmonary thromboembolism.
- Management: anticoagulation and appropriate mechanical approaches.

13.5.3 Bleeding In Cancer: Causes And Risk Management

Causes of Bleeding in Cancer

- Tumor-related causes: local invasion of vessels, tumor necrosis, fragile angiogenic vessels, mucosal ulceration.
- Treatment-related causes: radiotherapy-induced mucosal damage, chemotherapy-induced thrombocytopenia, targeted therapy (e.g., bevacizumab), surgery/biopsy complications.
- Systemic/host factors: coagulopathies (DIC, liver metastases), bone marrow suppression, anticoagulant use, infections.

Risk Factors

- High tumor burden or vascular invasion
- Hematologic malignancies (AML, ALL, lymphoma)
- Platelet count <50,000/µL
- Coagulopathy or liver dysfunction (INR >1.5)
- Renal failure with platelet dysfunction
- Recent surgery, radiation, invasive procedures
- Use of anticoagulants, NSAIDs, antiplatelets
- Fragile mucosal surfaces (GI, GU, respiratory tract)

13.5.4 Risk Management & Preventive Strategies

Primary Prevention

- Optimize blood counts before invasive procedures
- Careful anticoagulant selection in high-risk patients
- Gastroprotective agents in GI cancers
- Gentle oral/skin care to minimize mucosal trauma
- Avoid unnecessary IM injections

Monitoring & Early Detection

- Regular CBC monitoring during chemotherapy
- Clinical surveillance for bleeding symptoms
- Screening for coagulopathy in high-risk patients

Acute Management

- Stabilization: airway, IV fluids, hemodynamic support
- Blood products: platelet transfusion, FFP, cryoprecipitate
- Local measures: pressure, cauterization, hemostatic agents, endoscopic/IR embolization
- Systemic therapy: antifibrinolytics (tranexamic acid), correction of coagulopathy
- Definitive tumor control: palliative RT, surgery, embolization

Palliative Considerations

- Symptom control and comfort measures
- Dark towels/sheets to reduce visual distress
- Pre-emptive discussion with family about catastrophic hemorrhage risk

Table 13.4: Causes, Risks, and Management of Bleeding In Oncology

Category	Examples	Management
Tumor-related	Vessel invasion, necrosis, mucosal ulceration	Hemostatic RT, surgery, embolization
Treatment-related	Chemo-induced thrombocytopenia, RT mucositis, anti-angiogenics	Supportive transfusions, drug modification, local measures
Systemic factors	DIC, liver metastases, marrow suppression, anticoagulant use	Correct coagulopathy, transfusions, antifibrinolytics

Management

1. Identify bleeding site & stabilize patient (Airway-Breathing-Circulation)
2. Evaluate severity: mild, moderate, massive bleeding
3. Initiate supportive care: IV fluids, blood products
4. Apply local measures (pressure, cautery, endoscopy/IR)
5. Correct systemic causes (coagulopathy, thrombocytopenia)
6. Definitive tumor control (RT, embolization, surgery)
7. Palliative care if advanced/end-stage

13.6 Genitourinary Emergencies

13.6.1 Hemorrhagic Cystitis:

- Haematuria Management: gentle bladder irrigation, prostaglandins E2 or F2, 1% alum, or formalin can be instilled.
- Formalin instillation is painful and requires general or spinal anesthesia.
- To correct continued bleeding, some patients require surgery, hypogastric artery embolization, or open surgical intervention.

13.6.2 Urinary Tract Obstruction:

- Involve ureters, or urethra
- Management: Relieve obstruction by catheterization and stenting as appropriate.

13.6.3 Acute Kidney Injury (AKI):

- This is characterized by sudden decrease in glomerular filtration rate, leading to an acute rise in blood urea nitrogen (BUN) and serum creatinine (Cr) levels.
- It is a serious complication of many malignancies, which causes important morbidity and mortality.
- **Presentation-** Symptoms related to ARF will differ based on the underlying cause (Pre, intra and post renal causes) and mechanism.

- The diagnosis is based on laboratory results, with a decrease in glomerular filtration rate, and a rise in BUN and serum Cr levels.
- Treatment is established according to the underlying cause.

13.7 Anaphylaxis and other Acute Infusion Reactions

13.7.1 Causes

- Anaphylaxis, though rare, is a potentially catastrophic reaction seen in oncology practice.
  - **Biologics & Chemotherapy:** Serum products, L-asparaginase, Paclitaxel (Cremophor), Monoclonal antibodies (e.g., Rituximab), Carboplatin (delayed hypersensitivity)
  - **Other drugs:** Antibiotics (Penicillin), Iodine-based contrast media, virtually any drug (rare idiosyncratic reactions)

- Note: Monoclonal antibody infusion reactions may be due to cytokine release rather than IgE-mediated allergy. Risk is higher in patients with high tumor burden (e.g., CLL, lymphoma).

13.7.2 Clinical Manifestations

- Early/general symptoms: Anxiety, dyspnea, presyncope, urge to defecate
- Skin: Urticaria, generalized itching, flushing, pallor
- Respiratory: Bronchospasm, angioedema, laryngeal edema
- Cardiovascular: Hypotension, syncope, tachycardia/bradycardia
- Delayed onset: Sometimes hours later (especially with carboplatin)

13.7.3 Management (Guideline Approach)

1. Immediate Assessment

- Ensure airway patency
- Provide high-flow oxygen
- Prepare for intubation if airway compromise
- Surgical airway (cricothyrotomy/tracheostomy) if severe edema

2. Circulatory Support

- Rapid IV fluids (normal saline / Ringer's lactate)
- Vasopressors if refractory hypotension

3. Medications

- Epinephrine (first-line for severe anaphylaxis)
- Beta-agonist inhalation (salbutamol/albuterol) for bronchospasm
- Antihistamines: H1 blocker (diphenhydramine), H2 blocker (cimetidine, adjunctive)

- Corticosteroids (hydrocortisone/methylprednisolone) – not for acute rescue but useful for preventing late reactions

Table 13.5: Management Of Anaphylaxis And Other Acute Infusion Reactions

Aspect	Key Points
Causes	Biologics, chemo (taxanes, platinum), monoclonals, antibiotics, contrast media
Mechanism	IgE-mediated (true anaphylaxis) OR cytokine-release (infusion reactions)
Onset	Immediate or delayed (esp. carboplatin)
Symptoms	Anxiety, dyspnea, urticaria, angioedema, hypotension, bronchospasm
Airway Management	Oxygen → Intubation → Cricothyrotomy if needed
Circulation	IV fluids → Vasopressors if persistent shock
Medications	Epinephrine, Beta-agonist inhaler, H1/H2 blockers, Steroids (late-phase only)
Special Notes	Avoid epinephrine in minor pruritus-only reactions

Flowchart for Acute Management

1. Recognize reaction → Assess airway
2. If compromised: secure airway (O2, intubation, cricothyrotomy)
3. Start IV fluids
4. Severe symptoms: Epinephrine (preferred route)
5. Mild symptoms: Antihistamines ± H2 blocker
6. Add beta-agonist inhaler if bronchospasm
7. Consider corticosteroids for late-phase reaction
8. Continuous monitoring & supportive care

13.8 References

1. Provencio P., M (2016), ESMO handbook of Oncological emergencies, Madrid, Spain https://repositorio-aberto.up.pt/bitstream/10216/112860/2/273345.pdf
2. Hansen, E. K., & Mack Roach III. (2018). Handbook of Evidence-Based Radiation Oncology. In **Springer eBooks**. Springer Nature. https://doi.org/10.1007/978-3-319-62642-0
3. Kim JT, Park JY, Lee HJ, Cheon YJ. Guidelines for the management of extravasation. Vol. 17, Journal of Educational Evaluation for Health Professions. 2020. p. 1–6.
4. Thomas T, Clark C, Backler C, Bohlke K, Centofanti D, Jotwani AC. ASCO Special Articles ONS / ASCO Guideline on the Management of Antineoplastic Extravasation [Internet]. 2025. Available from: https://ascopubs.org/doi/pdf/10.1200/OP-25-00579

Chapter 14

Safety Standard for Administration of Antineoplastic Therapy

Background Justification

Cancer treatment involves highly toxic drugs with a **narrow therapeutic index**, meaning the margin between a therapeutic dose and a lethal dose is extremely small. Even minor errors such as incorrect dosing, wrong route of administration, or failure to recognize hypersensitivity can lead to irreversible patient harm, prolonged hospitalization, or death. Consequently, putting standards is crucial to standardize care across all facility levels, reduce operational variability, and protect the healthcare provider from hazardous drug exposure through mandated Personal Protective Equipment (PPE) and closed-system transfer devices. The mandotry requirement include :

Nursing Action Area	Mandatory Requirement
Dosing Verification	**Nurses need to** measure patient height/weight in metric units (kg/cm) and ensure **two clinicians** verify calculations for new treatment plans.
Safety Verification	Implement a **four-step verification process** (pre-preparation, post-preparation, pre-administration, and patient bedside) to ensure the right drug, dose, and rate.
Patient Education	**Nurses need to** educate patients in their preferred language about toxicities, home safety, handling of waste, and 24/7 triage contact numbers.
Patient Assessment	**Nurses need to** document functional status, vital signs, toxicities, and pain at **every encounter** before administering therapy.

See appendix 10 (page 291): Systemic Antineoplastic Therapy Administration Safety Protocol for further details.

References

1. **Ministry of Health, Kenya.** (2019). *Kenya National Cancer Treatment Protocols*. Nairobi: Ministry of Health. (This document establishes the necessity of evidence-based, standardized care).
2. **National Council for Cancer Prevention and Control.** (2023). *The Cancer Prevention and Control (Standards of Care) Rules, 2023*. Kenya Gazette Supplement. (This regulatory document explicitly highlights the requirement for training staff on chemotherapy safety and managing oncological emergencies).

Appendices

Appendix 1: Patient Performance Status for Children

Based on the Lansky performance measure for childhood cancer patients.(12)

- 100 – fully active, normal
- 90 – minor restrictions in strenuous physical activity
- 80 – active, but gets tired more quickly
- 70 – greater restriction of play and less time spent in play activity
- 60 – up and around, but active play minimal; keeps busy by being involved in quieter activities
- 50 – lying around much of the day, but gets dressed; no active playing participates in all quiet play and activities
- 40 – mainly in bed; participates in quiet activities
- 30 – bedbound; needing assistance even for quiet play
- 20 – sleeping often; play entirely limited to very passive activities
- 10 – doesn't play; does not get out of bed
- 0 – unresponsive

Appendix 2 : Performance Scales: Karnofsky and Eastern Co-operative Oncology Group (ECOG) Scores

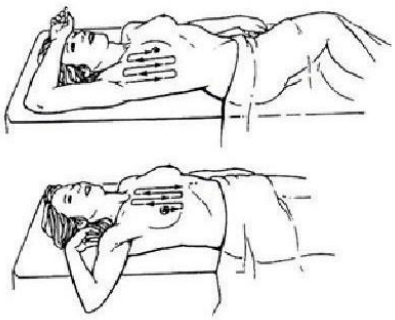
Karnofsky Status	Karnofsky Grade	ECOG Grade	ECOG Status
Normal, no complaints	100	0	Fully active, able to carry on all pre-disease performance without restriction
Able to carry on normal activities. Minor signs or symptoms of disease	90	1	Restricted in physically strenuous activity but ambulatory and able to carry out work of a light or sedentary nature, e.g., light house work, office work
Normal activity with effort	80	1	Restricted in physically strenuous activity but ambulatory and able to carry out work of a light or sedentary nature, e.g., light house work, office work
Care for self. Unable to carry on normal activity or to do active work	70	2	Ambulatory and capable of all selfcare but unable to carry out any work activities. Up and about more than 50% of waking hours
Requires occasional assistance, but able to care for most of his needs	60	2	Ambulatory and capable of all selfcare but unable to carry out any work activities. Up and about more than 50% of waking hours
Requires considerable assistance and frequent medical care	50	3	Capable of only limited selfcare, confined to bed or chair more than 50% of waking hours
Disabled. Requires special care and assistance	40	3	Capable of only limited selfcare, confined to bed or chair more than 50% of waking hours
Severly disabled. Hospitalisation indicated though death nonimminent	30	4	Completely disabled. Cannot carry on any selfcare. Totally confined to bed or chair
Very sick. Hospitalisation necessary. Active supportive treatment necessary	20	4	Completely disabled. Cannot carry on any selfcare. Totally confined to bed or chair
Moribund	10	4	Completely disabled. Cannot carry on any selfcare. Totally confined to bed or chair
Dead	0	5	Dead

As published in Am J Clin. Oncol.: Oken, M.M., Creech, R.H., Tormey, D.C., Horton, J., Davis, T.E., McFadden, E.T., Carbone, P.P.: Toxicity And Response Criteria Of The Eastern Cooperative Oncology Group. Am J Clin Oncol 5:649-655, 1982. The Eastern Cooperative Oncology Group, Robert Comis M.D., Group Chair. Adapted from ESMO Guidelines

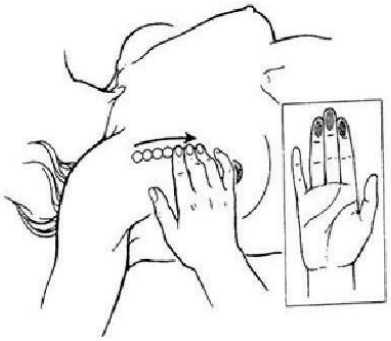
Appendix 3 : Clinical Breast Examination

Evaluation: Physical Exam

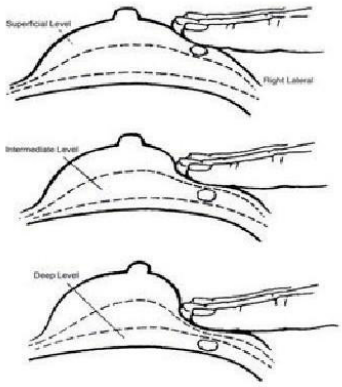
- Clinical Breast Exam:



Position the patient in the direction of palpation for the CBE.



Use pads of the index, third, and fourth fingers (*inset*) make small circular motions



Make three circles with the finger pads, increasing the level of pressure (subcutaneous, mid-level, and down to the chest wall) with each circle

Sanslow, D, et. al. Clinical breast examination" practical recommendations for optimizing performance and reporting. CA Cancer J Clin. 2004 Nov-Dec; 54(6): 327-44

Appendix 4 : Cotswold Modified ANN ARBO Staging of Hodgkins Lymphoma

Stage I	Involvement of a single lymph node region or lymphoid structure(e.g. spleen,thymus)
Stage II	Involvement of two or more lymph node regions on the same side of the diaphragm
Stage III	Involvement of lymph node regions on both sides of the diaphragm
Stage IV	Involvement of extranodal site(s)
A	No constitutional symptoms
B	Constitutional symptoms: Unexplained fever (>38°C), weight loss (≥10% total body weight in six months), drenching night sweats
X	Bulky disease: >1/3 the width of the mediastinum; >10cm maximal diameter of nodal mass

Appendix 5 : Sample Medical Referral Form

Use this form to refer patients to a cancer treatment facility. Fill in all the areas as completely and accurately as possible.

<table>
<tr><td colspan="4">Name (last) (first) (middle)</td></tr>
<tr><td colspan="2">Hospital name</td><td>Hospital No. /Ward</td><td>Ward</td></tr>
<tr><td colspan="2">Residence Address</td><td colspan="2">Contact information</td></tr>
<tr><td>Date of birth</td><td>Gender (Tick)
Male ☐
Female ☐</td><td>Height (cm)</td><td>Weight (Kg)</td></tr>
<tr><td>Referral Type</td><td colspan="3">☐ Immediate referral (to be seen within 24hrs of referral)
☐ Urgent referral (to be seen within 2 weeks of the referral)
☐ Non-urgent referral (can be seen after 2 weeks at referral facility)</td></tr>
<tr><td colspan="4">Presenting clinical features (include duration of symptoms, physical examinationfindings)</td></tr>
<tr><td colspan="4">Family and social history</td></tr>
<tr><td colspan="4">Immunization status(for children): Include the dates when given</td></tr>
<tr><td colspan="2">Vaccine</td><td colspan="2">Date (month and year if actual dates not known)</td></tr>
</table>

Investigation results *(include dates where known)*		
Probable diagnosis/differentials **OR**		
Diagnosis and stage *(if known)*		
Comorbidities *(include treatments for this)*		
Treatments given *(all anticancer treatments must be indicated)*		
Date	***Dose and route***	***duration***
Discharge medications *(list al)*		
Date	***Dose and route***	***duration***
Reason for referral **Person referring** (name)		**Date**

Appendix 6: Imaging Protocols

Breast Imaging Protocols

Bilateral Mammography

The optimal mammogram images will include:

1. Bilateral MLO and CC views
    a. MLO views: Pectoralis muscle well demonstrated with caudal level in line with the nipple convex outwards with inframammary fold and nipple in profile. No skin folds
    b. CC views: Pectoralis muscle demonstrated posteriorly in both breasts. Nipple in profile with inframammary folds. No skin folds.
2. Supplementary views as recommended by the radiologist

All mammograms should be read by a radiologist with experience in mammography reading.

Breast MRI

An ideal MRI should have:

1. High field (Bo) magnet (>1.5T) that is bilateral with complete breast and axillary coverage.
2. Breast dedicated only surface coils (High SNR)
3. Dynamic Contrast Imaging
4. Multiphasic T1
5. Minimum two post contrast T1W1 – the 1st within 4 minutes of contrast injections
6. DWI

Head & Neck Cancer Imaging Protocols

CT Scan Neck

This is the imaging modality of choice for most head and neck (except for Thyroid tumors where ultrasound is the first line imaging modality).

Protocol:

The field covered should extend from the middle cranial fossa to the thoracic inlet to show the arch of the aorta and the lung apices. This enables the demonstration of the brain stem which is home of the cranial nerve nuclei thus enabling demonstration of peri- neural spread if present.
This will also enable assessment of lymph node (i.e. levels I to VII) involvement by tumor at the same sitting for nodal staging. Pre and post I.V contrast scans should be acquired.
A radiologist's report should always be part of the examination.

Ultrasound

Ultrasound scan should be used as complementary to the CT scan for example:

- To characterize a neck mass as solid (lymph node) or cystic (cyst). But one should be cautious as a malignant necrotic lymph node may appear cystic.
- To guide FNA or Lymph node biopsy.

MRI (Magnetic Resonance Imaging)

This imaging modality plays a problem solving role i.e.:

- To assess for perineural tumour spread
- To plan for surgery where CT SCAN findings are equivocal

It has even more limited role in imaging of the laryngeal lesions due to image degradation by breathing and swallowing movements

Summary of Head & Neck Cancer Imaging

1. **Nasopharyngeal Cancer**
   This is a commonly symptom poor region in early disease and is not easily accessible to the clinician. The disease can first present with enlarged neck lymph nodes and reduced hearing. Imaging thus plays an important role in early detection of the disease. So a patient who presents to the primary level of health with suspicious enlarged neck lymph nodes should expeditiously be offered CT Scan Neck by the managing clinician.

2. **Oropharyngeal/Hypopharyngeal Cancer.**
   Commonly presents with pain or difficulty on swallowing. It is easily accessible to the clinician for assessment visually. The role of imaging is to assess for deep extent of the tumour and staging. Nodal involvement is common at time of presentation. CT scan is the imaging modality of choice.

3. **Oral Cavity Cancer**
   This includes tumours of the anterior 2/3rds of the tongue, gingiva and buccal mucosa.It is easily visualized by the clinician on oral clinical inspection. MRI is the imaging modality of choice for tongue tumours because CT scan has poor soft tissue contrast. CT SCAN with puffed mouth protocol is sufficient for gingival and buccal tumours assessment.

4. **Laryngeal Cancer**
   Early symptoms include change of voice e.g. hoarseness of voice. Tumour progression will cause air way obstruction which may need emergency tracheostomy. CT scan is the imaging modality of choice commonly done after laryngoscopy for tumour and nodal staging.

5. **Sinonasal Cancer**
   This is a tumour involving the nasal cavity and paranasal sinuses. Imaging modality of choice is CT scan. Images should be printed in axial and coronal series.

6. **Thyroid Cancer**
   May present with visible thyroid gland enlargement or hoarseness of voice. High resolution (linear Probe) can be used for the initial assessment. It is also useful for guiding tissue biopsy and lymph nodes involvement assessment.

7. **Tracheal Cancer.**
   This is rare compared to the other head and neck cancers. It may present with difficulties in breathing due to airway obstruction. CT scan is the imaging modality of choice

Clinical Indications for Pet CT Scan

CANCER	PET CT INDICATIONS
Breast	**FDG PET CT** • Assessment of multifocal disease or suspected recurrence where other imaging is negative • Characterization of indeterminate lesions on conventional imaging • Differentiation of treatment induced brachial plexopathy form tumor infiltration in symptomatic patients
Prostate	• Ga68-PSMA PET CT and F-Choline PET CT to assess for biochemical recurrence
Gynaecological	**FDG PET C** • Staging or restaging patients with uterine (cervical/endometrial) malignancies considered for radical chemo-radiotherapy or exenterative surgery • Suspected recurrence where conventional imaging is negative for ovarian cancer
Haematological	**FDG PET CT** **Lymphoma:** • Staging, • Radiotherapy planning • Remission assessment • Suspected relapse • Suspected post-transplant lymphoproliferative disorder **Myeloma:** • Staging and follow up to assess treatment response • Identify patients with smouldering myeloma with high risk of progression to symptomatic disease requiring treatment • Assessment of patients with apparently solitary plasmacytoma to exclude other sites of disease
GIT	**FDG PET CT** **Oesophago-gastric cancers** • Staging and assessment post neo-adjuvant therapy **Hepato-pancreatic tumors** • Staging of selected patients where conventional imaging is equivocal **Colorectal cancer** • Suspected recurrence, staging of patients with synchronous metastases at presentation
Head & Neck	**FDG PET CT**

Appendix 7 : Cancer Treatment Job Aid

Assess 	1) Evaluate the information the patient currently has about their diagnosis. Ask about: • What has the patient been told? • Who spoke with the patient? • Who was there with the patient? • What written/other information was offered to support the patient to learn more about: ▪ The specific Cancer, ▪ What to expect ▪ Treatment pathway (Chemotherapy, Radiotherapy, Surgery, etc.) ▪ Self-care ▪ What follow-up is required 2) Psychological status, financial implications and support structures • Assess the psychological and general financial implications to the patient and document the same in the patient file. • Assess the patient support system i.e. family? Friend(s)? 3) Specialist assessment and intervention summary • Explain the implications of the assessment and what it means for the patient 4) Treatment plan summary • Discuss the treatment plan with the patient • Address any concerns the patient and their family may have concerning the treatment *options* • Explain to the patient and their family the intent of the treatment Is the treatment curative? Why? Is the treatment palliative? Why? • *Any amendments must be made in consultation with the patient*
Advise 	1) Next steps – when treatment will be started • Discuss the treatment schedule with the patient and their family • Address any concerns the patient and their family may have concerning the treatment *schedule* 2) Summary of medications and alterations to medication • Discuss the summary of medication with the patient and their family • Outline what the patient should expect in the course of treatment • Highlight any potential side effects and how they will be managed

Agree 	1) Jointly set a treatment *start date* with the patient and their family 2) Agree on critical lifestyle modifications for the patients to enhance their quality of life e.g. cessation of tobacco and alcohol use, nutrition, rest, exercise, etc. 3) Agree on critical decisions in the treatment pathway namely / e.g. treatment, monitoring during survivorship, etc. This will help the patient to prepare for their recovery journey and enhance adherence to treatment 4) Agree on other support mechanisms to enhance quality of life e.g. linkages to palliative care, psychosocial support groups; Counseling; health talks and build these into the patient treatment plan and schedule *Note: The patient is the central focus of the multidisciplinary team and must be given full information in order to make informed choices about their treatment. Their agreement and participation in decisions about their treatment enhance adherence to plan and self-efficacy.*
Assist 	1) Assist the patient and their family to plan for the treatment journey by: • Providing them with a written summary of their treatment plan • Helping the patient and their family to schedule their next appointments • Giving clear and documented referrals to other entities that may / will support their treatment e.g. Laboratory, Nutrition, Counselling, Psychosocial support groups, insurance, etc. • Linking patients and their families to critical support entities ○ *NHIF:* For registration and/or further information about medical cover ○ *Foundations and Philanthropic entities* that may assist with various aspects of treatment ○ *Palliative care* 2) Skills and tools for self-management, adherence
Arrange 	1) At every visit, arrange and record next appointment date, including family (as appropriate) 2) Arrange and record care activities/support between visits 3) Provide the patient and family the contact details of the clinic for further information/discussion 4) Assure the patient that they can call at any time 5) Survivorship care planning

Appendix 8 : MDT Reporting Form Template

MINISTRY OF HEALTH

MDT REPORTING FORM TEMPLATE

HOSPITAL NAME: _____

Name		**I.P Number**	
Age		**Phone**	
Menopausal status		**SHA status**	
Marital status		**Family history of malignancy (specify type and individual a1ected.)**	
Comorbidities		**Residence**	

Clinical history summary:				**Anatomy Map:**
Diagnosis (Pathology):				
IHC				
Imaging				
Metastatic workup:				
Treatment given:				
MDT PLAN:				

SIGNATURE:

Surgery	**Oncology:**	**Radiology:**	**Pathology:**	**Nutrition:**	**Palliative:**

Summarized by:

Appendix 9 : Standard Operating Procedures on Use and Handling of Narcotic Medicines/ Opioids

MINISTRY OF HEALTH

STANDARD OPERATING PROCEDURES ON USE AND HANDLING OF NARCOTIC MEDICINES/OPIOIDS

Narcotic medicines/opioids are used to treat moderate or severe pain in adults and children.

Dispensing of Narcotic Medicines to outpatients

- The authorized person shall receive the prescription for the opioid medicine (it can be a new or a repeat prescription availed in duplicate).
- The prescription shall be checked for validity (should have prescriber's name and registration number. Check that the generic name, dosage, frequency and duration of therapy are written in words and numbers).
- If the prescription does not meet the requirements in bullet 2 above, the prescriber shall be contacted and patient sent back for a valid prescription. If the prescriber cannot be reached, the patient can be referred to another prescriber of the same or higher level of prescribing.
- The availability of the medicine shall be confirmed. If the medicine is available, bill, and send the patient to make payment (if applicable).
- Upon payment (if applicable), the authorized person shall retrieve the medicine and record in the narcotics register.
- Appropriate medicine use, counseling and education shall be done to the patient and/or caregiver.
- The Narcotic prescription shall be filed and kept for a minimum of 5 years

Dispensing Narcotic Medicines to inpatients and other dispensing points

- The person ordering for narcotics presents a patient treatment sheet/S-11 to the pharmacy
- The authorized person issues the narcotics based on their consumption and the available stocks. For individual narcotic medicines order, the patient shall be billed in the system depending on the available stocks (consider the facility billing system)
- The authorized person will then make the necessary debit in the pharmacy narcotic register and corresponding credit in the ward or clinic register and sign.
- The authorized person receiving the narcotic medicines shall confirm the quantities issued and entries made on the narcotic register and signs the S-11.

Method for handling Patients' Own Medications

- At the first encounter, the healthcare worker should ask the patient if they have brought in their own medicines.
- The pharmacist/assigned designate/healthcare worker at first encounter shall assess the suitability of POMs for use.
- If suitable and clinically appropriate, document the medicines and store them appropriately labeled as "Patient Own Medicine" (POM) on the medicine pack and on the treatment sheet.
- Inform the patient if the Patient Own Medicines are not physically and/or clinically suitable for use and advise them accordingly. If the medicines are expired, inform the patient/caregiver. Set them aside and label for disposal as per hospital policy.
- Seek verbal consent and assent to use Patient Own Medicines. POMs must be used for that patient only.

Storage of Narcotic Medicines

- All narcotic medicines should be locked in a designated cupboard
- Oral morphine solution should not be exposed to light
- The narcotics register should be up to date and should include – name of the medicine; strength; formulation; transaction entries for every receipt and issue including date, source of medicine receipt, destination of medicine issue; quantity in stock for each date; expiry date
- The register quantity should always tally with the physical stock
- Only authorized persons should access the storage area/cupboard for receipts or issues of narcotic medicines

Appendix 10: Systemic Antineoplastic Therapy Administration Safety Protocol

Adapted from ASCO-ONS Standards, 2024

Section	Number	Standard Description	Requirement / Nursing & Clinical Team Action
I. Environment & Training	**1.1**	Staff Qualifications	**Nurses and the clinical team need to** ensure documentation of initial education, competencies, and annual continuing education for all staff preparing/administering therapy.
	1.2	Comprehensive Education	The organization must utilize a comprehensive education program for initial and ongoing training for the **clinical team**.
	1.3	BLS Certification	At least one **nurse** with current (age-appropriate) basic life support certification must be present during administration.
	1.4	Practitioner Availability	A licensed practitioner must be on-site and immediately available to **nurses** during administration.
II. Pre-Treatment Documentation	**2.1**	Pathologic Confirmation	**Nurses need to** verify pathologic confirmation of the diagnosis in the medical record before the first administration.
	2.2	Cancer Staging	Current cancer stage or status must be documented in the record.
	2.3	History & Physical	**Nurses and the clinical team need to** ensure a complete H&P is documented, including fertility and pregnancy status.
	2.4	Fertility & Pregnancy Policy	Organization policy must address pregnancy testing, risk assessment during treatment, and fertility preservation counseling/referrals.
	2.5	Allergies & Hypersensitivity	Presence or absence of allergies and history of hypersensitivity reactions must be documented.
	2.6	Consent & Comprehension	**Nurses need to** assess and document patient/caregiver comprehension of the treatment plan and psychosocial needs, alongside obtaining informed consent.
	2.7	Therapy Plan Details	The plan must include diagnosis, drugs, doses, route, duration, and goals (palliative vs. curative).
	2.8	Monitoring Plan	Planned frequency of assessments and monitoring appropriate for the agent(s) must be documented.
	2.9	Social Determinants of Health	**Nurses need to** assess and document financial/logistical barriers to accessing required medications.

Section	Number	Standard Description	Requirement / Nursing & Clinical Team Action
III. Clinical Encounters	**3.1**	Daily Assessment	**Nurses need to** perform and document an assessment at every encounter, including functional status, vital signs, toxicities, and pain.
	3.2	Dosing Verification	**Nurses need to** verify height and weight in metric units (kg/cm) with a second **nurse** before initiating a new treatment plan.
	3.3	Psychosocial Screening	The **clinical team** must screen for and document psychosocial concerns with each cycle or as indicated.
	3.4	Medication Reconciliation	**Nurses need to** update and review all prescribed, OTC, herbal, and supplement medications at every encounter.
	3.5	Missed Visits	The organization must have a policy for follow-up on missed or canceled appointments.
	3.6	Triage Access	**Nurses need to** ensure patients have 24/7 access to a licensed practitioner for emergencies.
IV. Informed Consent & Education	**4.1**	Standardized Process	The organization must have a policy documenting a standardized process for obtaining informed consent/assent.
	4.2	Patient Education	**Nurses need to** provide verbal and written education before the first administration, documented in the patient's preferred language.
	4.3	Education Content	Education must cover: diagnosis, goals, duration, schedule, missed dose plan, drug interactions, adverse effects, fertility, and pregnancy prevention.
	4.4	Home Safety	Education must include safe handling of medications, waste disposal, bodily fluids, and spill cleanup at home.
	4.5	Contact Information	**Nurses need to** provide clear contact information for who to call and when.
V. Ordering & Preparation	**5.1**	Standardized Regimens	Organizations must define standard regimens by diagnosis with references.
	5.2	Order Signature	Orders must be signed manually or electronically by qualified licensed practitioners.
	5.3	Exception Orders	A policy for variations from standard regimens must require a reference and/or second licensed practitioner authorization.
	5.4	Verbal Orders	Verbal orders are not allowed except to hold or stop administration.
	5.5	Order Elements	**Nurses need to** verify orders contain patient name, DOB, identifier, date, prescriber, protocol, cycle/day, generic drug names, doses, route, rate, and supportive care.

Section	Number	Standard Description	Requirement / Nursing & Clinical Team Action
	5.6	Dose Calculation	Order must include calculation methodology, variables, and re-evaluation frequency.
	5.7	Oral Order Specifics	Oral orders must include drug quantity, refills (default zero), and administration schedule.
	5.8	Preparation Competency	Medications must be prepared by licensed personnel (**nurses** or pharmacists) with documented training and annual competency.
	5.9	Labeling	**Nurses need to** verify labels include patient name, DOB, prescriber, date, drug name, dose, route, and "HAZARDOUS DRUG" label if applicable.
VI. Dispensing & Admin.	**6.1**	Pharmacist Verification	A licensed pharmacist must verify all parenteral orders before dispensing.
	6.2	Four-Step Verification	**Nurses need to** perform four distinct verifications (pre-prep, post-prep, pre-admin, and patient bedside) documenting in the medical record.
	6.3	Bedside Verification	Two **nurses** must verify patient identification (two identifiers), drug name, dose, rate, and route at the bedside.
	6.4	Patient Confirmation	**Nurses need to** confirm the drug name, infusion time, route, and symptoms to report with the patient before initiation.
VII. Special Handling	**7.1**	Intrathecal Prep	Intrathecal agents must be prepared separately, labeled uniquely, and stored in an isolated location.
	7.2	Intrathecal Admin	Administered only after a "time out" and double-check procedure by two **nurses** or licensed practitioners.
	7.3	Vinca Alkaloids	IV Vinca Alkaloids must be administered only via infusion (mini-bag).
VIII. Monitoring & Adherence	**8.1**	Treatment Response	Standard, disease-specific processes must be used to monitor response based on guidelines.
	8.2	Emergency Management	Policies must address the availability of rescue agents and procedures for escalation of care.
	8.3	Laboratory Monitoring	Policies must determine appropriate time intervals for regimen-specific laboratory and organ function tests.
	8.4	Toxicity Tracking	**Nurses need to** assess/document toxicity at each encounter, including dose modifications and cumulative tracking of toxic agents.
	8.5	Barriers to Adherence	Ongoing assessment of social determinants of health and financial constraints must be documented.

Appendix 11: List of Individuals, Organisations and Institutions that Participated in the Guidelines Review Process

Name	Institution / Organization
Dr. Patrick Amoth	MOH- Director General for Health
Dr. David Soti	MOH- Office of the Director General
Dr. Isaak Bashir	MOH- Director, Directorate of Family Health
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Dr. Joan-Paula Bor	Ag Head of NCCP/MOH
Prof. Daniel Ojuka	KNH / UON
Dr. Abeid Athman	KUTRRH
Dr. Abdulrahman Said	Mombasa County
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Dr. Beatrice Melly	MTRH
Mr. Charles Langat	NCI-K
Ms. Christine Momanyi	NCI-K
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Dr. Collins Masolo	NCCP/MOH
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Dr. Dulcie Wanda	Nakuru
Dr. Edwin Walong	Maseno University
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Dr. Eunice Gathitu	MOH – DHPT/MOH
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Dr. Fiona Adagi	JOOTRH
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Dr. Helen Musau	RFH Specialist Hospital
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Mr. Isaac Kamau	NCCP/MOH
Dr. Irene Nzamu	KNH
Dr. Job Mwale	KNH
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Name	Institution / Organization
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Prof. Daniel Ojuka	KNH / UON
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Mr. Joseph Nyatete	NCI-K
Mr. Joseph Otieno	SHA
Dr. Maina Simon	Machakos County
Ms. Mary Wanjiru	CGTRH
Dr. Marjorie Watetu	Nakuru County Teaching & Referral Hospital
Dr. Mercy Gatua	KUTRRH
Dr. Michael Mwachiro	Chair- Surgical Society Of Kenya.
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Dr. Omar Abdihamid	Garissa Regional Cancer Center
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Felice Amoo	NCCP

Appendix 12: List of Organizations and Reviewers

List of professional organizations/Institutions

- Kenya Society of Hematology and Oncology (KESHO),
- Kenva Medical Practitioners and Dentists Council (KMPDC),
- Kenya Cancer Association (KENCO),
- Gynecologic Oncology Society of Kenya (GOSK),
- Clinical and Radiation Oncologists Society of Kenya (CROSK-Kenya)
- Kenva Hospices and Palliative Care Association (KEHPCA)
- Surgical Society of Kenya (SSK)
- National Cancer Institute-Kenya (NCIK).
- World Health Organisation (WHO.
- Oncology Nurses Society Kenya (ONS-K)
- Oncology Pharmacists Society of Kenya

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